

CLASS 12 · NCERT ALIGNED · NEET + IIT JEE MAIN

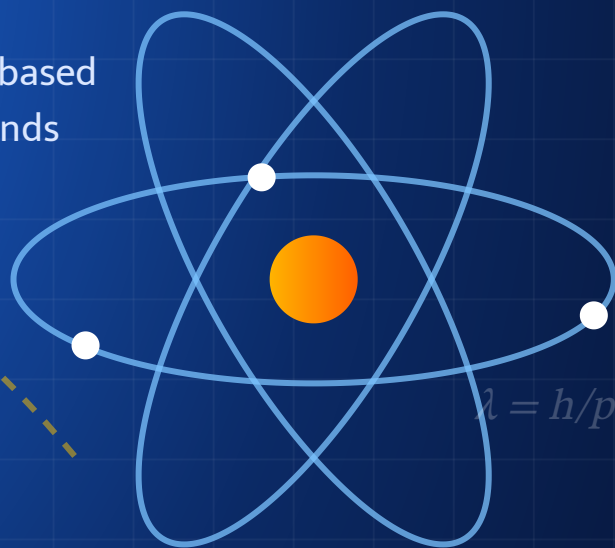
$$1/f = 1/v - 1/u$$

NEET + JEE PHYSICS

30-DAY RAPID REVISION

Concepts · Formulas · Diagrams · Traps · Pattern-based practice from NEET & JEE Main previous-year trends

CLASS 12



$$E = h\nu$$

14 NCERT Chapters

300+ MCQs with Solutions

Full Mock Test

Board + NEET + JEE Main

100+ Diagrams & Formula Boxes

$$\varepsilon = -$$



VS SIR

Vijay Kumar Sharma
20+ years · NEET/JEE Physics · Kota

NEET & JEE Main Physics: 30-Day Rapid Revision (Class 12)

Class 12 Physics · NCERT-aligned · For NEET & IIT JEE Main

by VS Sir (Vijay Kumar Sharma)

AI-Enabled Educator · 20+ years teaching NEET/JEE Physics, Kota

First edition: 2026 · Language: English

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About the questions: All practice questions in this book are original, pattern-based questions written by the author and modelled on the trends of NEET and JEE Main previous-year papers. They are not reproductions of official papers. Syllabus follows the rationalised NCERT Class 12 Physics textbook (Parts I & II), the NMC NEET-UG syllabus and the NTA JEE Main syllabus.

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ABOUT THE AUTHOR

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- 20+ years of teaching Physics to NEET and JEE aspirants in Kota
- AI-Enabled Educator – trains teachers to use AI tools in the classroom
- Author of "NEET Physics: 30-Day Rapid Revision (Class 11)" in English and Hindi, "Electrostatics" and "Practical Guide to AI for Educators"
- Known for turning long chapters into one-page concept maps and formula boxes

A message from VS Sir

Dear student,

In the last month before NEET or JEE Main, you do not need another 800-page book. You need a **clear, short, reliable revision** of everything NCERT says, the formulas that actually get used, and the traps where toppers lose marks.

This book is built exactly for that. One chapter per day or two, a formula box you can revise in five minutes, diagrams that make the idea obvious, and practice questions in the pattern NEET and JEE Main have followed year after year – each with a quick solution.

Follow the 30-day plan honestly, tick each box, and keep an error log. Consistency beats intensity. You can do this!

— VS Sir

How to use this book

Every chapter follows the same rhythm so your brain knows exactly where to look.

1. **Read** the chapter-at-a-glance map (2 min).
2. **Revise** concepts, diagrams and formula boxes (40–60 min).
3. **Solve** the 20 practice questions with a timer – 1 min each.
4. **Check** with the answer key; read the solution of every wrong/guessed question.
5. **Log** each mistake in your error log (page at the end). Re-solve on Day 29.

Colour code of the boxes

 Key Concept Core idea explained simply	 Formula Box Formulas to revise in 5 minutes
 Common Trap Where students lose marks	 VS Sir's Exam Tip Exam strategy & shortcuts
 NCERT Line to Remember Exact NCERT facts asked in NEET	 Memory Trick Tricks to remember
 JEE Main Extension Extra depth for JEE Main	

Question tags

NCERT based direct from textbook lines **NEET pattern** typical NEET level **JEE Main pattern** one step deeper
Assertion–Reason

 Honest note on PYQs The questions are original and written in the style of NEET and JEE Main previous-year questions (PYQ trends): same concepts, same difficulty, same traps. Solve your official PYQ papers too – this book prepares you to crack them.

Syllabus & references

This book covers the complete Class 12 Physics syllabus common to NEET-UG and JEE Main. Topics asked only in JEE Main are marked with a JEE box. Standard books were used only to cross-check results – all explanations, diagrams and questions in this book are original.

NCERT Physics Class XII, Parts I & II (rationalised edition)

Primary reference for concepts, definitions and standard values.

NEET-UG Physics syllabus – National Medical Commission (NMC); exam conducted by NTA · neet.nta.nic.in

Topic coverage and depth for NEET.

JEE Main Physics syllabus – National Testing Agency (NTA) · jeemain.nta.nic.in

Topic coverage and depth for JEE Main; JEE-only extras are marked.

H. C. Verma – Concepts of Physics, Vol. 2

Cross-check of derivations, sign conventions and standard results.

D. C. Pandey – Understanding Physics (Electricity & Magnetism; Optics & Modern Physics)

Cross-check of problem types asked in NEET and JEE Main.

Always check the latest official syllabus

NTA and NMC publish the syllabus and information bulletin for every exam year. Before your exam, confirm the current syllabus on the official websites.

Your 30-Day Revision Plan

Daily target: 2–3 hours for Physics. Tick the box when a day is complete. Weekly tests are built into Days 8 and 17.

Day 1 Ch 1 Charges, Coulomb, electric field	Day 2 Ch 1 Dipole, flux, Gauss's law	Day 3 Ch 2 Potential, equipotentials, energy	Day 4 Ch 2 Capacitors, dielectrics	Day 5 Ch 3 Drift, Ohm's law, resistivity
Day 6 Ch 3 Cells, Kirchhoff, Wheatstone	Day 7 Ch 4 Biot–Savart, Ampere's law	Day 8 Ch 4 Forces, galvanometer + Test (Ch 1–4)	Day 9 Ch 5 Magnetism and matter	Day 10 Ch 6 Electromagnetic induction
Day 11 Ch 7 AC: phasors, R, L, C circuits	Day 12 Ch 7 LCR, resonance, power, transformer	Day 13 Ch 8 Electromagnetic waves	Day 14 Ch 9 Reflection, refraction, TIR	Day 15 Ch 9 Lenses, prism, instruments
Day 16 Ch 10 Huygens, interference, YDSE	Day 17 Ch 10 Diffraction, polarisation + Test (Ch 5–10)	Day 18 Ch 11 Photoelectric effect, de Broglie	Day 19 Ch 12 Atoms: Bohr model, spectra	Day 20 Ch 13 Nuclei: binding energy, decay
Day 21 Ch 14 Semiconductors, p–n junction	Day 22 Ch 14 Diode circuits, rectifiers, logic gates	Day 23 Final Formula sheet: Electricity & Magnetism	Day 24 Final Formula sheet: Optics & Modern Physics	Day 25 Final Experimental skills (NEET/JEE)
Day 26 Final Traps list + important graphs	Day 27 Final FULL MOCK TEST (45 Q, 60 min)	Day 28 Final Mock analysis + error log	Day 29 Final Re-attempt all wrong questions	Day 30 Final Exam-day plan + final revision

Golden rules

- Never skip the practice set – revision without questions is only reading.
- Sleep 7 hours. Memory is stored while you sleep.

If you fall behind

Do the formula box + 10 questions of the missed day, then continue with today. Use Days 28–29 as buffer.

Contents

1	Electric Charges and Fields Charge & quantisation · Coulomb's law · Electric field · Electric dipole	Day 1–2	8
2	Electrostatic Potential and Capacitance Potential · Equipotentials · Potential energy · Conductors & dielectrics	Day 3–4	18
3	Current Electricity Drift velocity · Ohm's law · Resistivity & temperature · Power	Day 5–6	28
4	Moving Charges and Magnetism Lorentz force · Circular & helical motion · Biot–Savart law · Ampère's law	Day 7–8	39
5	Magnetism and Matter Bar magnet · Dipole in field · Gauss's law · M , H , χ , μ	Day 9	51
6	Electromagnetic Induction Magnetic flux · Faraday's law · Lenz's law · Motional EMF	Day 10	61
7	Alternating Current Peak & rms values · Phasors · R , L , C circuits · Series LCR & resonance	Day 11–12	69
8	Electromagnetic Waves Displacement current · Maxwell's idea · Nature of EM waves · $c = 1/\sqrt{(\mu_0\epsilon_0)}$	Day 13	79
9	Ray Optics and Optical Instruments Spherical mirrors · Refraction & TIR · Thin lenses · Lens maker's formula	Day 14–15	87
10	Wave Optics Huygens principle · Coherent sources · YDSE · Single-slit diffraction	Day 16–17	98
11	Dual Nature of Radiation and Matter Electron emission · Photoelectric effect · Stopping potential · Einstein's equation	Day 18	107
12	Atoms α -scattering · Nuclear model · Closest approach · Hydrogen spectrum	Day 19	116
13	Nuclei Size & density · Isotopes, isobars · Mass defect · B.E. per nucleon	Day 20	124
14	Semiconductor Electronics Energy bands · n - and p -type · p - n junction · Diode & rectifiers	Day 21–22	133
★	Last-Week Toolkit: Formulas, Experiments, Traps & Strategy 14 formula sheets · 9 experiments · 50 traps · 14 key graphs · exam-day plan · error log	Day 23–26	144
✍	Full Mock Test 45 questions · 60 minutes · NEET marking · Class 12 full syllabus	Day 27–29	162

CHAPTER

1

Electric Charges and Fields

17

Day 1-2

Weightage ★★★

NCERT Class 12

Charge & quantisation

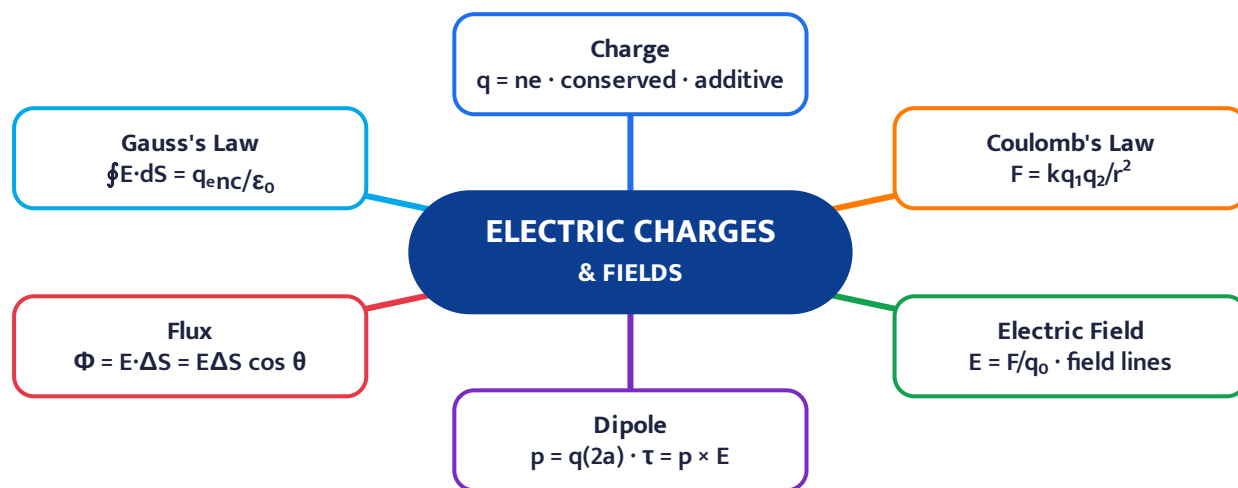
Coulomb's law

Electric field

Electric dipole

Electric flux

Gauss's law



Chapter at a glance

1.1 Electric Charge and its Properties

Electric charge is an intrinsic property of matter due to which it experiences electric and magnetic forces. There are two kinds – **positive** and **negative**; like charges repel, unlike charges attract. Charging by rubbing happens by **transfer of electrons**: the body that loses electrons becomes positive. SI unit: **coulomb (C)**.

3 basic properties

- **Additivity** – total charge = algebraic sum:
 $q = q_1 + q_2 + \dots$ (charge is a scalar).
- **Conservation** – total charge of an isolated system never changes; charge is neither created nor destroyed.
- **Quantisation** – $q = \pm ne$, $n = 0, 1, 2, \dots$;
 $e = 1.6 \times 10^{-19} \text{ C}$.

NCERT Line to Remember

- At the macroscopic level (charges of μC) quantisation can be ignored and charge treated as **continuous**.
- $1 \text{ C} \approx 6.25 \times 10^{18}$ electrons. Mass is transferred too, but negligibly.
- A charge is always attached to mass; charge is invariant with speed.

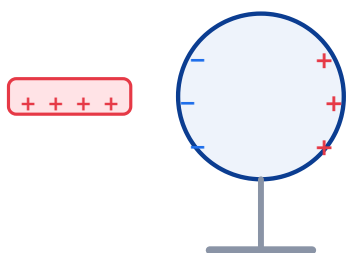
Charging by rubbing (NCERT pairs)

Glass rod rubbed with silk → glass +, silk –. Plastic/ebonite rod rubbed with wool or fur → rod –, wool +. Both bodies always get **equal and opposite** charges.

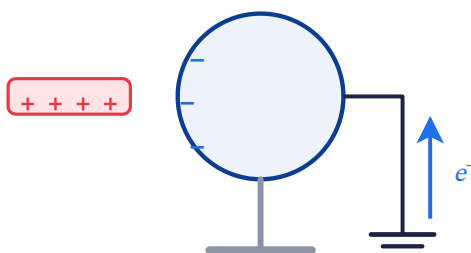
1.2 Conductors, Insulators and Charging by Induction

Conductors (metals, human body, earth) have free electrons and let charge move; charge given to a conductor spreads over its **outer surface**. **Insulators** (glass, plastic, nylon) hold charge where it is placed. **Earthing** (grounding) shares charge with the huge earth, so the body is neutralised. In **induction**, a charged body rearranges charges in a nearby conductor **without touching** it – the conductor finally gets a charge **opposite** to the inducing charge.

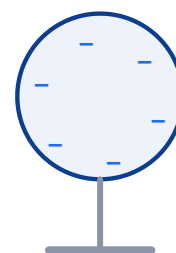
1. Bring a +ve rod near



2. Earth the far side



3. Remove earth, then rod



sphere gets opposite (-) charge

Charging a metal sphere by induction – the rod never touches the sphere

! Common Trap

- A charged body attracts a **neutral** body too (induced opposite charge sits nearer). So **attraction does not prove** the other body is charged – **repulsion is the sure test**.
- Remove the earth connection **before** removing the rod; otherwise the induced charge flows back and the sphere stays neutral.

1.3 Coulomb's Law and Superposition

The force between two stationary point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them, acting **along the line joining them**.

! Coulomb's law

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2} = k \frac{|q_1 q_2|}{r^2}$$

$$k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}, \epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

$$\vec{F}_{21} = k \frac{q_1 q_2}{r_{21}^2} \hat{r}_{21}, \quad \vec{F}_{12} = -\vec{F}_{21}$$

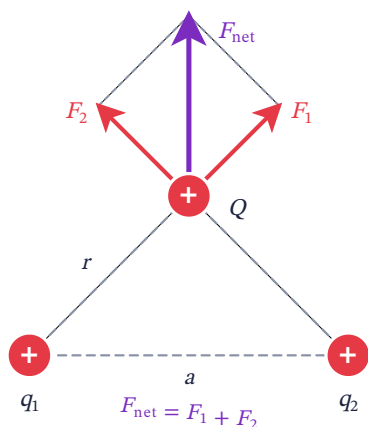
Vector form: force on 2 due to 1; $\hat{r}_{21}$ points from 1 to 2 (obeys Newton's third law)

$$F_{\text{medium}} = \frac{F_{\text{vacuum}}}{K}, \quad K = \frac{\epsilon}{\epsilon_0} = \epsilon_r$$

K = dielectric constant (relative permittivity) of the medium

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \dots + \vec{F}_{1n}$$

Superposition: each pair acts independently; forces add as vectors



Superposition of Coulomb forces

Solved Example

Charges $q_1 = q_2 = +1 \mu\text{C}$ sit at A and B, 0.2 m apart. A charge $Q = +1 \mu\text{C}$ is at a point 0.1 m directly above the midpoint of AB. Find the net force on Q.

Solution: $r = \sqrt{0.1^2 + 0.1^2} = 0.1\sqrt{2} \text{ m}$,
 $F_1 = F_2 = \frac{9 \times 10^9 \times 10^{-12}}{0.02} = 0.45 \text{ N}$ at 45° to the vertical.
 Horizontal parts cancel: $F_{\text{net}} = 2(0.45)\cos 45^\circ \approx \mathbf{0.64 \text{ N}}$,
vertically up.

Common Trap

- Coulomb's law is for **point** charges at rest. Force between two charges is **unchanged** by a third charge nearby (superposition) – only the net force changes.
- Touching two identical conducting spheres makes each charge $\frac{q_1 + q_2}{2}$ (with signs!).

Memory Trick

Electric vs gravitational force between two protons:
 $F_e / F_g \approx 10^{36}$. Electric force can attract **or** repel;
 gravity only attracts. Both obey the inverse-square law and are central forces.

1.4 Electric Field and Field Lines

The **electric field** at a point is the force per unit positive test charge placed there: $\vec{E} = \lim_{q_0 \rightarrow 0} \vec{F} / q_0$ (the test charge must be tiny so it does not disturb the source). Unit: $\text{N C}^{-1} = \text{V m}^{-1}$. A charge q in a field feels $\vec{F} = q\vec{E}$ – a positive charge along $\vec{E}$, a negative charge (electron) **opposite** to $\vec{E}$.

Electric field

$$\vec{E} = k \frac{q}{r^2} \hat{r}$$

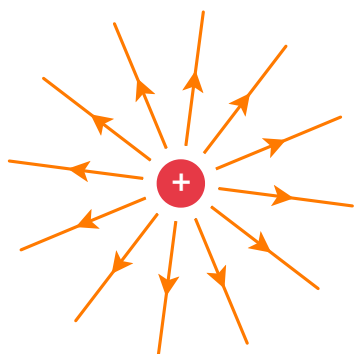
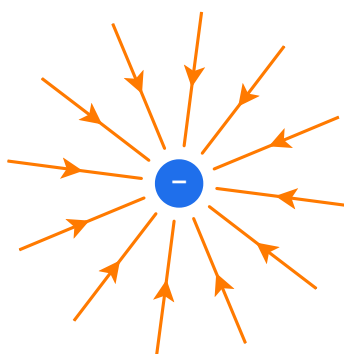
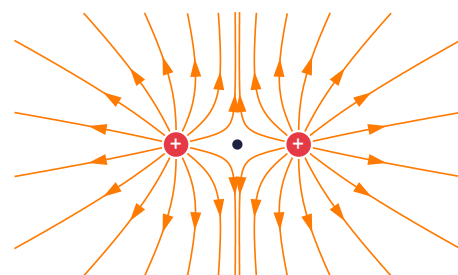
Point charge: radially outward for $+q$, inward for $-q$; $E \propto 1/r^2$

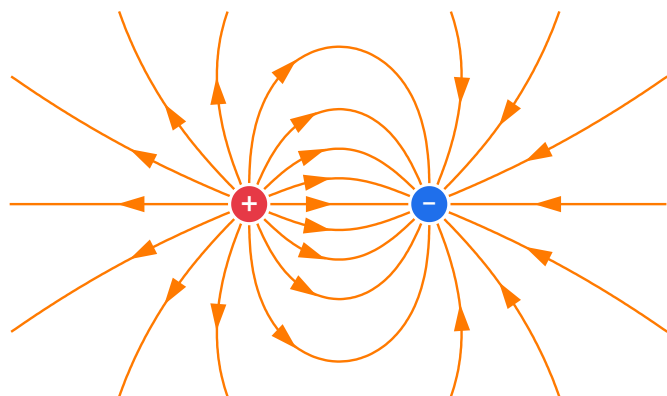
$$\vec{E} = \sum_i k \frac{q_i}{r_i^2} \hat{r}_i$$

System of charges – superposition

$$a = \frac{qE}{m}$$

Acceleration of a charged particle in a uniform field (path parabolic if it enters $\perp$ to E)

Isolated $+q$: lines go outIsolated $-q$: lines come inTwo equal $+q$: neutral point (dot) midway



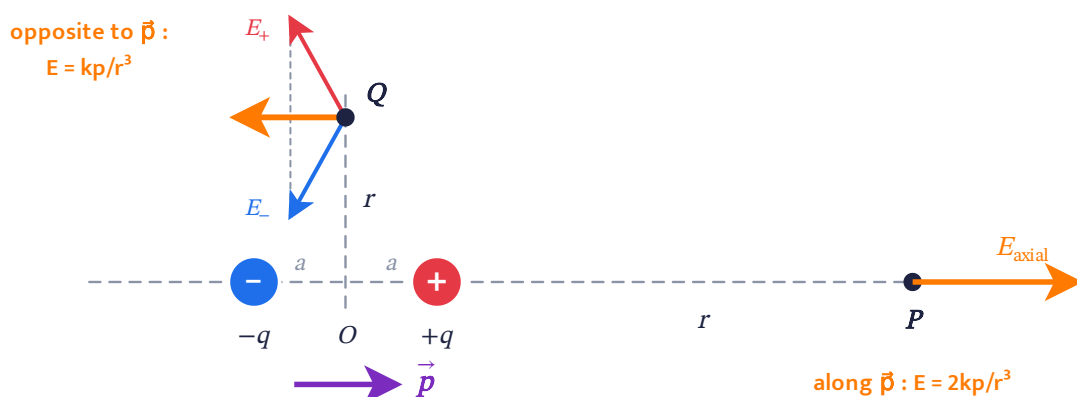
Electric dipole (+q, -q): lines start on +q and end on -q

Properties of field lines

- Start on +ve charge, end on -ve charge (or at infinity).
- Continuous curves in a charge-free region; **never form closed loops** (field is conservative).
- **Two lines never cross** – E has one direction at a point.
- Tangent gives direction of E; **crowding** shows strength.

1.5 Electric Dipole

Two equal and opposite charges $\pm q$ separated by a small distance $2a$ form an **electric dipole**. Its **dipole moment** $\vec{p} = q(2\vec{a})$ points **from -q to +q** (unit C m). Total charge is zero, but the field is not – it falls as $1/r^3$, faster than a point charge. Polar molecules such as H_2O have a permanent dipole moment.



Field of a dipole at an axial point P and an equatorial point Q

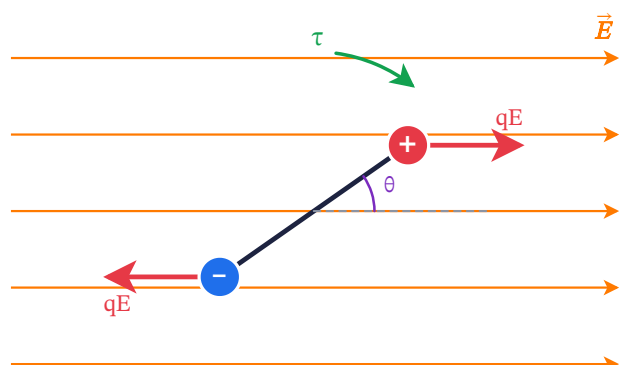
Dipole Formulas

$$E_{\text{axial}} = \frac{2kpr}{(r^2 - a^2)^2} \approx \frac{2kp}{r^3} \quad \text{On the axis, along } \vec{p} \text{ (for } r \gg a)$$

$$E_{\text{eq}} = \frac{kp}{(r^2 + a^2)^{3/2}} \approx \frac{kp}{r^3} \quad \text{On the equatorial plane, opposite to } \vec{p}$$

$$E_{\text{axial}} : E_{\text{eq}} = 2:1 \quad \text{At the same large distance}$$

$$\vec{\tau} = \vec{p} \times \vec{E}, \quad \tau = pE \sin \theta \quad \text{Torque in a uniform field; net force = 0}$$



Net force = 0; $\tau = pE \sin \theta$ turns $\vec{p}$ towards $\vec{E}$

Dipole in a uniform field – a couple acts

Solved Example

A dipole of charges $\pm 10 \mu\text{C}$ separated by 5 mm. Find E at 15 cm from its centre on (i) the axis, (ii) the equatorial line.

Solution: $p = 10^{-5} \times 5 \times 10^{-3} = 5 \times 10^{-8} \text{ C m}$. (i)

$$E = \frac{2kp}{r^3} = \frac{2(9 \times 10^9)(5 \times 10^{-8})}{(0.15)^3} \approx 2.67 \times 10^5 \text{ N/C along } \vec{p}.$$

Half: $1.33 \times 10^5 \text{ N/C}$, opposite to $\vec{p}$.

VS Sir's Exam Tip

- Uniform field $\Rightarrow \mathbf{F} = \mathbf{0}$, $\tau \neq \mathbf{0}$ (only rotation). Non-uniform field $\Rightarrow$ a net force also acts; when $\vec{p} \parallel \vec{E}$ the dipole moves towards the **stronger** field.
- τ is maximum ($= pE$) at $\theta = 90^\circ$; zero at $\theta = 0^\circ$ (**stable**) and 180° (**unstable** equilibrium).

1.6 Continuous Charge Distributions

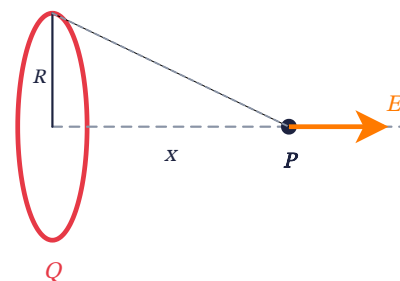
Distribution	Density	Unit	Element
Line (wire, ring)	$\lambda = \frac{dq}{dl}$	C m^{-1}	$dq = \lambda dl$
Surface (sheet, shell)	$\sigma = \frac{dq}{dS}$	C m^{-2}	$dq = \sigma dS$
Volume (solid sphere)	$\rho = \frac{dq}{dV}$	C m^{-3}	$dq = \rho dV$

Field of a continuous body: split it into elements dq , treat each as a point charge and add (integrate) vectorially:

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{r}$$

JEE Main Extension

- Ring (charge Q , radius R) on its axis: $E = \frac{kQx}{(R^2 + x^2)^{3/2}}$; $E = 0$ at centre, **maximum at** $x = R / \sqrt{2}$; for $x \gg R$ it acts like a point charge.

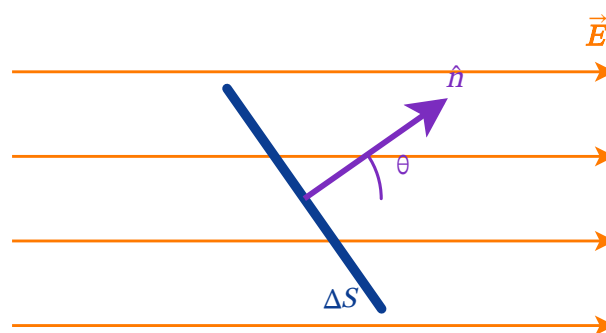


Uniformly charged ring – field on the axis

1.7 Electric Flux

Electric flux through a surface measures the number of field lines crossing it. For a small area ΔS with outward normal $\hat{n}$: $\Delta\Phi = \vec{E} \cdot \vec{\Delta S} = E \Delta S \cos\theta$. It is a **scalar**; unit $\text{N m}^2 \text{C}^{-1}$ ($= \text{V m}$).

- $\theta = 0^\circ$: $\Phi = E\Delta S$ (maximum); $\theta = 90^\circ$: $\Phi = 0$ (field grazes the surface).
- Closed surface: outward flux +ve, inward flux -ve.
- Uniform $\vec{E}$ through any closed surface: net flux = 0.



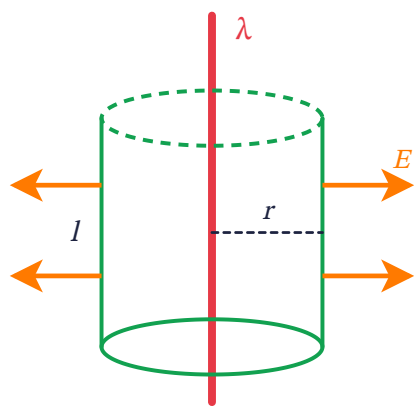
$$\Phi = E \Delta S \cos \theta \quad (\theta: \text{between } \vec{E} \text{ and normal})$$

Flux through a tilted area

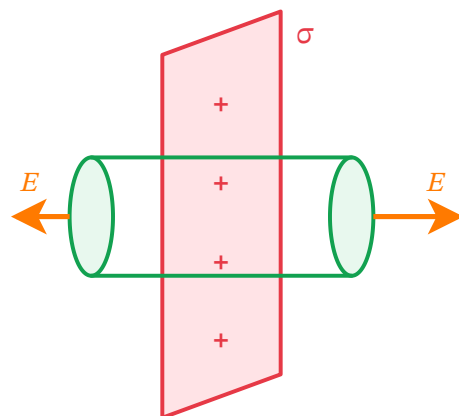
1.8 Gauss's Law and its Applications

The total flux through **any closed surface** equals $1 / \epsilon_0$ times the net charge enclosed. Choose a **Gaussian surface** that matches the symmetry so that E is constant on it and $\vec{E} \parallel d\vec{S}$ (or $\perp$) – then $E \times (\text{area}) = q_{\text{enc}} / \epsilon_0$.

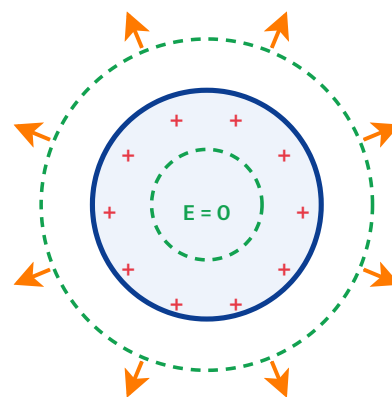
$$\oint \vec{E} \cdot d\vec{S} = \frac{q_{enc}}{\epsilon_0}$$



Line: cylinder



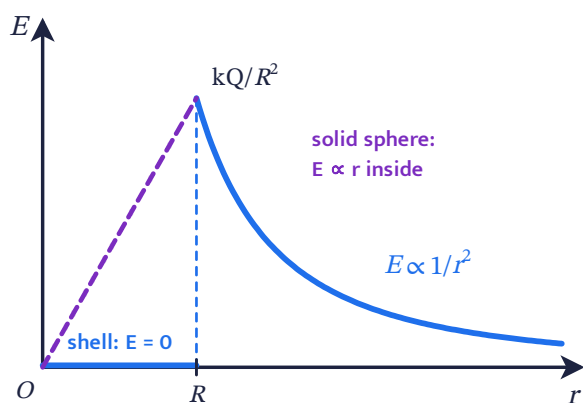
Sheet: pill-box



Shell: concentric spheres

Gaussian surfaces for the three standard cases

Charge system	Gaussian surface	Result	Dependence
Infinite line charge λ	Coaxial cylinder (r, l)	$E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2k\lambda}{r}$	$E \propto 1/r$
Infinite plane sheet σ	Pill-box through sheet	$E = \frac{\sigma}{2\epsilon_0}$	independent of r
Two sheets $+\sigma, -\sigma$	–	$\frac{\sigma}{\epsilon_0}$ between, 0 outside	uniform
Thin shell Q, radius R: $r > R$	Concentric sphere	$E = \frac{kQ}{r^2}$	like point charge
Thin shell: $r = R$	–	$E = \frac{kQ}{R^2} = \frac{\sigma}{\epsilon_0}$	maximum
Thin shell: $r < R$	Sphere inside	$E = 0$	no charge enclosed



E versus r for a charged shell (solid) and solid sphere (dashed)

Solved Example

An infinite line charge produces a field of 9×10^4 N/C at 2 cm. Find λ .

Solution: $E = \frac{2k\lambda}{r} \Rightarrow \lambda = \frac{Er}{2k} = \frac{9 \times 10^4 \times 0.02}{2 \times 9 \times 10^9} = 10^{-7}$ C/m (0.1 μ C/m).

Solved Example

A charge q sits at the centre of a cube. Find the flux through one face.

Solution: Total flux = q / ϵ_0 ; by symmetry the six faces share it equally: $q / 6\epsilon_0$.

JEE Main Extension

Uniformly charged **solid** (non-conducting) sphere, charge Q, radius R: inside $E = \frac{kQr}{R^3} = \frac{\rho r}{3\epsilon_0}$ ($\propto r$), outside $E = kQ / r^2$. Just outside any charged **conductor**: $E = \sigma / \epsilon_0$, normal to the surface.

 Common Trap

- $\vec{E}$ in Gauss's law is the field due to **all** charges (inside + outside), but the **flux** depends only on charges **inside**.
- Flux does **not** depend on the shape/size of the closed surface or the position of the charge inside it.
- Zero flux does not mean zero E on the surface.

 Memory Trick

Symmetric-charge trick: charge at the **centre** of a cube $\rightarrow q / 6\epsilon_0$ per face; at a **corner** $\rightarrow q / 8\epsilon_0$ through the cube; at the centre of a **face** $\rightarrow q / 2\epsilon_0$ through the cube.

 VS Sir's Exam Tip

Line $\rightarrow 1 / r$, sheet $\rightarrow r^0$, point/shell outside $\rightarrow 1 / r^2$, dipole $\rightarrow 1 / r^3$. Most NEET graph questions are just this ladder.

 Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1** **NCERT based** The number of electrons that make up a charge of -1 C is
 (a) 1.6×10^{-19} (b) 1.6×10^{19} (c) 6.25×10^{18} (d) 6.25×10^{19}
- 2** **NCERT based** Which of the following cannot be the charge on a body?
 (a) $1.6 \times 10^{-18}\text{ C}$ (b) $2.4 \times 10^{-19}\text{ C}$ (c) $3.2 \times 10^{-19}\text{ C}$ (d) $4.8 \times 10^{-19}\text{ C}$
- 3** **NEET pattern** Two point charges repel with force F . If each charge is doubled and the distance between them is halved, the force becomes
 (a) $4F$ (b) $8F$ (c) $16F$ (d) F
- 4** **NEET pattern** Two identical conducting spheres carrying $+8\text{ }\mu\text{C}$ and $-2\text{ }\mu\text{C}$ attract with force F at separation r . They are touched and put back at the same separation. The new force is
 (a) $9F/8$, repulsive (b) $F/16$, repulsive
 (c) $9F/16$, repulsive (d) $9F/16$, attractive
- 5** **NEET pattern** The force between two charges in air is 20 N . Keeping the distance same, the space between them is filled with a medium of dielectric constant 4 . The force becomes
 (a) 80 N (b) 10 N (c) 5 N (d) 20 N
- 6** **NEET pattern** Charges $+4q$ and $+q$ are fixed a distance d apart. The net electric field is zero at a point on the line joining them at a distance
 (a) $d/2$ from either (b) $d/3$ from $+4q$
 (c) $2d/3$ from $+4q$ (d) $d/5$ from $+q$
- 7** **NCERT based** Which statement about electrostatic field lines is FALSE?
 (a) They form closed loops
 (b) Crowded lines mean a stronger field
 (c) They start on positive charges
 (d) Two lines never intersect

- 8 **NCERT based** Charges $\pm 2 \mu\text{C}$ are 3 cm apart. The electric dipole moment is
 (a) $6 \times 10^{-8} \text{ C m}$, from $+q$ to $-q$ (b) $6 \times 10^{-8} \text{ C m}$, from $-q$ to $+q$
 (c) $1.2 \times 10^{-7} \text{ C m}$, from $-q$ to $+q$ (d) $6 \times 10^{-6} \text{ C m}$, from $-q$ to $+q$
- 9 **NEET pattern** For a short dipole, the ratio of the electric field at an axial point to that at an equatorial point at the same distance is
 (a) 2 : 1 (b) 1 : 1 (c) 1 : 2 (d) 4 : 1
- 10 **NEET pattern** A dipole of moment $4 \times 10^{-9} \text{ C m}$ is placed at 30° to a uniform field of $5 \times 10^4 \text{ N/C}$. The torque on it is
 (a) $1 \times 10^{-4} \text{ N m}$ (b) $1.73 \times 10^{-4} \text{ N m}$ (c) $2 \times 10^{-4} \text{ N m}$ (d) $0.5 \times 10^{-4} \text{ N m}$
- 11 **NCERT based** An electric dipole is placed in a uniform electric field at an angle θ ($0 < \theta < 180^\circ$). It experiences
 (a) zero net force and a torque (b) a net force and no torque
 (c) neither force nor torque (d) a net force and a torque
- 12 **JEE Main pattern** A dipole is placed in a NON-uniform field with $\vec{p}$ parallel to $\vec{E}$. It experiences
 (a) a net force towards the weaker field and a torque
 (b) only a torque
 (c) no force and no torque
 (d) a net force towards the stronger field, no torque
- 13 **NCERT based** A point charge q is placed at the centre of a cube. The electric flux through one face of the cube is
 (a) $q / 8\epsilon_0$ (b) $6q / \epsilon_0$ (c) q / ϵ_0 (d) $q / 6\epsilon_0$
- 14 **NEET pattern** A charge q is placed at one corner of a cube. The flux through the whole cube is
 (a) $q / 8\epsilon_0$ (b) $q / 24\epsilon_0$ (c) q / ϵ_0 (d) $q / 6\epsilon_0$
- 15 **NEET pattern** A uniform field $\vec{E} = (3\hat{i} + 4\hat{j}) \text{ N/C}$ passes through a square of area 10 m^2 lying in the y - z plane. The flux through it is
 (a) $40 \text{ N m}^2/\text{C}$ (b) $50 \text{ N m}^2/\text{C}$ (c) $70 \text{ N m}^2/\text{C}$ (d) $30 \text{ N m}^2/\text{C}$
- 16 **NEET pattern** An infinitely long line charge of density $2 \mu\text{C/m}$. The electric field at 10 cm from it is
 (a) $3.6 \times 10^6 \text{ N/C}$ (b) $3.6 \times 10^5 \text{ N/C}$ (c) $1.8 \times 10^5 \text{ N/C}$ (d) $7.2 \times 10^5 \text{ N/C}$
- 17 **JEE Main pattern** Two large parallel plates carry charge densities $+\sigma$ and $-\sigma$. The field in the region between them and outside them is respectively
 (a) σ/ϵ_0 and 0 (b) $\sigma/2\epsilon_0$ and $\sigma/2\epsilon_0$
 (c) $2\sigma/\epsilon_0$ and 0 (d) 0 and σ/ϵ_0

- 18 **JEE Main pattern** For a uniformly charged thin spherical shell of radius R , the graph of E versus distance r from the centre is
 (a) rises linearly up to R , then falls as $1/r^2$
 (b) constant up to R , then falls as $1/r$
 (c) falls as $1/r^2$ everywhere
 (d) zero up to R , jumps to maximum at R , then falls as $1/r^2$
- 19 **JEE Main pattern** Three equal charges q are at the vertices of an equilateral triangle of side a . The magnitude of the force on any one charge is
 (a) $2kq^2 / a^2$ (b) kq^2 / a^2 (c) $\sqrt{3} kq^2 / a^2$ (d) $\sqrt{2} kq^2 / a^2$
- 20 **JEE Main pattern** On the axis of a uniformly charged ring of radius R , the electric field is maximum at a distance from the centre of
 (a) $R\sqrt{2}$ (b) 0 (c) $R / 2$ (d) $R / \sqrt{2}$
- 21 **NEET pattern** An electron ($m = 9.1 \times 10^{-31}$ kg) is placed in a uniform field of 2×10^4 N/C. The magnitude of its acceleration is about
 (a) 3.5×10^{15} m/s² (b) 1.8×10^{15} m/s² (c) 1.9×10^{12} m/s² (d) 3.5×10^{11} m/s²
- 22 **NEET pattern** An oil drop of mass 4.8×10^{-15} kg is held stationary in a vertical field of 10^5 N/C ($g = 10$ m/s²). The number of excess electrons on it is
 (a) 30 (b) 2 (c) 5 (d) 3
- 23 **Assertion-Reason** Assertion: Two electric field lines can never intersect. Reason: At any point the electric field has a unique direction.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 24 **Assertion-Reason** Assertion: The net electric flux through a closed surface depends on charges placed outside it. Reason: The electric field at a point on a Gaussian surface is due to all charges, inside and outside.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

✓ Answer Key & Quick Solutions

- 1(c) 2(b) 3(c) 4(c) 5(c) 6(c) 7(a) 8(b) 9(a) 10(a) 11(a) 12(d) 13(d) 14(a) 15(d) 16(b)
 17(a) 18(d) 19(c) 20(d) 21(a) 22(d) 23(a) 24(c)

1. (c) $n = q / e = 1 / (1.6 \times 10^{-19}) = 6.25 \times 10^{18}$.
 2. (b) Charge must be an integral multiple of e ; $2.4 \times 10^{-19} / 1.6 \times 10^{-19} = 1.5$ – not an integer.
 3. (c) $F \propto q_1 q_2 / r^2 \Rightarrow (2 \times 2) / (1/2)^2 = 16 \Rightarrow 16F$.
 4. (c) Each gets $(8 - 2)/2 = +3 \mu\text{C}$. $F' / F = (3 \times 3) / (8 \times 2) = 9 / 16$; like charges $\Rightarrow$ repulsive.

5. (c) $F_m = F / K = 20 / 4 = 5 \text{ N}$.
6. (c) $\frac{4q}{x^2} = \frac{q}{(d-x)^2} \Rightarrow 2(d-x) = x \Rightarrow x = 2d / 3$ from $+4q$.
7. (a) Electrostatic field is conservative, so its lines never form closed loops.
8. (b) $p = q(2a) = 2 \times 10^{-6} \times 0.03 = 6 \times 10^{-8} \text{ C m}$, directed from $-q$ to $+q$.
9. (a) $E_{ax} = 2kp / r^3, E_{eq} = kp / r^3 \Rightarrow 2 : 1$.
10. (a) $\tau = pE \sin 30^\circ = 4 \times 10^{-9} \times 5 \times 10^4 \times 0.5 = 10^{-4} \text{ N m}$.
11. (a) Forces qE and $-qE$ cancel but form a couple: $\tau = pE \sin \theta \neq 0$.
12. (d) $\vec{p} \times \vec{E} = 0 \Rightarrow$ no torque; the force on the end in the stronger field wins $\Rightarrow$ net force towards increasing E .
13. (d) Total q / ϵ_0 shared equally by 6 faces.
14. (a) 8 such cubes surround the corner; each gets $\frac{1}{8} \cdot \frac{q}{\epsilon_0}$.
15. (d) Area vector along x : $\vec{\Phi} = \vec{E} \cdot \vec{S} = 3 \times 10 = 30 \text{ N m}^2/\text{C}$.
16. (b) $E = 2k\lambda / r = 2(9 \times 10^9)(2 \times 10^{-6}) / 0.1 = 3.6 \times 10^5 \text{ N/C}$.
17. (a) Each sheet gives $\sigma/2\epsilon_0$; the fields add between the plates and cancel outside.
18. (d) Inside: $q_{enc} = 0 \Rightarrow E = 0$; at $r = R$: $E = kQ/R^2$ (max); outside: kQ/r^2 . (Linear rise is for a solid sphere.)
19. (c) Two forces of kq^2 / a^2 at 60° : $F = 2F_0 \cos 30^\circ = \sqrt{3} F_0$.
20. (d) $E \propto x / (R^2 + x^2)^{3/2}$; $dE / dx = 0 \Rightarrow R^2 = 2x^2 \Rightarrow x = R / \sqrt{2}$.
21. (a) $a = eE / m = (1.6 \times 10^{-19})(2 \times 10^4) / (9.1 \times 10^{-31}) \approx 3.5 \times 10^{15} \text{ m/s}^2$, opposite to E .
22. (d) $qE = mg \Rightarrow q = 4.8 \times 10^{-14} / 10^5 = 4.8 \times 10^{-19} \text{ C} = 3e$.
23. (a) At an intersection there would be two tangents, i.e. two directions of E – impossible. R is the correct reason.
24. (c) Lines from an outside charge enter and leave – net flux zero. So A is false; R is true.

CHAPTER

2

Electrostatic Potential and Capacitance

Day 3–4

Weightage ★★★

NCERT Class 12

Potential

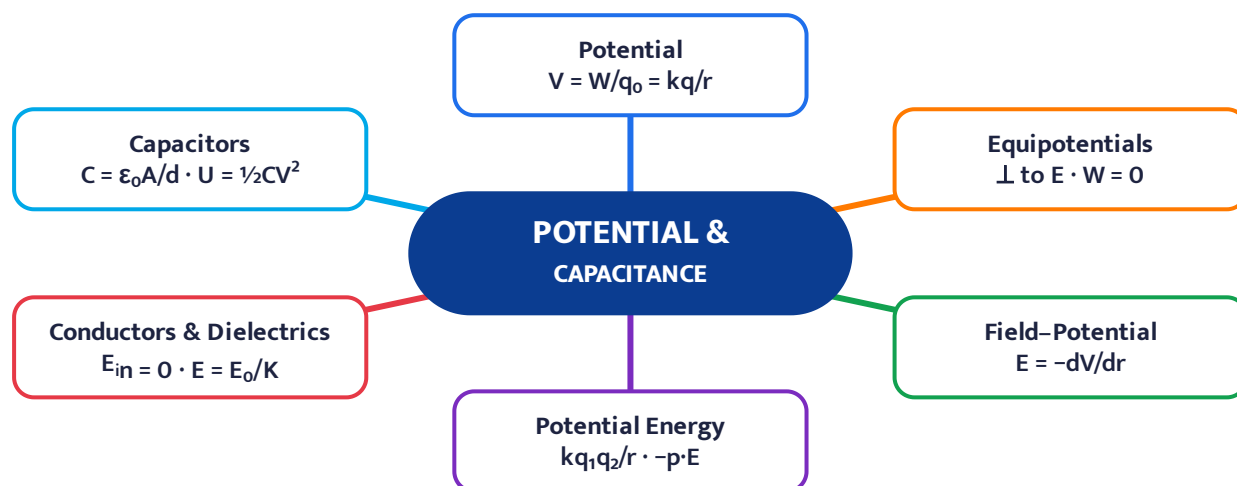
Equipotentials

Potential energy

Conductors & dielectrics

Capacitors

Energy stored



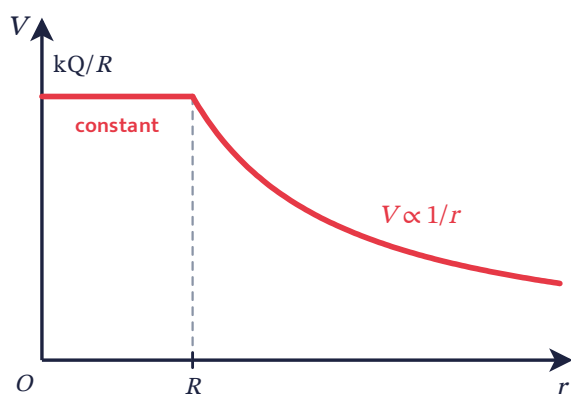
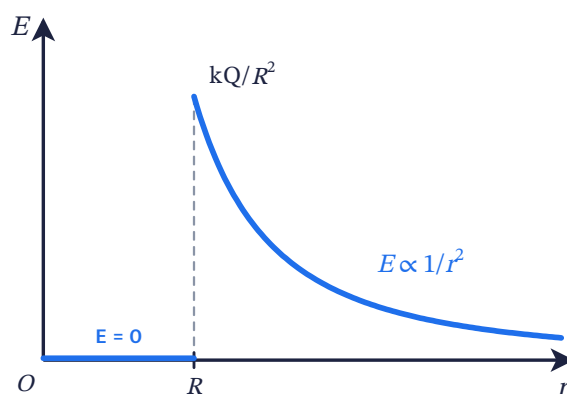
Chapter at a glance

2.1 Electrostatic Potential and Potential Difference

The electrostatic force is **conservative** – the work done in moving a charge between two points does not depend on the path. So we can define a **potential energy**, and per unit charge, a **potential**. The **potential at a point** is the work done by an external agent in bringing a unit positive test charge **slowly** (no acceleration) from infinity to that point. It is a **scalar**; unit volt (V) = J C⁻¹.

Potential formulas

$V = \frac{W_{\infty \rightarrow P}}{q_0}, \quad V_B - V_A = \frac{W_{A \rightarrow B}}{q_0}$	Potential and potential difference (work by external agent)
$V = \frac{kq}{r}$	Point charge – sign of q included; $V \propto 1/r$
$V = k \sum_i \frac{q_i}{r_i}$	System of charges – simple algebraic (scalar) sum
$V = \frac{kpcos\theta}{r^2}$	Short dipole: axial kp/r^2 , equatorial zero ; $V \propto 1/r^2$
$V_{out} = \frac{kQ}{r}, \quad V_{in} = V_{surface} = \frac{kQ}{R}$	Charged shell / conducting sphere – constant inside

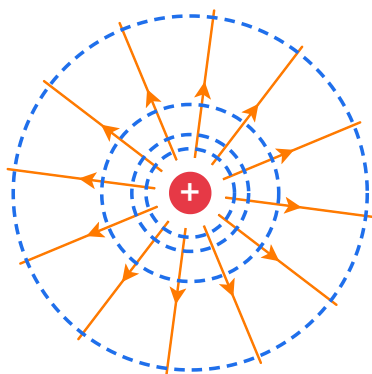
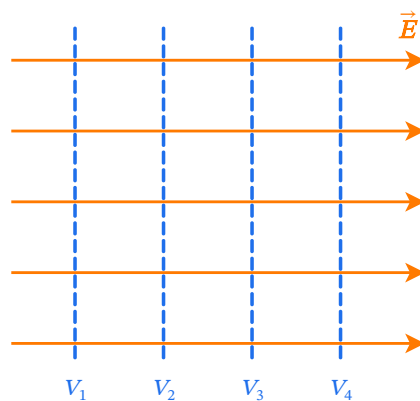
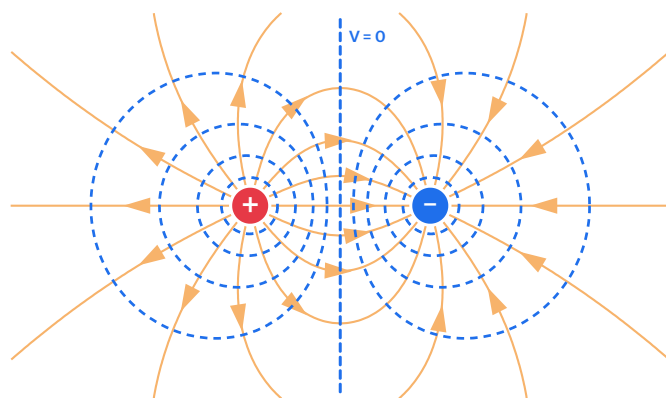
 V - r : constant inside, $1/r$ outside E - r : zero inside, $1/r^2$ outside**Solved Example**

Charges $3 \times 10^{-8} \text{ C}$ and $-2 \times 10^{-8} \text{ C}$ are 15 cm apart. At what points on the line joining them is the potential zero?

Solution: Let x (cm) be the distance from the positive charge. **Between** them: $\frac{3}{x} = \frac{2}{15-x} \Rightarrow x = 9 \text{ cm}$. **Outside**, beyond the negative charge: $\frac{3}{x} = \frac{2}{x-15} \Rightarrow x = 45 \text{ cm}$. (V can be zero where E is not!)

2.2 Equipotential Surfaces and $E = -dV/dr$

An **equipotential surface** has the same potential at every point. Point charge $\rightarrow$ concentric spheres; uniform field $\rightarrow$ planes perpendicular to $\vec{E}$; dipole $\rightarrow$ closed curves around each charge with the perpendicular bisector as the $V = 0$ plane.

Point charge: spheres crowd near q Uniform field: parallel planes, $V_1 > V_2 > V_3 > V_4$ Dipole: equipotentials (blue, dashed) cut field lines at 90° **Properties of equipotentials**

- $\vec{E}$ is always **normal** to an equipotential surface.
- No work** is done in moving a charge on it.
- Two equipotentials **never intersect**.
- Closer spacing (for equal ΔV) $\Rightarrow$ stronger field.

Field-potential relation

$$E = -\frac{dV}{dr}$$

Field = negative potential gradient; E points towards **decreasing** V (steepest fall)

$$\vec{E} = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right)$$

Components in 3-D

$$V_A - V_B = \int_A^B \vec{E} \cdot d\vec{l}, \quad E = \frac{V}{d}$$

Potential from field; uniform field between plates

Solved Example

$V = (5x^2 + 3)$ volt, x in metre. Find $\vec{E}$ at $x = 2$ m.

Solution: $E_x = -\frac{dV}{dx} = -10x = -20 \text{ V/m} \Rightarrow 20 \text{ V/m along } -x.$

Common Trap

- $V = 0$ does **not** mean $E = 0$ (dipole midpoint), and $E = 0$ does **not** mean $V = 0$ (inside a charged shell).
- A +ve charge moves from high V to low V; an electron from low V to high V.

2.3 Potential Energy of Charges and of a Dipole

The potential energy of a system is the work done by an external agent in assembling it from infinity. For a charge q in an external potential V: $U = qV$. Energy unit used for particles: **1 eV = 1.6×10^{-19} J** (energy gained by an electron through 1 V).

Potential energy

$$U = \frac{kq_1q_2}{r_{12}}$$

Two charges (put signs); + for like, - for unlike

$$U = k\left(\frac{q_1q_2}{r_{12}} + \frac{q_2q_3}{r_{23}} + \frac{q_1q_3}{r_{13}}\right)$$

Three charges – sum over every pair

$$U = q_1V(\vec{r}_1) + q_2V(\vec{r}_2) + \frac{kq_1q_2}{r_{12}}$$

Two charges in an external field

$$U = -\vec{p} \cdot \vec{E} = -pE\cos\theta$$

Dipole in uniform field: min $-pE$ at $\theta = 0^\circ$ (stable), max $+pE$ at 180° (unstable)

$$W_{\theta_1 \rightarrow \theta_2} = pE(\cos\theta_1 - \cos\theta_2)$$

Work to rotate a dipole; $0^\circ \rightarrow 180^\circ$ needs $2pE$

Solved Example

Three charges of $1 \mu\text{C}$ each are at the corners of an equilateral triangle of side 10 cm. Find the potential energy of the system.

Solution: Three pairs: $U = 3\frac{kq^2}{a} = 3 \times \frac{9 \times 10^9 \times 10^{-12}}{0.1} = 0.27 \text{ J}.$

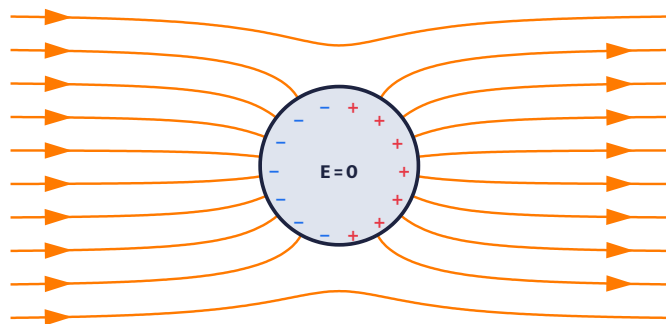
VS Sir's Exam Tip

- Work done by the **field** = $-\Delta U$; by an **external agent** (slowly) = $+\Delta U = q\Delta V$.
- Particle through ΔV : $\frac{1}{2}mv^2 = q\Delta V \Rightarrow v = \sqrt{2q\Delta V/m}$.

2.4 Conductors in Electrostatics

Six key results for a conductor

- $E = 0$ inside the material (free charges rearrange until it vanishes).
- Just outside, $\vec{E}$ is **normal** to the surface:
 $E = \sigma / \epsilon_0$.
- Net charge inside is zero; **excess charge resides on the surface**.
- The whole conductor is **one equipotential**.
- σ is larger at sharper points (smaller radius of curvature).
- Field inside an empty **cavity** is zero – **electrostatic shielding**.



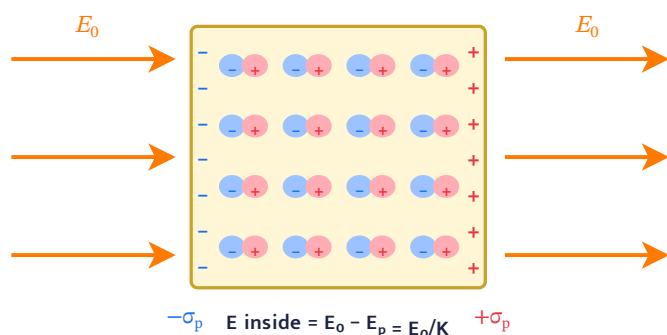
Neutral conductor in a uniform field: lines end normally on induced charges, $E = 0$ inside

NCERT Line to Remember

During a thunderstorm it is safest to stay **inside a car** or a closed metal body rather than under a tree – the metal body shields you (field inside a closed conductor is zero). Sensitive instruments are shielded the same way.

2.5 Dielectrics and Polarisation

Dielectrics are insulators with no free charges. In an external field $\vec{E}_0$ their molecules develop or align dipoles: **non-polar** molecules (O_2, N_2, H_2, CO_2) get **induced** dipoles; **polar** molecules (H_2O, HCl, NH_3) have **permanent** dipoles that partly align. The dipole moment per unit volume is the **polarisation** $\vec{P} = \chi_e \vec{E}$. The net effect is bound surface charges $\pm\sigma_p$ that set up a field opposing $\vec{E}_0$, so the field inside is reduced.



Polarised dielectric slab: bound charges $\pm\sigma_p$ on the faces

Dielectric

$$E = \frac{E_0}{K}$$

Field inside a dielectric

$$K = \frac{\epsilon}{\epsilon_0} = 1 + \frac{\chi_e}{\epsilon_0}$$

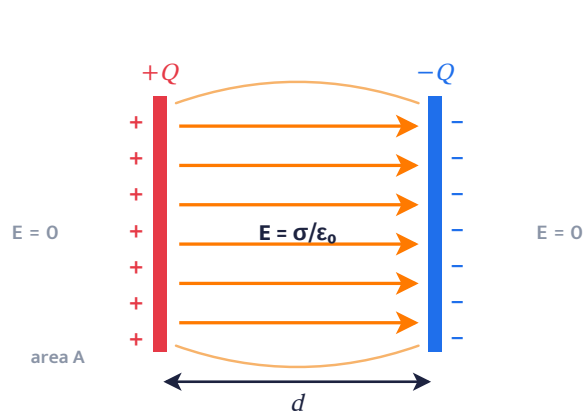
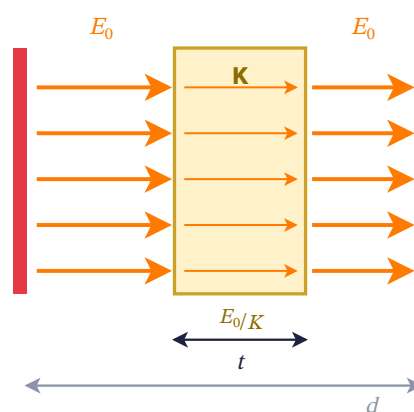
Dielectric constant ($K \geq 1$; vacuum 1, water ≈ 80 , metal $\rightarrow \infty$)

$$\sigma_p = P$$

Bound surface charge density

2.6 Capacitors and the Parallel-Plate Capacitor

A **capacitor** is a pair of conductors carrying charges $+Q$ and $-Q$, separated by an insulator. $Q \propto V$, so $C = Q/V$ – the **capacitance** depends only on geometry and the medium, **not** on Q or V . Unit: farad (F) = $C V^{-1}$; practical units $\mu F, nF, pF$.

Parallel-plate capacitor: uniform E between platesDielectric slab of thickness $t < d$

Capacitance formulas

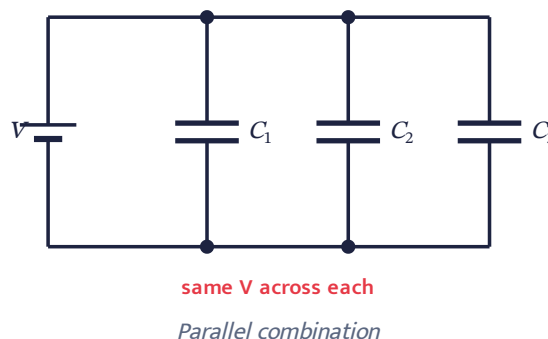
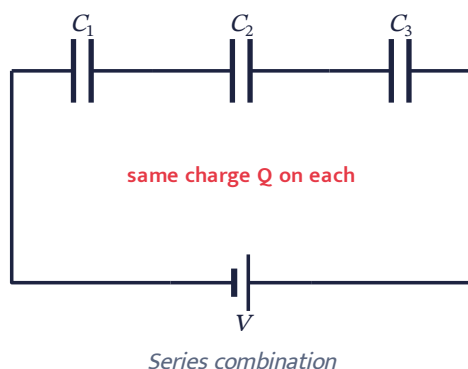
$C_0 = \frac{\epsilon_0 A}{d}$	Air/vacuum between plates (A = plate area, d = separation)
$C = \frac{K\epsilon_0 A}{d} = KC_0$	Fully filled with dielectric K
$C = \frac{\epsilon_0 A}{d - t + \frac{t}{K}}$	Dielectric slab of thickness t ($t < d$)
$C = \frac{\epsilon_0 A}{d - t}$	Metal (conducting) slab of thickness t ($K \rightarrow \infty$)
$C = 4\pi\epsilon_0 R$	Isolated sphere of radius R (Earth $\approx 711 \mu\text{F}$)

Solved Example

Plates of area 0.1 m^2 are 8.85 mm apart in air. Find C . If a slab of $K = 4$ and thickness 4.425 mm (half the gap) is inserted, find the new C .

Solution: $C_0 = \frac{8.85 \times 10^{-12} \times 0.1}{8.85 \times 10^{-3}} = 10^{-10} \text{ F} = 100 \text{ pF}$. With slab: $C = \frac{\epsilon_0 A}{d/2 + d/8} = \frac{8}{5} C_0 = 160 \text{ pF}$.

2.7 Combination of Capacitors



Feature	Series	Parallel
Equivalent	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$	$C = C_1 + C_2 + \dots$
Same for all	charge Q	potential difference V
Divides	$V: V_1 = \frac{C_2}{C_1 + C_2}V$ (smaller C gets larger V)	$Q: Q_1 = \frac{C_1}{C_1 + C_2}Q$ ($\propto C$)
Two capacitors	$C = \frac{C_1 C_2}{C_1 + C_2}$	$C = C_1 + C_2$
n identical (C each)	C / n	nC

Memory Trick

Capacitors combine **opposite** to resistors: series capacitors add like parallel resistors, and vice versa. Series C is **less than the smallest**; parallel C is more than the largest.

Solved Example

$2\ \mu\text{F}$, $3\ \mu\text{F}$ and $6\ \mu\text{F}$ are joined in series across $12\ \text{V}$. Find C and the charge on each.

Solution: $\frac{1}{C} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1 \Rightarrow C = 1\ \mu\text{F}$; $Q = CV = 12\ \mu\text{C}$ on each ($V = 6, 4, 2\ \text{V}$).

2.8 Energy Stored in a Capacitor

Charging a capacitor needs work against the field of the charge already on the plates; this work is stored as **electrostatic energy in the field** between the plates.

Energy

$$U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV$$

Energy stored (use the form where the constant quantity appears)

$$u = \frac{1}{2}\epsilon_0 E^2$$

Energy density (J m^{-3}); in a dielectric $\frac{1}{2}K\epsilon_0 E^2$

Dielectric K inserted	C	Q	V	E	U
Battery connected (V fixed)	KC_0	KQ_0	V_0	E_0	KU_0
Battery disconnected (Q fixed)	KC_0	Q_0	V_0 / K	E_0 / K	U_0 / K

Solved Example

A $10\ \mu\text{F}$ capacitor is charged to $100\ \text{V}$. Find the energy stored.

Solution: $U = \frac{1}{2}CV^2 = \frac{1}{2} \times 10^{-5} \times 10^4 = 0.05\ \text{J}$.

Common Trap

- Battery disconnected $\Rightarrow$ **Q stays constant**; battery connected $\Rightarrow$ **V stays constant**. Decide this first!
- When a battery charges a capacitor it supplies QV , but only $\frac{1}{2}QV$ is stored – the other half is lost as heat/radiation.

JEE Main Extension

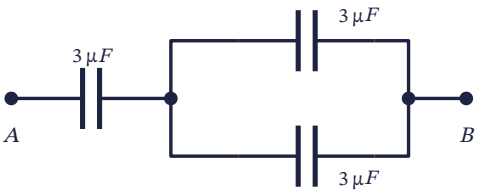
- Charge sharing:** joining C_1 (at V_1) and C_2 (at V_2): common $V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$; energy lost $= \frac{C_1 C_2 (V_1 - V_2)^2}{2(C_1 + C_2)}$.
- Force between plates: $F = \frac{Q^2}{2\epsilon_0 A}$ (field of one plate $\times Q$).



Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1 **NCERT based** The electric potential at a distance of 0.3 m from a point charge of 2 nC is
 (a) 6 V (b) 200 V (c) 60 V (d) 600 V
- 2 **NCERT based** The angle between an equipotential surface and the electric field line through it is
 (a) 45° (b) 0° (c) 90° (d) 180°
- 3 **NEET pattern** The work done in moving a $5\ \mu\text{C}$ charge between two points on an equipotential surface at 50 V is
 (a) 250 μJ (b) 0 J (c) 10 μJ (d) 2.5 μJ
- 4 **NEET pattern** Charges $+q$ and $-q$ are placed at A and B. At the midpoint of AB
 (a) $V \neq 0, E = 0$ (b) $V = 0, E \neq 0$ (towards A)
 (c) $V = 0, E \neq 0$ (towards B) (d) $V = 0, E = 0$
- 5 **JEE Main pattern** The potential in a region is $V = (3x^2 + 2y)$ volt (x, y in m). The magnitude of the electric field at (1 m, 1 m) is
 (a) 5 V/m (b) $\sqrt{13}$ V/m (c) 8 V/m (d) $\sqrt{40}$ V/m
- 6 **NEET pattern** Three charges of $1\ \mu\text{C}$ each are placed at the vertices of an equilateral triangle of side 10 cm. The electrostatic potential energy of the system is
 (a) 0.81 J (b) 0.18 J (c) 0.09 J (d) 0.27 J
- 7 **NEET pattern** An electron starting from rest is accelerated through 100 V. Its final speed is about ($m = 9.1 \times 10^{-31}$ kg)
 (a) 3.5×10^{13} m/s (b) 1.9×10^6 m/s (c) 5.9×10^5 m/s (d) 5.9×10^6 m/s
- 8 **NCERT based** A parallel-plate capacitor has plate area $0.1\ \text{m}^2$ and separation 8.85 mm in air. Its capacitance is
 (a) 1 nF (b) 100 nF (c) 10 pF (d) 100 pF
- 9 **NEET pattern** A capacitor remains connected to a battery while a dielectric of $K = 5$ completely fills the gap. The charge on the plates
 (a) becomes 25 times (b) becomes $1/5$
 (c) becomes 5 times (d) remains the same
- 10 **JEE Main pattern** A parallel-plate capacitor is charged and then disconnected from the battery. A dielectric slab of constant K now fills the gap. Which is correct?
 (a) E unchanged; U becomes $1/K$ times
 (b) Q unchanged; V and U both become $1/K$ times
 (c) V unchanged; U becomes K times
 (d) Q becomes K times; U unchanged
- 11 **NEET pattern** Capacitors of $2\ \mu\text{F}$, $3\ \mu\text{F}$ and $6\ \mu\text{F}$ are connected in series. The equivalent capacitance is
 (a) $1\ \mu\text{F}$ (b) $11\ \mu\text{F}$ (c) $2\ \mu\text{F}$ (d) $0.5\ \mu\text{F}$

- 12 **NEET pattern** The energy stored in a $10\ \mu\text{F}$ capacitor charged to $100\ \text{V}$ is
 (a) $0.5\ \text{J}$ (b) $5\ \text{mJ}$ (c) $0.1\ \text{J}$ (d) $0.05\ \text{J}$
- 13 **NCERT based** The energy density in air where the electric field is $3 \times 10^6\ \text{V/m}$ ($\epsilon_0 = 8.85 \times 10^{-12}\ \text{SI}$) is about
 (a) $4\ \text{J/m}^3$ (b) $13\ \text{J/m}^3$ (c) $40\ \text{J/m}^3$ (d) $80\ \text{J/m}^3$
- 14 **JEE Main pattern** A slab of dielectric constant 2 and thickness $d/2$ is inserted between the plates (separation d) of a capacitor of capacitance C_0 . The new capacitance is
 (a) $3C_0/2$ (b) $2C_0$ (c) C_0 (d) $4C_0/3$
- 15 **JEE Main pattern** A metal plate of thickness $d/2$ is inserted parallel to the plates of an air capacitor (separation d , capacitance C_0). The capacitance becomes
 (a) ∞ (b) C_0 (c) $2C_0$ (d) $C_0/2$
- 16 **JEE Main pattern** A $2\ \mu\text{F}$ capacitor charged to $100\ \text{V}$ is connected across an uncharged $3\ \mu\text{F}$ capacitor. The energy lost in the process is
 (a) $6\ \text{mJ}$ (b) $4\ \text{mJ}$ (c) 0 (d) $10\ \text{mJ}$
- 17 **NCERT based** The potential due to a short electric dipole at a point on its equatorial line is
 (a) $k p / r^3$ (b) $2k p / r^2$ (c) $k p / r^2$ (d) zero
- 18 **NEET pattern** For a charged conducting sphere of radius R , the graph of potential V versus distance r from the centre is
 (a) zero up to R , then falls as $1/r$
 (b) rises linearly up to R , then falls as $1/r$
 (c) falls as $1/r^2$ everywhere
 (d) constant up to R , then falls as $1/r$
- 19 **NCERT based** Which of the following molecules has a permanent electric dipole moment?
 (a) H_2O (b) O_2 (c) CO_2 (d) N_2
- 20 **NEET pattern** The equivalent capacitance between A and B in the network shown is

 (a) $4.5\ \mu\text{F}$ (b) $1\ \mu\text{F}$ (c) $2\ \mu\text{F}$ (d) $9\ \mu\text{F}$
- 21 **NEET pattern** The work done by an external agent in slowly moving a $2\ \mu\text{C}$ charge from a point at $10\ \text{V}$ to a point at $60\ \text{V}$ is
 (a) $1.4 \times 10^{-4}\ \text{J}$ (b) $-1 \times 10^{-4}\ \text{J}$ (c) $1 \times 10^{-4}\ \text{J}$ (d) $1.2 \times 10^{-4}\ \text{J}$

22 **Assertion-Reason** Assertion: In electrostatics, the excess charge on a conductor resides on its outer surface.

Reason: The electric field inside a conductor is zero in electrostatics.

- (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

23 **Assertion-Reason** Assertion: If the potential at a point is zero, the electric field there must also be zero. Reason: The electric field is the negative gradient of potential, $E = -dV/dr$.

- (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

24 **NEET pattern** A dipole of moment 10^{-8} C m is in a uniform field of 2×10^4 N/C. The work needed to rotate it from the stable to the unstable equilibrium position is

- (a) 0 (b) 4×10^{-4} J (c) 2×10^{-4} J (d) 1×10^{-4} J

✓ Answer Key & Quick Solutions

- 1(c) 2(c) 3(b) 4(c) 5(d) 6(d) 7(d) 8(d) 9(c) 10(b) 11(a) 12(d) 13(c) 14(d) 15(c)
 16(a) 17(d) 18(d) 19(a) 20(c) 21(c) 22(a) 23(c) 24(b)

- (c) $V = kq/r = 9 \times 10^9 \times 2 \times 10^{-9} / 0.3 = 60$ V.
- (c) E is always normal to equipotentials (no work along them).
- (b) $W = q\Delta V = q \times 0 = 0$.
- (c) Potentials cancel (scalars); fields from both point from + to -, i.e. towards B, and add.
- (d) $E_x = -6x = -6$, $E_y = -2 \Rightarrow |E| = \sqrt{36+4} = \sqrt{40} \approx 6.3$ V/m.
- (d) $U = 3kq^2/a = 3(9 \times 10^9)(10^{-12})/0.1 = 0.27$ J.
- (d) $\frac{1}{2}mv^2 = eV = 1.6 \times 10^{-17}$ J $\Rightarrow v = \sqrt{2 \times 1.6 \times 10^{-17} / 9.1 \times 10^{-31}} \approx 5.9 \times 10^6$ m/s.
- (d) $C = \epsilon_0 A/d = 8.85 \times 10^{-12} \times 0.1 / 8.85 \times 10^{-3} = 10^{-10}$ F.
- (c) V fixed, $C \rightarrow 5C \Rightarrow Q = CV$ becomes 5 times.
- (b) Q fixed, $C \rightarrow KC \Rightarrow V = Q/C$ and E fall by K; $U = Q^2/2C$ falls by K.
- (a) $1/C = 1/2 + 1/3 + 1/6 = 1 \Rightarrow C = 1$ μ F.
- (d) $U = \frac{1}{2}CV^2 = \frac{1}{2}(10^{-5})(10^4) = 0.05$ J.
- (c) $u = \frac{1}{2}\epsilon_0 E^2 = \frac{1}{2}(8.85 \times 10^{-12})(9 \times 10^{12}) \approx 39.8$ J/m³.
- (d) $C = \frac{\epsilon_0 A}{d/2 + d/4} = \frac{4}{3}C_0$.
- (c) $C = \epsilon_0 A/(d-t) = \epsilon_0 A/(d/2) = 2C_0$.
- (a) Common V = 200/5 = 40 V; $U_i = 10$ mJ, $U_f = \frac{1}{2}(5 \mu F)(40)^2 = 4$ mJ $\Rightarrow$ loss 6 mJ.
- (d) $V = kpcos\theta/r^2$ with $\theta = 90^\circ \Rightarrow 0$.
- (d) E = 0 inside $\Rightarrow$ V constant = kQ/R inside; outside V = kQ/r.
- (a) H₂O is bent (polar). O₂, N₂ and linear CO₂ are non-polar.
- (c) Parallel pair = 6 μ F; in series with 3 μ F: $\frac{3 \times 6}{3+6} = 2$ μ F.

21. (c) $W = q(V_B - V_A) = 2 \times 10^{-6} \times 50 = 10^{-4} \text{ J}.$

22. (a) $E = 0$ inside $\Rightarrow$ by Gauss's law any closed surface inside encloses zero charge $\Rightarrow$ charge only on the surface. R explains A.

23. (c) E depends on the slope of V , not its value – at a dipole's midpoint $V = 0$ but $E \neq 0$. A false, R true.

24. (b) $W = pE(\cos 0^\circ - \cos 180^\circ) = 2pE = 2(10^{-8})(2 \times 10^4) = 4 \times 10^{-4} \text{ J}.$

CHAPTER

3

Current Electricity

17

Day 5–6

Weightage ★★★

NCERT Class 12

Drift velocity

Ohm's law

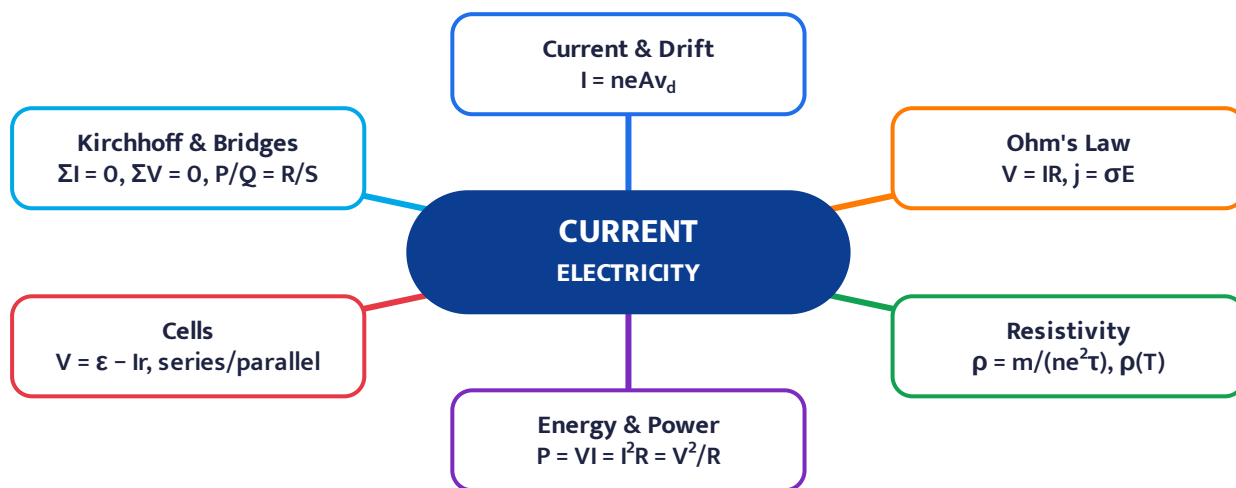
Resistivity & temperature

Power

Cells, emf, r

Kirchhoff's laws

Wheatstone & metre bridge



Chapter at a glance

3.1 Electric Current and Current Density

Electric current is the rate of flow of charge through a cross-section: $I = \frac{\Delta q}{\Delta t}$ (steady) or $I = \frac{dq}{dt}$ (instantaneous). SI unit: **ampere** (1 A = 1 C/s). By convention its direction is the direction of flow of **positive** charge – opposite to the flow of electrons. Current has a direction but adds like a **scalar** (it does not obey vector addition).



Current & current density

$$I = \frac{dq}{dt}, \quad q = ne \quad \text{Current; charge is quantised}$$

$$\vec{j} = \frac{I}{A} \hat{n} \quad \text{Current density (vector, A m}^{-2}\text{)}$$

$$I = \int \vec{j} \cdot d\vec{A} \quad \text{Current = flux of } \vec{j}$$



NCERT Line to Remember

- Typical currents: lightning $\sim 10^4$ A; household circuits ~ 1 –10 A; nerve signals $\sim \mu\text{A}$.
- 1 A means 6.25×10^{18} electrons crossing a section every second.

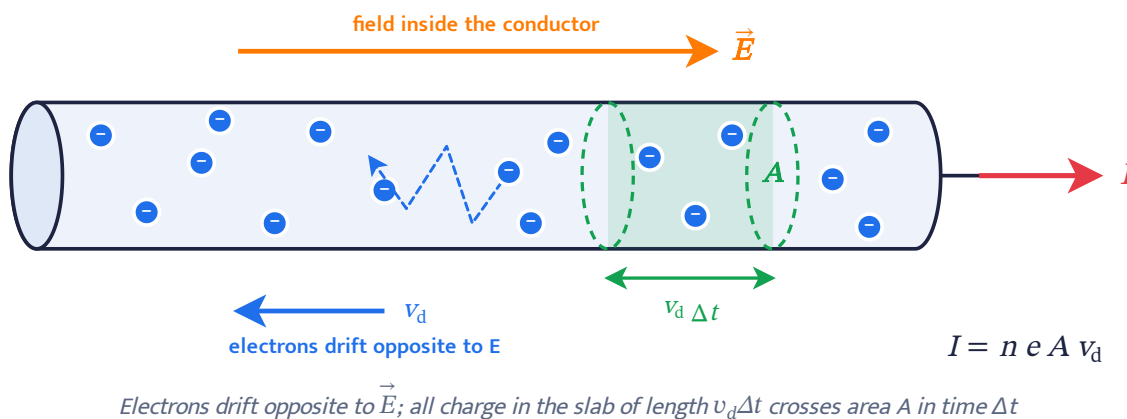
3.2 Drift Velocity, Relaxation Time and Mobility

Free electrons in a metal move randomly at thermal speeds ($\sim 10^5$ m/s) and collide with the ions, so with no field their **average velocity is zero**. When a field $\vec{E}$ is applied, each electron gains velocity $-\vec{e}\vec{E}t / m$ between collisions. Averaging over the **relaxation time** τ (mean time between collisions, $\sim 10^{-14}$ s) gives a small net **drift velocity** opposite to $\vec{E}$.



Three very different speeds

Thermal (random) speed $\sim 10^5$ m/s $\gg$ drift speed $\sim 10^{-4}$ m/s (a fraction of a mm per second). But the **signal** (electric field) travels along the wire at nearly 3×10^8 m/s.

**Drift and mobility**

$$\vec{v}_d = -\frac{e\vec{E}}{m}\tau$$

Drift velocity

$$I = neAv_d$$

 n = free-electron density (m^{-3})

$$\mu = \frac{v_d}{E} = \frac{e\tau}{m}$$

Mobility ($\text{m}^2 \text{V}^{-1} \text{s}^{-1}$), always positive

$$\vec{j} = \frac{ne^2\tau}{m}\vec{E} = \sigma\vec{E}$$

Ohm's law (microscopic form)

$$\sigma = \frac{ne^2\tau}{m} = ne\mu$$

Conductivity

$$\rho = \frac{1}{\sigma} = \frac{m}{ne^2\tau}$$

Resistivity

**Solved Example**

A copper wire of cross-section 1.0 mm^2 carries 2.0 A . Taking $n = 8.5 \times 10^{28} \text{ m}^{-3}$, find the drift speed and the time an electron takes to drift 1 m .

Solution: $v_d = \frac{I}{neA} = \frac{2.0}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-6}} \approx 1.5 \times 10^{-4} \text{ m/s}$ (0.15 mm/s). Time for $1 \text{ m} \approx 6.8 \times 10^3 \text{ s} \approx \mathbf{1.9 \text{ h}}$

**Common Trap**

- $\vec{j}$ is a vector but current I is a **scalar** – in a bent wire I is the same everywhere; never resolve I into components.
- For a wire of **non-uniform** cross-section carrying current I : I is the same everywhere, but j and v_d are larger where A is smaller ($v_d \propto 1/A$).
- At fixed V , $v_d = \frac{eV\tau}{ml}$ – doubling the length **halves** v_d ; changing only the radius does **not** change v_d .

3.3 Ohm's Law, Resistance and Resistivity

Ohm's law: at constant temperature (and other physical conditions), the current through a conductor is proportional to the potential difference across it: $V = IR$. Resistance R (unit ohm, Ω) depends on the material and the dimensions: $R = \frac{\rho l}{A}$, where ρ is the **resistivity** ($\Omega \text{ m}$) – a property of the material alone.

**Solved Example**

A wire is stretched so that its length increases by 10% . Find the percentage change in its resistance.

Solution: Volume constant $\Rightarrow R \propto l^2$. $R'/R = (1.1)^2 = 1.21 \Rightarrow \mathbf{21\% \text{ increase}}$. (For small stretching, $\frac{\Delta R}{R} \approx 2\frac{\Delta l}{l} = 20\%$.)



Resistance

$$R = \frac{\rho l}{A} \quad \text{Resistance of a uniform wire}$$

$$\sigma = \frac{1}{\rho} \quad \text{Conductivity (S m}^{-1}\text{)}$$

$$R' = n^2 R \quad \text{Wire stretched to } n \text{ times its length (volume constant)}$$

$$R' = \frac{R}{n^2} \quad \text{Wire cut into } n \text{ equal parts, all joined in parallel}$$

Class

 ρ (Ω m)

Conductors (Ag, Cu, Al)

 $10^{-8} - 10^{-6}$

Alloys (nichrome, manganin)

 $\sim 10^{-6}$

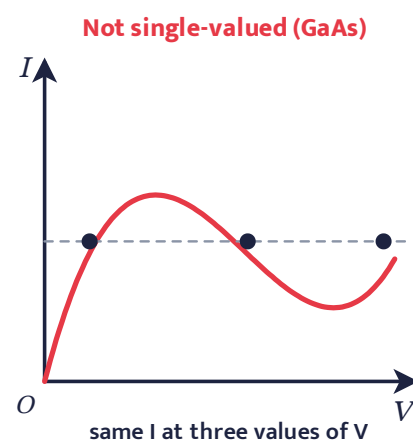
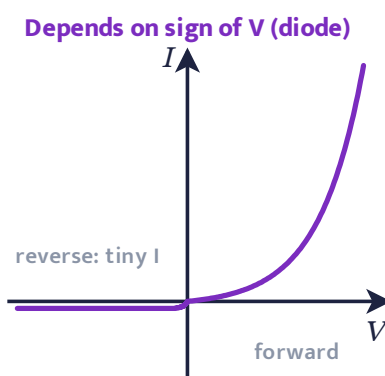
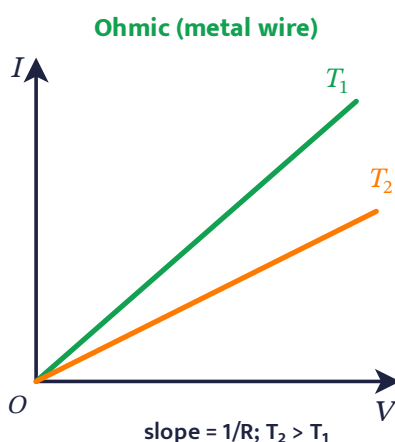
Semiconductors (Si, Ge)

 $\sim 10^{-5} - 10^3$

Insulators (glass, rubber)

 $10^{14} - 10^{16}$

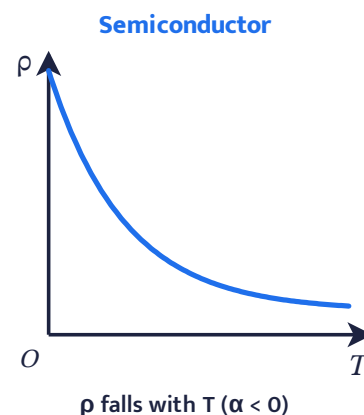
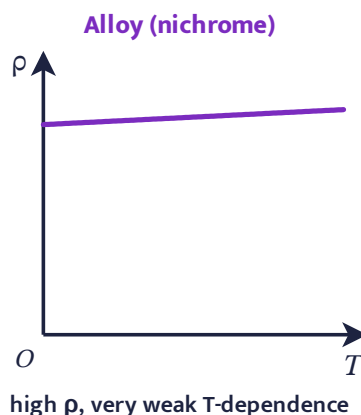
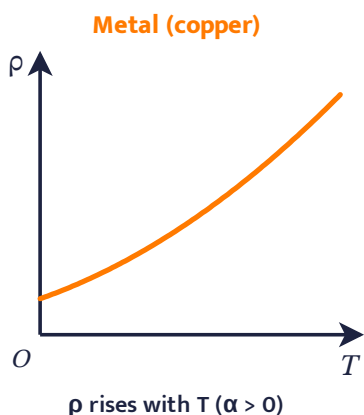
Limitations of Ohm's law – it is **not** a fundamental law. It fails when (i) V is not proportional to I , (ii) the relation depends on the **sign** of V (diode), or (iii) the relation is **not unique** – more than one V for the same I (GaAs). Such devices are **non-ohmic**.



V - I characteristics: linear (ohmic) and three kinds of non-linear behaviour

3.4 Temperature Dependence of Resistivity

Over a limited range, $\rho_T = \rho_0[1 + \alpha(T - T_0)]$, where α is the **temperature coefficient of resistivity**. Since $\rho = \frac{m}{ne^2\tau}$: in a **metal**, n is fixed but collisions become more frequent (τ falls) $\rightarrow \rho$ **rises**. In a **semiconductor**, n increases sharply with T and outweighs the fall in $\tau \rightarrow \rho$ **falls** (α negative).



Resistivity versus temperature for a metal, an alloy and a semiconductor

NCERT Line to Remember

- Alloys like **nichrome, manganin, constantan** have high ρ and very small α – used for **standard resistors**, heating elements and in bridges.
- For copper ρ vs T is linear only over a limited range; at low T it curves and tends to a finite value.

VS Sir's Exam Tip

For two temperatures use the ratio directly:

$$\frac{R_2}{R_1} = \frac{1 + \alpha t_2}{1 + \alpha t_1} \quad (t \text{ in } ^\circ\text{C, reference } 0^\circ\text{C}). \text{ Use}$$

$$R_2 \approx R_1[1 + \alpha(t_2 - t_1)] \text{ only when } \alpha \Delta t \text{ is small.}$$

Solved Example

A nichrome toaster element has resistance $75.3 \, \Omega$ at 27.0°C . On $230 \, \text{V}$ it settles at a steady current of $2.68 \, \text{A}$. With $\alpha = 1.70 \times 10^{-4} \, ^\circ\text{C}^{-1}$, find the steady temperature of the element.

Solution: Hot resistance $R = 230 / 2.68 = 85.8 \, \Omega$. $T - 27 = \frac{85.8 - 75.3}{75.3 \times 1.70 \times 10^{-4}} = 820^\circ\text{C} \Rightarrow T \approx 847^\circ\text{C}$.

3.5 Electrical Energy and Power

When charge q moves through a potential drop V , it loses energy qV , which appears as heat (Joule heating) in a resistor. Power dissipated = rate of energy transfer.

Energy & power

$$P = VI = I^2R = \frac{V^2}{R} \quad \text{Power in a resistor (W)}$$

$$H = I^2Rt \quad \text{Joule's law of heating}$$

$$1 \, \text{kWh} = 3.6 \times 10^6 \, \text{J} \quad \text{Commercial unit of energy}$$

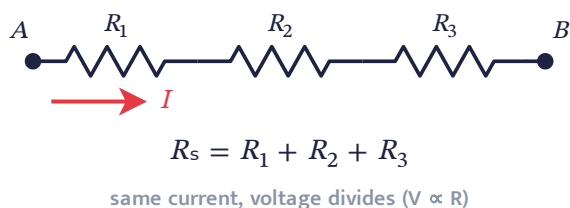
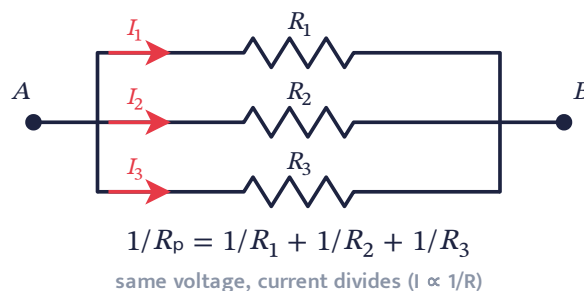
$$P_c = \frac{P^2 R_c}{V^2} \quad \text{Line loss when power } P \text{ is sent at voltage } V$$

Rated bulbs (P, V)

- $R = V_{\text{rated}}^2 / P_{\text{rated}}$: a $60 \, \text{W}$ bulb has **more** resistance than a $100 \, \text{W}$ bulb (same V).
- Series** (same I): $P \propto R \rightarrow$ lower-wattage bulb glows **brighter**.
- Parallel** (same V): $P \propto 1/R \rightarrow$ higher-wattage bulb glows brighter.
- At voltage V : $P = P_{\text{rated}}(V / V_{\text{rated}})^2$.

NCERT Line to Remember

Power is transmitted over long distances at **very high voltage** (and low current) because the loss in the transmission cables $P_c = P^2 R_c / V^2$ falls as $1 / V^2$.

3.6 Combination of Resistors**Series****Parallel**

Series and parallel combinations of resistors

**Series & parallel**

$$R_s = R_1 + R_2 + \dots \quad \text{Series: } R_s > \text{largest } R$$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots \quad \text{Parallel: } R_p < \text{smallest } R$$

$$R_p = \frac{R_1 R_2}{R_1 + R_2} \quad \text{Two in parallel (product/sum)}$$

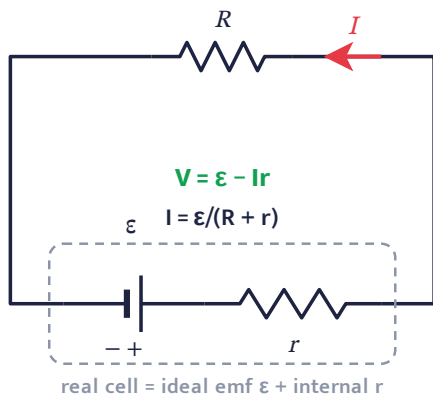
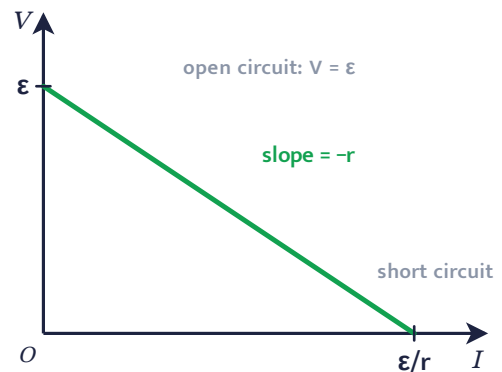
$$I_1 = I \frac{R_2}{R_1 + R_2} \quad \text{Current divider (two branches)}$$

**Memory Trick**

- n equal resistors R: series **nR**, parallel **R/n**; ratio $R_s / R_p = n^2$.
- "Series Same current, Parallel same Potential."
- Parallel branch with the smaller R takes the **larger** current.

3.7 Cells: emf, Internal Resistance, Terminal Voltage

The **emf** ϵ of a cell is the potential difference between its terminals when **no current** is drawn (open circuit) – the work done by the cell per unit charge. A real cell has an **internal resistance** r (of its electrolyte), so when it delivers current I the **terminal voltage** drops below ϵ .

Cell of emf ϵ and internal resistance r driving R Terminal voltage vs current: $V = \epsilon - Ir$ **Cell formulas**

$$I = \frac{\epsilon}{R + r} \quad \text{Current in the circuit}$$

$$V = \epsilon - Ir = \frac{\epsilon R}{R + r} \quad \text{Terminal voltage while **discharging**}$$

$$V = \epsilon + Ir \quad \text{Terminal voltage while **charging**}$$

$$r = R \left(\frac{\epsilon}{V} - 1 \right) \quad \text{Internal resistance from } \epsilon \text{ and } V$$

$$P_{\max} = \frac{\epsilon^2}{4r} \text{ at } R = r \quad \text{Maximum power to external } R \text{ (max power transfer)}$$

**Common Trap**

- emf is **not** a force – it is work per unit charge (volt). $V = \epsilon$ only on open circuit; on short circuit $V = 0$ and $I_{\max} = \epsilon / r$.
- At maximum power transfer the efficiency is only **50%** ($\eta = \frac{R}{R + r}$).

3.8 Cells in Series and in Parallel

Combination	Equivalent emf	Equivalent r
Series (+ to -)	$\varepsilon_1 + \varepsilon_2$	$r_1 + r_2$
Series, one reversed	$\varepsilon_1 - \varepsilon_2$	$r_1 + r_2$
Parallel (like terminals joined)	$\frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$	$\frac{r_1 r_2}{r_1 + r_2}$
n identical in series	$n\varepsilon$	nr
m identical in parallel	ε	r / m

VS Sir's Exam Tip

- In **series** with one cell reversed among n : $\text{emf} = (n - 2)\varepsilon$, but r stays nr .
- Parallel: $\frac{\varepsilon_{eq}}{r_{eq}} = \sum \frac{\varepsilon_i}{r_i}$ – quick for any number of cells.

JEE Main Extension

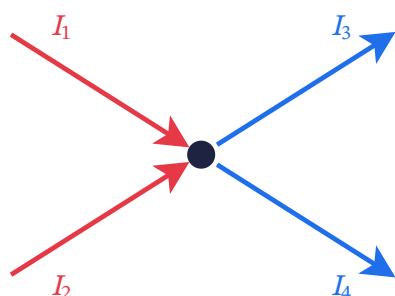
Mixed grouping: n cells in series per row, m such rows in parallel: $I = \frac{n\varepsilon}{R + nr/m}$. Current is **maximum** when $R = \frac{nr}{m}$ (external = internal resistance).

3.9 Kirchhoff's Rules

Junction rule (KCL): at any junction, the sum of currents entering = sum of currents leaving ($\sum I = 0$) – **conservation of charge**. **Loop rule (KVL):** the algebraic sum of changes in potential around any closed loop is zero ($\sum \Delta V = 0$) – **conservation of energy**.

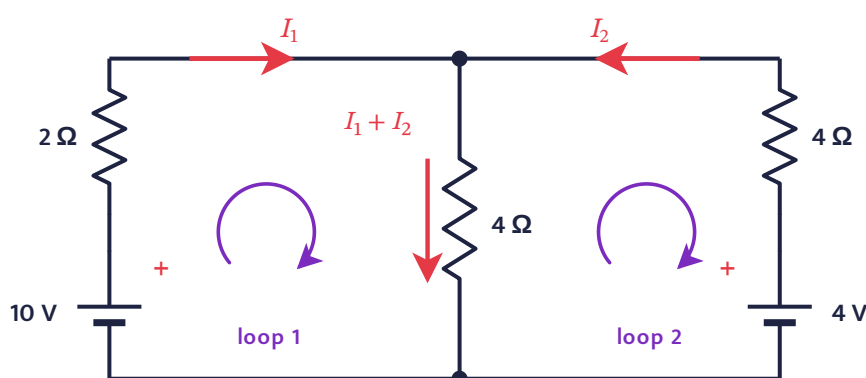
Sign convention for the loop rule

- Across a resistor **along** the assumed current: $-IR$; against it: $+IR$.
- Across a cell from $-$ to $+$ terminal: $+\varepsilon$; from $+$ to $-$: $-\varepsilon$.
- Assume any direction; a **negative** answer just means the current flows the other way.



Junction rule

$$I_1 + I_2 = I_3 + I_4$$



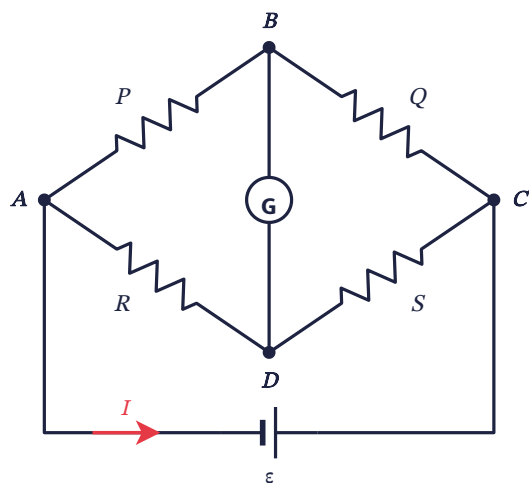
Loop rule: $\Sigma(\text{emf}) = \Sigma(IR)$ round any closed loop

Kirchhoff's junction rule and a two-loop circuit (used in the example below)

Solved Example

In the two-loop circuit above (10 V cell with $2\ \Omega$, 4 V cell with $4\ \Omega$, middle branch $4\ \Omega$, cells of negligible internal resistance), find I_1, I_2 and the current in the middle $4\ \Omega$.

Solution: Loop 1: $10 - 2I_1 - 4(I_1 + I_2) = 0 \Rightarrow 6I_1 + 4I_2 = 10$. Loop 2: $4 - 4I_2 - 4(I_1 + I_2) = 0 \Rightarrow 4I_1 + 8I_2 = 4$. Solving: $I_1 = 2\ \text{A}, I_2 = -0.5\ \text{A}$ (the 4 V cell is actually being **charged**), middle current = **1.5 A**.

3.10 Wheatstone Bridge

Wheatstone bridge: balanced when $I_g = 0$

Four resistances P, Q, R, S form a loop ABCD; a cell is connected across A–C and a galvanometer across B–D. The bridge is **balanced** when the galvanometer shows **no deflection** ($I_g = 0$), i.e. B and D are at the same potential.

$$\frac{P}{Q} = \frac{R}{S}$$

NCERT Line to Remember

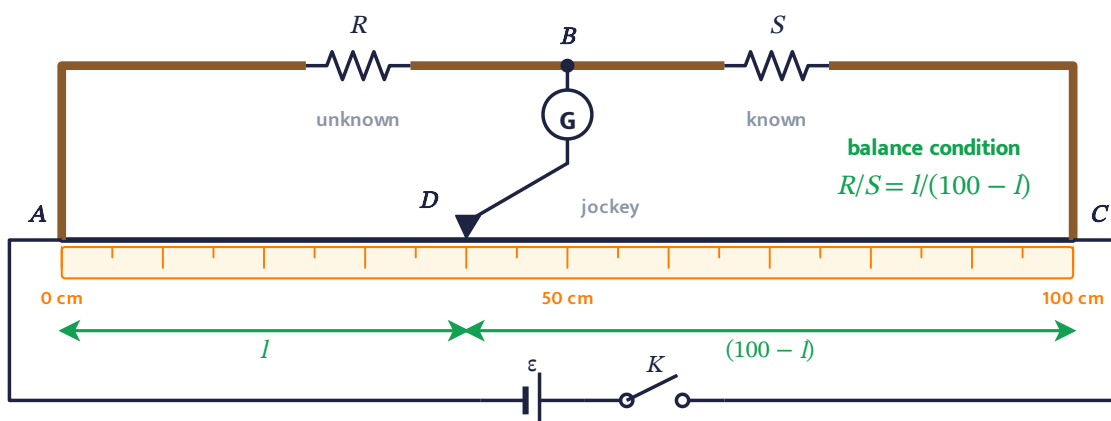
In the balanced state the galvanometer arm can be **removed, shorted or replaced by any resistor** – nothing changes. Interchanging the cell and the galvanometer keeps the bridge balanced. The bridge is most sensitive when P, Q, R, S are of the same order.

Common Trap

Before using a series–parallel shortcut on a "diamond" network, **check $P/Q = R/S$ first**. If balanced, drop the middle resistor: $R_{eq} = \frac{(P+Q)(R+S)}{P+Q+R+S}$. If not balanced, you must use Kirchhoff's rules.

3.11 Metre Bridge (Experimental Skill)

A metre bridge is a practical Wheatstone bridge: a 1 m uniform wire AC (usually constantan/manganin) on a metre scale, with thick copper strips of negligible resistance. The unknown R and a known S (resistance box) sit in the two gaps; a jockey slides on the wire to find the **null point** D. Since the wire is uniform, resistance $\propto$ length.



Metre bridge: unknown R in the left gap, known S in the right gap, null point D at distance l from A

**Metre-bridge formulas**

$$\frac{R}{S} = \frac{l}{100 - l} \quad \text{Balance (} l \text{ in cm, from end A)}$$

$$R = S \frac{l}{100 - l} \quad \text{Unknown resistance}$$

$$\rho = \frac{\pi r^2 R}{L} \quad \text{Resistivity of a wire of length } L, \text{ radius } r$$

**VS Sir's Exam Tip**

- Choose S so the null point is near the **middle** (40–60 cm) – percentage error is least there.
- End errors** (from contact resistance at A and C) are removed by interchanging R and S and averaging.
- Press the jockey gently – do not slide it while pressed.

**Solved Example**

In a metre bridge the null point is at 33.7 cm from A. When a 12Ω resistor is connected in parallel with S, the null point moves to 51.9 cm. Find R and S.

Solution: $\frac{R}{S} = \frac{33.7}{66.3}$ and $\frac{R}{S_{eq}} = \frac{51.9}{48.1}$ with $S_{eq} = \frac{12S}{S+12}$. Dividing: $\frac{S+12}{12} = \frac{51.9 \times 66.3}{48.1 \times 33.7} = 2.123 \Rightarrow S \approx 13.5 \Omega, R \approx 6.85 \Omega$.

**Memory Trick**

"Charge-Junction, Energy-Loop" – KCL counts charge, KVL counts energy. Bridge balance: **"top ratio = bottom ratio"** – P/Q (upper arms) = R/S (lower arms).

**Practice Questions**

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1 **NCERT based** A current of 1 A flows in a wire. The number of electrons crossing a cross-section per second is
 (a) 6.25×10^{18} (b) 1.6×10^{19} (c) 1.6×10^{-19} (d) 6.25×10^{19}

- 2 **NCERT based** A copper wire of cross-section $1.0 \times 10^{-7} \text{ m}^2$ carries 1.5 A. If $n = 8.5 \times 10^{28} \text{ m}^{-3}$, the drift speed is about
 (a) 1.1 m/s (b) 1.1 cm/s (c) 0.11 mm/s (d) 1.1 mm/s

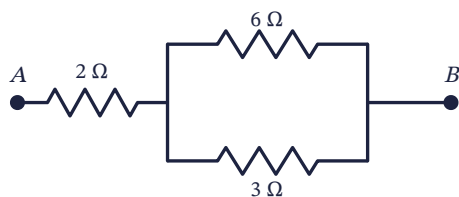
- 3 **NEET pattern** The SI unit of mobility of charge carriers is
 (a) $\text{m V}^{-1} \text{ s}^{-1}$ (b) $\text{m}^2 \text{ V}^{-1} \text{ s}^{-1}$
 (c) $\text{m}^2 \text{ V s}^{-1}$ (d) $\text{V m}^{-1} \text{ s}^{-1}$

- 4 **NEET pattern** A potential difference V is applied across a metal wire. If the length of the wire is doubled keeping V the same, the drift velocity
 (a) is unchanged (b) becomes half
 (c) becomes one-fourth (d) is doubled

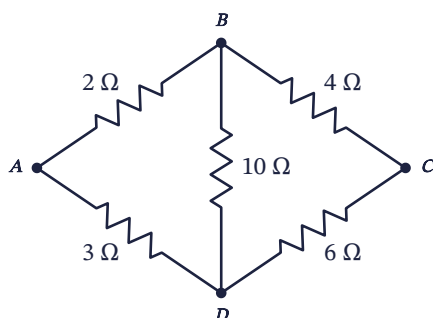
- 5 **NEET pattern** A wire of resistance R is stretched uniformly to three times its original length. Its new resistance is
 (a) $R/3$ (b) $9R$ (c) $3R$ (d) $27R$

- 6 **NCERT based** For a metallic conductor the resistivity is given by $\rho = m / (ne^2\tau)$. On heating the metal, ρ increases mainly because
 (a) n increases (b) m increases (c) n decreases (d) τ decreases
- 7 **NCERT based** With increase in temperature, the resistivity of a pure semiconductor
 (a) remains unchanged (b) first increases then decreases
 (c) decreases (d) increases
- 8 **NEET pattern** The resistance of a wire is $20\ \Omega$ at 20°C and $60\ \Omega$ at 500°C . At what temperature is its resistance $25\ \Omega$? (Use $R = R_0(1 + \alpha t)$.)
 (a) 140°C (b) 80°C (c) 100°C (d) 60°C
- 9 **JEE Main pattern** An electric field of what magnitude is needed to produce a current density of $3 \times 10^6\ \text{A m}^{-2}$ in a metal of conductivity $6 \times 10^7\ \text{S m}^{-1}$?
 (a) $0.05\ \text{V/m}$ (b) $20\ \text{V/m}$ (c) $0.5\ \text{V/m}$ (d) $1.8 \times 10^{14}\ \text{V/m}$
- 10 **NEET pattern** A $220\ \text{V}$, $100\ \text{W}$ bulb is operated at $110\ \text{V}$. The power consumed is
 (a) $50\ \text{W}$ (b) $75\ \text{W}$ (c) $25\ \text{W}$ (d) $100\ \text{W}$
- 11 **NEET pattern** Bulbs of $100\ \text{W}$ and $60\ \text{W}$ (both rated $220\ \text{V}$) are connected in series across $220\ \text{V}$. Then
 (a) neither glows (b) the $60\ \text{W}$ bulb glows brighter
 (c) both glow equally (d) the $100\ \text{W}$ bulb glows brighter
- 12 **NEET pattern** The coil of a heater is cut into two equal halves and only one half is used on the same supply. The heat produced per second becomes
 (a) half (b) one-fourth (c) four times (d) double

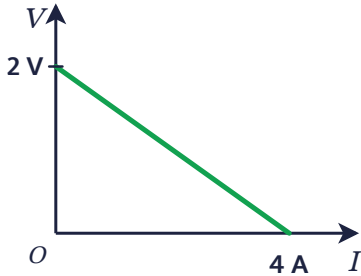
- 13 **NEET pattern** The equivalent resistance between A and B of the network shown is



- 14 **NEET pattern** In the network shown, the equivalent resistance between A and C is



- (a) $15\ \Omega$ (b) $2.8\ \Omega$ (c) $3.6\ \Omega$ (d) $5.0\ \Omega$

- 15 **NEET pattern** A cell of emf 2 V and internal resistance $0.5\ \Omega$ is connected to a $3.5\ \Omega$ resistor. The terminal voltage of the cell is
 (a) 0.25 V (b) 1.5 V (c) 2.0 V (d) 1.75 V
- 16 **NEET pattern** The terminal voltage V of a cell versus the current I drawn from it is shown. The internal resistance of the cell is
- 
- (a) $0.5\ \Omega$ (b) $0.25\ \Omega$ (c) $8\ \Omega$ (d) $2\ \Omega$
- 17 **JEE Main pattern** A battery of emf 12 V and internal resistance $2\ \Omega$ is connected to a variable resistor. The maximum power that can be delivered to the resistor is
 (a) 9 W (b) 36 W (c) 18 W (d) 72 W
- 18 **JEE Main pattern** Ten cells, each of emf 1.5 V and internal resistance $0.5\ \Omega$, are joined in series, but one cell is connected with reversed polarity. The combination is connected to a $1\ \Omega$ resistor. The current is
 (a) 2 A (b) 1.8 A (c) 2.5 A (d) 3 A
- 19 **JEE Main pattern** Two cells of emf 2 V and 4 V with internal resistances $1\ \Omega$ and $2\ \Omega$ are connected in parallel (like terminals together). The equivalent emf is
 (a) $8/3\ \text{V}$ (b) 6 V (c) 3 V (d) 2 V
- 20 **NCERT based** Kirchhoff's junction rule and loop rule are respectively based on the conservation of
 (a) momentum and charge (b) charge and momentum
 (c) energy and charge (d) charge and energy
- 21 **JEE Main pattern** In a metre bridge, an unknown resistance X is in the left gap and $6\ \Omega$ in the right gap. The null point is at 40 cm from the left end. X is
 (a) $4\ \Omega$ (b) $9\ \Omega$ (c) $6\ \Omega$ (d) $2.4\ \Omega$
- 22 **JEE Main pattern** In the circuit of Section 3.9 (10 V with $2\ \Omega$, 4 V with $4\ \Omega$, middle branch $4\ \Omega$), if the middle branch is removed, the current in the circuit is
 (a) 1.67 A (b) 1 A (c) 2.33 A (d) 0.5 A
- 23 **Assertion-Reason** Assertion: Manganin is used for making standard resistance coils. Reason: Manganin has a high resistivity and a very small temperature coefficient of resistance.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

24 **Assertion-Reason** Assertion: Ohm's law is applicable to all conducting elements. Reason: The V-I graph of a p-n junction diode is non-linear and depends on the sign of V.

- (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

✓ Answer Key & Quick Solutions

- 1(a) 2(d) 3(b) 4(b) 5(b) 6(d) 7(c) 8(b) 9(a) 10(c) 11(b) 12(d) 13(c) 14(c) 15(d)
 16(a) 17(c) 18(a) 19(a) 20(d) 21(a) 22(b) 23(a) 24(c)

- (a) $n = I / e = 1 / (1.6 \times 10^{-19}) = 6.25 \times 10^{18}$ per second.
- (d) $v_d = I / (neA) = 1.5 / (8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-7}) = 1.1 \times 10^{-3}$ m/s.
- (b) $\mu = v_d / E = (\text{m/s}) / (\text{V/m}) = \text{m}^2 \text{V}^{-1} \text{s}^{-1}$.
- (b) $v_d = eE\tau / m$ with $E = V / l \Rightarrow v_d \propto 1 / l \rightarrow$ halves.
- (b) Volume constant $\Rightarrow A$ becomes $A/3$; $R' = \rho(3l) / (A / 3) = 9R$.
- (d) In metals n is nearly constant; more frequent collisions reduce the relaxation time τ .
- (c) Carrier density n rises sharply with T , so ρ falls (negative α).
- (b) $3(1 + 20\alpha) = 1 + 500\alpha \Rightarrow \alpha = 1/220 \text{ } ^\circ\text{C}^{-1}$, $R_0 = 18.33 \text{ } \Omega$; $25 = 18.33(1 + t / 220) \Rightarrow t = 80 \text{ } ^\circ\text{C}$.
- (a) $j = \sigma E \Rightarrow E = 3 \times 10^6 / 6 \times 10^7 = 0.05 \text{ V/m}$.
- (c) R fixed $\Rightarrow P \propto V^2$: $100 \times (110 / 220)^2 = 25 \text{ W}$.
- (b) Series $\Rightarrow$ same I , $P = I^2 R$; the 60 W bulb has larger R (V^2 / P) so it dissipates more.
- (d) $R \rightarrow R/2$ at the same $V \Rightarrow P = V^2 / R$ doubles.
- (c) $6 \parallel 3 = 2 \text{ } \Omega$; in series with $2 \text{ } \Omega \Rightarrow 4 \text{ } \Omega$.
- (c) $2/4 = 3/6 \Rightarrow$ balanced; drop $10 \text{ } \Omega$: $(2+4) \parallel (3+6) = 54/15 = 3.6 \text{ } \Omega$.
- (d) $I = 2/4 = 0.5 \text{ A}$; $V = 2 - 0.5 \times 0.5 = 1.75 \text{ V}$.
- (a) Intercept gives $\varepsilon = 2 \text{ V}$; short-circuit current $\varepsilon/r = 4 \text{ A} \Rightarrow r = 0.5 \text{ } \Omega$.
- (c) Maximum at $R = r$: $P = \varepsilon^2 / 4r = 144 / 8 = 18 \text{ W}$.
- (a) $\text{emf} = (10 - 2)(1.5) = 12 \text{ V}$; $r = 10 \times 0.5 = 5 \text{ } \Omega$; $I = 12 / (5 + 1) = 2 \text{ A}$.
- (a) $\varepsilon_{eq} = (\varepsilon_1 r_2 + \varepsilon_2 r_1) / (r_1 + r_2) = (4 + 4) / 3 = 8 / 3 \text{ V} \approx 2.67 \text{ V}$.
- (d) Charge does not accumulate at a junction; net work round a closed loop (electrostatic field) is zero.
- (a) $X/6 = 40/60 \Rightarrow X = 4 \text{ } \Omega$.
- (b) Cells now oppose each other in one loop: $I = (10 - 4) / (2 + 4) = 1 \text{ A}$.
- (a) Its resistance hardly changes with temperature – exactly what a standard resistor needs.
- (c) Diodes, GaAs etc. are non-ohmic, so A is false; R is a true statement.

CHAPTER

4

Moving Charges and Magnetism

17

Day 7-8

Weightage ★★★

NCERT Class 12

Lorentz force

Circular & helical motion

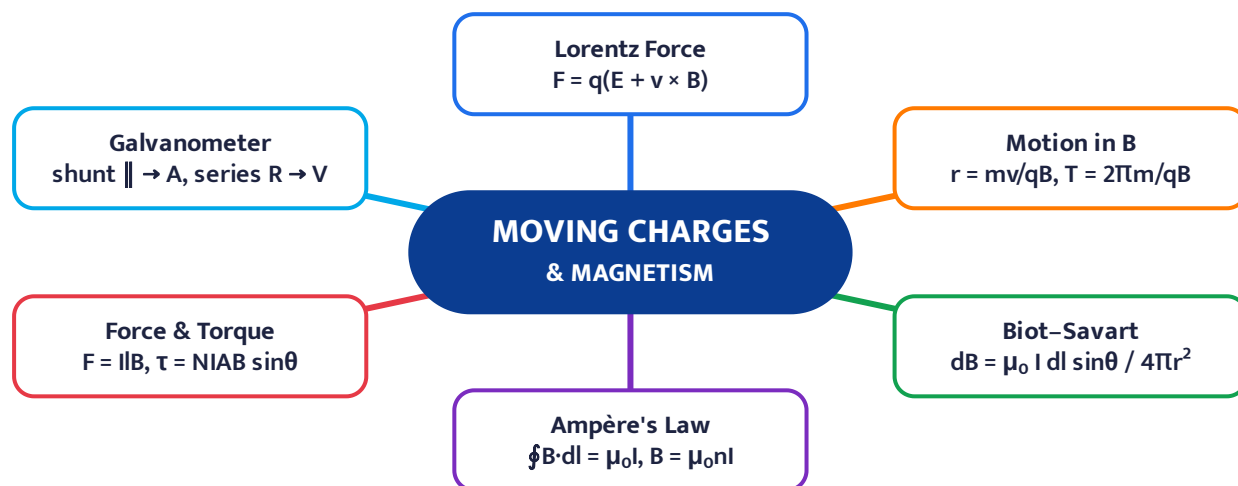
Biot-Savart law

Ampère's law

Force between currents

Torque on a loop

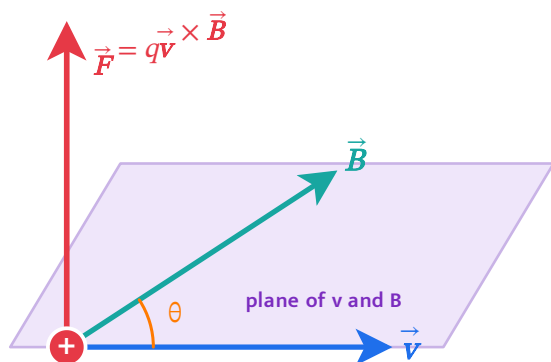
Galvanometer



Chapter at a glance

4.1 Magnetic Force and the Lorentz Force

Oersted (1820) found that a current-carrying wire deflects a nearby compass needle: **moving charges (currents) produce magnetic fields**. A charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ feels the **magnetic force** $\vec{F} = q(\vec{v} \times \vec{B})$. If an electric field is also present, the total force is the **Lorentz force**.

 $\vec{F} \perp \vec{v}$ and $\vec{F} \perp \vec{B} \Rightarrow$ no work $\vec{F}$ is perpendicular to the plane containing $\vec{v}$ and $\vec{B}$ 

Force on a moving charge

$$\vec{F} = q \left[\vec{E} + \vec{v} \times \vec{B} \right] \quad \text{Lorentz force}$$

$$F_B = qvB \sin\theta$$

Magnitude; θ = angle between $\vec{v}$ and $\vec{B}$

$$1 \text{ T} = 1 \text{ N s C}^{-1} \text{ m}^{-1} \quad \text{SI unit tesla; } 1 \text{ G} = 10^{-4} \text{ T}$$

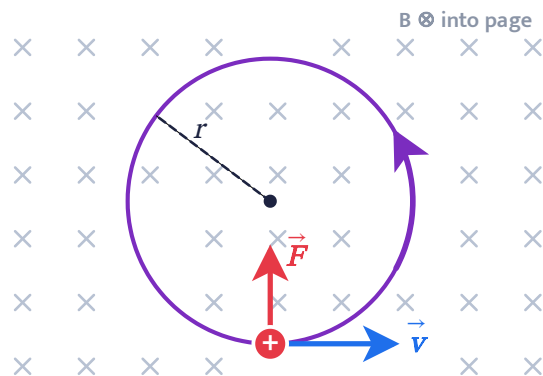


Key properties of the magnetic force

- Zero if the charge is at rest, or if $\vec{v} \parallel \vec{B}$ ($\theta = 0^\circ$ or 180°). Maximum qvB when $\vec{v} \perp \vec{B}$.
- Always $\perp$ to $\vec{v} \Rightarrow$ it does **no work**: speed and kinetic energy stay constant; only the direction of motion changes.
- For a **negative** charge (electron) the force is **opposite** to $\vec{v} \times \vec{B}$.

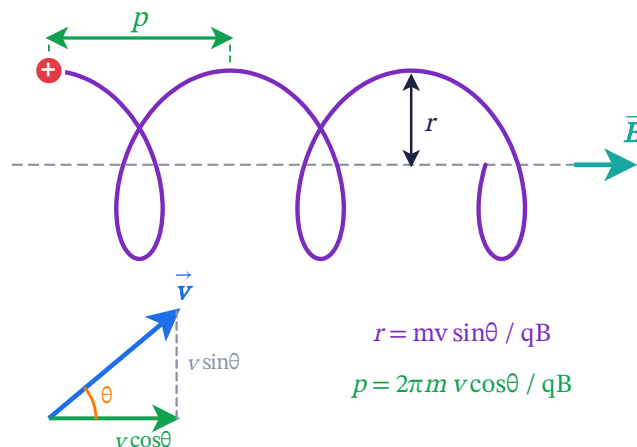
4.2 Motion of a Charge in a Uniform Magnetic Field

If $\vec{v} \perp \vec{B}$, the magnetic force qvB acts as the centripetal force and the charge moves in a **circle**. If $\vec{v}$ makes an angle θ with $\vec{B}$, the component $v \cos \theta$ (along $\vec{B}$) is unaffected while $v \sin \theta$ produces circular motion: the path is a **helix**.



$$r = mv/qB, T = 2\pi m/qB$$

$\vec{v} \perp \vec{B}$: uniform circular motion (positive charge, B into page)



$$r = mv \sin \theta / qB$$

$$p = 2\pi m v \cos \theta / qB$$

$\vec{v}$ at angle θ to $\vec{B}$: helical path of radius r and pitch p

Circular and helical motion

$$r = \frac{mv}{qB} = \frac{p}{qB}$$

Radius of circle (p = momentum)

$$r = \frac{\sqrt{2mK}}{qB} = \frac{\sqrt{2mqV}}{qB}$$

In terms of kinetic energy K , or accelerating p.d. V

$$T = \frac{2\pi m}{qB}, \quad \nu = \frac{qB}{2\pi m}$$

Period and frequency – **independent of v and r**

$$\omega = \frac{qB}{m}$$

Angular frequency

$$r = \frac{mv \sin \theta}{qB}$$

Radius of helix

$$p = \frac{2\pi m v \cos \theta}{qB}$$

Pitch = distance along $\vec{B}$ in one revolution

Solved Example

An electron ($m = 9 \times 10^{-31}$ kg) moves at 3×10^7 m/s perpendicular to a field of 6×10^{-4} T. Find the radius of its path, its frequency and its kinetic energy in keV.

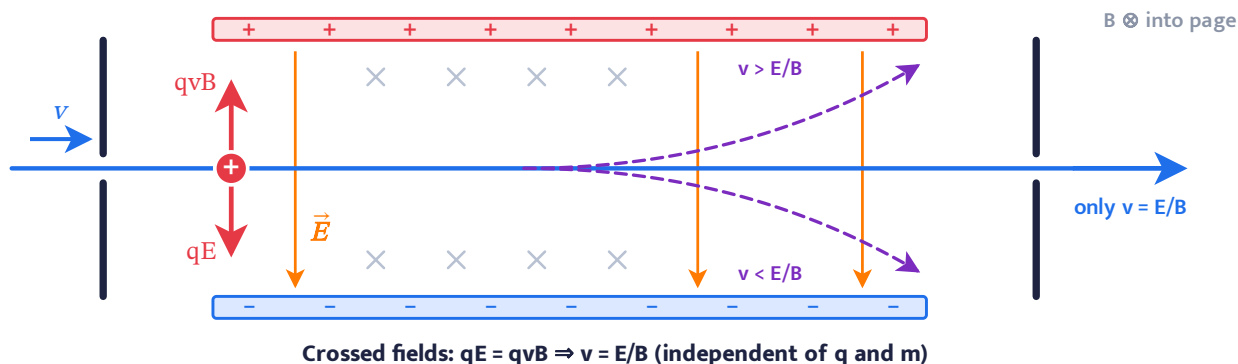
Solution: $r = \frac{mv}{eB} = \frac{9 \times 10^{-31} \times 3 \times 10^7}{1.6 \times 10^{-19} \times 6 \times 10^{-4}} = 0.28 \text{ m (28 cm)}$. $\nu = \frac{v}{2\pi r} \approx 1.7 \times 10^7 \text{ Hz} = 17 \text{ MHz}$. $K = \frac{1}{2}mv^2 = 4.05 \times 10^{-16} \text{ J} \approx 2.5 \text{ keV}$.

Common Trap

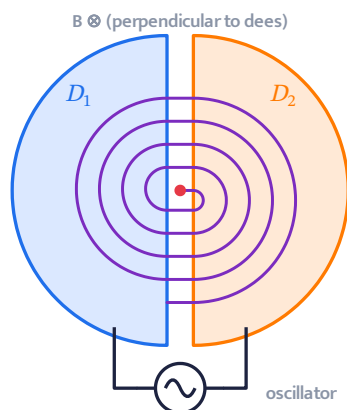
- **Same K :** $r \propto \frac{\sqrt{m}}{q}$ – proton and α -particle give **equal** radii. **Same V (accelerated from rest):** $r \propto \sqrt{m/q}$. **Same p :** $r \propto 1/q$.
- Doubling the speed doubles r but leaves T **unchanged**. A magnetic field can never change the kinetic energy of a charge.
- Electron and proton with the same velocity curve in **opposite senses**, and the electron circle is far smaller ($r \propto m$).

4.3 Crossed Fields: Velocity Selector

When $\vec{E}$, $\vec{B}$ and $\vec{v}$ are mutually perpendicular, the electric force $q\vec{E}$ and magnetic force $q\vec{v} \times \vec{B}$ act in opposite directions. Only particles with $v = E/B$ pass **undeflected** – whatever their charge or mass. This **velocity selector** is used in mass spectrometers and in J.J. Thomson's e/m experiment.



Velocity selector: E down, B into the page – forces balance only for $v = E/B$



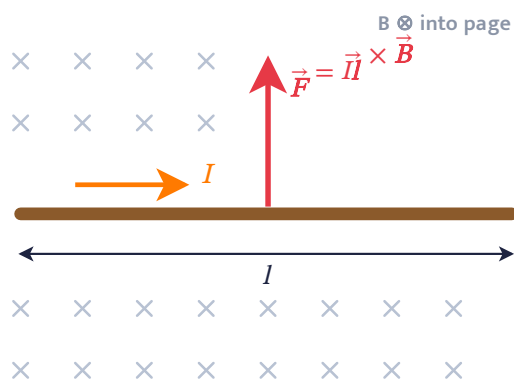
Cyclotron: ions spiral outwards in the dees



Cyclotron (short)

- Accelerates ions using **both** fields: a uniform $B \perp$ two hollow D-shaped electrodes (dees), and an AC voltage across the gap between them.
- Inside a dee there is no E ; the ion moves in a semicircle in time $T/2 = \pi m / qB$, independent of speed. **Resonance condition:** $v_a = v_c = \frac{qB}{2\pi m}$.
- At the exit radius R : $v = \frac{qBR}{m}$, $K_{max} = \frac{q^2 B^2 R^2}{2m}$ (does not depend on the AC voltage).
- Cannot accelerate electrons (mass grows at high speed, they fall out of step) or neutrons (no charge).

4.4 Force on a Current-Carrying Conductor



$F = IL \sin \theta$; direction: Fleming's left-hand rule

Force on a straight wire in a uniform field

A current is a stream of drifting charges, so a wire in a field feels the sum of their forces: $\vec{F} = I \vec{l} \times \vec{B}$, where $\vec{l}$ points along the current.



Force on a current

$$\vec{F} = I \vec{l} \times \vec{B}$$

Straight wire, uniform B

$$d\vec{F} = I d\vec{l} \times \vec{B}$$

Any shape (integrate)

VS Sir's Exam Tip

- In a **uniform** field, a curved wire feels the same force as the **straight line joining its ends**: a semicircle of radius R gives $F = 2IRB$.
- Hence a **closed loop** of any shape in a uniform field has **zero net force** (but may have a torque).

Solved Example

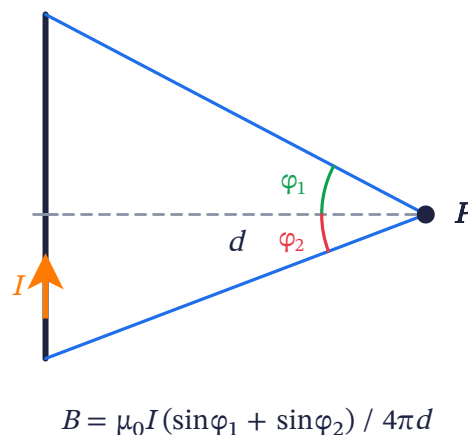
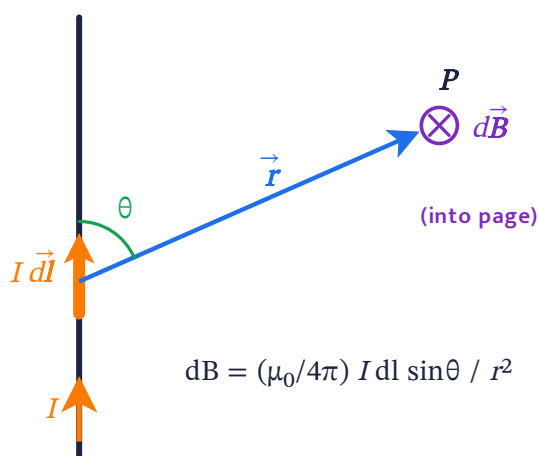
A horizontal rod of length 0.45 m and mass 60 g hangs from two vertical wires and carries 5.0 A. What magnetic field normal to the rod makes the tension in the wires zero? ($g = 9.8 \text{ m/s}^2$)

Solution: Need $IlB = mg \Rightarrow B = \frac{0.06 \times 9.8}{5.0 \times 0.45} = \mathbf{0.26 \text{ T}}$, horizontal and perpendicular to the rod, directed so that $\vec{I} \times \vec{B}$ points **upward**.

4.5 Biot–Savart Law

The field $d\vec{B}$ at a point P due to a small current element $I d\vec{l}$ is proportional to $I dl$ and $\sin\theta$, and inversely proportional to the square of the distance r . Its direction is that of $d\vec{l} \times \vec{r}$ (perpendicular to both).

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{r^3}, \quad dB = \frac{\mu_0}{4\pi} \frac{I dl \sin\theta}{r^2}$$



Left: a current element and $d\vec{B}$ at P. Right: angles used for a finite straight wire

Straight conductors

$$\frac{\mu_0}{4\pi} = 10^{-7} \text{ T m A}^{-1} \quad \mu_0 = \text{permeability of free space}$$

$$B = \frac{\mu_0 I}{4\pi d} (\sin\phi_1 + \sin\phi_2) \quad \text{Finite straight wire}$$

$$B = \frac{\mu_0 I}{2\pi d} \quad \text{Infinitely long wire (} \phi_1 = \phi_2 = 90^\circ \text{)}$$

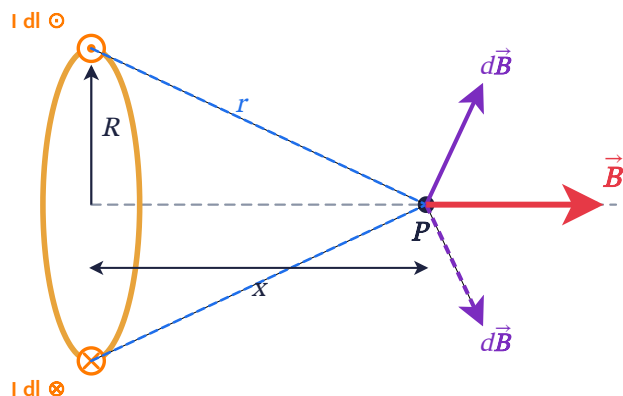
$$B = \frac{\mu_0 I}{4\pi d} \quad \text{Semi-infinite wire, P opposite its end}$$

Biot–Savart vs Coulomb

- Both are long-range, $\propto 1/r^2$, and obey superposition.
- Electric field: source is a **scalar** charge, field along $\vec{r}$. Magnetic: source is a **vector** $I d\vec{l}$, field $\perp$ to $d\vec{l}$ and $\vec{r}$; zero along the line of the element.
- $\frac{1}{\sqrt{\mu_0 \epsilon_0}} = c = \text{speed of light}$.

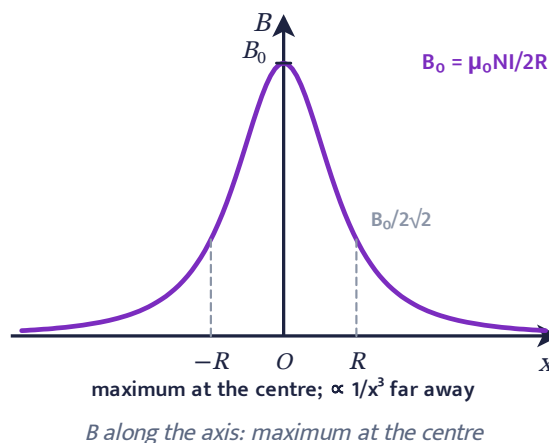
4.6 Circular Loop: Centre, Axis and Arc

For a loop of radius R , every element is $\perp$ to the line joining it to a point P on the axis. The components of $d\vec{B}$ perpendicular to the axis cancel in pairs (diametrically opposite elements) and only the **axial** components add.



$\perp$ components cancel; axial ones add

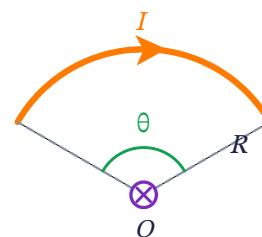
Field on the axis of a circular loop



B along the axis: maximum at the centre

Loop and arc

$B_{axis} = \frac{\mu_0 N I R^2}{2(R^2 + x^2)^{3/2}}$	On the axis at distance x (N turns)
$B_{centre} = \frac{\mu_0 N I}{2R}$	At the centre ($x = 0$)
$B \approx \frac{\mu_0 2m}{4\pi x^3}$	Far away ($x \gg R$), $m = NIA$ – like a dipole
$B_{arc} = \frac{\mu_0 I \theta}{4\pi R}$	Arc subtending θ (rad) at the centre



$$B = \frac{\mu_0 I \theta}{4\pi R}$$

(θ in radian)

Arc of angle θ : field at its centre O

Memory Trick

- **Right-hand thumb rule:** curl the fingers along the current in the loop – the thumb gives $\vec{B}$ at the centre (anticlockwise as seen $\Rightarrow B$ towards you).
- Arc family: full loop $\frac{\mu_0 I}{2R} \rightarrow$ semicircle $\frac{\mu_0 I}{4R} \rightarrow$ quarter circle $\frac{\mu_0 I}{8R}$. Straight parts pointing at the centre give **zero**.

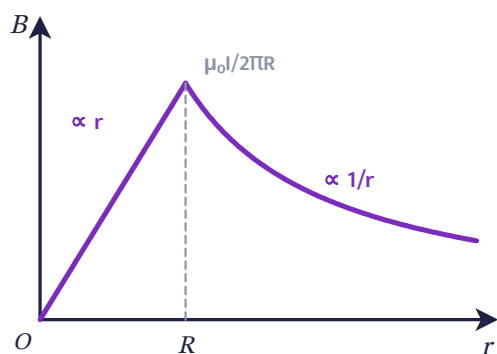
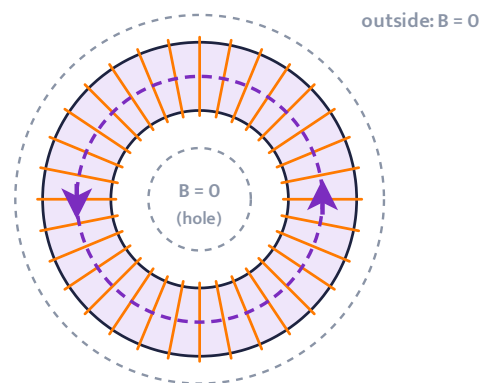
Solved Example

A 100-turn closely wound circular coil of radius 10 cm carries 3.2 A. Find B at its centre.

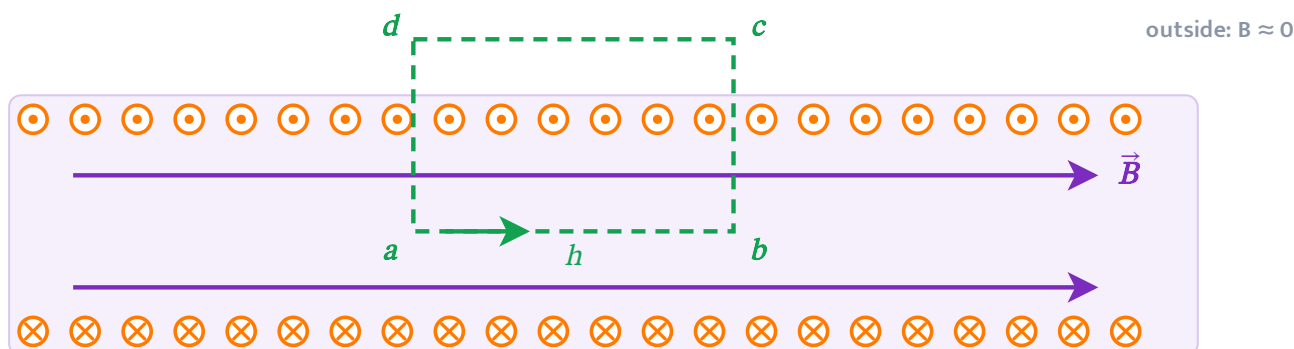
Solution: $B = \frac{\mu_0 N I}{2R} = \frac{4\pi \times 10^{-7} \times 100 \times 3.2}{2 \times 0.10} = 2.0 \times 10^{-3} \text{ T}$. (This coil is used again in Section 4.9.)

4.7 Ampère's Circital Law

The line integral of $\vec{B}$ around **any closed loop** equals μ_0 times the net current **threaded** by the loop: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$. Like Gauss's law, it is always true but is **useful** only when symmetry lets us take B out of the integral (B constant and tangential, or perpendicular, on parts of the loop).

Thick wire (radius R), uniform current B inside and outside a long thick wire $B = \mu_0 NI / 2\pi r$ (inside the core)

Toroid: dashed circles are Ampère loops

current $\odot$ out of page (top row), $\otimes$ into page (bottom row)

$$Bh = \mu_0(nh)I \Rightarrow B = \mu_0 nI$$

Long solenoid: Ampère rectangle $abcd$ with side $ab = h$ inside

Results of Ampère's law

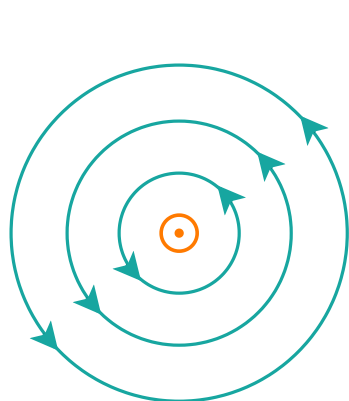
$B = \frac{\mu_0 I}{2\pi r}$	Long straight wire (outside, $r \geq R$)	$B = \frac{\mu_0 I r}{2\pi R^2}$	Inside a long thick wire ($r < R$), uniform current
$B = \mu_0 nI$	Inside a long solenoid; $n = N/L$ turns per metre	$B = \frac{\mu_0 nI}{2}$	At either end of a long solenoid
$B = \frac{\mu_0 NI}{2\pi r}$	Toroid (inside the core); zero in the hole and outside	$B = 0$	Inside a hollow pipe carrying current

NCERT Line to Remember

- The field inside a long solenoid is **uniform** and parallel to the axis; outside it is nearly **zero**. It does not depend on the radius or length – only on n and I .
- A solenoid behaves like a bar magnet; its field lines resemble those of a bar magnet.

4.8 Force Between Two Parallel Currents

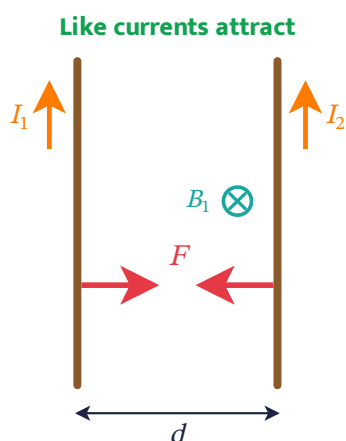
Wire 1 produces $B_1 = \frac{\mu_0 I_1}{2\pi d}$ at wire 2, which then feels $F = I_2 l B_1$. By Newton's third law wire 1 feels an equal and opposite force. **Parallel currents attract; antiparallel currents repel** – opposite to the rule for charges!



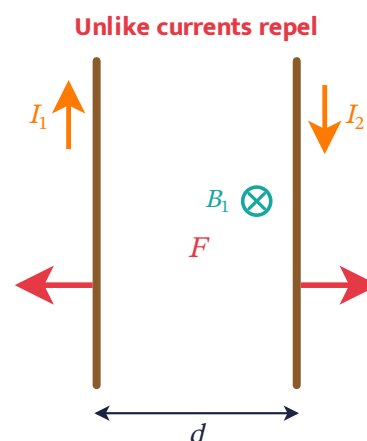
$I \odot$ out of page: B anticlockwise

$$B = \mu_0 I / 2\pi r$$

Field lines of a straight wire, and the force between parallel conductors



Like currents attract



Unlike currents repel

$$F/l = \mu_0 I_1 I_2 / 2\pi d$$

Parallel currents

$$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d} \quad \text{Force per unit length}$$

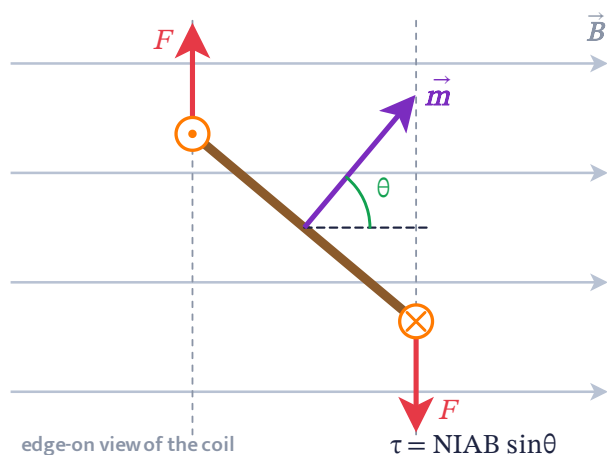
$$\frac{F}{l} = 2 \times 10^{-7} \text{ N m}^{-1} \quad \text{For 1 A in each wire, 1 m apart}$$

The ampere

One ampere is the steady current which, flowing in two very long, straight, parallel conductors of negligible cross-section placed 1 m apart in vacuum, produces a force of 2×10^{-7} N per metre of length on each. (Since 2019 the SI ampere is fixed through the value of e , but this force result still holds.)

4.9 Torque on a Current Loop; Magnetic Dipole

In a uniform field the net force on a loop is zero, but the forces on two opposite sides form a **couple**. For a rectangular loop (sides a, b) whose normal makes angle θ with $\vec{B}$, the arm of the couple is $b\sin\theta$, so $\tau = (IaB) b\sin\theta = IAB\sin\theta$. The loop behaves as a **magnetic dipole** of moment $\vec{m} = NI\vec{A}$ (direction by the right-hand rule).



edge-on view of the coil

$$\tau = NIAB \sin\theta$$

Couple on a coil: torque turns $\vec{m}$ towards $\vec{B}$

Dipole in a uniform field

$$\vec{m} = NI\vec{A} \quad \text{Magnetic dipole moment (A m}^2\text{)}$$

$$\vec{\tau} = \vec{m} \times \vec{B} \quad \tau = mB\sin\theta, \theta \text{ between } \vec{m} \text{ and } \vec{B}$$

$$U = -\vec{m} \cdot \vec{B} \quad \text{Potential energy: min at } \theta = 0 \text{ (stable), max at } 180^\circ$$

$$W_{\theta_1 \rightarrow \theta_2} = mB(\cos\theta_1 - \cos\theta_2) \quad \text{Work to rotate; } 0^\circ \rightarrow 180^\circ \text{ needs } 2mB$$

Common Trap

- θ is the angle between the **normal** ($\vec{m}$) and $\vec{B}$, not between the **plane** and $\vec{B}$. Plane of coil **parallel** to $B \Rightarrow \theta = 90^\circ \Rightarrow$ torque is **maximum** $NIAB$; plane $\perp B \Rightarrow$ torque zero.
- For a given wire length, a **circular** loop has the largest area and hence the largest m and torque.

Solved Example

The coil of Section 4.6 (100 turns, $R = 10$ cm, 3.2 A) is placed in a uniform horizontal field of 2 T, free to rotate. Find the torque when its axis is (a) along B, (b) perpendicular to B.

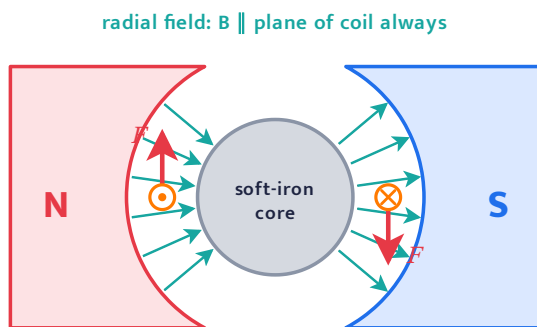
Solution: $m = NI\pi R^2 = 100 \times 3.2 \times \pi \times 0.01 = 10 \text{ A m}^2$. (a) $\theta = 0 \Rightarrow \tau = 0$. (b) $\theta = 90^\circ \Rightarrow \tau = mB = 20 \text{ N m}$.

Revolving electron as a magnetic dipole

An electron in a circular orbit is a current $I = \frac{ev}{2\pi r}$, so $\mu = I\pi r^2 = \frac{evr}{2} = \frac{e}{2m_e}L$ (gyromagnetic ratio $e / 2m_e$). With Bohr's $L = \frac{nh}{2\pi}$, the least value is the **Bohr magneton** $\mu_B = \frac{eh}{4\pi m_e} = 9.27 \times 10^{-24} \text{ A m}^2$.

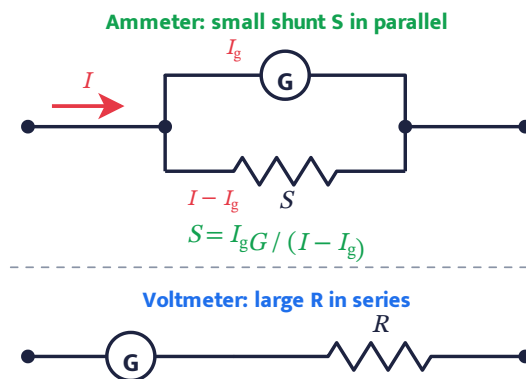
4.10 Moving-Coil Galvanometer, Ammeter and Voltmeter

A coil of N turns, area A , pivoted between **concave pole pieces** around a **soft-iron core**, sees a **radial** field: the plane of the coil is always parallel to B, so the deflecting torque is always $NIAB$. A spring of torsional constant k gives a restoring torque $k\phi$. In equilibrium $NIAB = k\phi \Rightarrow$ deflection $\phi \propto I$ (linear scale).



$$NIAB = k\phi \Rightarrow \phi \propto I$$

Radial field between concave poles and soft-iron core



$$R = V/I_g - G$$

Converting a galvanometer (resistance G , full-scale current I_g)

Galvanometer formulas

$\frac{\phi}{I} = \frac{NAB}{k}$	Current sensitivity (rad A^{-1} or div A^{-1})	$\frac{\phi}{V} = \frac{NAB}{kG}$	Voltage sensitivity (G = coil resistance)
$S = \frac{I_g G}{I - I_g}$	Shunt for an ammeter of range I (in parallel)	$R = \frac{V}{I_g} - G$	Series resistance for a voltmeter of range V
$R_A = \frac{GS}{G + S} \approx S$	Ammeter resistance – very small	$R_V = G + R$	Voltmeter resistance – very large

Solved Example

A galvanometer of resistance 12Ω shows full-scale deflection for 3 mA. How will you convert it into (a) a voltmeter of range 0–18 V, (b) an ammeter of range 0–6 A?

Solution: (a) $R = \frac{18}{3 \times 10^{-3}} - 12 = 5988 \Omega$ in series. (b) $S = \frac{3 \times 10^{-3} \times 12}{6 - 0.003} \approx 6 \times 10^{-3} \Omega$ (6 m Ω) in parallel.

! Common Trap

- Doubling N doubles the current sensitivity, but G also doubles $\Rightarrow$ **voltage sensitivity is unchanged.**
- A galvanometer cannot be used directly as an ammeter: it is too sensitive (μA) and its resistance would change the circuit current.

🧠 Memory Trick

- **Ammeter:** Add a small **Shunt Side-by-side** (parallel). **Voltmeter:** **Very** big R in series.
- Ideal ammeter $R = 0$; ideal voltmeter $R = \infty$.

🎯 Day 8 – Test (Chapters 1–4)

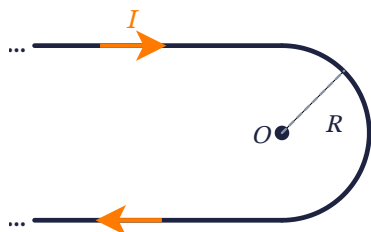
- First solve the 24 questions below in **30 minutes**, then re-attempt the practice questions of Chapters 1–3 that you got wrong.
- Log every mistake as concept / formula / calculation, and re-read the matching Trap box before Day 9.

📝 Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1** **NCERT based** The SI unit of magnetic field is the tesla. 1 T equals
 (a) 10^6 G (b) 10^4 G (c) 10^2 G (d) 10^{-4} G
- 2** **NCERT based** A charged particle moves in a uniform magnetic field. The magnetic force on it does no work because
 (a) the force is always parallel to the velocity
 (b) the force is zero inside a uniform field
 (c) the force is always perpendicular to the velocity
 (d) the charge of the particle changes
- 3** **NEET pattern** A proton ($m = 1.67 \times 10^{-27}$ kg) moves at 2×10^6 m/s perpendicular to a field of 0.1 T. The radius of its path is about
 (a) 21 m (b) 2.1 m (c) 0.21 m (d) 0.021 m
- 4** **NEET pattern** A proton and an α -particle with the same kinetic energy enter a uniform magnetic field perpendicularly. The ratio of their radii $r_p:r_\alpha$ is
 (a) 1:1 (b) 1:2 (c) $1:\sqrt{2}$ (d) 2:1
- 5** **NEET pattern** A charged particle moves in a circle in a uniform magnetic field. If its speed is doubled, its time period
 (a) remains the same (b) is doubled
 (c) is halved (d) becomes four times
- 6** **JEE Main pattern** A charged particle enters a uniform magnetic field with its velocity at 60° to the field. The ratio of the pitch of the helix to its radius is
 (a) $2\pi / \sqrt{3}$ (b) $2\sqrt{3} \pi$ (c) $\pi / \sqrt{3}$ (d) 2π

- 7 **NCERT based** In a velocity selector, $E = 3 \times 10^4$ V/m and $B = 0.02$ T are mutually perpendicular. The speed of particles that pass undeflected is
 (a) 6×10^2 m/s (b) 1.5×10^6 m/s (c) 1.5×10^4 m/s (d) 6×10^6 m/s
- 8 **JEE Main pattern** In a cyclotron with $B = 1$ T, the frequency of the oscillator needed to accelerate protons is about
 (a) 7.6 MHz (b) 30.5 MHz (c) 15.2 MHz (d) 95.8 MHz
- 9 **NEET pattern** A straight wire 0.5 m long carries 2 A and makes an angle of 30° with a uniform field of 0.4 T. The force on it is
 (a) 0.1 N (b) 0.4 N (c) 0.35 N (d) 0.2 N
- 10 **NEET pattern** A semicircular wire of radius R carrying current I lies in a plane perpendicular to a uniform field B . The net force on it is
 (a) πIRB (b) $2IRB$ (c) IRB (d) zero
- 11 **NCERT based** A circular coil of 100 turns and radius 10 cm carries 1 A. The magnetic field at its centre is
 (a) 6.28×10^{-5} T (b) 6.28×10^{-4} T (c) 3.14×10^{-4} T (d) 1.26×10^{-3} T
- 12 **NCERT based** The magnetic field at a distance of 10 cm from a long straight wire carrying 5 A is
 (a) 1×10^{-6} T (b) 1×10^{-5} T (c) 3.14×10^{-5} T (d) 5×10^{-5} T
- 13 **JEE Main pattern** A long wire is bent into a hairpin shape: two parallel semi-infinite straight parts joined by a semicircle of radius R (see figure). The field at the centre O is



- (a) $\frac{\mu_0 I}{4\pi R}(\pi + 2)$ (b) $\frac{\mu_0 I}{4R}$ (c) $\frac{\mu_0 I}{4\pi R}(\pi + 1)$ (d) $\frac{\mu_0 I}{4\pi R}(\pi - 2)$
- 14 **NEET pattern** The ratio of the magnetic field at the centre of a circular loop to that on its axis at a distance equal to its radius is
 (a) 2 : 1 (b) $2\sqrt{2} : 1$ (c) $\sqrt{2} : 1$ (d) $1 : 2\sqrt{2}$
- 15 **NCERT based** A solenoid 0.5 m long with 500 turns carries 5 A. The magnetic field inside it is about
 (a) 6.28×10^{-3} T (b) 3.14×10^{-3} T (c) 6.28×10^{-4} T (d) 1.26×10^{-2} T
- 16 **NEET pattern** A toroid of mean radius 10 cm has 500 turns and carries 2 A. The field in its core is
 (a) 1 mT (b) zero (c) 2 mT (d) 4 mT

- 17 **NEET pattern** Two long parallel wires 10 cm apart carry currents of 5 A and 10 A in the same direction. The force per unit length between them is
(a) 10^{-4} N/m, attractive (b) 10^{-5} N/m, attractive
(c) 2×10^{-4} N/m, attractive (d) 10^{-4} N/m, repulsive
- 18 **NEET pattern** A coil of 100 turns and area $2 \times 10^{-3} \text{ m}^2$ carries 2 A in a uniform field of 0.5 T. If the plane of the coil is parallel to the field, the torque on it is
(a) 0.4 N m (b) zero (c) 0.1 N m (d) 0.2 N m
- 19 **JEE Main pattern** A current loop of magnetic moment 0.5 A m^2 is aligned with a field of 0.4 T. The work needed to turn it through 180° is
(a) 0.2 J (b) 0.8 J (c) zero (d) 0.4 J
- 20 **NEET pattern** A galvanometer of resistance 100Ω gives full-scale deflection for 1 mA. To convert it into an ammeter of range 0–1 A, one must connect
(a) about 0.1Ω in series (b) about 999.9Ω in parallel
(c) about 0.1Ω in parallel (d) about 999.9Ω in series
- 21 **NEET pattern** A galvanometer of resistance 50Ω gives full-scale deflection for 2 mA. The resistance needed to convert it into a voltmeter of range 0–10 V is
(a) 4950Ω in series (b) 5000Ω in series
(c) 5050Ω in series (d) 4950Ω in parallel
- 22 **JEE Main pattern** The number of turns of a moving-coil galvanometer is doubled (so its resistance also doubles). Its voltage sensitivity
(a) is doubled (b) becomes four times
(c) is halved (d) remains unchanged
- 23 **Assertion-Reason** Assertion: A charged particle can move through a uniform magnetic field with constant velocity. Reason: The magnetic force on a charge moving parallel to the field is zero.
(a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 24 **Assertion-Reason** Assertion: The time period of a charged particle moving in a circle in a uniform magnetic field does not depend on its speed. Reason: The magnetic force does no work on the particle.
(a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false

✓ Answer Key & Quick Solutions

- 1(b) 2(c) 3(c) 4(a) 5(a) 6(a) 7(b) 8(c) 9(d) 10(b) 11(b) 12(b) 13(a) 14(b) 15(a)
16(c) 17(a) 18(d) 19(d) 20(c) 21(a) 22(d) 23(a) 24(b)

1. (b) $1 \text{ G} = 10^{-4} \text{ T}$, so $1 \text{ T} = 10^4 \text{ G}$.
2. (c) $\vec{F} = q\vec{v} \times \vec{B}$ is $\perp \vec{v}$, so $\vec{F} \cdot \vec{v} = 0$: power and work are zero.
3. (c) $r = mv / qB = 1.67 \times 10^{-27} \times 2 \times 10^6 / (1.6 \times 10^{-19} \times 0.1) = 0.21 \text{ m}$.
4. (a) $r = \sqrt{2mK} / qB \propto \sqrt{m} / q$; $r_p / r_\alpha = \frac{\sqrt{1} / 1}{\sqrt{4} / 2} = 1$.
5. (a) $T = 2\pi m / qB$ does not contain v (r doubles, path length doubles, time stays the same).
6. (a) $p / r = 2\pi v \cos \theta / (v \sin \theta) = 2\pi \cot 60^\circ = 2\pi / \sqrt{3} \approx 3.63$.
7. (b) $v = E / B = 3 \times 10^4 / 0.02 = 1.5 \times 10^6 \text{ m/s}$, the same for any q and m .
8. (c) $\nu = qB / 2\pi m = 1.6 \times 10^{-19} / (2\pi \times 1.67 \times 10^{-27}) \approx 1.52 \times 10^7 \text{ Hz}$. (95.8 MHz is $\omega/10^6$, not ν .)
9. (d) $F = IlB \sin \theta = 2 \times 0.5 \times 0.4 \times 0.5 = 0.2 \text{ N}$.
10. (b) In a uniform field the force equals that on the straight line joining the ends (length $2R$): $F = I(2R)B$.
11. (b) $B = \mu_0 NI / 2R = 4\pi \times 10^{-7} \times 100 \times 1 / 0.2 = 6.28 \times 10^{-4} \text{ T}$.
12. (b) $B = \mu_0 I / 2\pi d = 2 \times 10^{-7} \times 5 / 0.1 = 10^{-5} \text{ T}$.
13. (a) Semicircle $\frac{\mu_0 I}{4R}$ + two semi-infinite wires $2 \times \frac{\mu_0 I}{4\pi R}$, all into the page $\Rightarrow \frac{\mu_0 I}{4\pi R}(\pi + 2)$.
14. (b) $\frac{B_c}{B_{axis}} = \frac{(R^2 + x^2)^{3/2}}{R^3} = (2)^{3/2} = 2\sqrt{2}$ at $x = R$.
15. (a) $n = 500/0.5 = 1000 \text{ m}^{-1}$; $B = \mu_0 nI = 4\pi \times 10^{-7} \times 1000 \times 5 = 6.28 \times 10^{-3} \text{ T}$.
16. (c) $B = \mu_0 NI / 2\pi r = 2 \times 10^{-7} \times 500 \times 2 / 0.1 = 2 \times 10^{-3} \text{ T}$.
17. (a) $F / l = 2 \times 10^{-7} \times 5 \times 10 / 0.1 = 10^{-4} \text{ N/m}$; like currents attract.
18. (d) Plane $\parallel \vec{B} \Rightarrow \vec{m} \perp \vec{B} \Rightarrow \tau = NIAB = 100 \times 2 \times 2 \times 10^{-3} \times 0.5 = 0.2 \text{ N m}$ (maximum).
19. (d) $W = mB(\cos 0^\circ - \cos 180^\circ) = 2mB = 2 \times 0.5 \times 0.4 = 0.4 \text{ J}$.
20. (c) $S = I_g G / (I - I_g) = 10^{-3} \times 100 / 0.999 \approx 0.1 \Omega$, as a shunt in parallel.
21. (a) $R = V / I_g - G = 10 / (2 \times 10^{-3}) - 50 = 4950 \Omega$ in series.
22. (d) $\phi / V = NAB / (kG)$: N and G both double, so the ratio is unchanged (current sensitivity doubles).
23. (a) If $\vec{v} \parallel \vec{B}$, $qvB \sin 0^\circ = 0$, so the particle continues with constant velocity – R explains A .
24. (b) Both are true, but T is independent of v because $r \propto v$ ($T = 2\pi r / v = 2\pi m / qB$), not because the force does no work.

CHAPTER

5

Magnetism and Matter

17

Day 9

Weightage ★☆☆

NCERT Class 12

Bar magnet

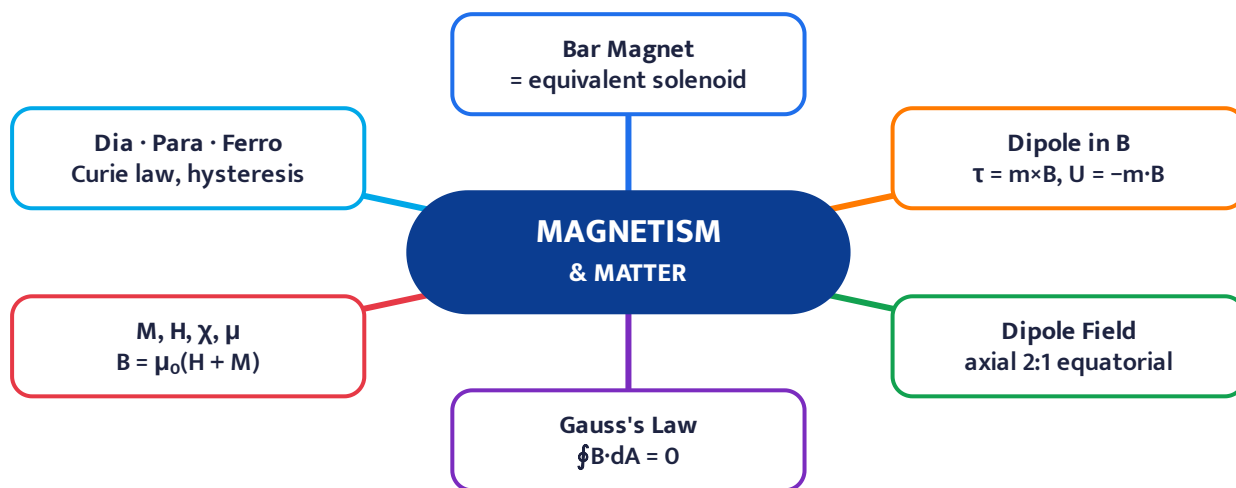
Dipole in field

Gauss's law

M, H, χ , μ

Dia • Para • Ferro

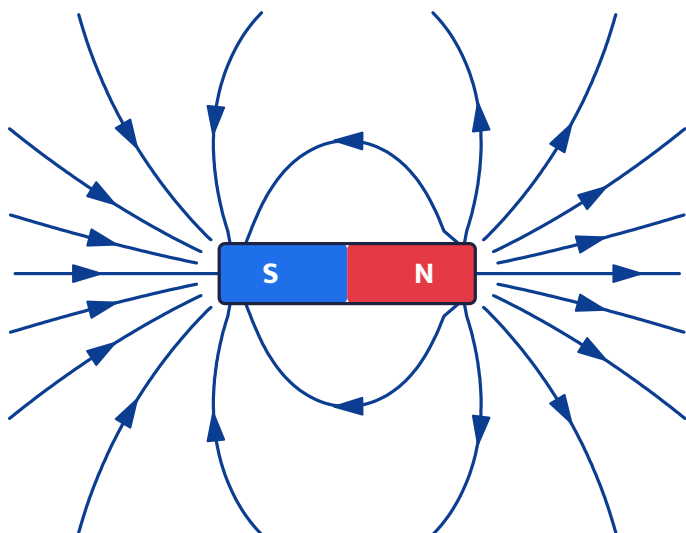
Hysteresis



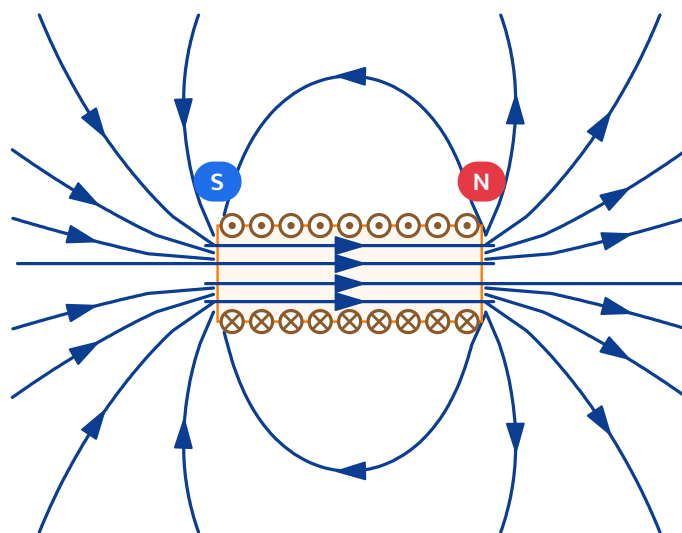
Chapter at a glance

5.1 The Bar Magnet and Magnetic Field Lines

A bar magnet has two poles, N and S, which **always occur in pairs** – cut a magnet into two and you get two smaller magnets, never an isolated pole. Its strength is described by the **magnetic dipole moment** $\vec{m}$ (SI unit $\text{A m}^2 = \text{J/T}$), directed from S to N inside the magnet.



Field lines of a bar magnet

Finite solenoid (current $\odot$ top, $\otimes$ bottom) – the same pattern

Use this **dipole analogy** (NCERT Table 5.1, summarised below): every electric-dipole result becomes a magnetic one by $\vec{p} \rightarrow \vec{m}, \vec{E} \rightarrow \vec{B}, 1/\epsilon_0 \rightarrow \mu_0$. The big difference: electric field lines begin and end on charges, magnetic field lines never end.

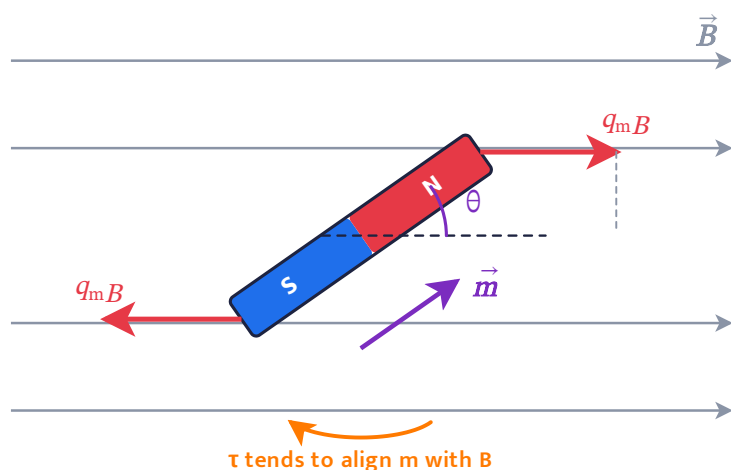
**Properties of B-field lines**

- They form **continuous closed loops** (outside N $\rightarrow$ S, inside S $\rightarrow$ N).
- Tangent at a point gives the direction of $\vec{B}$.
- Closer lines $\Rightarrow$ stronger field.
- Two lines **never intersect** (B would have two directions).

**Bar magnet = equivalent solenoid**

- Each turn of a solenoid is a current loop of moment IA ; for N turns $m = NIA$.
- At far axial points the solenoid field is $\frac{\mu_0 2m}{4\pi r^3}$ – identical to a bar magnet.
- So a bar magnet behaves as a collection of circulating (atomic) currents.

Electrostatics	Magnetism	Electrostatics	Magnetism
$1 / \epsilon_0$	μ_0	Dipole moment $\vec{p}$	Dipole moment $\vec{m}$
Axial $E = \frac{2p}{4\pi\epsilon_0 r^3}$	Axial $B = \frac{\mu_0 2m}{4\pi r^3}$	Torque $\vec{p} \times \vec{E}$	Torque $\vec{m} \times \vec{B}$
Equatorial $E = -\frac{p}{4\pi\epsilon_0 r^3}$	Equatorial $B = -\frac{\mu_0 m}{4\pi r^3}$	Energy $-\vec{p} \cdot \vec{E}$	Energy $-\vec{m} \cdot \vec{B}$

5.2 Magnetic Dipole in a Uniform Magnetic Field

$$\tau = mB \sin\theta$$

$$U = -mB \cos\theta$$

$$\theta = 0^\circ: U = -mB \text{ (stable)}$$

$$\theta = 180^\circ: U = +mB \text{ (unstable)}$$

$$\text{net force in uniform } \vec{B} = 0$$

Equal and opposite forces on the poles form a couple – torque but no net force

**Dipole in uniform $\vec{B}$**

$$\vec{\tau} = \vec{m} \times \vec{B}, \quad \tau = mB \sin\theta$$

Torque (max mB at $\theta = 90^\circ$)

$$U = -\vec{m} \cdot \vec{B} = -mB \cos\theta$$

Potential energy (zero at $\theta = 90^\circ$)

$$W_{\theta_1 \rightarrow \theta_2} = mB(\cos\theta_1 - \cos\theta_2)$$

Work done by external agent

$$W_{0 \rightarrow 180^\circ} = 2mB, \quad W_{0 \rightarrow 90^\circ} = mB$$

Flip completely / turn perpendicular

$$T = 2\pi \sqrt{\frac{I}{mB}}$$

Small oscillations of a freely suspended magnet (I = moment of inertia)

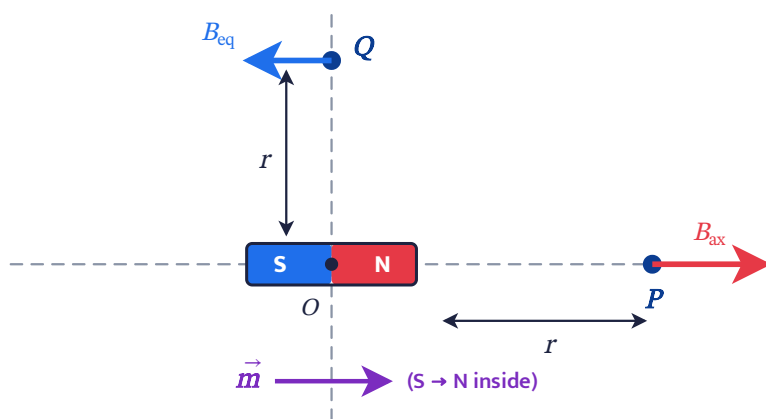
Solved Example

A short bar magnet placed with its axis at 30° to a uniform field of 0.25 T experiences a torque of 4.5×10^{-2} J. Find (i) its magnetic moment, (ii) the work needed to turn it from the most stable to the most unstable position.

Solution: (i) $m = \frac{\tau}{B \sin \theta} = \frac{0.045}{0.25 \times 0.5} = 0.36 \text{ J/T}$. (ii) $W = 2mB = 2(0.36)(0.25) = 0.18 \text{ J}$.

Common Trap

- Torque is **maximum at 90°** , but potential energy is **zero** there. U is minimum ($-mB$) at 0° , where torque is zero.
- In a **uniform** field net force on a dipole is zero (pure rotation); in a **non-uniform** field there is also a net force.

5.3 Field of a Magnetic Dipole – Axis and Equator

Short magnet ($r \gg l$)

$$B_{ax} = (\mu_0/4\pi) \cdot 2m/r^3 \text{ along } m$$

$$B_{eq} = (\mu_0/4\pi) \cdot m/r^3 \text{ opposite to } m$$

$$B_{ax} : B_{eq} = 2 : 1$$

At equal distance, the axial field is twice the equatorial field, and the two fields point in opposite directions

Dipole field ($\mu_0/4\pi = 10^{-7} \text{ T m/A}$)

$$B_{ax} = \frac{\mu_0}{4\pi} \frac{2mr}{(r^2 - l^2)^2} \approx \frac{\mu_0}{4\pi} \frac{2m}{r^3}$$

On the axis (magnet length $2l$)

$$B_{eq} = \frac{\mu_0}{4\pi} \frac{m}{(r^2 + l^2)^{3/2}} \approx \frac{\mu_0}{4\pi} \frac{m}{r^3}$$

On the equatorial line

$$B = \frac{\mu_0}{4\pi} \frac{m}{r^3} \sqrt{1 + 3\cos^2 \theta}$$

General point at angle θ from the axis

Solved Example

A short bar magnet has $m = 0.48 \text{ J/T}$. Find the field at 10 cm from its centre (i) on the axis, (ii) on the equatorial line.

Solution: (i) $B = 10^{-7} \times \frac{2(0.48)}{(0.1)^3} = 9.6 \times 10^{-5} \text{ T}$ along S→N. (ii) Half of it: $4.8 \times 10^{-5} \text{ T}$, directed N→S (opposite to $\vec{m}$).

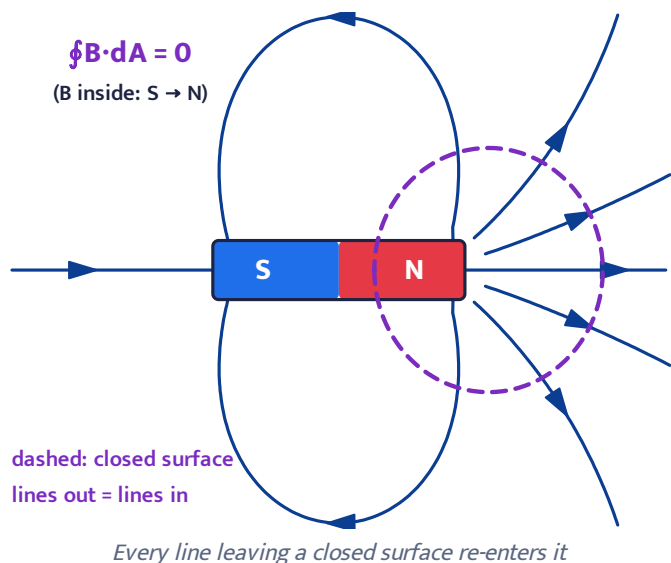
Memory Trick

"**Axis = 2, Equator = 1, both fall as $1/r^3$.**" Dipole fields fall as $1/r^3$; a single pole or charge would fall as $1/r^2$.

JEE Main Extension

- Magnet cut **perpendicular** to its length into 2 equal parts: each $m/2$ (length halves, pole strength same).
- Straight magnet bent into a semicircle: $m' = \frac{2m}{\pi}$ (pole distance becomes the diameter $2L/\pi$).

5.4 Gauss's Law in Magnetism



The net magnetic flux through **any** closed surface is zero:

$$\oint \vec{B} \cdot d\vec{A} = 0$$

NCERT Line to Remember

Gauss's law for magnetism says **isolated magnetic poles (monopoles) do not exist**. The simplest magnetic element is a dipole or current loop, so B-lines are closed loops.

5.5 Magnetisation, Intensity, Susceptibility, Permeability

Inside a material the atomic dipoles line up partly with the applied field. The **magnetisation** M is the net dipole moment per unit volume. The part of the field produced by external (free) currents is described by the **magnetic intensity** H .

The four material quantities

$M = \frac{m_{\text{net}}}{V}$	Magnetisation; unit A m^{-1} (vector)
$H = \frac{B}{\mu_0} - M, \quad B = \mu_0(H + M)$	Magnetic intensity; unit A m^{-1}
$M = \chi H$	χ = magnetic susceptibility (dimensionless)
$B = \mu_0(1 + \chi)H = \mu_0\mu_r H = \mu H$	Permeability μ ; relative permeability μ_r
$\mu_r = 1 + \chi = \frac{\mu}{\mu_0}$	Link between μ_r and χ
$H = nI, \quad B = \mu_0\mu_r nI$	Solenoid with a core: H depends only on the current

Solved Example

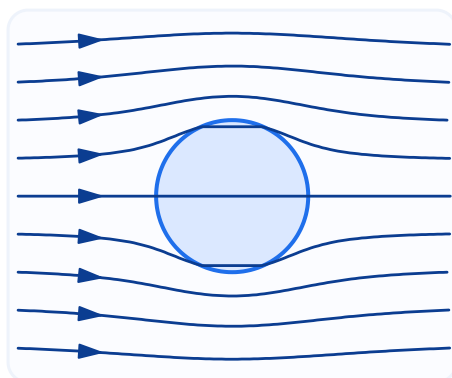
A solenoid with 1000 turns per metre carries 2 A and has an iron core of relative permeability 400. Find H , B and M in the core.

Solution: $H = nI = 2000 \text{ A/m}$. $B = \mu_0\mu_r H = 4\pi \times 10^{-7} \times 400 \times 2000 \approx 1.0 \text{ T}$. $M = (\mu_r - 1)H = 399 \times 2000 \approx 8 \times 10^5 \text{ A/m}$ – the core contributes almost all of B .

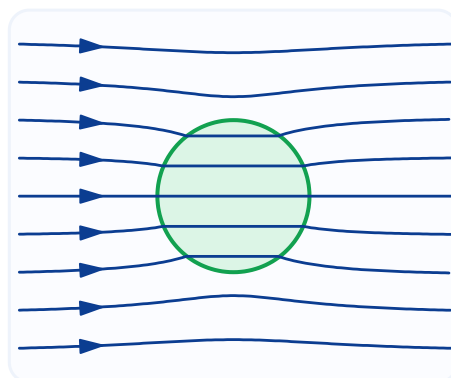
Common Trap

- H and M have the **same unit (A/m)**; B is in tesla. χ and μ_r are dimensionless, but μ has the unit of μ_0 ($\text{T m/A} = \text{H/m}$).
- Inserting a core does **not** change $H = nI$ – it changes M and hence B .

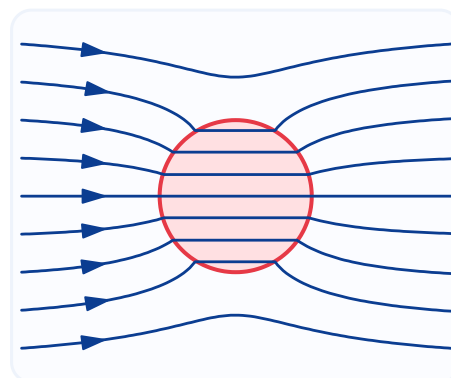
5.6 Dia-, Para- and Ferromagnetic Substances



Diamagnetic: lines pushed out



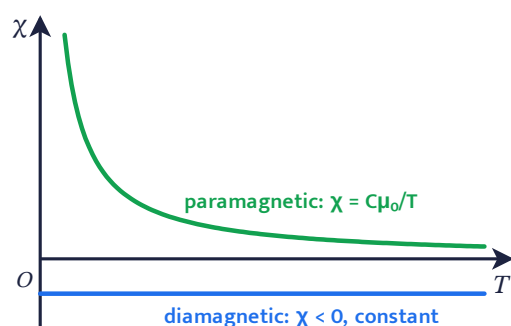
Paramagnetic: lines drawn in slightly



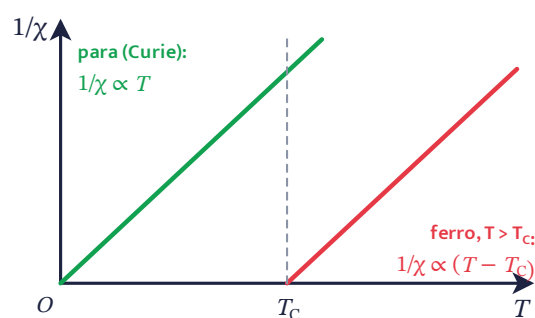
Ferromagnetic: lines crowd in strongly

Behaviour of the three classes of materials placed in a uniform external field

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Susceptibility χ	small, negative ($-1 \leq \chi < 0$)	small, positive ($0 < \chi < \epsilon$)	very large, positive ($\chi \gg 1$)
Relative permeability	$0 \leq \mu_r < 1$	$1 < \mu_r < 1 + \epsilon$	$\mu_r \gg 1$
Magnetisation M	weak, opposite to H	weak, along H	strong, along H
In a non-uniform field	moves strong $\rightarrow$ weak (repelled)	moves weak $\rightarrow$ strong (weakly attracted)	strongly attracted
Liquid in U-tube (one arm in field)	level falls	level rises	—
Effect of temperature	independent of T	Curie law $\chi \propto 1/T$	para above Curie temp. T_C
Origin	no permanent atomic moment; induced moment opposes B (Lenz-like)	permanent atomic moments, randomised by heat	domains – groups of aligned moments
Examples	Bi, Cu, Pb, Si, water, NaCl, N_2 (STP), superconductors	Al, Na, Ca, O_2 (STP), $CuCl_2$	Fe, Co, Ni, Gd, alnico



Para: $\chi \propto 1/T$; dia: $\chi < 0$, T -independent



$1/\chi$ vs T : straight lines (Curie and Curie-Weiss laws)



Temperature dependence

$$\chi = C \frac{\mu_0}{T} \quad \text{or} \quad M = C \frac{B_0}{T}$$

Curie law for paramagnets (C = Curie constant)

$$\chi = \frac{C}{T - T_C} \quad (T > T_C)$$

Ferromagnet above its Curie temperature

NCERT Line to Remember

- A **superconductor** is a perfect diamagnet: $\chi = -1$, $\mu_r = 0$ – it expels B completely (**Meissner effect**); used in magnetically levitated trains.
- Diamagnetism is present in **all** substances; it is masked in para/ferro materials.
- Curie temperatures: Fe 1043 K, Co 1394 K, Ni 631 K, Gd 317 K.

Memory Trick

"**Dia Dislikes, Para Pulls a little, Ferro Falls in love.**" Sign of χ : dia (-), para (+ small), ferro (+ huge). Only para **and** ferro (above T_C) care about temperature.

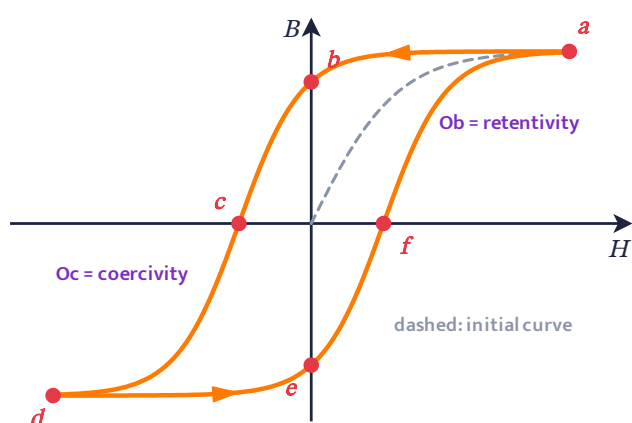
Solved Example

The susceptibility of a paramagnetic salt is 1.2×10^{-5} at 300 K. What is it at 200 K?

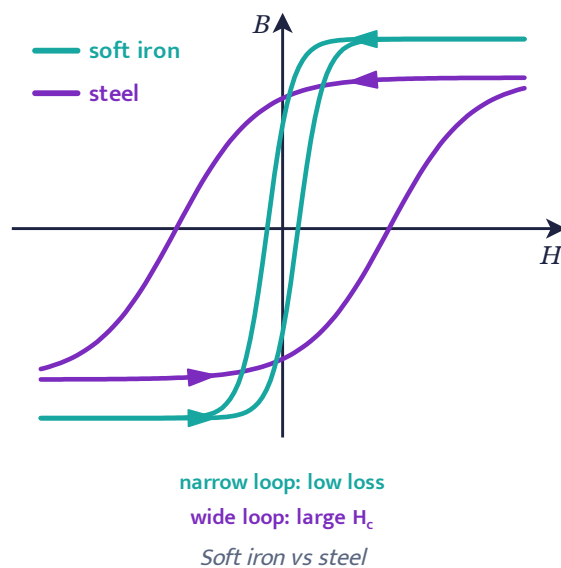
Solution: Curie law: $\chi_2 = \chi_1 \frac{T_1}{T_2} = 1.2 \times 10^{-5} \times \frac{300}{200} = 1.8 \times 10^{-5}$. (Always use kelvin!)

5.7 Hysteresis, Permanent Magnets and Electromagnets

When a ferromagnet is taken through a cycle of H, B does not retrace its path – **B lags behind H**. This is **hysteresis**; the closed B–H curve is the hysteresis loop.



Hysteresis loop $a \rightarrow b \rightarrow c \rightarrow d \rightarrow e \rightarrow f \rightarrow a$



- Retentivity (remanence) $B_r = Ob$:** B left when H is reduced to zero.
- Coercivity $H_c = Oc$:** reverse H needed to make B zero.
- Area of the loop** = energy dissipated as heat per unit volume per cycle.

Need	Desired properties	Material
Permanent magnet	high retentivity, high coercivity	steel, alnico, cobalt steel, ticonal
Electromagnet core	high permeability, low retentivity & coercivity	soft iron
Transformer / motor core	high μ , small loop area (low hysteresis loss), high resistivity	soft iron, silicon steel (laminated)

 VS Sir's Exam Tip

NEET loves one-liners: "**steel for permanent magnets, soft iron for electromagnets**". Choose steel because of its **large coercivity** (hard to demagnetise); choose soft iron because it magnetises and demagnetises easily with little loss.

 Common Trap

Retentivity alone does not decide a permanent magnet – soft iron's retentivity can even exceed that of steel. The deciding factor is **coercivity**.

 Earth's magnetism – quick facts

- Not in the rationalised NCERT text; know only the basics. Earth behaves like a huge dipole whose **geomagnetic S-pole is near the geographic North**; field $\approx 4 \times 10^{-5}$ T (0.4 G).
- Elements: **declination** D (angle between geographic and magnetic meridians), **dip** δ (angle of B with horizontal), $B_H = B \cos \delta$, $B_V = B \sin \delta$, $\tan \delta = B_V / B_H$.
- Dip is 0° at the magnetic equator and 90° at the magnetic poles.

 VS Sir's Exam Tip

Unit check that saves marks: m in $A\ m^2$ ($= J/T$); M and H in A/m ; B in T ; $\mu_0 = 4\pi \times 10^{-7}$ T m/A. If an answer for B comes out in A/m , you forgot the μ_0 .

 Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- NCERT based** Magnetic field lines of a bar magnet
 - intersect at the neutral point
 - are absent inside the magnet
 - start at N-pole and end at S-pole
 - form closed continuous loops
- NCERT based** Gauss's law in magnetism, $\oint \vec{B} \cdot d\vec{A} = 0$, implies that
 - B is zero inside any closed surface
 - B -lines never enter a surface
 - magnetic flux is always zero
 - isolated magnetic poles do not exist
- NEET pattern** A magnet of moment $2\ A\ m^2$ is placed at 30° to a uniform field of $0.5\ T$. The torque on it is
 - $0.87\ N\ m$
 - $0.25\ N\ m$
 - $0.5\ N\ m$
 - $1.0\ N\ m$
- NEET pattern** Work done in turning a magnet of moment $2\ J/T$ from the field direction to the opposite direction in a field of $0.1\ T$ is
 - $0.4\ J$
 - $0.1\ J$
 - $0.2\ J$
 - zero
- NEET pattern** For a short magnet, the ratio of field at a point on its axis to that at a point on its equatorial line at the same distance is
 - $2 : 1$
 - $4 : 1$
 - $1 : 2$
 - $1 : 1$

- 6 **NCERT based** A potential-energy minimum (stable equilibrium) of a magnetic dipole in a uniform field occurs when the angle between $\vec{m}$ and $\vec{B}$ is
(a) 45° (b) 180° (c) 0° (d) 90°
- 7 **NCERT based** Which of the following is diamagnetic?
(a) Nickel (b) Bismuth (c) Sodium (d) Aluminium
- 8 **NEET pattern** The susceptibility of a paramagnetic material is χ at 27°C . At 327°C it becomes
(a) 2χ (b) $\chi/2$ (c) $\chi/4$ (d) $\chi/12$
- 9 **NEET pattern** A material has susceptibility 599. Its permeability is ($\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$)
(a) $6.0 \times 10^2 \text{ T m/A}$ (b) $7.5 \times 10^{-5} \text{ T m/A}$ (c) $7.5 \times 10^{-4} \text{ T m/A}$ (d) $2.4 \times 10^{-4} \text{ T m/A}$
- 10 **NEET pattern** A long solenoid with 1000 turns/m carries 1 A and is filled with a core of $\mu_r = 1000$. The field inside is about
(a) 12.6 T (b) 0.126 T (c) 1.26 mT (d) 1.26 T
- 11 **NEET pattern** An iron bar of volume $5 \times 10^{-6} \text{ m}^3$ has magnetisation $8 \times 10^5 \text{ A/m}$. Its magnetic moment is
(a) 0.4 A m^2 (b) 40 A m^2 (c) 4 A m^2 (d) 1.6 A m^2
- 12 **NEET pattern** Above its Curie temperature a ferromagnetic substance becomes
(a) a superconductor (b) paramagnetic
(c) more strongly ferromagnetic (d) diamagnetic
- 13 **NEET pattern** The area enclosed by the hysteresis loop of a material represents
(a) energy lost per unit volume per cycle
(b) total magnetic moment
(c) retentivity
(d) permeability
- 14 **NEET pattern** The material best suited for making a permanent magnet should have
(a) high retentivity and low coercivity
(b) low retentivity and low coercivity
(c) high retentivity and high coercivity
(d) low retentivity and high permeability
- 15 **NCERT based** A small rod of a diamagnetic material is placed in a non-uniform magnetic field. It
(a) aligns along the field and is strongly attracted
(b) moves from weaker to stronger field region
(c) moves from stronger to weaker field region
(d) does not move at all
- 16 **JEE Main pattern** A bar magnet oscillates in a uniform field with period T . It is cut into two equal halves perpendicular to its length. Each half, oscillating in the same field, has period
(a) $T/\sqrt{2}$ (b) $2T$ (c) T (d) $T/2$

- 17 **JEE Main pattern** A thin straight bar magnet of moment m is bent into a semicircular arc. Its new magnetic moment is
 (a) m (b) m / π (c) $\pi m / 2$ (d) $2m / \pi$
- 18 **JEE Main pattern** A magnet with moment of inertia $2.5 \times 10^{-5} \text{ kg m}^2$ and moment 1 A m^2 oscillates in a uniform field of 0.1 T . Its time period is about
 (a) 0.01 s (b) 1 s (c) 0.3 s (d) 0.1 s
- 19 **Assertion-Reason** Assertion: Diamagnetism is present in all materials. Reason: Diamagnetism arises from orbital moments induced by the applied field that oppose it.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 20 **Assertion-Reason** Assertion: Magnetic field lines of a bar magnet start at its N-pole and end at its S-pole. Reason: Magnetic field lines are continuous closed curves.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

✓ Answer Key & Quick Solutions

- 1(d) 2(d) 3(c) 4(a) 5(a) 6(c) 7(b) 8(b) 9(c) 10(d) 11(c) 12(b) 13(a) 14(c) 15(c)
 16(d) 17(d) 18(d) 19(a) 20(c)

- (d) B-lines are closed loops: $N \rightarrow S$ outside and $S \rightarrow N$ inside the magnet.
- (d) Zero net flux through every closed surface $\Rightarrow$ no net "magnetic charge" (monopole) can be enclosed.
- (c) $\tau = mB \sin 30^\circ = 2 \times 0.5 \times 0.5 = 0.5 \text{ N m}$.
- (a) $W = mB(\cos 0^\circ - \cos 180^\circ) = 2mB = 0.4 \text{ J}$.
- (a) $B_{ax} = \frac{\mu_0 2m}{4\pi r^3}$, $B_{eq} = \frac{\mu_0 m}{4\pi r^3} \Rightarrow 2:1$.
- (c) $U = -mB \cos \theta$ is least ($-mB$) at $\theta = 0^\circ$.
- (b) Bi is diamagnetic; Al and Na are paramagnetic; Ni is ferromagnetic.
- (b) Curie law $\chi \propto 1/T$: $300 \text{ K} \rightarrow 600 \text{ K}$ halves χ . (Using $^\circ\text{C}$, $27/327 \approx 1/12$, is the classic mistake.)
- (c) $\mu_r = 1 + \chi = 600$; $\mu = 600 \times 4\pi \times 10^{-7} \approx 7.5 \times 10^{-4} \text{ T m/A}$.
- (d) $B = \mu_0 \mu_r nI = 4\pi \times 10^{-7} \times 1000 \times 1000 \times 1 \approx 1.26 \text{ T}$.
- (c) $m = MV = 8 \times 10^5 \times 5 \times 10^{-6} = 4 \text{ A m}^2$.
- (b) Thermal agitation destroys domain alignment; it then obeys $\chi = C / (T - T_C)$ – paramagnetic.
- (a) $\oint H \cdot dB$ over a cycle = heat dissipated per unit volume per cycle.
- (c) It must keep its magnetism (retentivity) and resist demagnetisation (coercivity) – e.g. steel, alnico.
- (c) Induced M is opposite to H , so energy is lowered by moving to weaker field – it is repelled.
- (d) Each half: $m' = m/2$, $I' = \frac{(M/2)(L/2)^2}{12} = I/8$. $T \propto \sqrt{I/m} \Rightarrow T' = T\sqrt{(1/8)/(1/2)} = T/2$.
- (d) Length $L = \pi R$; pole separation becomes $2R = 2L/\pi$; pole strength unchanged $\Rightarrow m' = 2m/\pi$.

18. (d) $T = 2\pi\sqrt{I/mB} = 2\pi\sqrt{2.5 \times 10^{-4}} = 2\pi(0.0158) \approx 0.099 \text{ s} \approx 0.1 \text{ s}.$

19. (a) Every atom has orbiting electrons, so every material shows this induced (Lenz-like) opposing moment – R explains A.

20. (c) Lines do not end at the poles – they continue inside from S to N. A is false, R is true.

CHAPTER

6

Electromagnetic Induction

17

Day 10

Weightage ★★☆☆

NCERT Class 12

Magnetic flux

Faraday's law

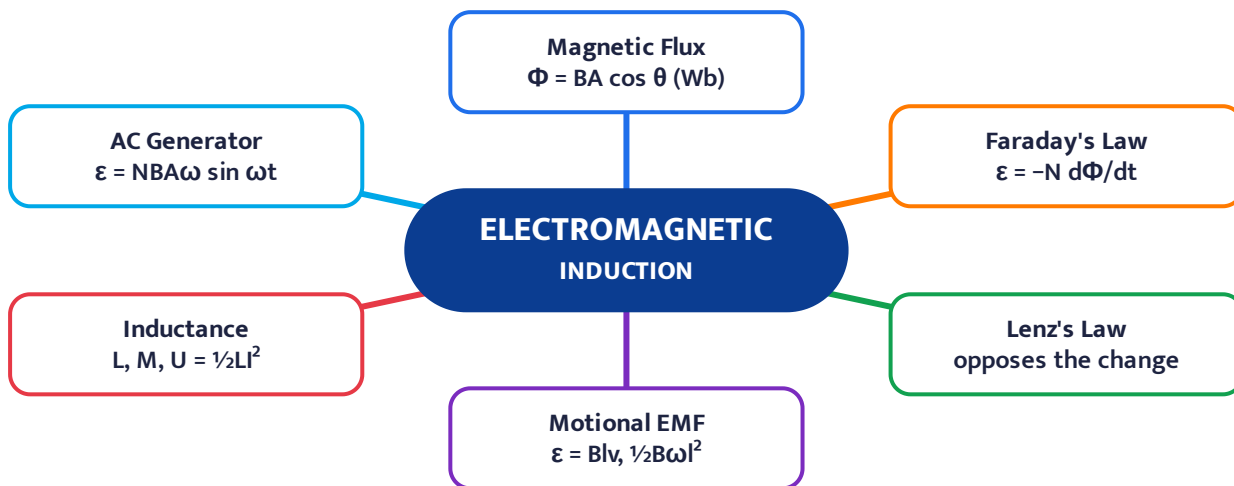
Lenz's law

Motional EMF

Eddy currents

Self & mutual inductance

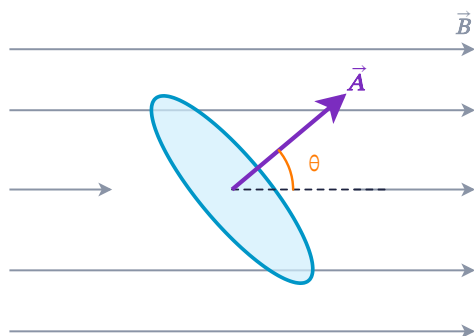
AC generator



Chapter at a glance

6.1 Faraday's Experiments and Magnetic Flux

Faraday (and independently Henry) showed that a **changing magnetic flux** through a coil drives a current in it – **electromagnetic induction**. No battery is needed; the EMF lasts only while the flux is changing.



$$\Phi = \vec{B} \cdot \vec{A} = BA \cos \theta$$

Flux through a plane loop

Magnetic flux

$$\Phi_B = \vec{B} \cdot \vec{A} = BA \cos \theta$$

Uniform B ; θ = angle between B and the area vector (normal)

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

Non-uniform B

$$1 \text{ Wb} = 1 \text{ T m}^2$$

SI unit weber; flux is a **scalar**

Common Trap

θ is measured from the **normal**, not from the plane. Coil plane $\perp B \Rightarrow \theta = 0, \Phi = BA$ (max). Coil plane $\parallel B \Rightarrow \Phi = 0$.

Experiment	What is done	Observation
1. Magnet–coil	Bar magnet pushed towards / pulled away from a coil with a galvanometer	Deflection only while moving; opposite for approach and recession; faster motion $\rightarrow$ larger deflection
2. Coil–coil	Magnet replaced by a current-carrying coil moved relative to the first	Same results – it is relative motion that matters
3. Key on/off	Both coils at rest; current in one switched on or off	Momentary deflection at make and break only (iron rod increases it)

6.2 Faraday's Law of Induction

The magnitude of the induced EMF in a circuit equals the time rate of change of magnetic flux through it. Flux can be changed by changing (i) B, (ii) the area A, or (iii) the orientation θ .

Faraday's law

$$\varepsilon = -N \frac{d\Phi_B}{dt}$$

N turns; minus sign = Lenz's law

$$I = \frac{\varepsilon}{R} = \frac{N d\Phi}{R dt}$$

Induced current in a circuit of resistance R

$$\Delta q = \frac{N \Delta \Phi}{R}$$

Induced charge – independent of how fast the flux changes

Solved Example

A 100-turn coil of area 0.01 m^2 is held perpendicular to a field of 0.5 T . The field is reduced uniformly to zero in 0.05 s . The coil resistance is 5Ω . Find the induced EMF, current and charge that flows.

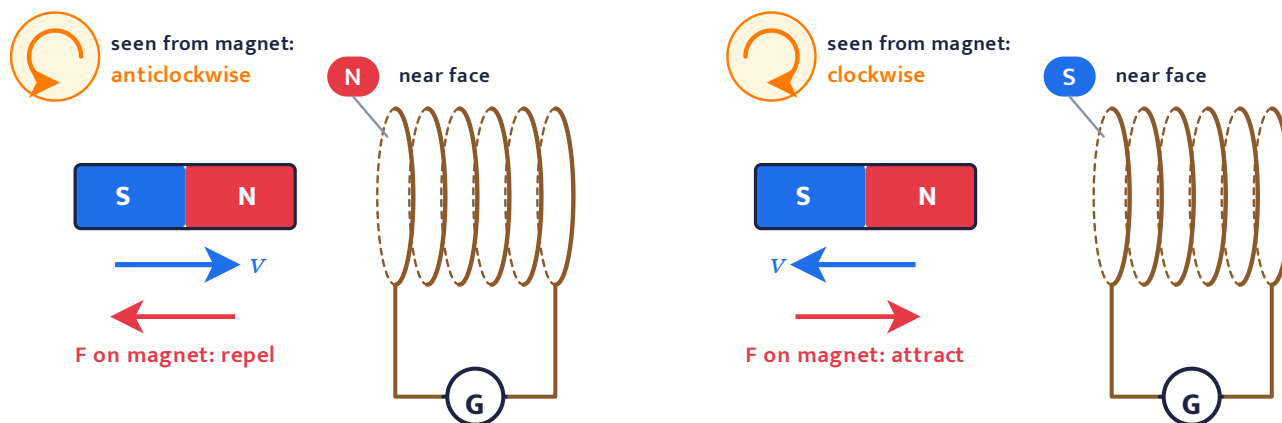
Solution: $\varepsilon = N \frac{\Delta(BA)}{\Delta t} = \frac{100 \times 0.5 \times 0.01}{0.05} = 10 \text{ V}$; $I = 10 / 5 = 2 \text{ A}$; $\Delta q = I \Delta t = \frac{N \Delta \Phi}{R} = \frac{100 \times 0.005}{5} = 0.1 \text{ C}$.

VS Sir's Exam Tip

If Φ is given as a function of time, just **differentiate**: $\Phi = 5t^2 + 3t + 2 \text{ mWb} \Rightarrow \varepsilon = 10t + 3 \text{ mV}$. Do not plug t into Φ itself – the EMF depends on the **rate**, not the value, of flux.

6.3 Lenz's Law and Conservation of Energy

The polarity of the induced EMF is such that it tends to produce a current which opposes the change in magnetic flux that produced it. This is the minus sign in Faraday's law.



N-pole approaching: near face becomes N – motion opposed

N-pole receding: near face becomes S – motion opposed

Why Lenz's law = energy conservation

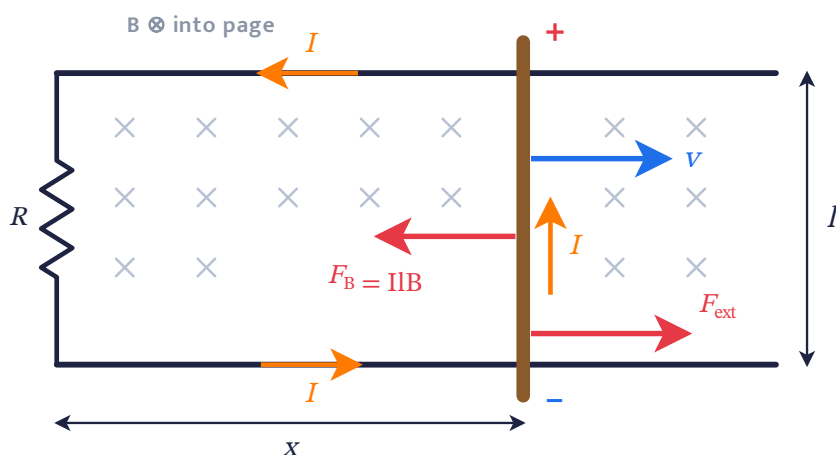
- If the induced current **helped** the change, the magnet would accelerate by itself and produce current for free – a perpetual-motion machine.
- Because it **opposes**, we must do mechanical work; that work appears as electrical energy (Joule heat).

Common Trap

- A magnet dropped through a closed metal ring (or copper tube) falls with $a < g$ – both while entering and leaving.
- Ring with a **cut** (open circuit): EMF is induced but no current $\Rightarrow$ no opposition $\Rightarrow a = g$.

6.4 Motional EMF

A conductor of length l moving with velocity v perpendicular to a uniform field B has an EMF Blv across it. Two views give the same answer: (i) the area of the circuit changes, $\Phi = Blx \Rightarrow \varepsilon = Bl \frac{dx}{dt} = Blv$; (ii) the Lorentz force $q\vec{v} \times \vec{B}$ pushes free charges along the rod.



$$\begin{aligned}\Phi &= Blx \\ \varepsilon &= Blv \\ I &= Blv / R \\ F_B &= B^2 l^2 v / R \\ P &= B^2 l^2 v^2 / R \\ &= \text{Joule heat in } R \\ &\quad (\text{anticlockwise current})\end{aligned}$$

Rod sliding on rails in a field into the page – the magnetic force IlB always opposes the motion

Motional EMF

$$\varepsilon = Blv \quad (\text{general: } \varepsilon = (\vec{v} \times \vec{B}) \cdot \vec{l}) \quad \text{Rod, } B, v \text{ mutually perpendicular}$$

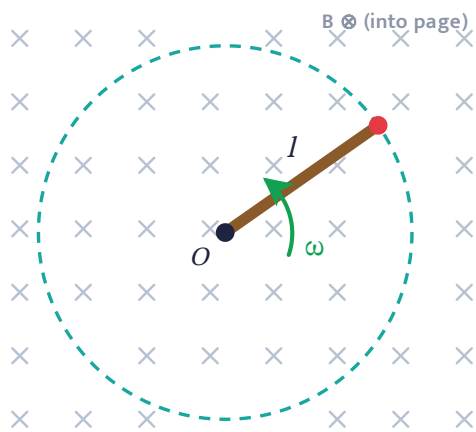
$$F = \frac{B^2 l^2 v}{R}, \quad P = Fv = \frac{B^2 l^2 v^2}{R} = I^2 R \quad \text{Force needed for constant } v; \text{ mechanical power} = \text{Joule heat}$$

$$v_t = \frac{FR}{B^2 l^2} \quad \text{Terminal speed under a constant pull } F$$

$$\varepsilon = \frac{1}{2} B \omega l^2 \quad \text{Rod of length } l \text{ rotating about one end (also a disc of radius } l)$$

Common Trap

- A loop moving **entirely inside** a uniform field has **no** induced EMF (flux constant) – EMF appears only while entering or leaving.
- Use only the component of v perpendicular to both B and the rod; if $v \parallel l$ or $v \parallel B$, $\varepsilon = 0$.



$$\varepsilon = \frac{1}{2} B\omega l^2$$

Rotating rod: average speed $\omega l/2$

Solved Example

A rod of length 0.5 m slides at 5 m/s on rails ($R = 2 \Omega$) in a field of 0.4 T. Find ε , I , the force needed and the power.

Solution: $\varepsilon = Blv = 0.4 \times 0.5 \times 5 = 1 \text{ V}$; $I = 0.5 \text{ A}$;

$F = IlB = 0.5 \times 0.5 \times 0.4 = 0.1 \text{ N}$; $P = Fv = 0.5 \text{ W} = I^2 R \checkmark$.

Solved Example

A 1 m metal rod rotates at 400 rad/s about one end in a field of 0.5 T normal to its plane of rotation. EMF between its ends?

Solution: $\varepsilon = \frac{1}{2} B\omega l^2 = \frac{1}{2} \times 0.5 \times 400 \times 1^2 = 100 \text{ V}$.

6.5 Eddy Currents

When a **bulk piece of metal** is in a changing flux, induced currents circulate inside it like whirlpools – **eddy (Foucault) currents**. By Lenz's law they oppose the motion (damping) and heat the metal.

Useful applications

- **Magnetic braking** in trains (electromagnets near the rails).
- **Electromagnetic damping** – dead-beat galvanometers (coil on a metal core).
- **Induction furnace** – melting metals; induction cooking.
- **Electric power meters** (analogue, rotating aluminium disc).

Minimising eddy losses

- Use **laminated cores** – thin sheets insulated by lacquer, planes parallel to B – in transformers and motors.
- Cutting **slots** in a swinging metal plate reduces the damping sharply.

6.6 Inductance – Mutual and Self

Flux linked with a coil is proportional to the current producing it. The constant of proportionality is the **inductance**; it depends only on **geometry and the medium (μ_r)**, not on the current. SI unit: henry ($1 \text{ H} = 1 \text{ Wb/A} = 1 \text{ V s/A}$).

Inductance formulas

$$N_2 \Phi_2 = M I_1, \quad \varepsilon_2 = -M \frac{dI_1}{dt}$$

Mutual inductance M ($M_{12} = M_{21}$)

$$M = \mu_0 n_1 n_2 \pi r_1^2 l \quad (\times \mu_r \text{ with a core})$$

Co-axial solenoids (r_1 = inner radius)

$$N\Phi = LI, \quad \varepsilon = -L \frac{dI}{dt}$$

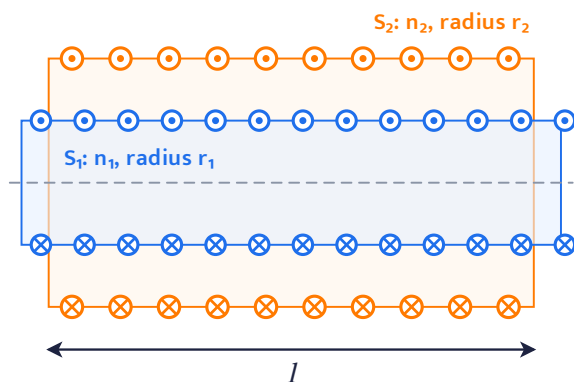
Self-inductance L ; the EMF is a "back EMF"

$$L = \mu_0 n^2 A l = \frac{\mu_0 N^2 A}{l} \quad (\times \mu_r)$$

Long solenoid ($n = N/l$)

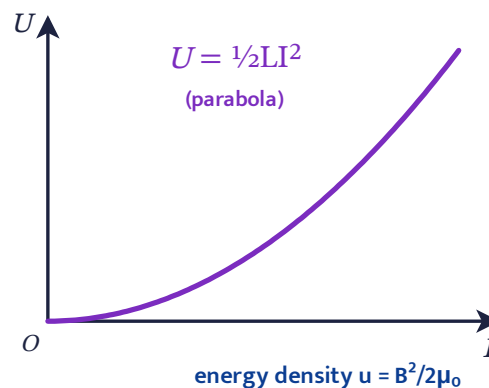
$$U = \frac{1}{2} L I^2, \quad u_B = \frac{B^2}{2\mu_0}$$

Energy stored; energy per unit volume



$$M = \mu_0 n_1 n_2 \pi r_1^2 l$$

Two long co-axial solenoids of length l



Energy stored in an inductor

Solved Example

A solenoid of 1000 turns, length 0.5 m and cross-section 10 cm^2 carries 2 A. Find its self-inductance and the energy stored.

Solution: $L = \frac{\mu_0 N^2 A}{l} = \frac{4\pi \times 10^{-7} \times 10^6 \times 10^{-3}}{0.5} \approx 2.5 \text{ mH}$; $U = \frac{1}{2} LI^2 = \frac{1}{2} (2.51 \times 10^{-3}) (4) \approx 5.0 \text{ mJ}$.

Memory Trick

L is electrical inertia: mass resists change in velocity, L resists change in current.

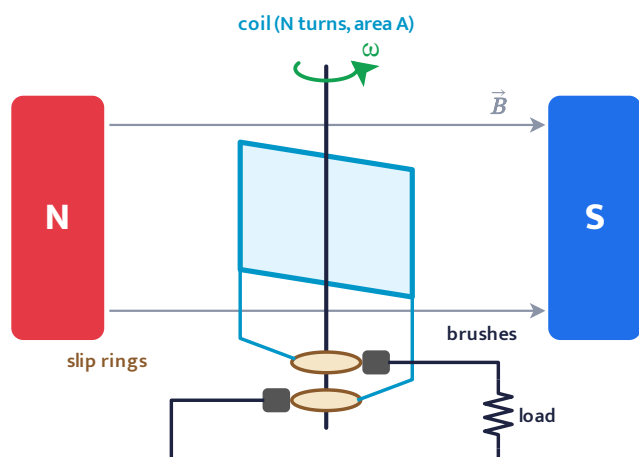
$$\frac{1}{2} LI^2 \leftrightarrow \frac{1}{2} mv^2, \text{ and } L \propto N^2 \text{ just as } m \text{ adds up.}$$

JEE Main Extension

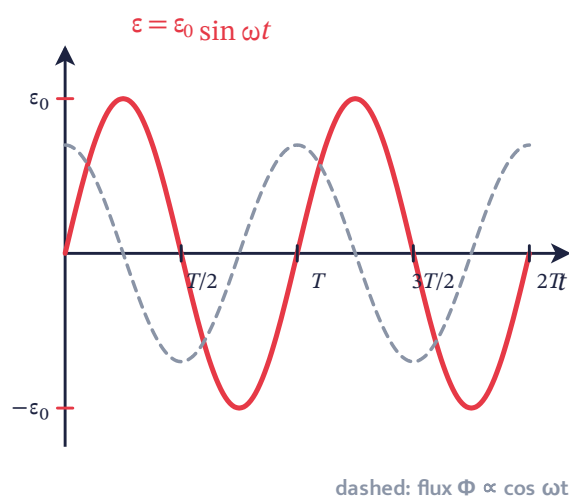
- Coupling: $M = k\sqrt{L_1 L_2}$, $0 \leq k \leq 1$.
- No coupling: series $L = L_1 + L_2$; parallel $\frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2}$.
- LR growth: $I = I_0(1 - e^{-t/\tau})$, decay $I = I_0 e^{-t/\tau}$, $\tau = L / R$.

6.7 AC Generator

A coil of N turns and area A is rotated with angular speed ω in a uniform field B . The angle between the area vector and B is $\theta = \omega t$, so $\Phi = BA \cos \omega t$ and the EMF is sinusoidal. **Slip rings** and **brushes** connect the rotating coil to the external circuit. Mechanical energy $\rightarrow$ electrical energy.



Coil rotating between the poles; slip rings + brushes



EMF is maximum when the flux is zero

AC generator

$$\varepsilon = NBA\omega \sin \omega t = \varepsilon_0 \sin \omega t$$

Instantaneous EMF

$$\varepsilon_0 = NBA\omega = NBA(2\pi\nu)$$

Peak EMF – when the coil plane is parallel to B

Solved Example

A 100-turn coil of area 0.1 m^2 rotates at 0.5 rev/s in a field of 0.01 T perpendicular to the axis of rotation. Find the maximum EMF.

Solution: $\varepsilon_0 = NBA(2\pi\nu) = 100 \times 0.01 \times 0.1 \times 2\pi \times 0.5 = \mathbf{0.314 \text{ V}}$. The average EMF over a full cycle is zero.

VS Sir's Exam Tip

Generator phase trap: coil plane $\perp \mathbf{B} \Rightarrow \Phi$ maximum, $\varepsilon = 0$; coil plane $\parallel \mathbf{B} \Rightarrow \Phi = 0$, ε maximum. EMF and flux are 90° out of phase – the dashed and solid curves above.

Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1 **NCERT based** The SI unit of magnetic flux is
 (a) T m^{-2} (b) tesla (c) henry (d) weber (T m^2)
- 2 **NEET pattern** The flux through a coil varies as $\Phi = (5t^2 + 3t + 2) \text{ mWb}$. The induced EMF at $t = 2 \text{ s}$ is
 (a) 23 mV (b) 20 mV (c) 28 mV (d) 13 mV
- 3 **NEET pattern** A 50-turn coil of area 0.02 m^2 is perpendicular to a field that rises from 0.1 T to 0.5 T in 0.1 s . The average induced EMF is
 (a) 0.08 V (b) 4 V (c) 8 V (d) 2 V
- 4 **NCERT based** Lenz's law is a consequence of the law of conservation of
 (a) charge (b) mass (c) momentum (d) energy
- 5 **NEET pattern** A bar magnet is dropped vertically through a closed horizontal copper ring. Its acceleration while passing through the ring is
 (a) more than g (b) equal to g (c) zero (d) less than g
- 6 **NEET pattern** A 1 m long rod moves at 2 m/s perpendicular to its length and to a field of 0.5 T . The EMF across its ends is
 (a) 4 V (b) 1 V (c) 0.25 V (d) 2 V
- 7 **NEET pattern** A 0.5 m rod rotates about one end at 20 rad/s in a perpendicular field of 0.2 T . The EMF between its ends is
 (a) 0.25 V (b) 2 V (c) 0.5 V (d) 1 V

- 8 **NEET pattern** The flux through each turn of a 100-turn coil ($R = 10\ \Omega$) changes by 2×10^{-3} Wb. The charge that flows is
(a) depends on the time taken (b) 0.2 C
(c) 0.02 C (d) 0.002 C
- 9 **NCERT based** Eddy current losses in a transformer core are reduced by
(a) using a laminated core (b) increasing the number of turns
(c) using a thicker iron block (d) using a solid copper core
- 10 **NEET pattern** If the number of turns of a solenoid is doubled keeping its length and area the same, its self-inductance becomes
(a) half (b) four times (c) double (d) unchanged
- 11 **NEET pattern** The current in a 50 mH coil falls uniformly from 4 A to 0 in 0.1 s. The induced EMF is
(a) 0.5 V (b) 0.2 V (c) 20 V (d) 2 V
- 12 **NEET pattern** The energy stored in an inductor of 0.2 H carrying 3 A is
(a) 1.8 J (b) 0.3 J (c) 0.6 J (d) 0.9 J
- 13 **JEE Main pattern** A long solenoid has 2000 turns/m. A coil of 100 turns and radius 1 cm is placed coaxially inside it. The mutual inductance is about
(a) 25 μH (b) 7.9 μH (c) 0.79 mH (d) 79 μH
- 14 **JEE Main pattern** A generator coil has 100 turns of area $0.05\ \text{m}^2$ and rotates at 50 rev/s in a field of 0.2 T. The peak EMF is about
(a) 628 V (b) 50 V (c) 100 V (d) 314 V
- 15 **JEE Main pattern** A rod of length 1 m on smooth horizontal rails ($R = 2\ \Omega$) in a vertical field of 0.5 T is pulled by a constant force of 0.5 N. Its terminal speed is
(a) 2 m/s (b) 4 m/s (c) 8 m/s (d) 1 m/s
- 16 **NEET pattern** The magnetic energy density in a region where $B = 0.2\ \text{T}$ is about
(a) $3.2 \times 10^4\ \text{J m}^{-3}$ (b) $8.0 \times 10^3\ \text{J m}^{-3}$ (c) $1.6 \times 10^{-2}\ \text{J m}^{-3}$ (d) $1.6 \times 10^4\ \text{J m}^{-3}$
- 17 **NCERT based** In an AC generator the EMF induced in the rotating coil is maximum when the plane of the coil is
(a) parallel to the magnetic field
(b) at 45° to the field
(c) perpendicular to the magnetic field
(d) at any angle – EMF is constant
- 18 **JEE Main pattern** A square loop of side 10 cm moves at 2 m/s through a region of uniform field 0.5 T that is 30 cm wide (field $\perp$ loop). The EMF while (i) entering and (ii) completely inside the field is
(a) 0, 0.1 V (b) 0.1 V, 0 (c) 0.2 V, 0 (d) 0.1 V, 0.1 V

19 **Assertion-Reason** Assertion: When a magnet is moved towards a closed coil, work has to be done against a repulsive force. Reason: The induced current opposes the change in flux, in accordance with conservation of energy.

- (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

20 **Assertion-Reason** Assertion: The self-inductance of a coil increases with the current through it. Reason: The flux linked with a coil is $N\Phi = LI$.

- (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

✓ Answer Key & Quick Solutions

- 1(d) 2(a) 3(b) 4(d) 5(d) 6(b) 7(c) 8(c) 9(a) 10(b) 11(d) 12(d) 13(d) 14(d) 15(b)
 16(d) 17(a) 18(b) 19(a) 20(c)

- (d) $\Phi = BA \Rightarrow \text{T m}^2 = \text{weber (Wb)}$.
- (a) $\varepsilon = d\Phi / dt = 10t + 3 = 23 \text{ mV}$ at $t = 2 \text{ s}$.
- (b) $\varepsilon = N\Delta B / \Delta t = 50 \times 0.02 \times 0.4 / 0.1 = 4 \text{ V}$.
- (d) Opposition means work must be done to produce the induced current – energy is conserved.
- (d) Induced current opposes the relative motion (Lenz) $\Rightarrow$ upward magnetic force $\Rightarrow a < g$.
- (b) $\varepsilon = Blv = 0.5 \times 1 \times 2 = 1 \text{ V}$.
- (c) $\varepsilon = \frac{1}{2}B\omega l^2 = \frac{1}{2}(0.2)(20)(0.25) = 0.5 \text{ V}$.
- (c) $\Delta q = N\Delta\Phi / R = 100 \times 2 \times 10^{-3} / 10 = 0.02 \text{ C}$ – independent of time.
- (a) Insulated thin laminations break the large eddy-current loops.
- (b) $L = \mu_0 N^2 A / l \propto N^2 \Rightarrow 4L$.
- (d) $|\varepsilon| = L\Delta I / \Delta t = 0.05 \times 4 / 0.1 = 2 \text{ V}$.
- (d) $U = \frac{1}{2}LI^2 = \frac{1}{2}(0.2)(9) = 0.9 \text{ J}$.
- (d) $M = \mu_0 nN\pi r^2 = 4\pi \times 10^{-7} \times 2000 \times 100 \times \pi \times 10^{-4} \approx 7.9 \times 10^{-5} \text{ H}$.
- (d) $\varepsilon_0 = NBA(2\pi\nu) = 100 \times 0.2 \times 0.05 \times 2\pi \times 50 \approx 314 \text{ V}$.
- (b) At terminal speed $F = B^2 l^2 v / R \Rightarrow v = FR / (B^2 l^2) = 0.5 \times 2 / 0.25 = 4 \text{ m/s}$.
- (d) $u = B^2 / 2\mu_0 = 0.04 / (2 \times 4\pi \times 10^{-7}) \approx 1.6 \times 10^4 \text{ J m}^{-3}$.
- (a) $\varepsilon = NBA\omega \sin\omega t$ is maximum when flux is zero, i.e. coil plane $\parallel B$.
- (b) Entering: $\varepsilon = Bav = 0.5 \times 0.1 \times 2 = 0.1 \text{ V}$. Fully inside: flux constant $\Rightarrow 0$.
- (a) The near face becomes a like pole (Lenz) and repels; that work becomes electrical energy – R explains A.
- (c) L depends only on geometry and $\mu_r (\mu_0 N^2 A / l)$; flux grows with I but L stays constant. A false, R true.

CHAPTER

7

Alternating Current

Day 11-12

Day 11-12

Weightage ★★☆☆

NCERT Class 12

Peak & rms values

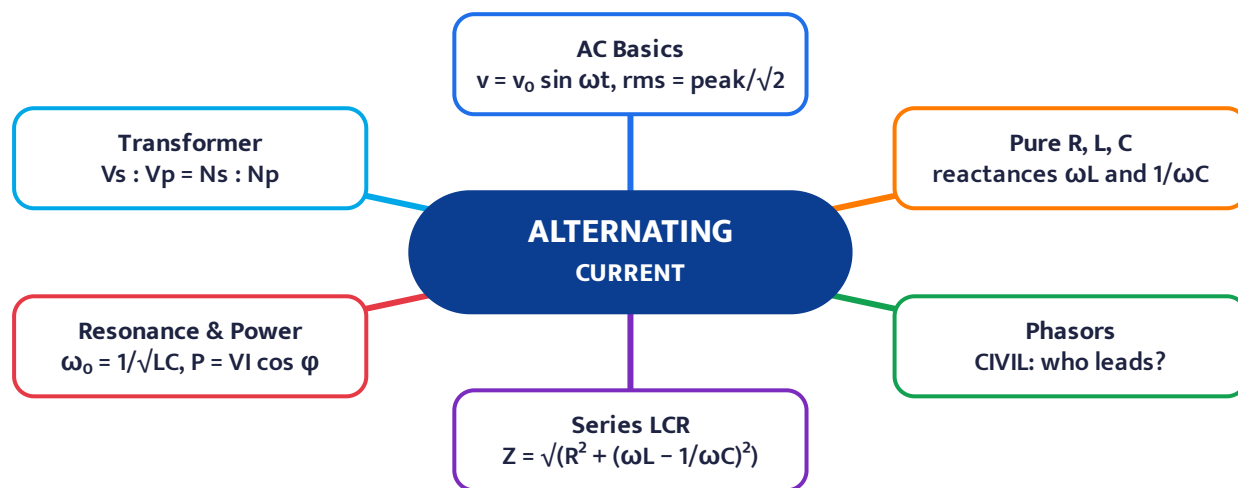
Phasors

R, L, C circuits

Series LCR & resonance

Power factor

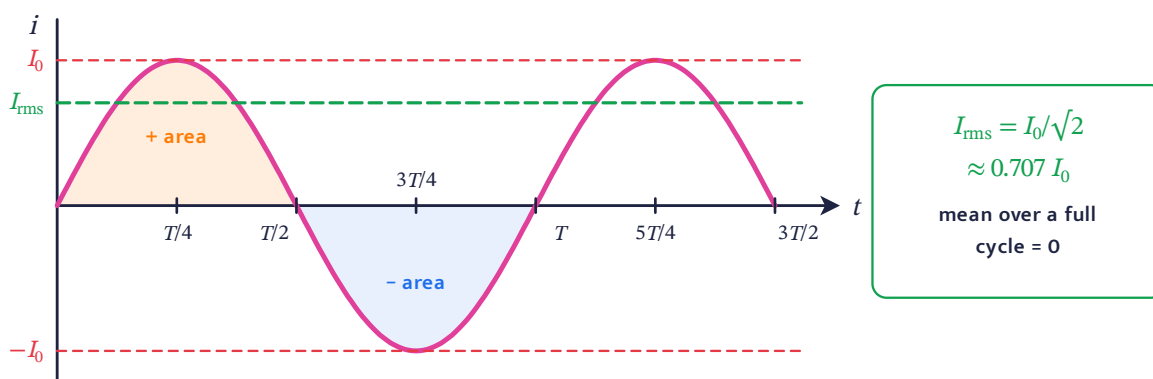
Transformer



Chapter at a glance

7.1 AC Voltage, Peak and rms Values

An **alternating voltage** changes its magnitude continuously and reverses its direction periodically. The simplest (and the one NCERT uses) is sinusoidal: $v = v_m \sin \omega t$, where v_m is the **peak value (amplitude)** and $\omega = 2\pi\nu$. Such a voltage is produced by an **AC generator** (a coil rotating in a magnetic field, $\varepsilon = NBA\omega \sin \omega t$). Indian mains: 220 V, **50 Hz**.



Sinusoidal AC: positive and negative half-cycles cancel, so the mean over a full cycle is zero – hence we use the rms value

The **rms (root-mean-square) or effective value** is the steady (DC) current that would produce the **same heat** in a resistor in the same time. Since the average of $\sin^2 \omega t$ over a cycle is $\frac{1}{2}$, the average power in a resistor is $\bar{P} = \frac{1}{2} i_m^2 R = I^2 R$.

**Peak, rms and average**

$$I = I_{rms} = \frac{i_m}{\sqrt{2}} \approx 0.707 i_m$$

rms (effective) current

$$V = V_{rms} = \frac{v_m}{\sqrt{2}}$$

rms voltage – what AC meters read

$$\bar{i}_T = 0, \quad \bar{i}_{T/2} = \frac{2i_m}{\pi} \approx 0.637 i_m$$

Average over a full cycle (T) and over a half cycle (T/2)

$$\frac{I_{rms}}{\bar{i}_{T/2}} = \frac{\pi}{2\sqrt{2}} \approx 1.11$$

Form factor of a sine wave**Solved Example**

The mains supply is 220 V, 50 Hz. Find (i) the peak voltage, (ii) ω , (iii) the time taken by the voltage to rise from zero to its peak.

Solution: (i) $v_m = \sqrt{2} \times 220 \approx 311 \text{ V}$. (ii) $\omega = 2\pi(50) \approx 314 \text{ rad/s}$. (iii) Zero $\rightarrow$ peak is a quarter cycle:
 $T/4 = \frac{1}{4 \times 50} = 5 \text{ ms}$.

**Common Trap**

- "220 V AC" is the **rms** value. The insulation must withstand the **peak** 311 V.
- Average of AC over a full cycle is **zero**, but average **power** is not.
- $I_{rms} = i_m / \sqrt{2}$ holds only for **sinusoidal** AC.

**JEE Main Extension**

For a general current $i(t)$: $I_{rms} = \sqrt{\text{mean of } i^2}$. Example: $i = I_1 + I_2 \sin \omega t \Rightarrow I_{rms} = \sqrt{I_1^2 + \frac{I_2^2}{2}}$. For a square wave of height i_0 , $I_{rms} = i_0$; for a half-wave rectified sine, $I_{rms} = i_m / 2$.

7.2 Phasors and AC through R, L and C

A **phasor** is a vector that rotates anticlockwise about the origin with angular speed ω . Its length is the peak value (v_m or i_m) and its **projection on the vertical axis** gives the instantaneous value. The angle between the V and I phasors is the **phase difference** ϕ . Phasors let us add voltages that are out of phase like vectors.

**Pure R, L and C (reactances in ohm)**

$$i = \frac{v_m}{R} \sin \omega t$$

R: current in phase with voltage; $\bar{P} = VI$

$$X_L = \omega L = 2\pi \nu L, \quad i = \frac{v_m}{X_L} \sin \left(\omega t - \frac{\pi}{2} \right)$$

L: inductive reactance; current **lags** by $\pi/2$; $\bar{P} = 0$

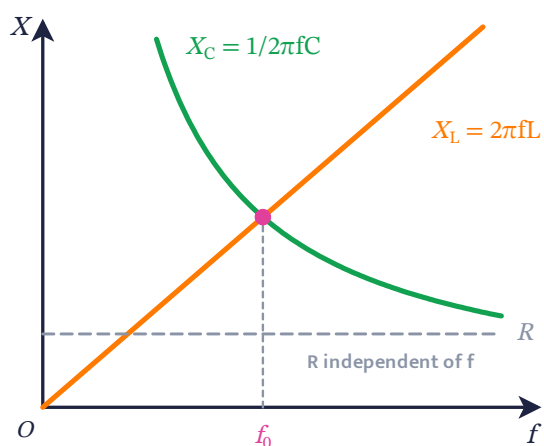
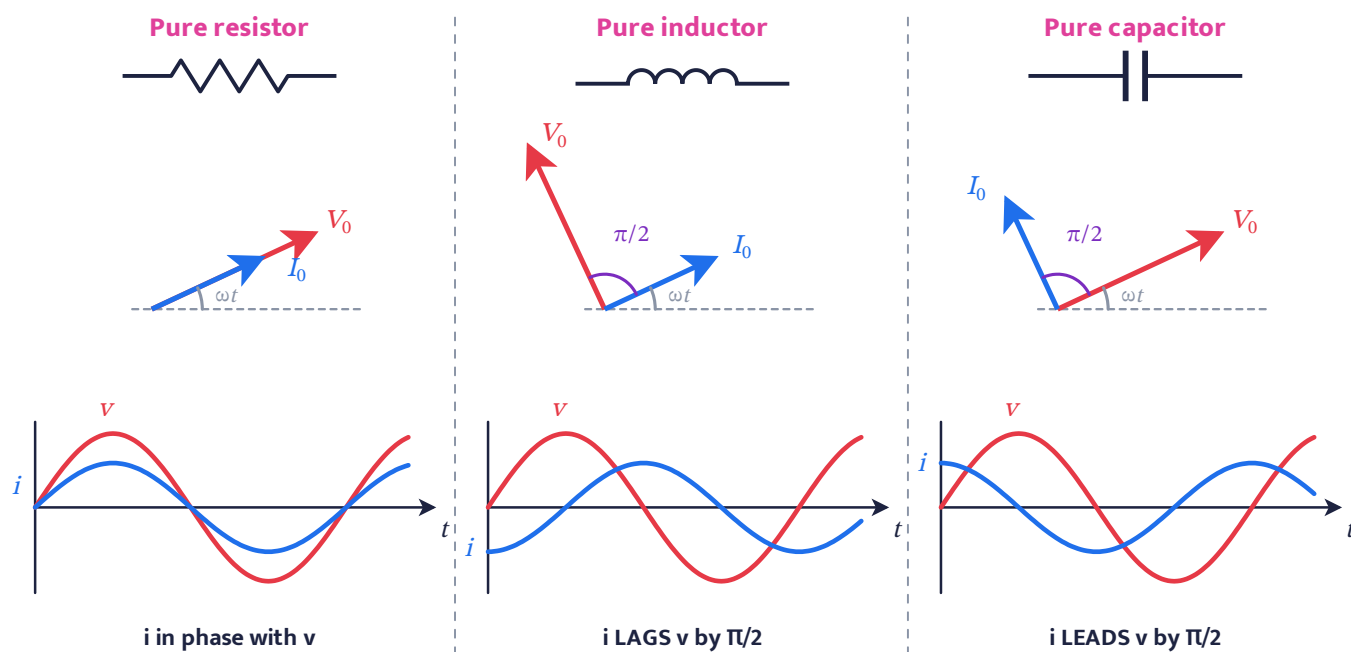
$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi \nu C}, \quad i = \frac{v_m}{X_C} \sin \left(\omega t + \frac{\pi}{2} \right)$$

C: capacitive reactance; current **leads** by $\pi/2$; $\bar{P} = 0$ **Solved Example**

A 44 mH inductor is connected to a 220 V, 50 Hz supply. Find the rms current. What would it be at 100 Hz?

Solution: $X_L = 2\pi(50)(0.044) \approx 13.8 \Omega \Rightarrow I = 220 / 13.8 \approx 15.9 \text{ A}$. Doubling ν doubles $X_L \Rightarrow$ current halves $\approx 8.0 \text{ A}$.

Property	Resistor R	Inductor L	Capacitor C
Opposition	R	$X_L = \omega L$	$X_C = 1 / \omega C$
Variation with ν	independent	$\propto \nu$	$\propto 1 / \nu$
For DC ($\nu = 0$)	R	0 (short circuit)	∞ (blocks DC)
Phase of i w.r.t. v	in phase	lags by $\pi/2$	leads by $\pi/2$
Average power	$I^2 R$	zero	zero



X_L rises linearly with f , X_C falls as $1/f$; they are equal at f_0

Memory Trick

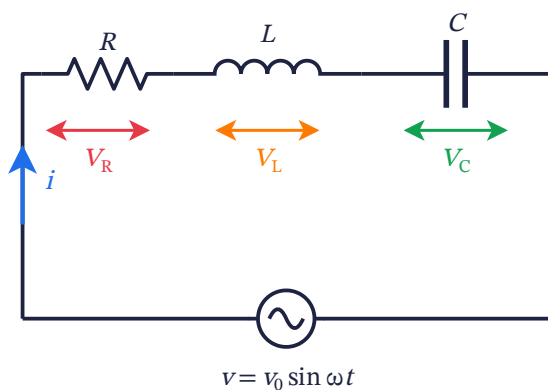
CIVIL – in **C**, **I** leads **V**; **V** leads **I** in **L**. (Also "ELI the ICE man": E before I in L, I before E in C.)

NCERT Line to Remember

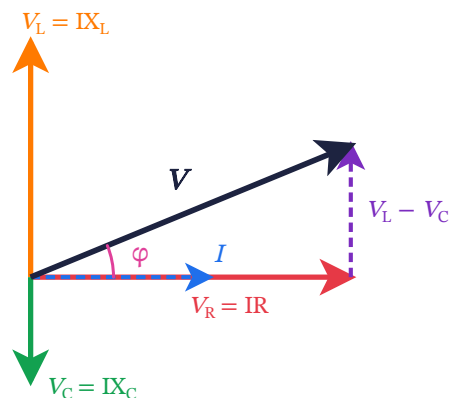
In a pure inductor or capacitor the energy stored during one quarter cycle is returned to the source in the next quarter – the **average power over a cycle is zero**.

7.3 Series LCR Circuit and Impedance

In series, the **same current** $i = i_m \sin(\omega t + \phi)$ flows through R, L and C. Take the current phasor as reference: V_R is along I , V_L is 90° ahead of I and V_C is 90° behind I . V_L and V_C are opposite, so they subtract; the result is added to V_R as a vector.



Series LCR circuit driven by an AC source

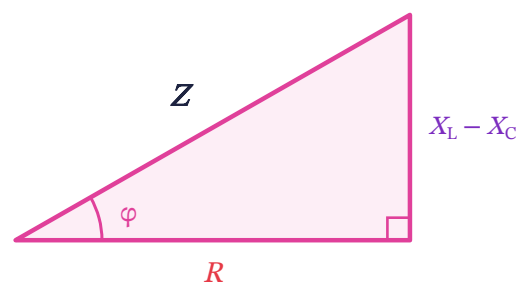
Phasor diagram (case $X_L > X_C$): V leads I by ϕ **Series LCR**

$$V = \sqrt{V_R^2 + (V_L - V_C)^2} \quad \text{Applied (rms) voltage}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \quad \text{Impedance } (\Omega); I = V / Z$$

$$\tan \phi = \frac{X_L - X_C}{R}, \quad \cos \phi = \frac{R}{Z} \quad \phi = \text{phase of } V \text{ ahead of } I$$

$$\cos \phi = R/Z, \quad \tan \phi = (X_L - X_C)/R$$



Impedance triangle

Condition	Nature of circuit	Phase
$X_L > X_C$ ($\omega > \omega_0$)	inductive	current lags voltage by ϕ
$X_L < X_C$ ($\omega < \omega_0$)	capacitive	current leads voltage by ϕ
$X_L = X_C$ ($\omega = \omega_0$)	purely resistive (resonance)	in phase, $\phi = 0$

Solved Example

A series LCR circuit has $R = 30 \, \Omega$, $X_L = 80 \, \Omega$, $X_C = 40 \, \Omega$ and is connected to 220 V (rms). Find Z , I , the power factor and the rms voltage across each element.

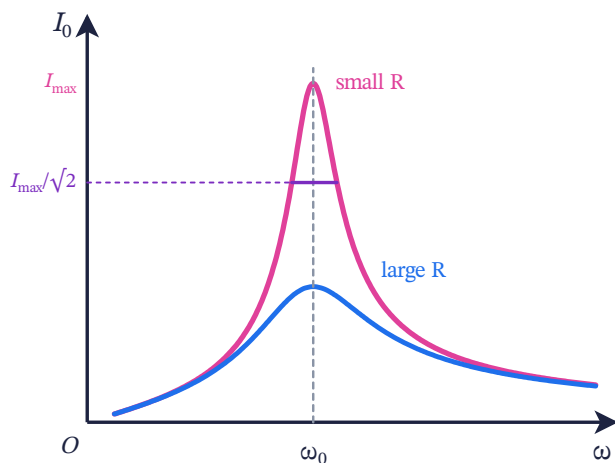
Solution: $Z = \sqrt{30^2 + 40^2} = 50 \, \Omega$; $I = 220 / 50 = 4.4 \, \text{A}$; $\cos \phi = 30 / 50 = 0.6$ (inductive). $V_R = 132 \, \text{V}$, $V_L = 352 \, \text{V}$, $V_C = 176 \, \text{V}$. Note $V_L > 220 \, \text{V}$, yet $\sqrt{132^2 + (352 - 176)^2} = 220 \, \text{V} \checkmark$.

Common Trap

- Never add rms voltages algebraically: $V \neq V_R + V_L + V_C$. Use $V = \sqrt{V_R^2 + (V_L - V_C)^2}$.
- The voltage across L or C alone can be **larger than the source voltage** – this is not a violation of anything.

7.4 Resonance

As ω is varied, X_L rises and X_C falls. At the **resonant frequency** they are equal, the impedance is **minimum** ($Z = R$) and the current is **maximum** ($i_m = v_m / R$), in phase with the voltage. V_L and V_C are equal and opposite and can each be many times the applied voltage.



Resonance curves: smaller $R \rightarrow$ taller, sharper peak at ω_0



Resonance

$$\omega_0 = \frac{1}{\sqrt{LC}}, \quad \nu_0 = \frac{1}{2\pi\sqrt{LC}} \quad \text{Resonant (angular) frequency}$$

$$Z_{\min} = R, \quad i_{\max} = \frac{v_m}{R} \quad \text{At resonance; power factor} = 1$$



NCERT Line to Remember

Resonance is possible **only if both L and C are present** (LC or LCR). It is used in the **tuning circuit of a radio or TV**: C is varied until ω_0 matches the station frequency.

Solved Example

A series LCR circuit has $L = 0.1 \text{ H}$, $C = 10 \mu\text{F}$, $R = 10 \Omega$ and a source of 100 V (rms) with variable frequency. Find the resonant frequency, the current at resonance and V_L at resonance.

Solution: $\omega_0 = 1 / \sqrt{0.1 \times 10^{-5}} = 1000 \text{ rad/s}$ ($\nu_0 \approx 159 \text{ Hz}$). $I = V / R = 10 \text{ A}$. $V_L = I\omega_0 L = 10 \times 100 = 1000 \text{ V} = V_C$ – ten times the source voltage!



Sharpness of resonance – Q factor and bandwidth

- Half-power frequencies ω_1, ω_2 : current falls to $i_{\max} / \sqrt{2}$ (power to half). **Bandwidth** $2\Delta\omega = \omega_2 - \omega_1 = \frac{R}{L}$.
- Quality factor** $Q = \frac{\omega_0}{2\Delta\omega} = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R} \sqrt{\frac{L}{C}}$. Large Q (small R) $\rightarrow$ sharp resonance, better tuning. At resonance $V_L = V_C = QV$.

7.5 Power in AC Circuits – Power Factor

With $v = v_m \sin \omega t$ and $i = i_m \sin(\omega t + \phi)$, the instantaneous power $p = vi$ averages over a cycle to

$$\bar{P} = \frac{v_m i_m}{2} \cos \phi = V_{\text{rms}} I_{\text{rms}} \cos \phi = I^2 Z \cos \phi = I^2 R$$

Circuit	ϕ	Power factor $\cos \phi$	Average power
Pure R (or LCR at resonance)	0	1	maximum, VI
Pure L or pure C	$\pi / 2$	0	zero (wattless)
Series LCR / RL / RC	$\tan^{-1} \frac{X_L - X_C}{R}$	R / Z	$VI \cos \phi$

 Wattless current

- Resolve I into $I\cos\phi$ (along V , does work) and $I\sin\phi$ ($\perp$ to V).
- $I\sin\phi$ contributes **no average power** – the **wattless current**.
- A **choke coil** (large L , tiny R) limits AC with almost no power loss.

 NCERT Line to Remember

Power is transmitted at a high power factor. A low power factor means a large current for the same power, so large I^2R loss in transmission lines. It is improved by connecting a suitable **capacitor in parallel** with inductive loads.

 Solved Example

For $v = 100\sin(100t)$ V and $i = 10\sin(100t + \pi/3)$ A, find the average power and the wattless component of the current.

Solution: $\bar{P} = \frac{100 \times 10}{2} \cos 60^\circ = \mathbf{250 \text{ W}}$. Wattless (rms) current $= \frac{10}{\sqrt{2}} \sin 60^\circ \approx \mathbf{6.1 \text{ A}}$ (peak $10\sin 60^\circ \approx 8.7 \text{ A}$).

 VS Sir's Exam Tip

Power questions: first find ϕ from the equations (difference of phase angles), then use $\bar{P} = \frac{v_{im}}{2} \cos\phi$ with **peak** values, or $VI\cos\phi$ with **rms** values – never mix the two. Only the resistor consumes power: $\bar{P} = I_{rms}^2 R$ is the quickest check.

 LC oscillations (JEE)

- A charged capacitor (q_0) connected across an inductor: charge oscillates as $q = q_0 \cos\omega_0 t$, $\omega_0 = 1/\sqrt{LC}$; $i_{max} = \omega_0 q_0$.
- Energy swings between electric $\frac{q^2}{2C}$ and magnetic $\frac{1}{2}Li^2$; total $\frac{q_0^2}{2C}$ stays constant (no R). Analogy: $L \leftrightarrow m$, $1/C \leftrightarrow k$, $q \leftrightarrow x$.

7.6 Transformer

A transformer changes an alternating voltage using **mutual induction**. Two coils – **primary** (N_p turns) and **secondary** (N_s turns) – are wound on a laminated soft-iron core so that (ideally) the same flux links every turn of both coils.

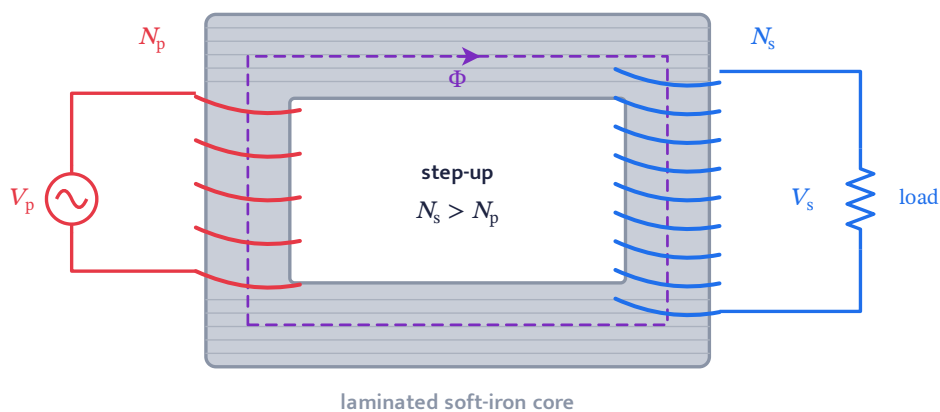
 Transformer equations

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} \quad \text{Ideal transformer } (V_p I_p = V_s I_s)$$

$$\eta = \frac{V_s I_s}{V_p I_p} \times 100\% \quad \text{Efficiency (real: 90–99\%)}$$

 Step-up vs step-down

- **Step-up:** $N_s > N_p \rightarrow V$ increases, I decreases.
- **Step-down:** $N_s < N_p \rightarrow V$ decreases, I increases.
- Frequency is **unchanged**; power is not stepped up.



Step-up transformer: the changing flux Φ in the core links both coils

Energy loss	Cause	Remedy
Flux leakage	not all flux of primary links the secondary	wind coils one over the other
Copper (Joule) loss	resistance of windings, I^2R	thick copper wire (for high-current coil)
Eddy currents	induced currents in the iron core	laminated core
Hysteresis	repeated magnetisation of the core	soft iron (low hysteresis loss)

Solved Example

A step-down transformer converts 2200 V to 220 V and supplies 20 A to a load. The primary has 4000 turns. Find (i) N_s , (ii) the primary current if the transformer is ideal, (iii) the primary current if the efficiency is 90%.

Solution: (i) $N_s = 4000 \times \frac{220}{2200} = 400$. (ii) $I_p = \frac{220 \times 20}{2200} = 2$ A. (iii) $I_p = \frac{4400}{0.9 \times 2200} \approx 2.22$ A.

Common Trap

- A transformer does **not** work on steady DC – no changing flux, no induced emf.
- Step-up raises **voltage**, not power or frequency.

VS Sir's Exam Tip

Why transmit at high voltage? For fixed power $P = VI$, raising V lowers I , and line loss I^2R falls as $1/V^2$. Power stations step up (≈ 11 kV $\rightarrow$ 220 kV or more); substations step down to 220 V.

Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- NCERT based** The rms value of the household supply in India is 220 V. Its peak value is about

(a) 311 V (b) 220 V (c) 156 V (d) 440 V
- NEET pattern** For $i = 5\sin(100\pi t)$ A, the rms current and frequency are

(a) 3.54 A, 50 Hz (b) 3.54 A, 100 Hz

(c) 5 A, 50 Hz (d) 2.5 A, 50 Hz
- NCERT based** In a purely inductive AC circuit, the current

(a) is in phase with the voltage (b) lags the voltage by π

(c) lags the voltage by $\pi/2$ (d) leads the voltage by $\pi/2$

- 4 **NEET pattern** An inductor of $\frac{1}{\pi}$ H is connected to a 220 V, 50 Hz supply. The rms current is
(a) 22 A (b) 0.7 A (c) 2.2 A (d) 1.1 A
- 5 **NEET pattern** The reactance of a $\frac{100}{\pi}$ μ F capacitor at 50 Hz is
(a) 50 Ω (b) 200 Ω (c) 10 Ω (d) 100 Ω
- 6 **NEET pattern** The graph of capacitive reactance X_C against frequency ν is
(a) a straight line parallel to the ν -axis
(b) a rectangular hyperbola
(c) a straight line through the origin
(d) a parabola
- 7 **NEET pattern** A series LCR circuit with $R = 30 \Omega$, $X_L = 80 \Omega$ and $X_C = 40 \Omega$ is connected to a 200 V (rms) source. The rms current is
(a) 2.5 A (b) 1.33 A (c) 4 A (d) 5 A
- 8 **NEET pattern** In a series LCR circuit the rms voltages across R, L and C are 80 V, 100 V and 40 V. The rms source voltage is
(a) 220 V (b) 20 V (c) 100 V (d) 140 V
- 9 **NCERT based** At resonance in a series LCR circuit
(a) impedance is maximum and current is minimum
(b) impedance is minimum and current is maximum
(c) voltage across L is zero
(d) current lags voltage by $\pi/2$
- 10 **NEET pattern** The resonant angular frequency of a series circuit with $L = 2$ mH and $C = 5 \mu$ F is
(a) 10^4 rad/s (b) 10^5 rad/s (c) 10^3 rad/s (d) 10^2 rad/s
- 11 **NCERT based** The power factor of a series LCR circuit at resonance is
(a) 0.5 (b) $1/\sqrt{2}$ (c) 1 (d) 0
- 12 **NEET pattern** For $v = 100\sin(100t)$ V and $i = 10\sin(100t + \pi/3)$ A, the average power is
(a) 433 W (b) 250 W (c) 500 W (d) 1000 W
- 13 **NCERT based** Current in a circuit is called wattless when the phase difference between current and voltage is
(a) 0° (b) 45° (c) 90° (d) 180°
- 14 **NEET pattern** An ideal transformer steps down 2200 V to 220 V and delivers 4.4 kW to a load. The current in the primary is
(a) 20 A (b) 200 A (c) 0.2 A (d) 2 A
- 15 **NCERT based** The core of a transformer is laminated to reduce
(a) eddy current loss (b) copper loss
(c) hysteresis loss (d) flux leakage

- 16 **JEE Main pattern** A series LCR circuit has $R = 10\ \Omega$, $L = 0.1\ \text{H}$ and $C = 10\ \mu\text{F}$. Its quality factor is
(a) 10 (b) 0.1 (c) 1 (d) 100
- 17 **JEE Main pattern** The rms value of the current $i = 3 + 4\sin\omega t$ (in A) is
(a) $\sqrt{17}\ \text{A}$ (b) 7 A (c) $\sqrt{8}\ \text{A}$ (d) 5 A
- 18 **JEE Main pattern** A series LCR circuit is driven at a frequency below its resonant frequency. Then
(a) it is inductive and the current lags the voltage
(b) it is purely resistive
(c) it is capacitive and the current lags the voltage
(d) it is capacitive and the current leads the voltage
- 19 **NEET pattern** A $100\ \Omega$ resistor and a capacitor of reactance $100\ \Omega$ are in series across an AC source. The phase relation between current and voltage is
(a) current leads by $\pi/2$ (b) in phase
(c) current leads by $\pi/4$ (d) current lags by $\pi/4$
- 20 **NEET pattern** A bulb in series with an inductor glows on an AC supply. If a soft-iron rod is inserted into the inductor, the bulb
(a) glows brighter (b) first brighter then dimmer
(c) glows the same (d) glows dimmer
- 21 **Assertion-Reason** Assertion: A capacitor blocks DC but allows AC to pass. Reason: The reactance $X_C = 1 / \omega C$ is infinite for $\omega = 0$.
(a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 22 **Assertion-Reason** Assertion: A transformer can be used to step up a DC voltage. Reason: A transformer works on mutual induction, which needs a changing magnetic flux.
(a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 23 **JEE Main pattern** For 50 Hz AC $i = i_m \sin\omega t$, the minimum time taken by the current to rise from zero to half its peak value is
(a) $\frac{1}{100}\ \text{s}$ (b) $\frac{1}{200}\ \text{s}$ (c) $\frac{1}{600}\ \text{s}$ (d) $\frac{1}{300}\ \text{s}$
- 24 **JEE Main pattern** An ideal LC circuit has $L = 20\ \text{mH}$ and $C = 50\ \mu\text{F}$. The capacitor initially carries 10 mC. The maximum current in the circuit is
(a) 100 A (b) 1 A (c) 0.1 A (d) 10 A



Answer Key & Quick Solutions

- 1(a) 2(a) 3(c) 4(c) 5(d) 6(b) 7(c) 8(c) 9(b) 10(a) 11(c) 12(b) 13(c) 14(d) 15(a) 16(a)
 17(a) 18(d) 19(c) 20(d) 21(a) 22(c) 23(c) 24(d)

1. (a) $v_m = \sqrt{2} \times 220 \approx 311$ V.
2. (a) $I = 5 / \sqrt{2} = 3.54$ A; $\nu = 100\pi / 2\pi = 50$ Hz.
3. (c) For L, $i = \frac{v_m}{X_L} \sin(\omega t - \pi/2)$: current lags by $\pi/2$.
4. (c) $X_L = 2\pi(50)(1/\pi) = 100 \Omega$; $I = 220/100 = 2.2$ A.
5. (d) $X_C = \frac{1}{2\pi(50)(10^{-4}/\pi)} = \frac{1}{10^{-2}} = 100 \Omega$.
6. (b) $X_C \propto 1/\nu \Rightarrow X_C \nu = \text{constant}$: rectangular hyperbola. (X_L vs ν is the straight line.)
7. (c) $Z = \sqrt{30^2 + 40^2} = 50 \Omega \Rightarrow I = 200/50 = 4$ A.
8. (c) $V = \sqrt{80^2 + (100 - 40)^2} = \sqrt{6400 + 3600} = 100$ V.
9. (b) $X_L = X_C \Rightarrow Z = R$ (minimum), $I = V/R$ (maximum), $\phi = 0$.
10. (a) $LC = 2 \times 10^{-3} \times 5 \times 10^{-6} = 10^{-8} \Rightarrow \omega_0 = 10^4$ rad/s.
11. (c) $\phi = 0 \Rightarrow \cos \phi = 1$ ($Z = R$).
12. (b) $\bar{P} = \frac{100 \times 10}{2} \cos 60^\circ = 250$ W.
13. (c) $\bar{P} = VI \cos 90^\circ = 0$ (pure L or pure C).
14. (d) $I_p = P/V_p = 4400/2200 = 2$ A (secondary carries 20 A).
15. (a) Thin insulated laminations increase the resistance to eddy-current paths.
16. (a) $\omega_0 = 1/\sqrt{10^{-6}} = 1000$ rad/s; $Q = \omega_0 L/R = 100/10 = 10$.
17. (a) $I_{rms} = \sqrt{3^2 + \frac{4^2}{2}} = \sqrt{9+8} = \sqrt{17} \approx 4.12$ A.
18. (d) $\omega < \omega_0 \Rightarrow X_C > X_L \Rightarrow$ capacitive, current leads.
19. (c) $\tan \phi = X_C/R = 1 \Rightarrow \phi = \pi/4$; RC circuit $\Rightarrow$ current leads.
20. (d) Iron raises L $\Rightarrow X_L$ and Z increase $\Rightarrow$ current falls $\Rightarrow$ dimmer.
21. (a) DC has $\omega = 0 \Rightarrow X_C \rightarrow \infty \Rightarrow$ no steady current; for AC, X_C is finite. R explains A.
22. (c) Steady DC gives constant flux $\Rightarrow$ no induced emf. A is false, R is true.
23. (c) $\sin \omega t = \frac{1}{2} \Rightarrow \omega t = \pi/6 \Rightarrow t = T/12 = \frac{1}{600}$ s.
24. (d) $\omega_0 = 1/\sqrt{10^{-6}} = 1000$ rad/s; $i_{max} = \omega_0 q_0 = 1000 \times 0.01 = 10$ A.

CHAPTER

8

Electromagnetic Waves

Day 13

Weightage ★★☆☆

NCERT Class 12

Displacement current

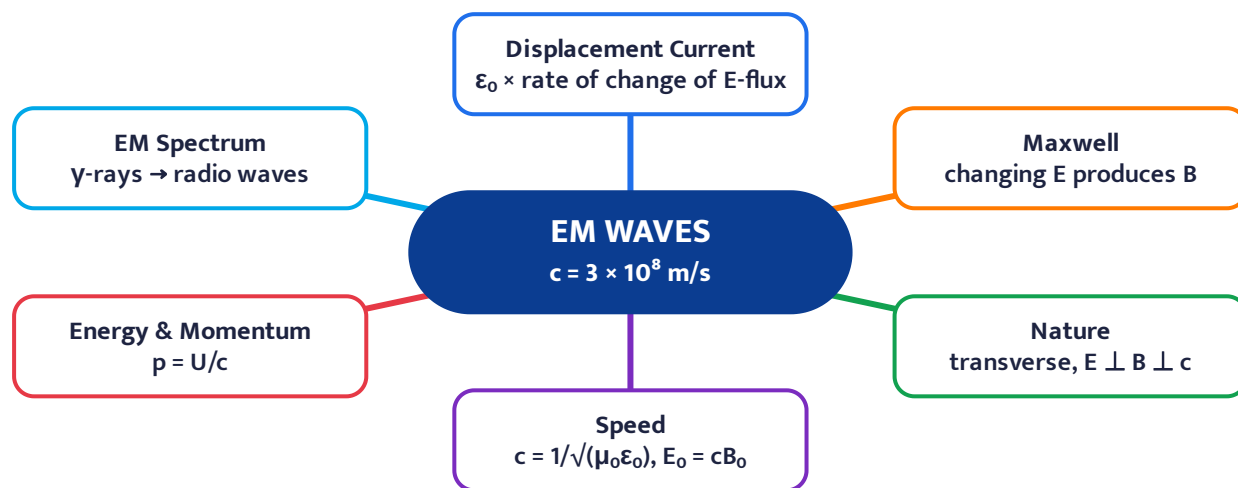
Maxwell's idea

Nature of EM waves

 $c = 1/\sqrt{(\mu_0\epsilon_0)}$

Energy & momentum

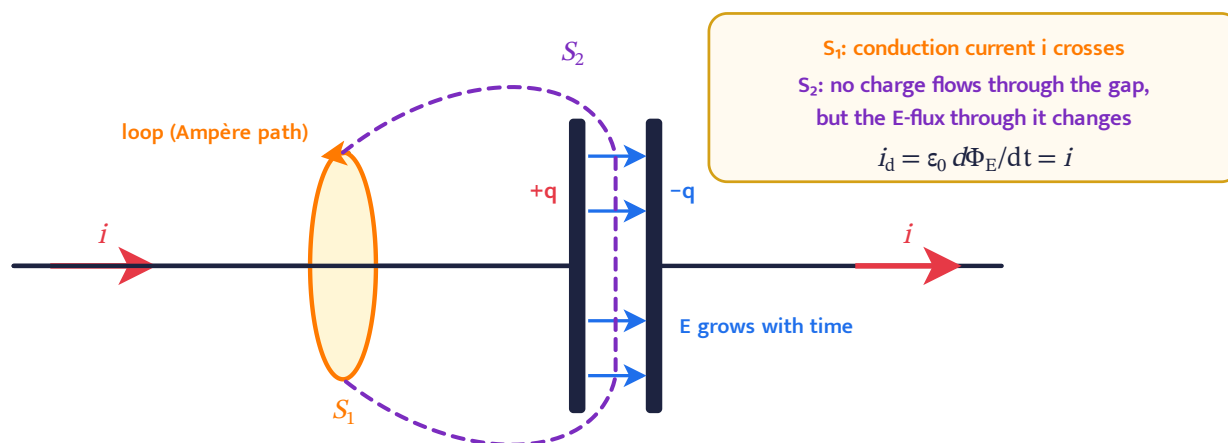
EM spectrum



Chapter at a glance

8.1 Displacement Current

Faraday showed that a **changing magnetic field produces an electric field**. Maxwell argued the converse: a **changing electric field produces a magnetic field**. He found that Ampère's circuital law is inconsistent for a charging capacitor: for the same boundary loop, a flat surface S_1 is crossed by the current i , but a pot-shaped surface S_2 passing between the plates is crossed by **no** current – so $\oint \vec{B} \cdot d\vec{l}$ would be both $\mu_0 i$ and 0.



Charging capacitor: the conduction current i in the wires continues as the displacement current i_d in the gap

The flux of E through S_2 is $\Phi_E = \frac{q}{\epsilon_0}$, so $\epsilon_0 \frac{d\Phi_E}{dt} = \frac{dq}{dt} = i$. Maxwell called this term the **displacement current** and added it to Ampère's law:

Displacement current

$$i_d = \epsilon_0 \frac{d\Phi_E}{dt}$$

Displacement current (A)

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_c + \epsilon_0 \frac{d\Phi_E}{dt} \right)$$

Ampère–Maxwell law

$$i_d = C \frac{dV}{dt} = \epsilon_0 A \frac{dE}{dt}$$

For a parallel-plate capacitor

NCERT Line to Remember

- Total current = conduction current + displacement current; the **total current is continuous** everywhere in the circuit.
- Inside the gap $i_c = 0$ and $i_d = i$; in the wires $i_d = 0$ and $i_c = i$. For steady (DC) fields $i_d = 0$.

Solved Example

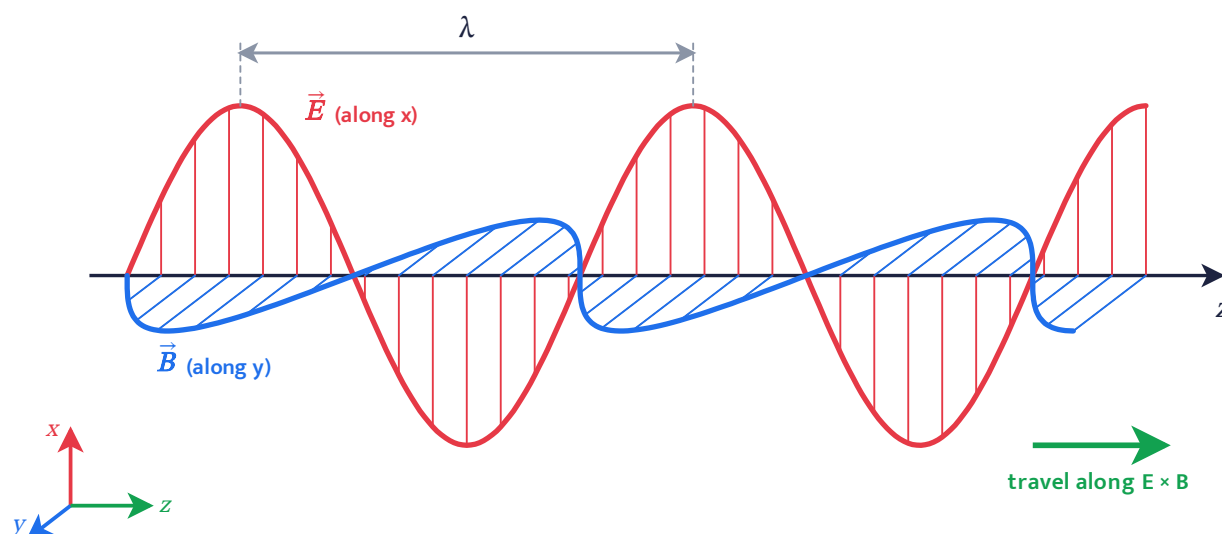
A capacitor with plate area 0.02 m^2 is charged by a current of 0.15 A . Find the displacement current and dE/dt between the plates.

Solution: $i_d = i = 0.15 \text{ A}$. $\frac{dE}{dt} = \frac{i_d}{\epsilon_0 A} = \frac{0.15}{8.85 \times 10^{-12} \times 0.02} \approx 8.5 \times 10^{11} \text{ V m}^{-1} \text{ s}^{-1}$.

Maxwell's equation (qualitative)	Form	Meaning
Gauss's law (electricity)	$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$	charges are sources of E
Gauss's law (magnetism)	$\oint \vec{B} \cdot d\vec{A} = 0$	no magnetic monopoles
Faraday's law	$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$	changing B $\rightarrow$ E
Ampère–Maxwell law	$\oint \vec{B} \cdot d\vec{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$	current and changing E $\rightarrow$ B

8.2 Electromagnetic Waves – Sources and Nature

A charge at rest produces only E; a charge moving with **constant velocity** produces steady E and B but **no wave**. An **accelerated charge** radiates: an oscillating charge produces an oscillating E, which produces an oscillating B, which regenerates E ... and the disturbance travels as an **electromagnetic wave**, needing no medium. **The frequency of the wave equals the frequency of oscillation of the charge.**



A plane EM wave travelling along z: E_x and B_y oscillate in phase, perpendicular to each other and to the direction of travel

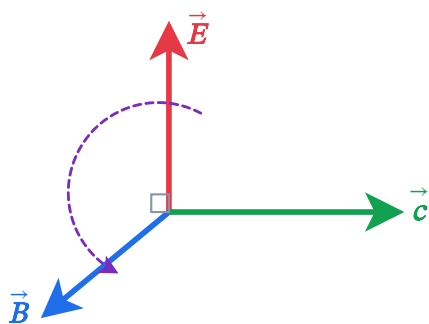
EM wave equations

$$E_x = E_0 \sin(kz - \omega t), \quad B_y = B_0 \sin(kz - \omega t) \quad \text{Plane wave along } +z \text{ (NCERT form)}$$

$$k = \frac{2\pi}{\lambda}, \quad \omega = 2\pi\nu, \quad c = \nu\lambda = \frac{\omega}{k} \quad \text{Wave relations}$$

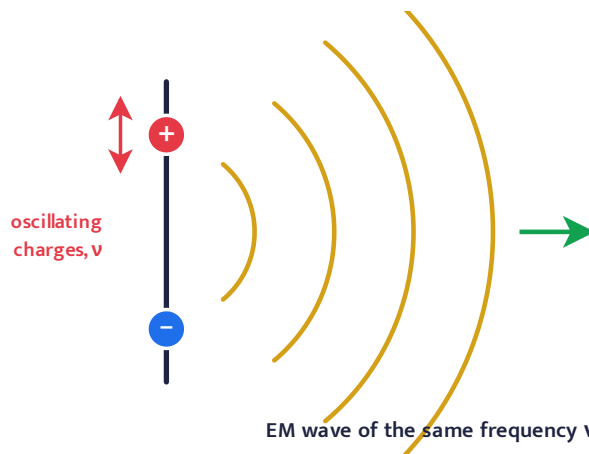
$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m s}^{-1} \quad \text{Speed in vacuum – light is an EM wave}$$

$$\frac{E_0}{B_0} = c, \quad v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{\sqrt{\mu_r \epsilon_r}} \quad \text{Field ratio; speed in a medium}$$



direction of travel = direction of $\vec{E} \times \vec{B}$

$\vec{E}$, $\vec{B}$ and the velocity form a right-handed set



EM wave of the same frequency ν

An oscillating (accelerated) charge is the source of EM waves

Properties of EM waves

- **Transverse:** $\vec{E} \perp \vec{B} \perp$ direction of propagation; direction of travel is along $\vec{E} \times \vec{B}$.
- E and B oscillate **in phase** (maxima and zeros together) with the same frequency.
- Need **no material medium**; all travel with the same speed c in vacuum, independent of wavelength.
- Not deflected by electric or magnetic fields (they are uncharged); they carry **energy and momentum** and can be polarised.
- In a medium, frequency stays the same while speed and wavelength decrease; refractive index $n = \sqrt{\mu_r \epsilon_r}$.

Solved Example

A plane EM wave of frequency 25 MHz travels in vacuum along x . At a point, $\vec{E} = 6.3 \hat{j}$ V/m. Find λ and $\vec{B}$ at that point.

Solution: $\lambda = c / \nu = 3 \times 10^8 / 25 \times 10^6 = 12 \text{ m}$.

$B = E / c = 6.3 / 3 \times 10^8 = 2.1 \times 10^{-8} \text{ T}$. Since $\hat{j} \times \hat{k} = \hat{i}$, $\vec{B}$ is along $+\hat{i}$.

History NEET lifts from NCERT

- **Hertz (1887)** first produced and detected EM waves in the lab.
- **J. C. Bose** (Kolkata) produced waves of **25 mm to 5 mm** wavelength.
- **Marconi** transmitted EM waves over many kilometres – the start of radio communication.

8.3 Energy, Intensity and Momentum

An EM wave carries energy in both its fields. The average energy is **shared equally** between the electric and magnetic fields. It also carries momentum, so it exerts a **radiation pressure** on a surface it falls on (e.g. the tail of a comet points away from the Sun).



Energy & momentum of EM waves

$$u_E = \frac{1}{2}\epsilon_0 E^2, \quad u_B = \frac{B^2}{2\mu_0}, \quad \bar{u}_E = \bar{u}_B \quad \text{Energy densities (J m}^{-3}\text{) – equal on average}$$

$$\bar{u} = \frac{1}{2}\epsilon_0 E_0^2 = \frac{B_0^2}{2\mu_0} \quad \text{Average total energy density}$$

$$I = \bar{u} c = \frac{1}{2}c\epsilon_0 E_0^2 \quad \text{Intensity (W m}^{-2}\text{); point source: } I = \frac{P}{4\pi r^2}$$

$$p = \frac{U}{c} \text{ (absorbed),} \quad p = \frac{2U}{c} \text{ (reflected)} \quad \text{Momentum delivered to a surface}$$



JEE Main Extension

- Radiation pressure (normal incidence): $P_{rad} = \frac{I}{c}$ (absorber), $\frac{2I}{c}$ (perfect reflector).
- rms fields: $E_{rms} = E_0 / \sqrt{2}$, and $I = c\epsilon_0 E_{rms}^2$.



Common Trap

- $E_0 = cB_0$ does **not** mean the electric field is "stronger" – energy densities are **equal**.
- Reflection doubles the momentum transfer ($2U/c$), not absorption.

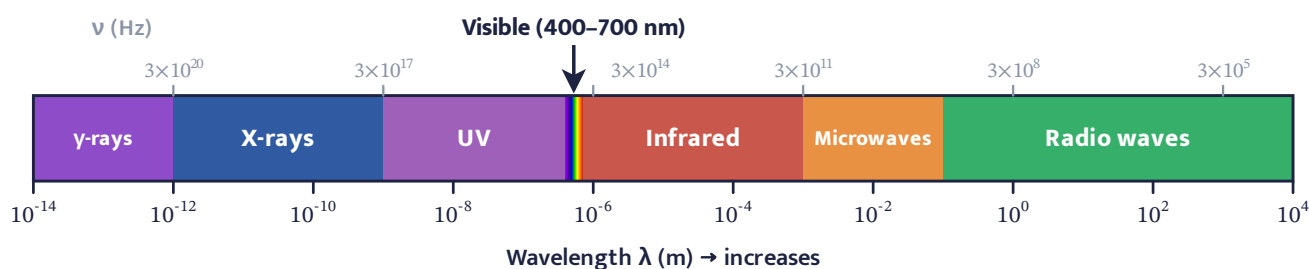
Solved Example

A wave is described by $E_x = 30\sin(2 \times 10^7 z - 6 \times 10^{15} t)$ V/m (SI units). Find λ , ν , B_0 and the region of the spectrum.

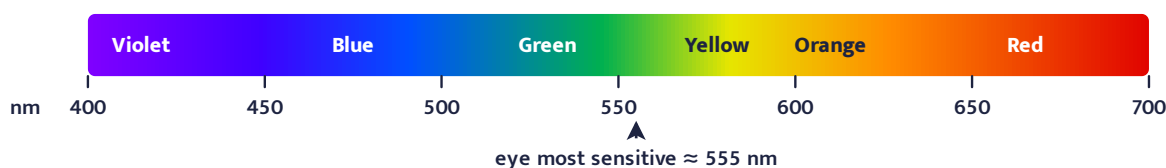
Solution: $\lambda = 2\pi / k = 2\pi / (2 \times 10^7) \approx 314 \text{ nm}$; $\nu = \omega / 2\pi \approx 9.5 \times 10^{14} \text{ Hz}$; check $\omega / k = 3 \times 10^8 \text{ m/s} \checkmark$.
 $B_0 = E_0 / c = 10^{-7} \text{ T}$, along y (wave moves along +z). $314 \text{ nm} < 400 \text{ nm} \Rightarrow \text{ultraviolet}$.

8.4 The Electromagnetic Spectrum

EM waves are classified by wavelength (or frequency). The bands overlap and have no sharp boundaries; all differ only in λ and ν – the nature and speed in vacuum are the same.



The EM spectrum on a logarithmic scale: wavelength increases to the right, frequency and energy to the left



The visible band: violet ($\approx 400 \text{ nm}$) to red ($\approx 700 \text{ nm}$), i.e. about $4 \times 10^{14} \text{ Hz}$ to $7.5 \times 10^{14} \text{ Hz}$

Band	Wavelength	Production	Detection
Radio	$> 0.1 \text{ m}$	rapid acceleration/deceleration of electrons in aerials	receiver's aerials
Microwave	$0.1 \text{ m} - 1 \text{ mm}$	klystron, magnetron, Gunn diode	point-contact diodes
Infrared	$1 \text{ mm} - 700 \text{ nm}$	vibration of atoms and molecules (hot bodies)	thermopiles, bolometer, IR photographic film
Visible	$700 \text{ nm} - 400 \text{ nm}$	electrons in atoms moving from a higher to a lower energy level	the eye, photocells, photographic film
Ultraviolet	$400 \text{ nm} - 1 \text{ nm}$	inner-shell electrons moving between levels; very hot bodies (Sun), UV lamps	photocells, photographic film
X-rays	$1 \text{ nm} - 10^{-3} \text{ nm}$	X-ray tubes (high-energy electrons hit a metal target); inner-shell electrons	photographic film, Geiger tubes, ionisation chamber
γ -rays	$< 10^{-3} \text{ nm}$	radioactive decay of nuclei; nuclear reactions	as for X-rays

Uses and NCERT facts – band by band

Band	Key uses / Facts
Radio	AM band 530–1710 kHz; short-wave up to 54 MHz; TV 54–890 MHz; FM 88–108 MHz ; cellular phones use the UHF band.
Microwave	Radar (aircraft navigation) and speed guns; microwave ovens – frequency matched to the resonant frequency of water molecules , which heat the food.
Infrared	" Heat waves " (readily absorbed by water, CO_2 , NH_3); keeps Earth warm by the greenhouse effect ; physical therapy lamps; TV remote (LEDs); IR detectors in satellites (crop growth, military), night vision.
Visible	The only part our eyes detect. Some animals (e.g. snakes) can detect infrared; bees can see some UV.
Ultraviolet	Mostly absorbed by the ozone layer ($\approx 40\text{--}50 \text{ km}$); large doses harmful. LASIK eye surgery (narrow beams), water purifiers (kill germs); welders wear special goggles.
X-rays	Medical diagnosis and treatment of certain cancers; damage living tissue – avoid over-exposure.
γ-rays	Highest frequency; used in medicine to destroy cancer cells ; produced in nuclear reactions.

Memory Trick

Increasing wavelength: **Get Xtra UnderValued Investments More Rapidly $\rightarrow$ γ , X, UV, Visible, IR, Micro, Radio**. Read backwards for increasing frequency. Inside visible: **VIBGYOR** (violet shortest λ , red longest).

VS Sir's Exam Tip

NEET asks "match the radiation with its use/source" almost every year. Learn the pairs: radar–microwave, remote–IR, LASIK/water purifier–UV, ozone–UV, greenhouse–IR, cancer– γ , bone fracture–X-ray, klystron–microwave.

! Common Trap

- All EM waves have the **same speed in vacuum**; in a medium, speed depends on λ (that is why a prism disperses light).
- X-rays and γ -rays can overlap in λ ; they are named by their **source** (electrons vs nucleus), not only by λ .
- Sound and ultrasonic waves are **not** EM waves – they need a medium.

Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- NCERT based** The displacement current is given by
 (a) $\mu_0 \frac{d\Phi_E}{dt}$ (b) $\epsilon_0 \frac{d\Phi_E}{dt}$ (c) $\epsilon_0 \frac{d\Phi_B}{dt}$ (d) $\frac{1}{\epsilon_0} \frac{d\Phi_E}{dt}$
- NEET pattern** A $2 \mu\text{F}$ capacitor is being charged so that the voltage across it changes at 10^6 V/s . The displacement current between its plates is
 (a) 0.5 A (b) $2 \times 10^{-6} \text{ A}$ (c) 2 A (d) 20 A
- NCERT based** In an electromagnetic wave, the electric and magnetic fields are
 (a) parallel to each other
 (b) along the direction of propagation
 (c) perpendicular to each other and to the direction of propagation, and in phase
 (d) perpendicular to each other but out of phase by $\pi/2$
- NEET pattern** The peak electric field of an EM wave in vacuum is 6 V/m. The peak magnetic field is
 (a) $5 \times 10^7 \text{ T}$ (b) $2 \times 10^{-6} \text{ T}$ (c) $2 \times 10^{-8} \text{ T}$ (d) $1.8 \times 10^9 \text{ T}$
- NEET pattern** An EM wave travels along +z with its electric field along +x. Its magnetic field is along
 (a) +y (b) +z (c) -x (d) -y
- NCERT based** The correct order of increasing wavelength is
 (a) radio, microwaves, IR, visible, UV, X-rays, γ -rays
 (b) γ -rays, X-rays, UV, visible, IR, microwaves, radio
 (c) X-rays, γ -rays, UV, visible, IR, radio, microwaves
 (d) UV, γ -rays, X-rays, visible, microwaves, IR, radio
- NCERT based** Radar systems used in aircraft navigation employ
 (a) infrared waves (b) γ -rays
 (c) microwaves (d) ultraviolet rays
- NCERT based** The radiation mainly responsible for the greenhouse effect is
 (a) visible (b) ultraviolet (c) infrared (d) X-rays

- 9 **NEET pattern** An EM wave has frequency 3 GHz. Its wavelength and the band it belongs to are
(a) 10 cm, microwave (b) 1 cm, ultraviolet
(c) 1 m, radio wave (d) 10 cm, infrared
- 10 **NEET pattern** The speed of EM waves in a medium with $\mu_r = 1$ and $\epsilon_r = 4$ is
(a) 0.75×10^8 m/s (b) 1.5×10^8 m/s (c) 6×10^8 m/s (d) 3×10^8 m/s
- 11 **NEET pattern** γ -rays are produced by
(a) radioactive decay of nuclei (b) outer-electron transitions
(c) klystron valves (d) molecular vibrations
- 12 **JEE Main pattern** A plane EM wave in vacuum has $E_0 = 100$ V/m. Its intensity is about ($\epsilon_0 = 8.85 \times 10^{-12}$ SI)
(a) 1.33 W/m² (b) 13.3 W/m² (c) 26.6 W/m² (d) 6.6 W/m²
- 13 **JEE Main pattern** In a plane EM wave, the ratio of the average electric energy density to the average magnetic energy density is
(a) c (b) 1 (c) $1/c$ (d) c^2
- 14 **JEE Main pattern** For $E = 60\sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)$ V/m (SI), the wavelength and direction of propagation are
(a) 0.5 mm, along $-x$ (b) 12.6 cm, along $+x$
(c) 1.26 cm, along $+x$ (d) 1.26 cm, along $-x$
- 15 **NCERT based** Water purifiers and LASIK eye surgery make use of
(a) infrared radiation (b) microwaves
(c) X-rays (d) ultraviolet radiation
- 16 **NEET pattern** The dimensions of E / B (ratio of electric to magnetic field) are those of
(a) acceleration (b) it is dimensionless
(c) force (d) velocity
- 17 **Assertion-Reason** Assertion: Electromagnetic waves can travel through vacuum. Reason: They consist of mutually perpendicular electric and magnetic fields that regenerate each other and need no material medium.
(a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 18 **Assertion-Reason** Assertion: X-rays are deflected by a magnetic field. Reason: X-rays are electromagnetic waves.
(a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false

- 19 **JEE Main pattern** Light carrying a total energy of 30 J falls normally on a perfectly absorbing surface. The momentum delivered to the surface is
 (a) 9×10^9 kg m/s (b) 5×10^{-8} kg m/s (c) 1×10^{-7} kg m/s (d) 2×10^{-7} kg m/s
- 20 **NEET pattern** Electromagnetic waves are produced by
 (a) an accelerated charge (b) a charge at rest
 (c) a uniformly moving charge (d) a neutral particle at rest

✓ Answer Key & Quick Solutions

- 1(b) 2(c) 3(c) 4(c) 5(a) 6(b) 7(c) 8(c) 9(a) 10(b) 11(a) 12(b) 13(b) 14(d) 15(d)
 16(d) 17(a) 18(c) 19(c) 20(a)

- (b) $i_d = \epsilon_0 d\Phi_E / dt$ – rate of change of electric flux.
- (c) $i_d = C dV / dt = 2 \times 10^{-6} \times 10^6 = 2$ A.
- (c) EM waves are transverse; $E \perp B \perp$ direction of travel, with E and B in phase.
- (c) $B_0 = E_0 / c = 6 / (3 \times 10^8) = 2 \times 10^{-8}$ T.
- (a) Direction of travel = $\hat{E} \times \hat{B}$: $\hat{i} \times \hat{j} = \hat{k} \Rightarrow B$ along +y.
- (b) λ increases from γ -rays to radio waves.
- (c) Short wavelength microwaves are suitable for radar.
- (c) IR emitted by the Earth is trapped by greenhouse gases.
- (a) $\lambda = c / \nu = 3 \times 10^8 / 3 \times 10^9 = 0.1$ m = 10 cm (1 mm – 0.1 m $\Rightarrow$ microwave).
- (b) $v = c / \sqrt{\mu_r \epsilon_r} = 3 \times 10^8 / 2 = 1.5 \times 10^8$ m/s.
- (a) γ -rays come from nuclei (radioactive decay, nuclear reactions).
- (b) $I = \frac{1}{2} c \epsilon_0 E_0^2 = 0.5 \times 3 \times 10^8 \times 8.85 \times 10^{-12} \times 10^4 \approx 13.3$ W/m².
- (b) $\frac{1}{2} \epsilon_0 E^2 = \frac{B^2}{2\mu_0}$ since $E = cB$ and $c^2 = 1 / \mu_0 \epsilon_0 \Rightarrow$ ratio 1.
- (d) $\lambda = 2\pi / k = 2\pi / 500 \approx 1.26$ cm; " $kx + \omega t$ " $\Rightarrow$ travels along -x ($\omega/k = 3 \times 10^8$ ✓).
- (d) UV kills germs and can be focused into very narrow beams.
- (d) $E = cB$ (also $F = qE = qvB \Rightarrow E/B$ has dimensions of velocity, $[LT^{-1}]$).
- (a) Changing E makes B and changing B makes E – self-sustaining, so no medium is needed. R explains A.
- (c) EM waves carry no charge, so they are not deflected by E or B fields. A is false, R is true.
- (c) $p = U / c = 30 / (3 \times 10^8) = 10^{-7}$ kg m/s (it would be double for a perfect reflector).
- (a) Only accelerated (e.g. oscillating) charges radiate EM waves.

CHAPTER

9

Ray Optics and Optical Instruments

17

Day 14–15

Weightage ★★★

NCERT Class 12

Spherical mirrors

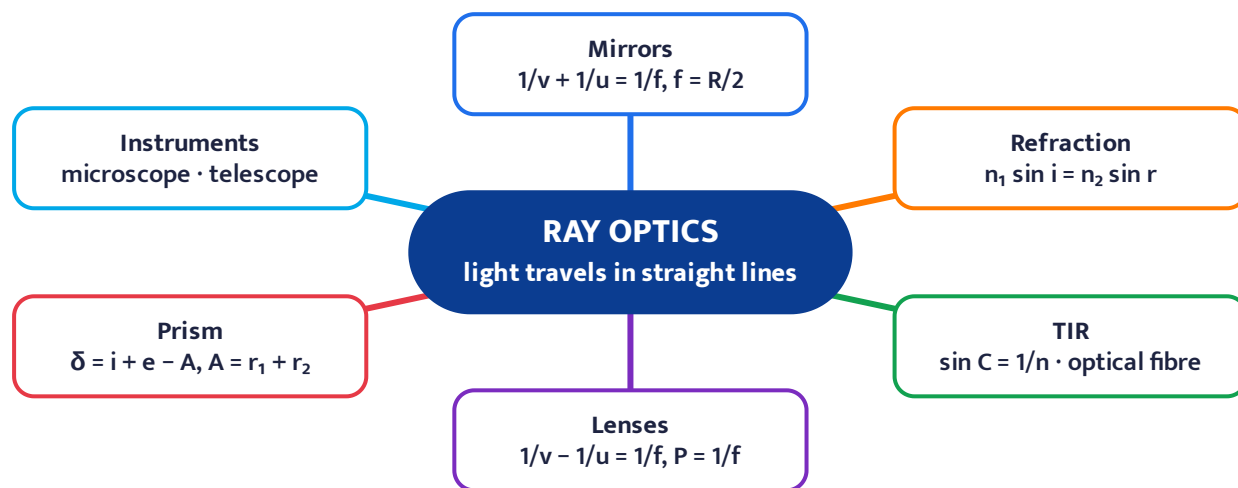
Refraction & TIR

Thin lenses

Lens maker's formula

Prism

Microscope & telescope



Chapter at a glance

9.1 Reflection by Spherical Mirrors

Laws of reflection: the angle of incidence equals the angle of reflection, and the incident ray, reflected ray and the **normal** at the point of incidence lie in one plane. For a spherical mirror the normal is along the **radius** through that point. Near the axis (paraxial rays), $f = R/2$.



Cartesian sign convention

- All distances measured **from the pole** (or optical centre).
- Along the **incident light**: positive; against it: negative.
- Heights **above** the axis +ve, below -ve.
- Real object: $u < 0$. Concave mirror: $f < 0$; convex mirror: $f > 0$.



Mirror formulas

$$f = \frac{R}{2}$$

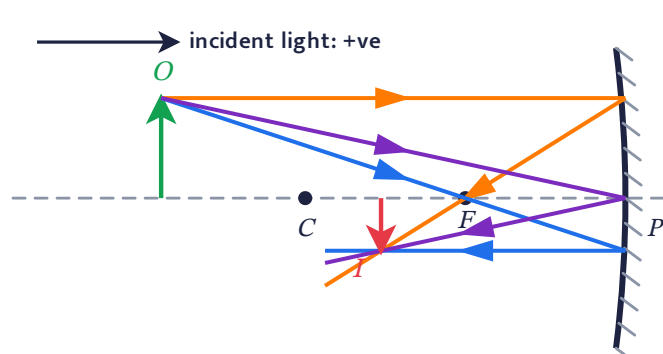
Focal length (paraxial rays)

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

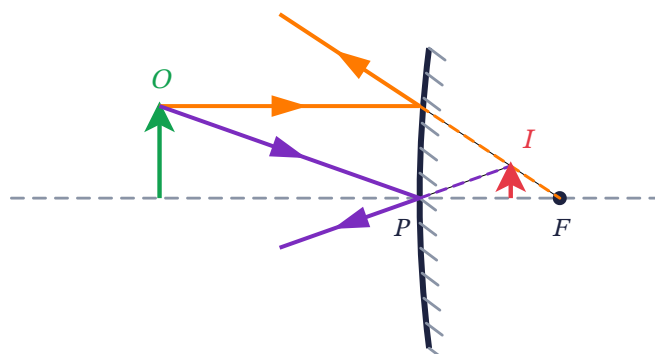
Mirror formula

$$m = \frac{h'}{h} = -\frac{v}{u}$$

Magnification: $m < 0$ real & inverted, $m > 0$ virtual & erect



Concave mirror; object beyond C: image between F and C – real, inverted, diminished



Convex mirror: image always virtual, erect, diminished, between P and F

Object (concave mirror)	Image position	Nature and size
At infinity	At F	real, inverted, point-sized
Beyond C	Between F and C	real, inverted, diminished
At C	At C	real, inverted, same size ($m = -1$)
Between C and F	Beyond C	real, inverted, magnified
At F	At infinity	real, inverted, highly magnified
Between F and P	Behind the mirror	virtual, erect, magnified

Solved Example

An object 2 cm tall is placed 30 cm in front of a concave mirror of focal length 20 cm. Find the position, size and nature of the image.

Solution: $u = -30, f = -20: \frac{1}{v} = \frac{1}{-20} - \frac{1}{-30} = -\frac{1}{60} \Rightarrow v = -60 \text{ cm}$ (in front, real). $m = -v/u = -2 \Rightarrow h' = -4 \text{ cm}$: real, inverted, twice as large.

Common Trap

- Convex mirror (real object) **always** gives a virtual, erect, diminished image – that is why it is used as a rear-view mirror (wide field of view).
- Mirror focal length does **not** depend on the medium or the colour of light (no refraction involved).

VS Sir's Exam Tip

- Object moving along the axis: image speed $v_I = m^2 v_O$ (longitudinal magnification $= -m^2$ for a mirror).
- Plane mirror: $f = \infty, m = +1$; rotating it by θ turns the reflected ray by 2θ .

9.2 Refraction

When light crosses from one medium to another, its **speed and wavelength change but frequency does not**. Snell's law: $\frac{\sin i}{\sin r} = n_{21} = \frac{n_2}{n_1} = \frac{v_1}{v_2}$. A ray bends **towards** the normal on entering a denser medium ($n_{21} > 1$). Also $n_{12} = 1/n_{21}$ and $n_{32} = n_{31} \times n_{12}$.

Refraction Formulas

$$n = \frac{c}{v}, \quad \lambda_{\text{medium}} = \frac{\lambda_0}{n}$$

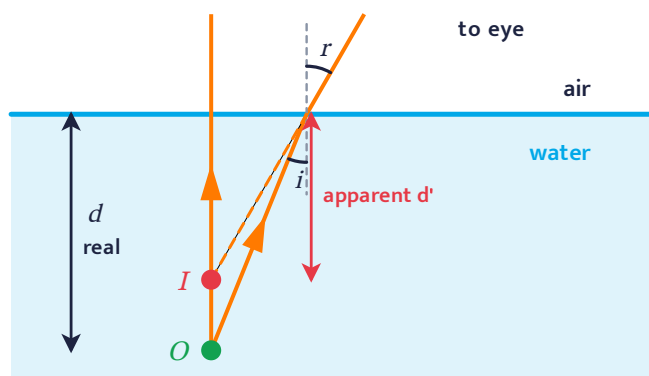
Absolute refractive index; wavelength in the medium

$$d' = \frac{d}{n_{21}}$$

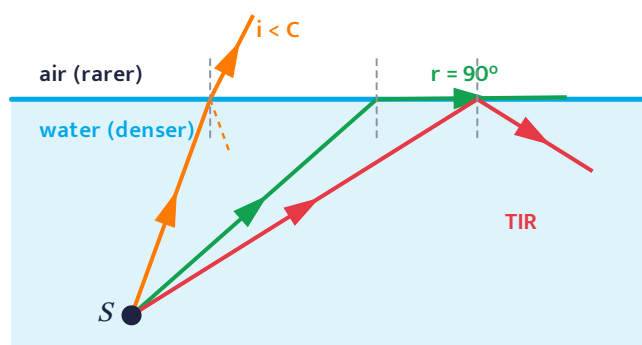
Apparent depth of an object in a denser medium seen from air (near-normal viewing)

$$\Delta d = d \left(1 - \frac{1}{n} \right)$$

Normal shift (apparent rise); same shift for a glass slab of thickness d



Real and apparent depth: the bottom of a tank appears raised



As i increases past the critical angle C , light is totally reflected

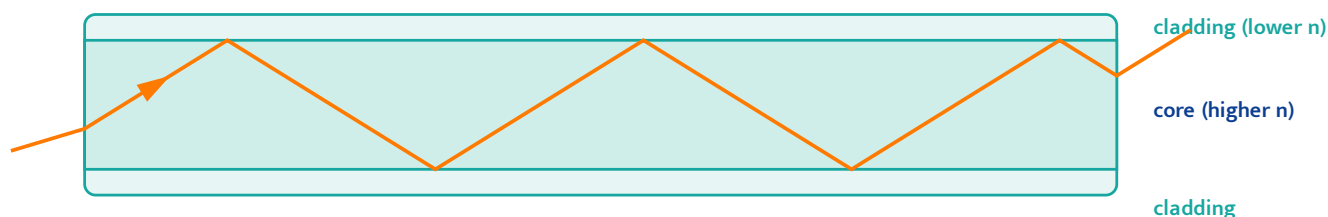
NCERT Line to Remember

A glass slab produces **lateral shift** of the ray but no deviation – the emergent ray is parallel to the incident ray. The Sun is visible ~2 min before actual sunrise and ~2 min after actual sunset because of **atmospheric refraction**.

9.3 Total Internal Reflection (TIR)

When light goes from a **denser to a rarer** medium, the refracted ray bends away from the normal. At the **critical angle C** the angle of refraction is 90° . For $i > C$ no refraction is possible – all light is reflected back: **total internal reflection**. Two conditions: (i) denser $\rightarrow$ rarer, (ii) $i > C$.

$$\sin C = \frac{1}{n_{21}} = \frac{n_{\text{rarer}}}{n_{\text{denser}}} \quad (\text{water } 48.6^\circ, \text{ glass } 41.8^\circ, \text{ diamond } 24.4^\circ)$$



Light is guided by repeated total internal reflection (angle at core–cladding boundary $> C$)

Optical fibre: a glass/quartz core surrounded by cladding of lower refractive index

Applications of TIR

- **Optical fibres** – telecommunication, endoscopy (light pipes); negligible loss.
- **Totally reflecting prisms** (45° – 90° – 45°) bend light by 90° or 180° (periscopes, binoculars).
- **Sparkling of diamond** (very small $C \approx 24.4^\circ$) and **mirage** in hot deserts.

Common Trap

- TIR is impossible when light goes from rarer to denser, whatever the angle.
- For a glass–water boundary use the **relative** index: $\sin C = n_w / n_g = 8/9$, not $2/3$.
- Red light has the larger C (smaller n) of all colours.

9.4 Refraction at Spherical Surfaces and Lenses

For refraction at a single spherical surface of radius R separating media n_1 (object side) and n_2 , and then applying it twice for a thin lens, gives the lens maker's formula. Same sign convention: R is +ve if the centre of curvature lies on the side of the outgoing light; $n_{21} = n_{\text{lens}} / n_{\text{medium}}$.

Lens formulas

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R} \quad \text{Refraction at a spherical surface}$$

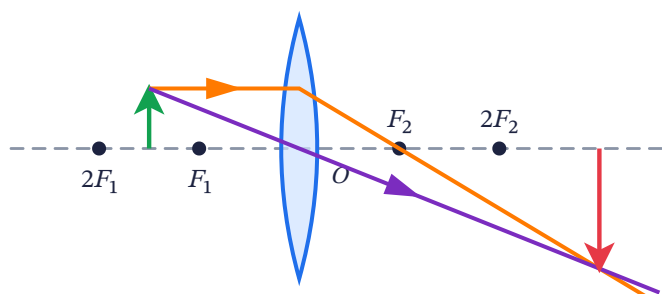
$$\frac{1}{f} = (n_{21} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \text{Lens maker's formula}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad \text{Thin lens formula (convex } f > 0, \text{ concave } f < 0)$$

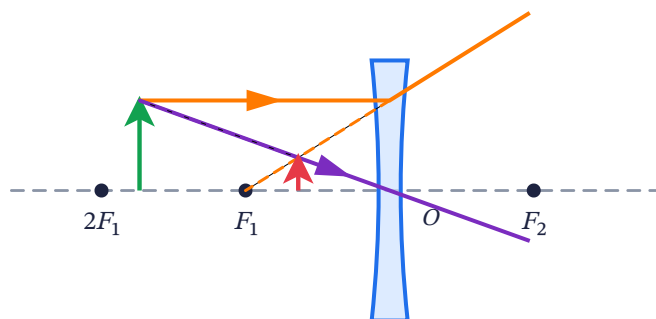
$$m = \frac{h'}{h} = \frac{v}{u} \quad \text{Linear magnification of a lens}$$

$$P = \frac{1}{f \text{ (in m)}} \text{ dioptre (D)} \quad \text{Power: converging +, diverging -}$$

$$P = P_1 + P_2 + \dots, \quad m = m_1 m_2 \dots \quad \text{Thin lenses in contact}$$



Convex lens, object between F_1 and $2F_1$: real, inverted, magnified image beyond $2F_2$



Concave lens: parallel ray appears to diverge from F_1 ; image always virtual, erect, diminished

Solved Example

A glass ($n = 1.5$) convex lens has $f = 20$ cm in air. Find its focal length when immersed in water ($n = 4/3$).

Solution: $\frac{f_w}{f_a} = \frac{n_g - 1}{n_g / n_w - 1} = \frac{0.5}{1.125 - 1} = 4 \Rightarrow f_w = 80$ cm (still converging, but weaker).

Solved Example

A convex lens of focal length 25 cm is kept in contact with a concave lens of focal length 20 cm. Find the power and nature of the combination.

Solution: $P = \frac{1}{0.25} - \frac{1}{0.20} = 4 - 5 = -1$ D $\Rightarrow f = -100$ cm, a **diverging** combination.

! Common Trap

- Lens in a liquid of **equal** n : $f \rightarrow \infty$ (lens "disappears"); in a **denser** liquid a convex lens **diverges**.
- Covering half of a lens: the **full** image still forms, only its brightness drops.
- Cutting an equiconvex lens perpendicular to its axis: each half has focal length **$2f$** .

🧠 Memory Trick

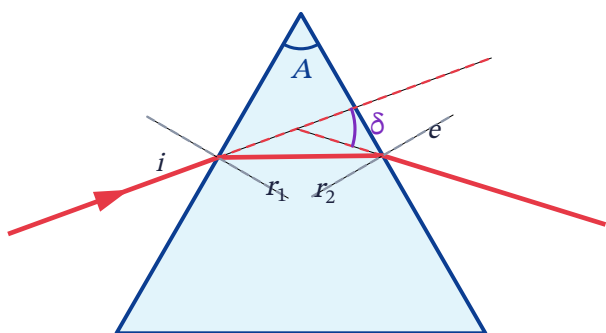
- Mirror: $\frac{1}{v} + \frac{1}{u}, m = -\frac{v}{u}$. Lens: $\frac{1}{v} - \frac{1}{u}, m = +\frac{v}{u}$.
- Minimum distance between object and **real** image for a convex lens = **$4f$** (at $u = -2f$).

🚀 JEE Main Extension

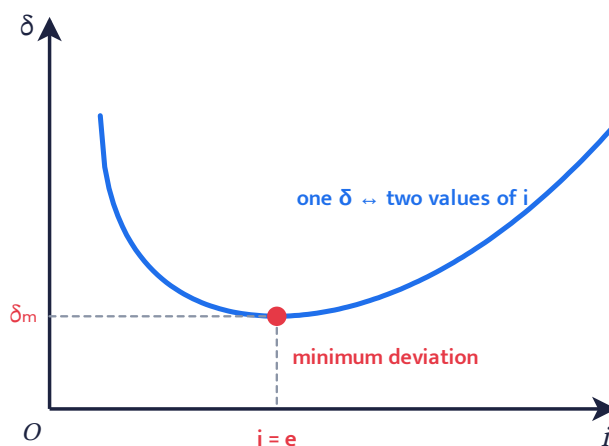
Silvered lens acts as a mirror: $\frac{1}{F} = \frac{2}{f_L} + \frac{1}{f_M}$ (powers: $P = 2P_L + P_M$). Plano-convex lens silvered on the plane face: $F = \frac{R}{2(n-1)}$; on the curved face: $F = \frac{R}{2n}$. **Displacement method**: for fixed object–screen distance $D > 4f$, two lens positions give sharp images; $f = \frac{D^2 - x^2}{4D}$ and $h = \sqrt{h_1 h_2}$.

9.5 Refraction through a Prism

A ray refracts twice – at the two faces inclined at the **angle of prism A** – and bends towards the base. The angle between the incident and emergent rays is the **angle of deviation δ** .



Path through a prism ($A = 60^\circ$, $n = 1.5$, $i = 50^\circ$): $\delta \approx 37^\circ$



δ - i graph for the same prism: minimum $\delta_m \approx 37.2^\circ$ at $i = e \approx 48.6^\circ$

📐 Prism formulas

$$r_1 + r_2 = A, \quad \delta = i + e - A \quad \text{General relations}$$

$$i = e, \quad r_1 = r_2 = \frac{A}{2} \quad \text{At minimum deviation (ray inside } \parallel \text{ base for an isosceles prism)}$$

$$n_{21} = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} \quad \text{Refractive index from minimum deviation}$$

$$\delta \approx (n_{21} - 1)A \quad \text{Thin prism (A small): } \delta \text{ independent of } i$$

Solved Example

For a prism of angle 60° , the angle of minimum deviation is 30° . Find its refractive index and the angle of incidence at minimum deviation.

Solution: $n = \frac{\sin 45^\circ}{\sin 30^\circ} = \sqrt{2} \approx 1.414$. At δ_m : $i = \frac{A + \delta_m}{2} = 45^\circ$.

VS Sir's Exam Tip

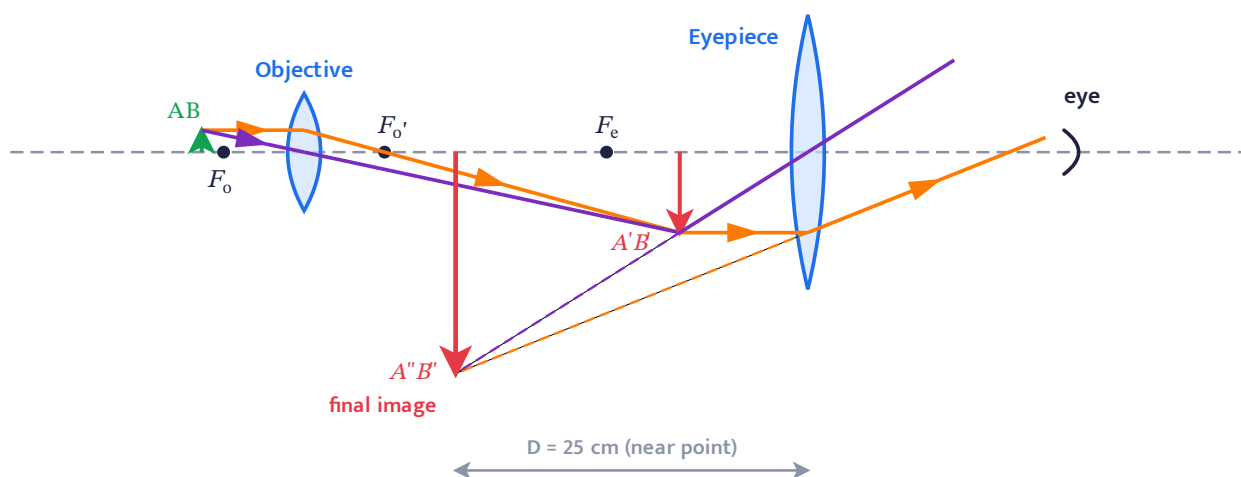
- Maximum deviation occurs at grazing incidence ($i = 90^\circ$) – one δ (other than δ_m) corresponds to two values of i (i and e are interchangeable).
- No emergence (TIR at 2nd face) if $A > 2C$.

Dispersion (JEE Main)

- n depends on λ : $n_v > n_r \Rightarrow$ violet deviates most. Angular dispersion $\theta = \delta_v - \delta_r = (n_v - n_r)A$.
- Dispersive power $\omega = \frac{n_v - n_r}{n - 1}$, $n = \frac{n_v + n_r}{2}$.

9.6 Optical Instruments

Simple microscope (magnifier): a convex lens of short focal length with the object inside F gives a virtual, erect, magnified image. The angular size of an object is largest when it is at the **near point $D = 25$ cm**; the magnifier lets us bring it much closer.



Compound microscope: the objective forms a real, inverted, magnified image A' inside F_e ; the eyepiece magnifies it further into a virtual final image

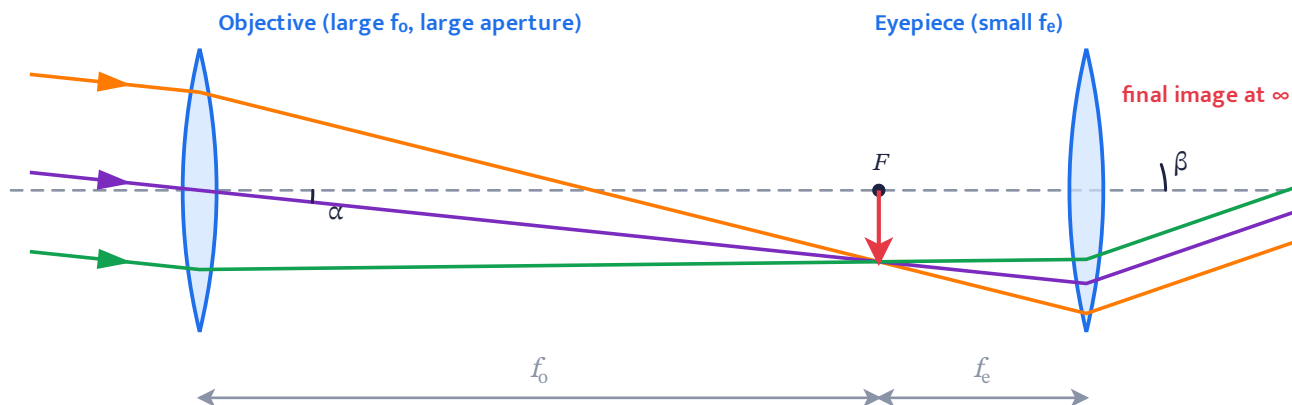
Instrument	Image at near point (D)	Normal adjustment (image at ∞)
Simple microscope	$m = 1 + \frac{D}{f}$	$m = \frac{D}{f}$
Compound microscope	$m = \frac{v_o}{ u_o } \left(1 + \frac{D}{f_e} \right)$	$m \approx \frac{L}{f_o} \cdot \frac{D}{f_e}$
Astronomical telescope	$m = \frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$	$m = \frac{f_o}{f_e}$, $L = f_o + f_e$

In the compound microscope L is the **tube length** (distance between the second focal point of the objective and the first focal point of the eyepiece). Both f_o and f_e are small, with $f_o < f_e$.

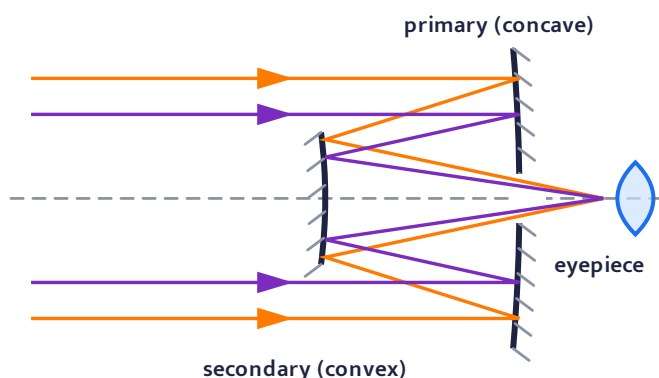
 Solved Example

A compound microscope has $f_o = 1$ cm, $f_e = 5$ cm and tube length $L = 20$ cm. Find the magnifying power in normal adjustment. What if the final image is at 25 cm (take $v_o \approx L + f_o = 21$ cm)?

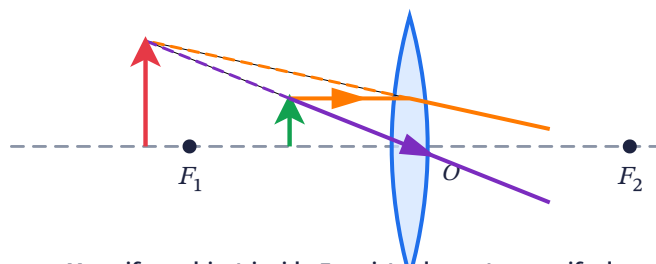
Solution: Normal: $m = \frac{20}{1} \times \frac{25}{5} = 100$. Near point: $m_o = \frac{v_o}{f_o} - 1 = 20$, $m_e = 1 + \frac{25}{5} = 6 \Rightarrow m = 120$.



Astronomical (refracting) telescope in normal adjustment: $m = \beta / \alpha = f_o / f_e$; the final image is inverted



Cassegrain reflecting telescope: concave primary + convex secondary



Magnifier: object inside $F \rightarrow$ virtual, erect, magnified

Simple microscope (magnifying glass)

 Why reflecting telescopes?

- No **chromatic aberration** (a mirror does not disperse light).
- A **parabolic** mirror removes spherical aberration.
- A mirror is lighter and supported over its whole back – large apertures possible $\rightarrow$ more light-gathering and better resolution.

 Common Trap

- Telescope: f_o **large**, f_e small. Microscope: **both** small, $f_o < f_e$. Swapping lenses in a telescope gives $m < 1$.
- Magnifying power of a telescope does not depend on the object distance; it is angular magnification.

 Solved Example

An astronomical telescope has an objective of focal length 100 cm and an eyepiece of 5 cm. Find the magnifying power and tube length in normal adjustment, and the magnifying power when the final image is at 25 cm.

Solution: $m = \frac{100}{5} = 20$, $L = 100 + 5 = 105$ cm. At D: $m = 20 \left(1 + \frac{5}{25} \right) = 24$.

 Memory Trick

"Telescope wants a **tall** objective (large f_o), **Microscope** wants **mini** lenses (small f_o, f_e)."
Telescope in normal adjustment: $m = f_o / f_e, L = f_o + f_e$.

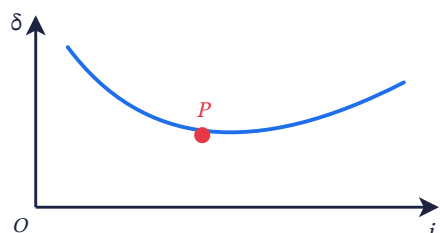
 Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1 **NCERT based** An object is placed 10 cm in front of a concave mirror of focal length 15 cm. The image is
 - (a) 30 cm behind the mirror, virtual, erect, 3 times
 - (b) 30 cm in front of the mirror, real, inverted, 3 times
 - (c) 6 cm behind the mirror, virtual, erect, 0.6 times
 - (d) 30 cm behind the mirror, virtual, inverted, 3 times
- 2 **NCERT based** The critical angle for glass of refractive index 1.5 (glass–air boundary) is about
 - (a) 60°
 - (b) 30°
 - (c) 48.6°
 - (d) 41.8°
- 3 **NCERT based** Optical fibres transmit light over long distances with little loss because of
 - (a) dispersion
 - (b) diffraction
 - (c) total internal reflection
 - (d) refraction
- 4 **NCERT based** A biconvex lens ($n = 1.5$) has both radii of curvature 20 cm. Its focal length in air is
 - (a) 30 cm
 - (b) 40 cm
 - (c) 10 cm
 - (d) 20 cm
- 5 **NEET pattern** A coin lies at the bottom of a tank filled with water ($n = 4/3$) to a depth of 20 cm. Seen from above (near-normal), the coin appears raised by
 - (a) 2.5 cm
 - (b) 5 cm
 - (c) 6.67 cm
 - (d) 15 cm
- 6 **NCERT based** A thin prism of angle 5° and refractive index 1.5 produces a deviation of
 - (a) 3.3°
 - (b) 7.5°
 - (c) 2.5°
 - (d) 5°
- 7 **NCERT based** The focal length of a concave mirror in air is f . When the mirror is immersed in water ($n = 4/3$), its focal length becomes
 - (a) f
 - (b) $3f/4$
 - (c) $4f$
 - (d) $4f/3$
- 8 **NEET pattern** A convex lens of focal length 25 cm and a concave lens of focal length 20 cm are placed in contact. The combination is
 - (a) diverging, $P = -1$ D
 - (b) diverging, $P = -9$ D
 - (c) converging, $P = +1$ D
 - (d) converging, $P = +9$ D
- 9 **NEET pattern** A convex lens forms a real image of the same size as the object; the distance between object and image is 40 cm. The power of the lens is
 - (a) +2.5 D
 - (b) +10 D
 - (c) +5 D
 - (d) +20 D
- 10 **NEET pattern** A glass convex lens ($n = 1.5$) has focal length 10 cm in air. Its focal length in water ($n = 4/3$) is
 - (a) 13.3 cm
 - (b) 10 cm
 - (c) 20 cm
 - (d) 40 cm

- 11 **NEET pattern** The angle of minimum deviation of an equilateral prism is 30° . The refractive index of the prism is
 (a) $\sqrt{3}$ (b) $\sqrt{2}$ (c) $\frac{1}{\sqrt{2}}$ (d) 1.5

- 12 **NEET pattern** For the δ - i curve of a prism shown, at the lowest point P



- (a) $e = 90^\circ$ (grazing emergence)
 (b) $\delta = A$
 (c) $i = 90^\circ$ (grazing incidence)
 (d) $i = e$ and the ray inside is symmetric ($r_1 = r_2 = A/2$)
- 13 **NEET pattern** An astronomical telescope has $f_o = 100$ cm and $f_e = 5$ cm. In normal adjustment, its magnifying power and length are
 (a) 500, 105 cm (b) 20, 105 cm (c) 20, 95 cm (d) 0.05, 105 cm
- 14 **NEET pattern** A compound microscope has $f_o = 1$ cm, $f_e = 5$ cm and tube length 20 cm. Its magnifying power in normal adjustment ($D = 25$ cm) is
 (a) 100 (b) 20 (c) 120 (d) 125
- 15 **NEET pattern** The maximum magnifying power of a simple microscope of focal length 5 cm ($D = 25$ cm) is
 (a) 5 (b) 6 (c) 25 (d) 4
- 16 **NEET pattern** Light of wavelength 600 nm in air enters glass ($n = 1.5$). Which statement is correct?
 (a) Speed 2×10^8 m/s, wavelength 600 nm, frequency decreases
 (b) Speed becomes 2×10^8 m/s, wavelength 400 nm, frequency unchanged
 (c) Speed unchanged, wavelength 400 nm, frequency increases
 (d) Speed 4.5×10^8 m/s, wavelength 900 nm, frequency unchanged
- 17 **Assertion-Reason** Assertion (A): A diamond sparkles much more than a glass piece cut in the same shape. Reason (R): The critical angle for diamond ($\approx 24^\circ$) is much smaller than that for glass, so light suffers repeated total internal reflection inside it.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

- 18 **Assertion-Reason** Assertion (A): A convex glass lens ($n = 1.5$) behaves as a diverging lens when immersed in carbon disulphide ($n = 1.63$). Reason (R): In the lens maker's formula the factor $(n_{\text{lens}} / n_{\text{medium}} - 1)$ becomes negative in a denser medium.
- (a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 19 **Assertion-Reason** Assertion (A): The magnifying power of an astronomical telescope in normal adjustment increases if an eyepiece of smaller focal length is used. Reason (R): The resolving power of a telescope depends on the diameter of its objective.
- (a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 20 **JEE Main pattern** An object moves towards a concave mirror of focal length 20 cm along its axis at 1 cm/s. When the object is 30 cm from the mirror, the image moves at
- (a) 0.25 cm/s, towards the mirror
(b) 4 cm/s, away from the mirror
(c) 2 cm/s, away from the mirror
(d) 4 cm/s, towards the mirror
- 21 **JEE Main pattern** A point object in air is 30 cm in front of a convex spherical glass surface ($n = 1.5$, $R = 20$ cm). The image is formed
- (a) 90 cm inside the glass (real)
(b) 180 cm from the surface on the object side (virtual)
(c) 60 cm inside the glass (real)
(d) 120 cm on the object side (virtual)
- 22 **JEE Main pattern** An equiconvex lens of focal length f is cut into two identical halves by a plane perpendicular to its principal axis. The focal length of each half is
- (a) $f/2$
(b) $2f$
(c) $4f$
(d) f
- 23 **JEE Main pattern** A plano-convex lens ($n = 1.5$, $R = 20$ cm) is silvered on its plane face. It behaves like a concave mirror of focal length
- (a) 40 cm
(b) 6.67 cm
(c) 10 cm
(d) 20 cm
- 24 **JEE Main pattern** The critical angle for light going from glass ($n = 1.5$) into water ($n = 4/3$) is
- (a) $\sin^{-1}(1/2)$
(b) $\sin^{-1}(3/4)$
(c) $\sin^{-1}(8/9)$
(d) $\sin^{-1}(2/3)$

✓ Answer Key & Quick Solutions

- 1(a) 2(d) 3(c) 4(d) 5(b) 6(c) 7(a) 8(a) 9(b) 10(d) 11(b) 12(d) 13(b) 14(a) 15(b)
16(b) 17(a) 18(a) 19(b) 20(b) 21(b) 22(b) 23(d) 24(c)

1. (a) $\frac{1}{v} = \frac{1}{-15} + \frac{1}{10} = \frac{1}{30} \Rightarrow v = +30$ cm (behind); $m = -v/u = +3$: virtual, erect, magnified.
2. (d) $\sin C = 1/1.5 = 0.667 \Rightarrow C \approx 41.8^\circ$. (48.6° is for water.)

3. (c) Light striking the core–cladding boundary at angles greater than C is totally internally reflected again and again.
4. (d) $\frac{1}{f} = 0.5 \left(\frac{1}{20} + \frac{1}{20} \right) = \frac{1}{20} \Rightarrow f = 20 \text{ cm}.$
5. (b) Shift = $d(1 - 1/n) = 20(1 - 3/4) = 5 \text{ cm}$ (apparent depth 15 cm).
6. (c) $\delta = (n - 1)A = 0.5 \times 5^\circ = 2.5^\circ.$
7. (a) $f = R/2$ depends only on the geometry of the mirror – reflection does not depend on the medium.
8. (a) $P = 4 - 5 = -1 \text{ D} \Rightarrow f = -100 \text{ cm}$, diverging.
9. (b) Same size real image $\Rightarrow u = -2f, v = 2f \Rightarrow 4f = 40 \text{ cm} \Rightarrow f = 10 \text{ cm} \Rightarrow P = +10 \text{ D}.$
10. (d) $f_w / f_a = 0.5 / (1.125 - 1) = 4 \Rightarrow 40 \text{ cm}.$
11. (b) $n = \sin 45^\circ / \sin 30^\circ = \sqrt{2}.$
12. (d) The minimum of δ occurs when the ray passes symmetrically: $i = e, r_1 = r_2 = A/2.$
13. (b) $m = f_o/f_e = 20; L = f_o + f_e = 105 \text{ cm}.$
14. (a) $m = (L/f_o)(D/f_e) = 20 \times 5 = 100.$
15. (b) Maximum when the image is at the near point: $m = 1 + D/f = 1 + 5 = 6.$
16. (b) $v = c/n = 2 \times 10^8 \text{ m/s}; \lambda = 600/1.5 = 400 \text{ nm};$ frequency is fixed by the source.
17. (a) Both true; small C traps light by multiple TIR, which causes the sparkle.
18. (a) $1.5/1.63 - 1 < 0 \Rightarrow f < 0$: the lens diverges. R explains A.
19. (b) A: $m = f_o/f_e$ – true. R is also true but concerns resolution, not magnification.
20. (b) $v = -60 \text{ cm}, m = -2 \Rightarrow$ image speed = $m^2(1) = 4 \text{ cm/s}$; as the object approaches F the image recedes (away from the mirror).
21. (b) $\frac{1.5}{v} - \frac{1}{-30} = \frac{0.5}{20} \Rightarrow \frac{1.5}{v} = -\frac{1}{120} \Rightarrow v = -180 \text{ cm}$: virtual, on the object side.
22. (b) Each half is plano-convex: $\frac{1}{f} = (n - 1)\frac{1}{R} = \frac{1}{2f} \Rightarrow f = 2f.$
23. (d) $f_L = R / (n - 1) = 40 \text{ cm}; \frac{1}{F} = \frac{2}{f_L}$ (plane mirror adds no power) $\Rightarrow F = 20 \text{ cm} = R/[2(n - 1)].$
24. (c) $\sin C = n_w / n_g = (4/3) / 1.5 = 8/9 (\approx 62.7^\circ).$

CHAPTER

10

Wave Optics

Day 16–17

Day 16–17

Weightage ★★☆☆

NCERT Class 12

Huygens principle

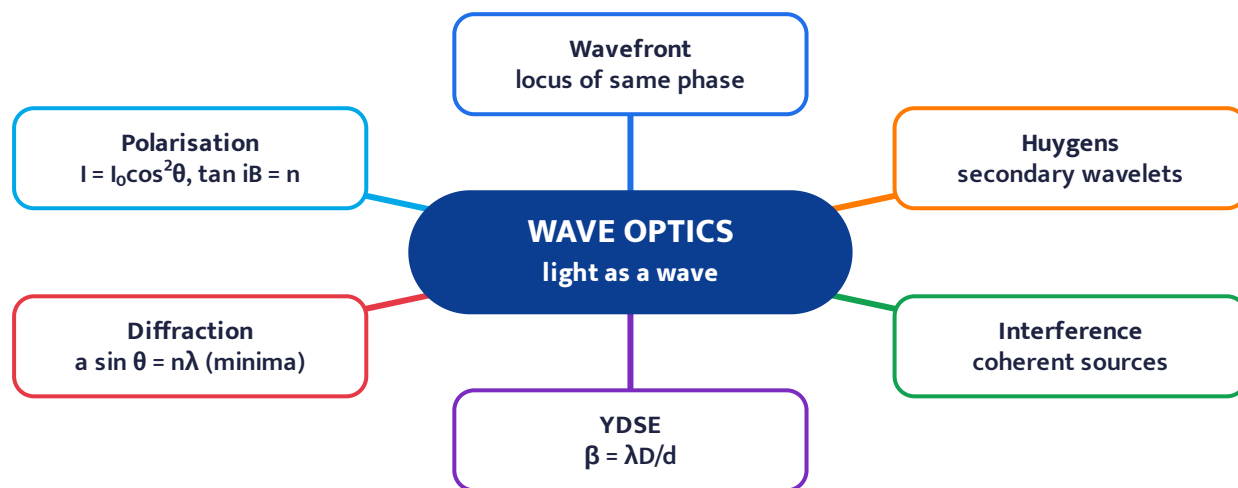
Coherent sources

YDSE

Single-slit diffraction

Polarisation

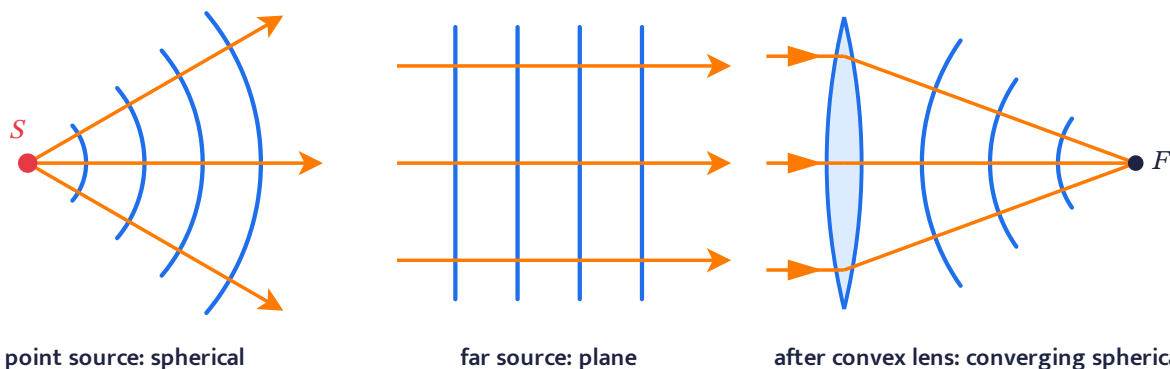
Malus & Brewster



Chapter at a glance

10.1 Wavefront and Huygens' Principle

A **wavefront** is the locus of all points that oscillate **in the same phase**. Energy (and the ray) travels **perpendicular to the wavefront**. A point source gives spherical wavefronts; at large distances a small part of a sphere is a **plane wavefront**.

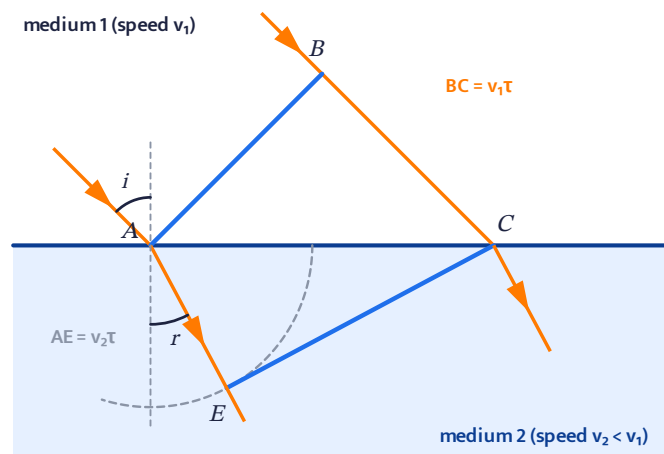


Wavefronts (blue) and rays (orange) – rays are always normal to wavefronts

Huygens' principle

- Every point of a wavefront acts as a source of **secondary wavelets** that spread out in the forward direction with the speed of the wave.
- The **forward envelope** (common tangent) of these wavelets after time τ gives the new wavefront. (The backwave is absent.)

Refraction and reflection using Huygens' principle



Plane wave AB enters a denser medium: wavelet from A has radius $v_2\tau$ while B travels $v_1\tau$ to C



Deriving Snell's law

$$\sin i = \frac{BC}{AC} = \frac{v_1\tau}{AC} \quad \text{From triangle ABC}$$

$$\sin r = \frac{AE}{AC} = \frac{v_2\tau}{AC} \quad \text{From triangle AEC}$$

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1} \quad \text{Snell's law}$$



Note

Reflection: the wavelet from A grows in the **same** medium, so $AE = BC \Rightarrow i = r$. Frequency is unchanged; $\lambda_2 = \lambda_1 v_2 / v_1$.



Common Trap

- Wave theory predicts light is **slower** in a denser medium ($v_2 < v_1$ when bending towards the normal) – confirmed by Foucault; the corpuscular theory wrongly predicted it faster.
- A convex lens / concave mirror turns a plane wavefront into a **converging spherical** one; a prism only tilts a plane wavefront.

10.2 Coherent Sources and Interference

Coherent sources emit waves of the same frequency with a **constant (time-independent) phase difference**. Two independent bulbs or sodium lamps are **incoherent** – their phase difference changes randomly every $\sim 10^{-10}$ s, so intensities simply add and no pattern is seen. In practice coherent sources are derived from one source (two slits of YDSE).



Superposition of two waves

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi \quad \text{Resultant intensity (coherent), } \phi = \text{phase difference}$$

$$\phi = \frac{2\pi}{\lambda} \Delta x$$

Phase difference from path difference Δx

$$\text{Bright: } \Delta x = n\lambda, \quad \phi = 2n\pi \quad \text{Constructive: } I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$\text{Dark: } \Delta x = (2n - 1)\frac{\lambda}{2} \quad \text{Destructive: } I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

$$I_1 = I_2 = I_0: I = 4I_0 \cos^2 \frac{\phi}{2} \quad \text{Equal intensities: } I_{\max} = 4I_0, I_{\min} = 0$$

$$I = I_1 + I_2$$

Incoherent sources: average $\langle \cos \phi \rangle = 0$

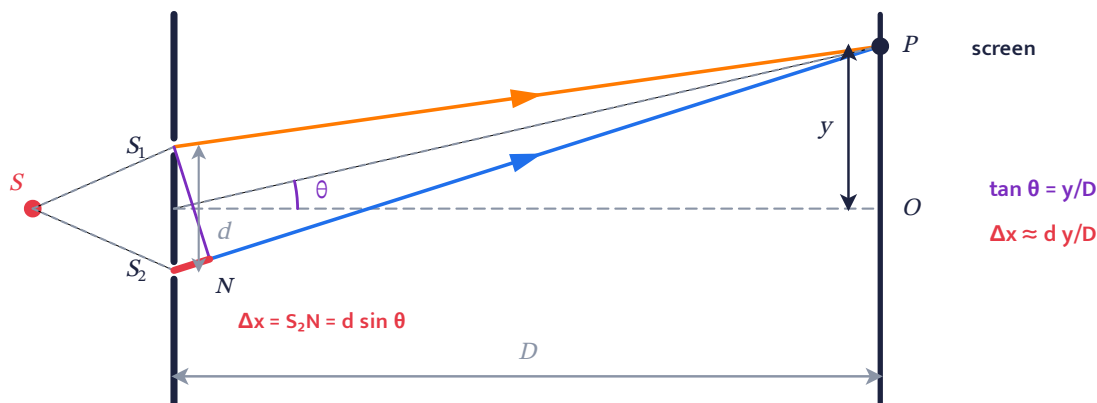


Note

Interference only **redistributes** energy: the average intensity over the pattern is still $I_1 + I_2$ ($= 2I_0$ for equal sources). Energy is conserved.

10.3 Young's Double-Slit Experiment (YDSE)

Light from a narrow slit S falls on two close slits S_1, S_2 (separation d) that act as coherent sources. On a screen at distance D ($D \gg d$), the path difference at a point P at height y is $\Delta x = S_2P - S_1P \approx d \sin \theta \approx \frac{yd}{D}$.

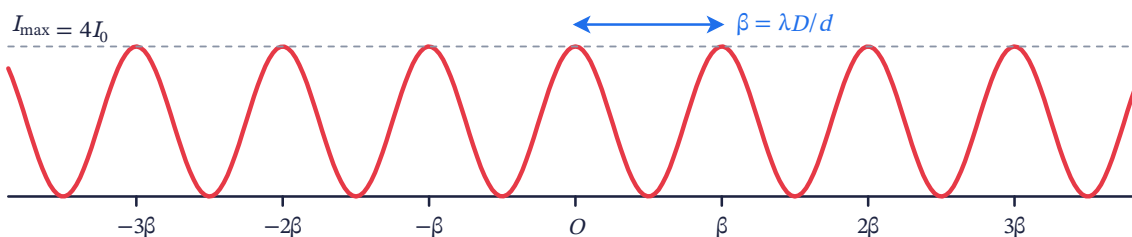
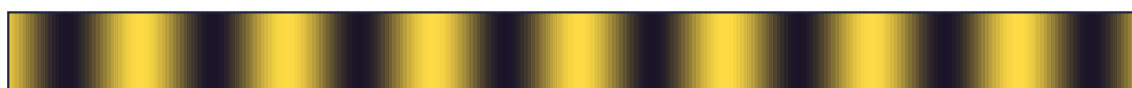


YDSE geometry: path difference $S_2N = d \sin \theta$

YDSE formulas

$y_n = \frac{n\lambda D}{d}$	n -th bright fringe ($n = 0, 1, 2 \dots$)	$y'_n = \frac{(2n-1)\lambda D}{2d}$	n -th dark fringe ($n = 1, 2 \dots$)
$\beta = \frac{\lambda D}{d}$	Fringe width (bright = dark width)	$\theta_0 = \frac{\lambda}{d}$	Angular fringe width (independent of D)
$\beta_{\text{medium}} = \frac{\beta}{n}$	Whole set-up in a medium of refractive index n	$y_0 = \frac{(\mu-1)tD}{d}$	Shift of pattern when a thin sheet (μ, t) covers one slit – towards that slit

central bright fringe (O)



Equally spaced bright and dark fringes and the $\cos^2$ intensity distribution ($I = 4I_0 \cos^2 \frac{2\pi y}{\beta}$)

Solved Example

In a YDSE, $d = 0.5 \text{ mm}$, $D = 1 \text{ m}$ and $\lambda = 600 \text{ nm}$. Find (i) the fringe width, (ii) the distance of the 3rd dark fringe from the centre, (iii) the fringe width if the whole apparatus is immersed in water ($n = 4/3$).

Solution: (i) $\beta = \frac{600 \times 10^{-9} \times 1}{0.5 \times 10^{-3}} = 1.2 \text{ mm}$. (ii) $y'_3 = \frac{5}{2}\beta = 3.0 \text{ mm}$. (iii) $\beta / n = 1.2 \times \frac{3}{4} = 0.9 \text{ mm}$.

! Common Trap

- Covering one slit with a glass sheet **shifts** the pattern but does **not** change β .
- With **white light** the central fringe is white; nearby fringes are coloured (blue/violet edge nearer the centre, red farther, since $\beta \propto \lambda$).
- If the source slit S is made too wide, or $s / S > \lambda / d$ (s = source size, S = distance to slits), the fringes disappear.

🎯 VS Sir's Exam Tip

- Intensity ratio problems: write $\frac{I_{max}}{I_{min}} = \left(\frac{r+1}{r-1}\right)^2$ with $r = \sqrt{I_1/I_2}$ = amplitude ratio.
- Number of fringes in a fixed width $\propto 1/\lambda$:
 $n_1\lambda_1 = n_2\lambda_2$.

🔧 Solved Example

Two coherent sources have intensities in the ratio 9 : 1. Find I_{max} / I_{min} in the interference pattern.

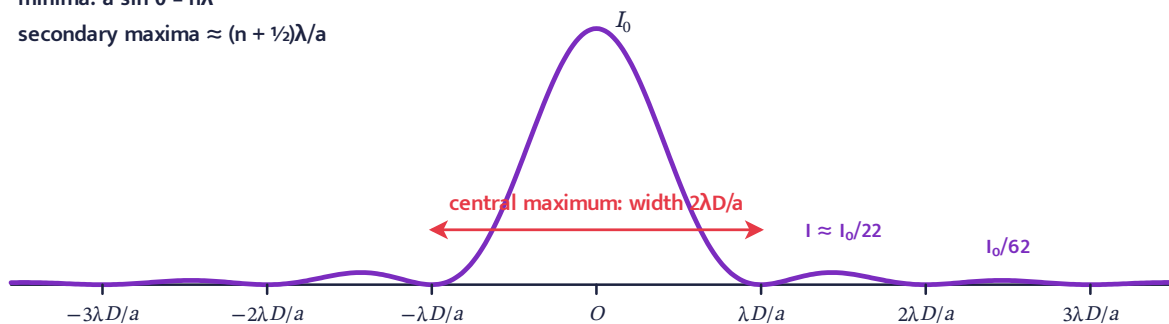
Solution: Amplitude ratio $r = \sqrt{9} = 3 \Rightarrow \frac{I_{max}}{I_{min}} = \left(\frac{3+1}{3-1}\right)^2 = 4 : 1$.

10.4 Diffraction at a Single Slit

Diffraction is the bending of light around the edges of an obstacle or aperture whose size is comparable to λ . For a single slit of width a illuminated by a plane wave, divide the slit into halves (or into $2n$ parts): wavelets from corresponding points cancel in pairs at angles where $a \sin \theta = n\lambda$.

minima: $a \sin \theta = n\lambda$

secondary maxima $\approx (n + \frac{1}{2})\lambda/a$



Single-slit intensity: a bright, wide central maximum and much weaker secondary maxima

🔧 Single-slit formulas

$a \sin \theta = n\lambda$ ($n = \pm 1, \pm 2 \dots$) Minima (dark)

$a \sin \theta \approx \left(n + \frac{1}{2}\right)\lambda$ Secondary maxima ($n = 1, 2 \dots$), intensity falls rapidly

$2\theta = \frac{2\lambda}{a}$, $W = \frac{2\lambda D}{a}$ Angular and linear width of the central maximum

$W = \frac{2\lambda f}{a}$ Pattern focused by a lens of focal length f

Feature	Interference (YDSE)	Single-slit diffraction
Cause	superposition of two wavefronts from two coherent sources	superposition of wavelets from different parts of the same wavefront
Fringe width	all fringes equal	central max twice the others
Intensity	all bright fringes equally bright	falls rapidly with order
Number of visible fringes	large	few (intensity dies out quickly)

Solved Example

Light of wavelength 500 nm falls normally on a slit of width 0.1 mm. Find the width of the central maximum on a screen 1 m away.

Solution: $W = \frac{2\lambda D}{a} = \frac{2 \times 500 \times 10^{-9} \times 1}{10^{-4}} = 10^{-2} \text{ m} = \mathbf{10 \text{ mm}}$ (angular width $2\lambda / a = 0.01 \text{ rad}$).

Common Trap

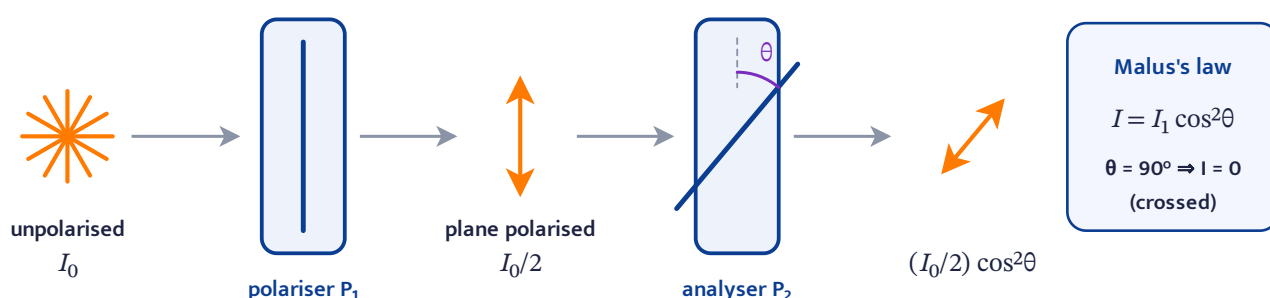
- Narrower slit $\Rightarrow$ **wider** central maximum ($W \propto 1/a$). Larger λ (red) $\Rightarrow$ wider pattern.
- The condition $a \sin \theta = n\lambda$ is for **minima** in diffraction but for **maxima** in YDSE ($d \sin \theta = n\lambda$) – do not mix them.

Beyond the current syllabus (quick reference)

- Fresnel distance $z_F = a^2 / \lambda$: ray optics is valid for distances $\ll z_F$.
- Angular resolution of a telescope $\Delta \theta \approx 1.22\lambda / D$ (D = aperture): bigger objective $\Rightarrow$ better resolution.

10.5 Polarisation

Light is a **transverse** EM wave: its electric vector vibrates perpendicular to the direction of propagation. In **unpolarised** light (sun, bulb) the vibrations occur in all directions in the transverse plane, changing randomly. If the vibrations are confined to a single direction, the light is **plane (linearly) polarised**. Only transverse waves can be polarised – sound (longitudinal) cannot.



A polaroid transmits the component of E along its pass axis: unpolarised $\rightarrow I_0 / 2$; then Malus's law

Polarisation laws

$$I = \frac{I_0}{2}$$

Unpolarised light through one ideal polaroid – any orientation

$$I = I_1 \cos^2 \theta$$

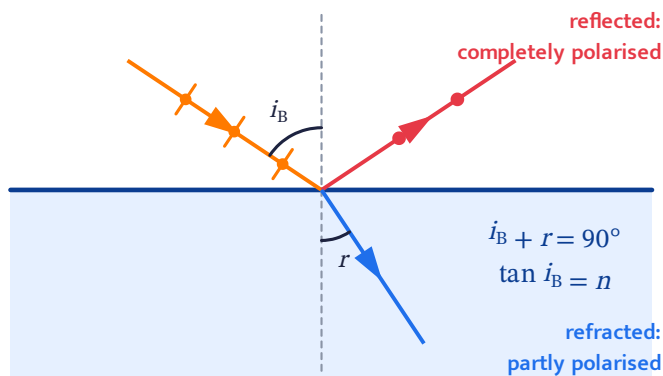
Malus's law: polarised light I_1 through a polaroid at angle θ

$$\tan i_B = n (= n_{21})$$

Brewster's law: reflected light is completely polarised

$$i_B + r = 90^\circ$$

At Brewster angle, reflected $\perp$ refracted ray



Polarisation by reflection at Brewster's angle (dots = vibrations $\perp$ page)

Uses of polaroids

- Polaroid **sunglasses** cut glare from reflecting surfaces (water, roads).
- Camera filters, window panes of trains/aircraft to control light intensity.
- **3-D movies** (two pictures with crossed polarisation).
- LCD displays; testing whether light is polarised; scattered sunlight at 90° is polarised.

Solved Example

Unpolarised light of intensity I_0 passes through three polaroids. The first and third are crossed and the middle one is at 45° to the first. Find the transmitted intensity. What if the middle polaroid is removed?

Solution: $I = \frac{I_0}{2} \cos^2 45^\circ \cos^2 45^\circ = \frac{I_0}{8}$. Without the middle one, the two crossed polaroids give **zero**.

Memory Trick

- **Brewster** $\leftrightarrow$ Both rays at 90° : $\tan i_B = n$. Glass $n = 1.5 \Rightarrow i_B \approx 56.3^\circ$; $n = \sqrt{3} \Rightarrow 60^\circ$.
- "Half at the first, $\cos^2$ after that" – the polaroid chain rule.

VS Sir's Exam Tip

- Rotating an analyser through 360° in polarised light gives **two maxima and two minima (zero)**; in unpolarised light the intensity stays constant.
- Partially polarised light shows maxima and **non-zero** minima.

Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1 **NCERT based** The wavefront due to a point source at a small distance, and due to a source at infinity, are respectively

- (a) cylindrical, plane
(c) plane, spherical

- (b) spherical, cylindrical
(d) spherical, plane

- 2 **NCERT based** The angle between a ray of light and its wavefront is

(a) 45°

(b) 0°

(c) 90°

(d) 180°

- 3 **NCERT based** Two independent sodium lamps placed side by side do not produce an interference pattern because
 (a) their phase difference changes randomly with time
 (b) light from them is unpolarised
 (c) they emit different wavelengths
 (d) their intensities are unequal
- 4 **NEET pattern** In YDSE, $d = 0.5 \text{ mm}$, $D = 1 \text{ m}$ and $\lambda = 600 \text{ nm}$. The fringe width is
 (a) 1.2 mm (b) 12 mm (c) 0.6 mm (d) 0.12 mm
- 5 **NEET pattern** The YDSE apparatus of the previous question ($\beta = 1.2 \text{ mm}$ in air) is immersed in water ($n = 4/3$). The new fringe width is
 (a) 0.6 mm (b) 1.2 mm (c) 1.6 mm (d) 0.9 mm
- 6 **NEET pattern** Two coherent sources have intensities in the ratio 9 : 1. The ratio $I_{\max} : I_{\min}$ is
 (a) 9 : 1 (b) 3 : 1 (c) 16 : 1 (d) 4 : 1
- 7 **NEET pattern** In YDSE with equal slits, the maximum intensity is I_m . The intensity at a point where the path difference is $\lambda/4$ is
 (a) $I_m / 2$ (b) I_m (c) $I_m / 4$ (d) $3I_m / 4$
- 8 **JEE Main pattern** In YDSE, the distance between the 2nd dark fringe and the 5th bright fringe on the same side of the centre is
 (a) 3β (b) 4.5β (c) 3.5β (d) 2.5β
- 9 **JEE Main pattern** In YDSE, a thin glass sheet ($\mu = 1.5$) placed over one slit shifts the central fringe by 5 fringes ($\lambda = 600 \text{ nm}$). The thickness of the sheet is
 (a) $3 \mu\text{m}$ (b) $6 \mu\text{m}$ (c) $1.2 \mu\text{m}$ (d) $9 \mu\text{m}$
- 10 **NCERT based** The angular width of a fringe in YDSE with $\lambda = 589 \text{ nm}$ is 0.20° . If the whole apparatus is immersed in water ($n = 4/3$), the angular fringe width becomes
 (a) 0.20° (b) 0.15° (c) 0.27° (d) 0.10°
- 11 **NEET pattern** Light of wavelength 500 nm falls on a slit of width 0.1 mm . The linear width of the central maximum on a screen 1 m away is
 (a) 20 mm (b) 5 mm (c) 10 mm (d) 1 mm
- 12 **NEET pattern** In a single-slit experiment with $\lambda = 600 \text{ nm}$, the first minimum is observed at $\theta = 30^\circ$. The slit width is
 (a) $0.6 \mu\text{m}$ (b) $1.2 \mu\text{m}$ (c) $2.4 \mu\text{m}$ (d) $0.3 \mu\text{m}$
- 13 **NCERT based** In single-slit diffraction, if the slit width is halved, the width of the central maximum
 (a) doubles (b) becomes four times
 (c) halves (d) remains the same
- 14 **NEET pattern** Unpolarised light of intensity 32 W/m^2 passes through two polaroids whose pass axes are at 60° . The transmitted intensity is
 (a) 8 W/m^2 (b) 4 W/m^2 (c) 2 W/m^2 (d) 16 W/m^2

- 15 **NEET pattern** The Brewster angle for a medium of refractive index $\sqrt{3}$ is
(a) 90° (b) 60° (c) 45° (d) 30°
- 16 **NCERT based** At the Brewster angle, the angle between the reflected and refracted rays is
(a) 0° (b) 90° (c) 45° (d) 180°
- 17 **NEET pattern** Which phenomenon shows that light waves are transverse?
(a) Interference (b) Polarisation (c) Reflection (d) Diffraction
- 18 **NEET pattern** In YDSE using white light, the central fringe is
(a) white, with coloured fringes on either side
(b) violet
(c) red
(d) dark
- 19 **Assertion-Reason** Assertion (A): In YDSE the fringe width decreases when the apparatus is immersed in water. Reason (R): The wavelength of light decreases in a denser medium while its frequency remains unchanged.
(a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 20 **Assertion-Reason** Assertion (A): Interference pattern can be obtained using two separate identical bulbs. Reason (R): For sustained interference, the sources must have a constant phase difference.
(a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 21 **Assertion-Reason** Assertion (A): Sound waves cannot be polarised. Reason (R): Sound waves in air are longitudinal.
(a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 22 **JEE Main pattern** In YDSE, $d = 1 \text{ mm}$ and $D = 1 \text{ m}$. Light containing 600 nm and 480 nm is used. The least distance from the central maximum where bright fringes of both wavelengths coincide is
(a) 4.8 mm (b) 1.2 mm (c) 2.4 mm (d) 0.6 mm
- 23 **JEE Main pattern** Unpolarised light of intensity I_0 falls on three polaroids; the first and the last are crossed and the middle one makes 45° with the first. The emergent intensity is
(a) 0 (b) $I_0 / 8$ (c) $I_0 / 2$ (d) $I_0 / 4$

24. **JEE Main pattern** In YDSE with equal slits, the intensity at the central maximum is I_0 . The intensity at a point where the path difference is $\lambda/3$ is

(a) $I_0 / 4$

(b) $I_0 / 3$

(c) $3I_0 / 4$

(d) $I_0 / 2$



Answer Key & Quick Solutions

- 1(d) 2(c) 3(a) 4(a) 5(d) 6(d) 7(a) 8(c) 9(b) 10(b) 11(c) 12(b) 13(a) 14(b) 15(b)
16(b) 17(b) 18(a) 19(a) 20(c) 21(a) 22(c) 23(b) 24(a)

1. (d) A point source gives spherical wavefronts; far away a small part of the sphere is plane.
2. (c) Rays are normal to wavefronts (direction of energy flow).
3. (a) They are incoherent: phase difference fluctuates every $\sim 10^{-10}$ s, so the pattern averages out.
4. (a) $\beta = \lambda D/d = 6 \times 10^{-7} \times 1 / 5 \times 10^{-4} = 1.2 \times 10^{-3}$ m.
5. (d) λ decreases n times $\Rightarrow \beta' = \beta/n = 1.2 \times 3/4 = 0.9$ mm.
6. (d) Amplitudes $3 : 1 \Rightarrow (3+1)^2 : (3-1)^2 = 16 : 4 = 4 : 1$.
7. (a) $\phi = (2\pi/\lambda)(\lambda/4) = \pi/2 \Rightarrow I = I_m \cos^2(\pi/4) = I_m/2$.
8. (c) 5th bright at 5β , 2nd dark at $1.5\beta \Rightarrow 3.5\beta$.
9. (b) $(\mu - 1)t = 5\lambda \Rightarrow t = 5 \times 600 \text{ nm} / 0.5 = 6000 \text{ nm} = 6 \mu\text{m}$.
10. (b) $\theta = \lambda/d$ and $\lambda \rightarrow \lambda/n \Rightarrow 0.20^\circ \times 3/4 = 0.15^\circ$.
11. (c) $W = 2\lambda D/a = 2 \times 5 \times 10^{-7} / 10^{-4} = 10^{-2}$ m.
12. (b) $a \sin 30^\circ = \lambda \Rightarrow a = 2\lambda = 1.2 \mu\text{m}$.
13. (a) $W = 2\lambda D/a \propto 1/a$.
14. (b) $32/2 = 16$; $16 \cos^2 60^\circ = 16 \times 1/4 = 4 \text{ W/m}^2$.
15. (b) $\tan i_B = \sqrt{3} \Rightarrow i_B = 60^\circ$.
16. (b) $i_B + r = 90^\circ \Rightarrow$ reflected and refracted rays are perpendicular.
17. (b) Only transverse waves can be polarised; interference and diffraction occur for all waves.
18. (a) At the centre $\Delta x = 0$ for all wavelengths, so all colours are bright together $\rightarrow$ white.
19. (a) $\beta = \lambda D/d$ and $\lambda \rightarrow \lambda/n$. Both true; R explains A.
20. (c) Two separate bulbs are incoherent – A is false; R is the correct condition.
21. (a) Polarisation needs transverse vibrations; longitudinal sound cannot be polarised.
22. (c) $n_1(600) = n_2(480) \Rightarrow 4 \times 600 = 5 \times 480 = 2400 \text{ nm} \Rightarrow y = 2400 \text{ nm} \times 1 \text{ m} / 1 \text{ mm} = 2.4 \text{ mm}$.
23. (b) $(I_0/2)(\cos^2 45^\circ)(\cos^2 45^\circ) = I_0/8$.
24. (a) $\phi = 2\pi/3 \Rightarrow I = I_0 \cos^2(\pi/3) = I_0/4$.

CHAPTER

11

Dual Nature of Radiation and Matter

Day 18

Weightage ★★☆☆

NCERT Class 12

Electron emission

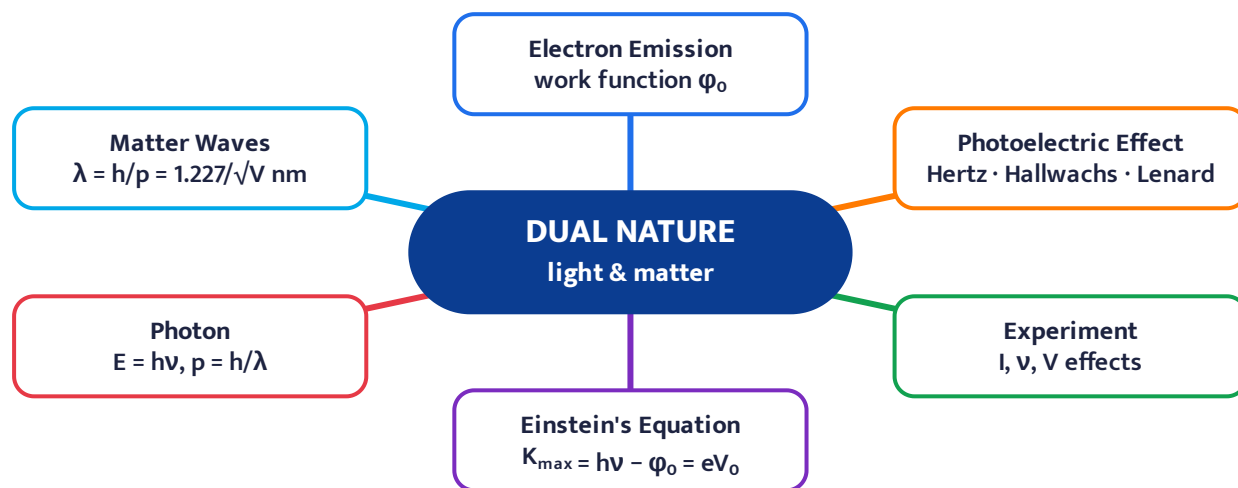
Photoelectric effect

Stopping potential

Einstein's equation

Photon

de Broglie waves



Chapter at a glance

11.1 Electron Emission and Work Function

Free electrons in a metal cannot normally leave its surface – the positive ions pull them back. The **minimum energy** an electron needs to escape from the metal surface is the **work function** ϕ_0 (unit: electron volt, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$). It depends on the **metal and the nature of its surface**.

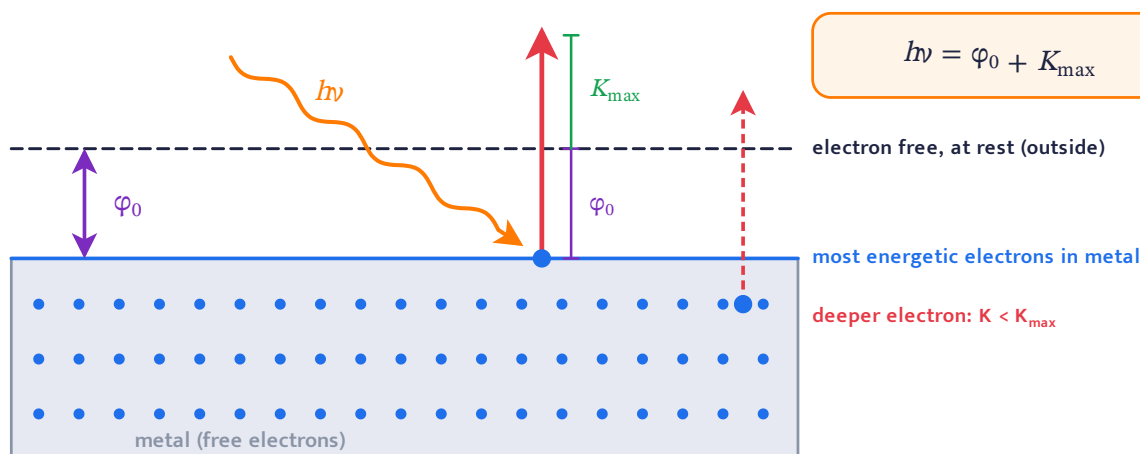
 Three ways to supply ϕ_0

- **Thermionic emission** – heating (e.g. filament of a vacuum tube).
- **Field emission** – a very strong electric field ($\sim 10^8 \text{ V/m}$) pulls electrons out.
- **Photoelectric emission** – light of suitable frequency falls on the surface.

Metal	ϕ_0 (eV)	Metal	ϕ_0 (eV)
Cs	2.14	Al	4.28
K	2.30	Cu	4.65
Na	2.75	Ag	4.70
Ca	3.20	Ni	5.15
Mo	4.17	Pt	5.65

NCERT Line to Remember

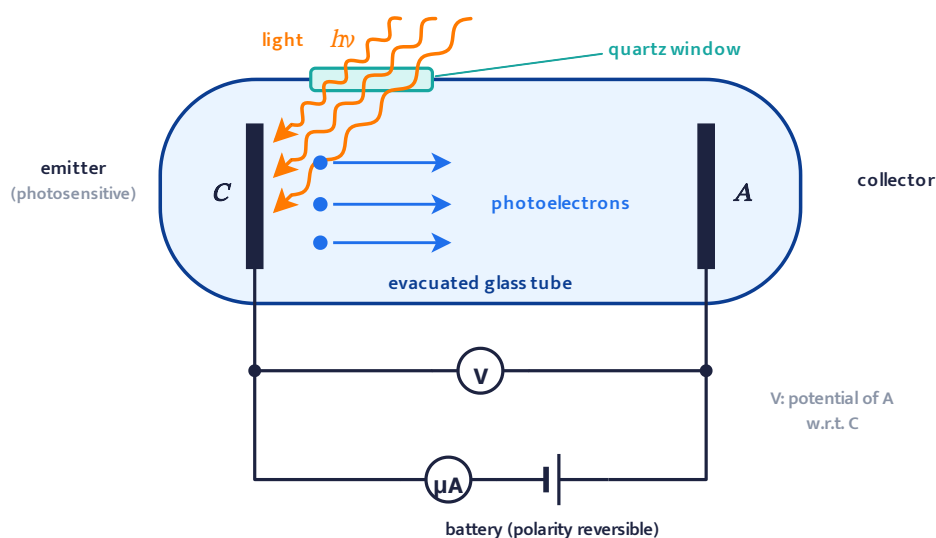
Caesium has the **lowest** work function (2.14 eV) and platinum the **highest** (5.65 eV) in the NCERT table. Alkali metals (Li, Na, K, Cs, Rb) show photoelectric emission even with **visible light**; zinc, cadmium, magnesium respond only to **ultraviolet**.



Energy picture of photoemission: the photon energy pays the work function first; the rest is kinetic energy

11.2 Photoelectric Effect – Hertz, Hallwachs and Lenard

Hertz (1887) found that a spark jumps across a detector gap more easily when its emitter plate is lit by **ultraviolet** light. **Hallwachs and Lenard (1886–1902)** showed that UV light falling on the emitter plate of an evacuated tube makes a current flow – negatively charged particles (**electrons**) are emitted. When light falls on a negatively charged zinc plate it loses charge; a neutral plate becomes **positively** charged. Emission of electrons from a metal surface by light is the **photoelectric effect**; the electrons are **photoelectrons** and the current is the **photocurrent**.



Experimental arrangement: light through a quartz window ejects electrons from C; the potential of A is varied and the photocurrent read on the microammeter

Note

The window is of **quartz** because ordinary glass absorbs ultraviolet. For a given metal there is a minimum (cut-off) frequency, the **threshold frequency** ν_0 , below which **no** emission occurs however intense the light is.

11.3 Experimental Study of the Photoelectric Effect

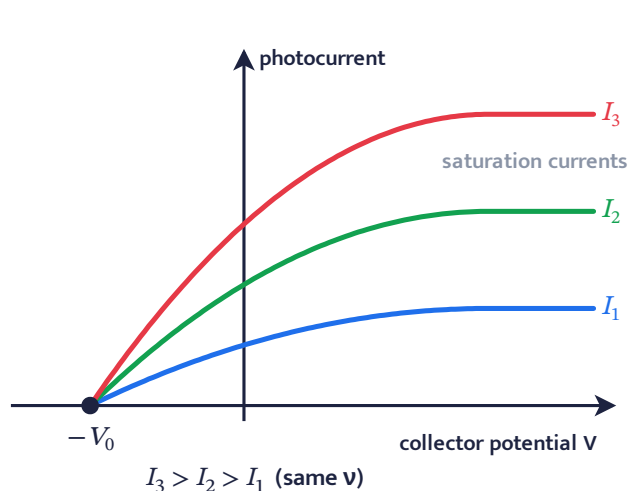
(a) Effect of intensity

With frequency and accelerating potential fixed, the photocurrent is **directly proportional to intensity** – the number of photoelectrons emitted per second $\propto$ intensity.

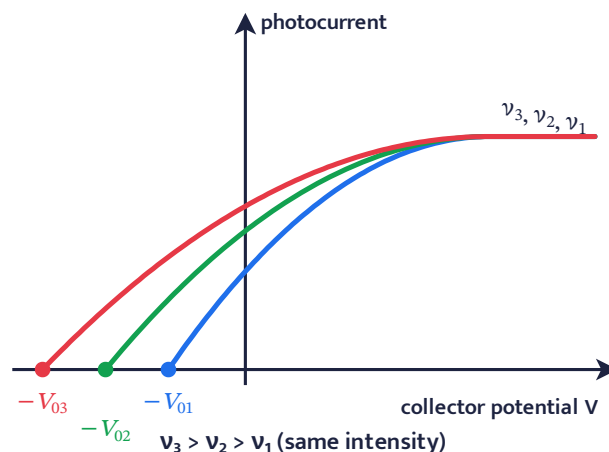
(b) Effect of potential – saturation current and stopping potential

As the positive potential of A increases, the current rises and then becomes constant – the **saturation current** (all emitted electrons reach A). Making A negative repels electrons; at a certain negative potential $-V_0$ even the fastest electron is stopped and the current becomes zero. V_0 is the **stopping (cut-off) potential**:

$$eV_0 = K_{\max} = \frac{1}{2}mv_{\max}^2$$



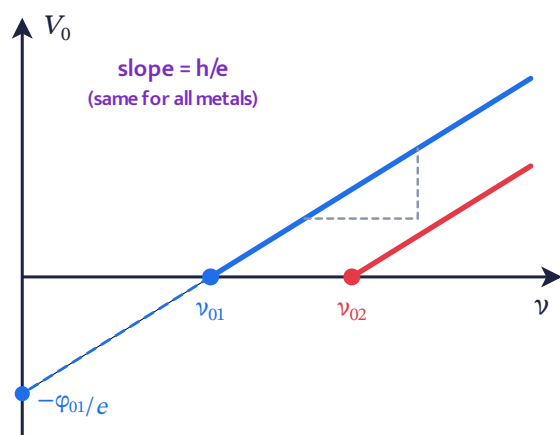
Different intensities, same ν : same V_0 , saturation current $\propto$ intensity



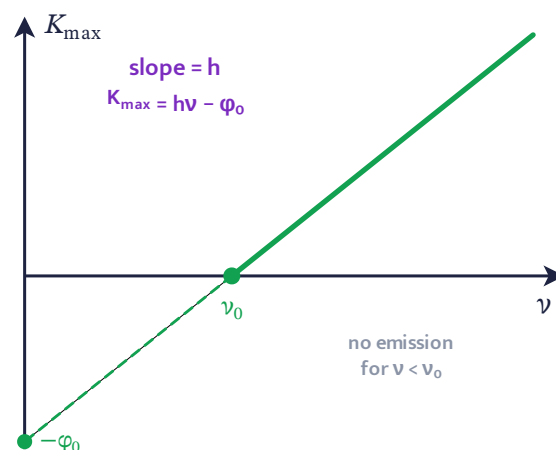
Different ν , same intensity: same saturation current, larger V_0 for larger ν

(c) Effect of frequency

The stopping potential (and hence $K_{\max}$) **increases linearly with frequency** and is **independent of intensity**. The V_0 - ν graphs for different metals are **parallel straight lines** of slope h/e ; each cuts the ν -axis at that metal's threshold frequency ν_0 .



V_0 vs ν for two metals: parallel lines, slope h/e



$K_{\max}$ vs ν : slope h , ν -intercept ν_0 , y-intercept $-\phi_0$

The four laws of photoelectric emission (NCERT summary)

- Photocurrent $\propto$ intensity (for $\nu > \nu_0$).
- Saturation current $\propto$ intensity; stopping potential is **independent of intensity**.
- Below a threshold frequency ν_0 no emission, whatever the intensity; above it $K_{\max}$ rises linearly with ν .
- Emission is **instantaneous** – time lag of order 10^{-9} s or less, even for very dim light.

! Why the wave theory fails

- Waves predict K_{max} should grow with **intensity** – experiment says it does not.
- Waves predict emission at **any** frequency if light is intense enough – experiment shows a sharp threshold.
- Waves predict a time lag (hours for dim light) to accumulate energy – emission is instantaneous.

11.4 Einstein's Photoelectric Equation

Einstein (1905): radiation is made of discrete energy packets – **quanta** (photons) of energy $h\nu$. An electron absorbs **one whole quantum** in a single collision. Part of it, ϕ_0 , is used to escape; the rest appears as kinetic energy. Electrons from the surface get the maximum, K_{max} .

📐 Einstein's equation – all forms

$$K_{max} = h\nu - \phi_0 = eV_0$$

Einstein's photoelectric equation

$$\phi_0 = h\nu_0 = \frac{hc}{\lambda_0}$$

Work function, threshold frequency and threshold wavelength

$$V_0 = \left(\frac{h}{e}\right)\nu - \frac{\phi_0}{e}$$

Straight line: slope h/e , same for all metals

$$K_{max} = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$$

In terms of wavelengths (emission only if $\lambda < \lambda_0$)

$$E(\text{eV}) = \frac{1240}{\lambda(\text{nm})}$$

Shortcut: $hc = 1240 \text{ eV nm}$

🔧 Solved Example

Light of frequency $6.0 \times 10^{14} \text{ Hz}$ falls on a caesium surface ($\phi_0 = 2.14 \text{ eV}$). Find (i) K_{max} , (ii) the stopping potential, (iii) the threshold frequency. ($h = 6.63 \times 10^{-34} \text{ J s}$)

Solution: $h\nu = \frac{6.63 \times 10^{-34} \times 6.0 \times 10^{14}}{1.6 \times 10^{-19}} = 2.49 \text{ eV}$. (i) $K_{max} = 2.49 - 2.14 = \mathbf{0.35 \text{ eV}}$. (ii) $V_0 = \mathbf{0.35 \text{ V}}$. (iii)

$$\nu_0 = \frac{\phi_0}{h} = \frac{2.14 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}} = 5.16 \times 10^{14} \text{ Hz}.$$

🔧 Solved Example

For a metal, the stopping potential is 1.20 V at $8.0 \times 10^{14} \text{ Hz}$ and 2.86 V at $1.2 \times 10^{15} \text{ Hz}$. Find Planck's constant and the work function.

Solution: Slope $\frac{h}{e} = \frac{2.86 - 1.20}{4.0 \times 10^{14}} = 4.15 \times 10^{-15} \text{ V s} \Rightarrow h = 4.15 \times 10^{-15} \times 1.6 \times 10^{-19} = 6.64 \times 10^{-34} \text{ J s}$.

$$\phi_0 = h\nu_1 - eV_1 = (4.15 \times 0.8) \text{ eV} - 1.20 \text{ eV} = 3.32 - 1.20 \approx \mathbf{2.12 \text{ eV}}.$$

! Common Trap

- Doubling **intensity** doubles the photocurrent but leaves V_0 and K_{max} **unchanged**.
- Doubling **frequency** does **not** double K_{max} :
 $K_2 = 2h\nu - \phi_0 > 2K_1$.

🧠 Memory Trick

"Intensity → How many; Frequency → How fast." Intensity controls the number of photoelectrons (current); frequency controls their maximum speed (V_0).

11.5 Particle Nature of Light – the Photon

- In interaction with matter, radiation behaves as if made of **photons**.
- Each photon has energy $E = h\nu$ and momentum $p = \frac{h\nu}{c} = \frac{h}{\lambda}$, and speed c in vacuum. All photons of a given frequency are identical.
- Photons are **electrically neutral**, not deflected by electric or magnetic fields; rest mass is zero.
- Increasing intensity (at fixed λ) increases the **number of photons** per second per unit area, not the energy of each.
- In photon-particle collisions (e.g. photon-electron), total **energy and momentum are conserved**, but the **number of photons need not be** – photons can be absorbed or created.



Photon formulas

$$N = \frac{P}{h\nu} = \frac{P\lambda}{hc}$$

Photons emitted per second by a source of power P

$$p = \frac{E}{c} = \frac{h}{\lambda}$$

Photon momentum



Solved Example

Monochromatic light of frequency 6.0×10^{14} Hz is produced by a laser of power 2.0×10^{-3} W. Find the energy of a photon and the number of photons emitted per second.

Solution: $E = h\nu = 6.63 \times 10^{-34} \times 6.0 \times 10^{14} = 3.98 \times 10^{-19}$ J. $N = \frac{P}{E} = \frac{2.0 \times 10^{-3}}{3.98 \times 10^{-19}} \approx 5.0 \times 10^{15}$ photons/s.



Radiation pressure (force of light)

- Light of power P falling **normally** on a perfectly **absorbing** surface exerts force $F = \frac{P}{c}$; on a perfectly **reflecting** surface $F = \frac{2P}{c}$ (momentum reverses).
- Pressure = $F/A = I/c$ (absorber) or $2I/c$ (reflector), where I is intensity (W m^{-2}).

11.6 Wave Nature of Matter – de Broglie Relation

Nature is symmetric: if light (a wave) can behave like particles, then **moving material particles** (electrons, protons, even cricket balls) should show wave properties. **de Broglie (1924)** proposed that a particle of momentum p has an associated wavelength – the **de Broglie wavelength**. These **matter waves** are not electromagnetic.



de Broglie wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

de Broglie relation

$$\lambda = \frac{h}{\sqrt{2mK}}$$

In terms of kinetic energy K

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

Charge q accelerated from rest through V

$$\lambda_e = \frac{1.227}{\sqrt{V}} \text{ nm}$$

Electron accelerated through V volts

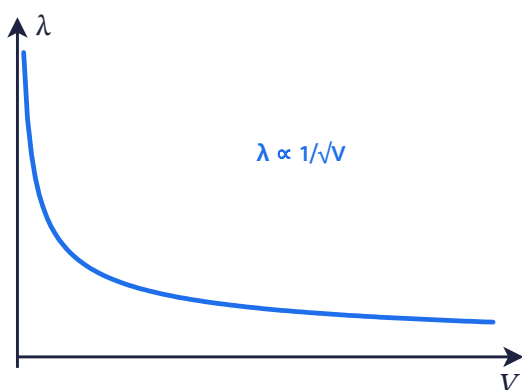
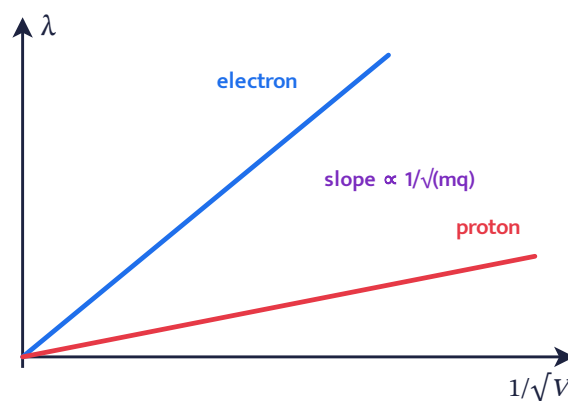


Solved Example

Find the de Broglie wavelength of (i) an electron accelerated through 100 V, (ii) a 0.12 kg ball moving at 20 m s^{-1} .

Solution: (i) $\lambda = \frac{1.227}{\sqrt{100}} = 0.123 \text{ nm}$ – comparable to atomic spacing, so electrons are diffracted by crystals. (ii)

$\lambda = \frac{6.63 \times 10^{-34}}{0.12 \times 20} = 2.8 \times 10^{-34} \text{ m}$ – far too small to detect; hence no wave effects for everyday objects.

(a) λ versus V (b) λ versus $1/\sqrt{V}$ – straight lines

de Broglie wavelength of charged particles accelerated through V ($\lambda \propto 1/\sqrt{V}$)

VS Sir's Exam Tip

- Same V : $\lambda \propto 1/\sqrt{mq}$, so $\lambda_p:\lambda_e = 2\sqrt{2} : 1$.
- Same K : $\lambda \propto \frac{1}{\sqrt{m}}$ – electron has the longest λ .
- Same $\lambda \Rightarrow$ same **momentum** (not same energy or speed).

Common Trap

- $\lambda \propto 1/\sqrt{K}$: making K four times **halves** λ (not quarters).
- For a **photon** $E = pc$ so $\lambda = hc/E$; for a **particle** $\lambda = h/\sqrt{2mK}$. Don't mix them.

Experimental proof (brief)

Electron diffraction by crystals (Davisson and Germer, 1927; 54 V electrons, $\lambda \approx 0.167$ nm) confirmed matter waves. The detailed experiment is no longer in the rationalised NCERT – just know the result. The **electron microscope** uses the short de Broglie wavelength of fast electrons for high resolution.

JEE Main Extension

- Particle in thermal equilibrium at T : $K = \frac{3}{2}k_B T \Rightarrow \lambda = \frac{h}{\sqrt{3mk_B T}}$.
- Photon and electron of the **same** λ : $E_{\text{photon}} = pc$ is much larger than $K_e = p^2/2m$.

Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1 **NCERT based** Photoelectric emission from a given metal surface takes place only when the frequency of incident light is

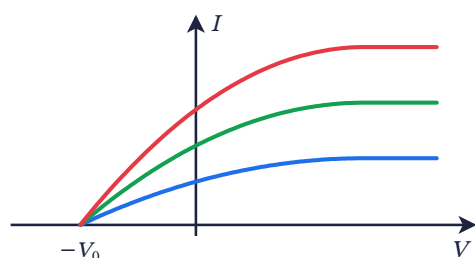
- exactly in the visible region
- greater than the threshold frequency
- any value, if the intensity is high enough
- less than the threshold frequency

- 2 **NCERT based** For a given photosensitive surface, the stopping potential depends on

- intensity of incident light only
- both frequency and intensity
- frequency of incident light only
- the potential difference between plates

- 3 **NEET pattern** Light of wavelength 400 nm falls on a metal of work function 2.0 eV. The maximum kinetic energy of photoelectrons is
(a) 5.1 eV (b) 2.0 eV (c) 1.1 eV (d) 3.1 eV
- 4 **NEET pattern** The threshold wavelength of a metal of work function 2.48 eV is
(a) 500 nm (b) 400 nm (c) 620 nm (d) 250 nm
- 5 **NEET pattern** In a photoelectric experiment the intensity of light is doubled keeping its frequency constant. Then
(a) both double
(b) saturation current unchanged, stopping potential doubles
(c) both remain unchanged
(d) saturation current doubles, stopping potential unchanged
- 6 **NEET pattern** The slope of the stopping potential versus frequency graph for a photosensitive metal is
(a) ϕ_0 / e (b) h (c) h / e (d) e / h
- 7 **NEET pattern** A 100 W source emits light of wavelength 663 nm. The number of photons emitted per second is about
(a) 6.6×10^{20} (b) 1.5×10^{20} (c) 3.3×10^{18} (d) 3.3×10^{20}
- 8 **JEE Main pattern** The wavelength of light falling on a photosensitive surface is changed from 400 nm to 300 nm. The stopping potential
(a) decreases by about 1.03 V (b) increases by about 1.03 V
(c) increases by about 0.33 V (d) remains unchanged
- 9 **NCERT based** The de Broglie wavelength of an electron accelerated from rest through 100 V is about
(a) 0.123 nm (b) 0.0123 nm (c) 12.3 nm (d) 1.23 nm
- 10 **NEET pattern** A proton and an α -particle are accelerated from rest through the same potential difference. The ratio of their de Broglie wavelengths $\lambda_p : \lambda_\alpha$ is
(a) $1:2\sqrt{2}$ (b) $\sqrt{2}:1$ (c) $2\sqrt{2}:1$ (d) $2:1$
- 11 **NEET pattern** An electron and a proton have the same de Broglie wavelength. Then they have the same
(a) momentum (b) kinetic energy
(c) total energy (d) speed
- 12 **NEET pattern** If the kinetic energy of a free particle is made four times, its de Broglie wavelength becomes
(a) one-fourth (b) four times (c) half (d) double
- 13 **JEE Main pattern** A photon and an electron both have wavelength 1 nm. Their energies are approximately
(a) photon 1.5 eV, electron 1240 eV
(b) photon 1240 eV, electron 1.5 eV
(c) both 1.5 eV
(d) both 1240 eV

- 14 **NEET pattern** Light of wavelength 400 nm is incident on metals A ($\phi_0 = 2.0$ eV) and B ($\phi_0 = 5.0$ eV). Photoelectrons are emitted from
(a) A only (b) B only
(c) both A and B (d) neither A nor B
- 15 **Assertion-Reason** Assertion (A): Photoelectric emission is practically instantaneous. Reason (R): An electron absorbs the energy of a whole photon in a single collision.
(a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 16 **Assertion-Reason** Assertion (A): The maximum kinetic energy of photoelectrons increases with the intensity of light. Reason (R): The number of photons per second increases with intensity.
(a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 17 **JEE Main pattern** Light of frequency $1.5\nu_0$ falls on a metal of threshold frequency ν_0 . If the frequency is doubled (to $3\nu_0$), the maximum kinetic energy of photoelectrons becomes
(a) 3 times (b) unchanged (c) 2 times (d) 4 times
- 18 **NCERT based** The momentum of a photon of energy 3.3×10^{-19} J is
(a) 9.9×10^{-11} kg m s⁻¹ (b) 3.3×10^{-27} kg m s⁻¹
(c) 1.1×10^{-27} kg m s⁻¹ (d) 1.1×10^{-19} kg m s⁻¹
- 19 **JEE Main pattern** A 30 W beam of light falls normally on a perfectly reflecting surface. The force exerted on the surface is
(a) 3×10^{-7} N (b) 1×10^{-7} N (c) 2×10^{-7} N (d) 0.5×10^{-7} N
- 20 **NCERT based** The photocurrent (I) versus collector potential (V) curves shown were obtained with the same photosensitive surface. They correspond to



- (a) same frequency and same intensity
(b) same intensity, different frequencies
(c) same frequency, different intensities
(d) different frequencies and different intensities

✓ Answer Key & Quick Solutions

- 1(b) 2(c) 3(c) 4(a) 5(d) 6(c) 7(d) 8(b) 9(a) 10(c) 11(a) 12(c) 13(b) 14(a) 15(a) 16(c)
 17(d) 18(c) 19(c) 20(c)

1. (b) Emission needs $h\nu \geq \phi_0$, i.e. $\nu \geq \nu_0$; intensity cannot compensate.
2. (c) $eV_0 = h\nu - \phi_0$: for fixed ϕ_0 it depends only on ν .
3. (c) $E = 1240/400 = 3.1$ eV; $K_{max} = 3.1 - 2.0 = 1.1$ eV.
4. (a) $\lambda_0 = 1240 / 2.48 = 500$ nm.
5. (d) Intensity sets the number of photons (current); V_0 depends only on ν .
6. (c) $V_0 = (h/e)\nu - \phi_0/e \Rightarrow \text{slope} = h/e$.
7. (d) $E = hc/\lambda = 3.0 \times 10^{-19}$ J; $N = 100/(3.0 \times 10^{-19}) \approx 3.3 \times 10^{20} \text{ s}^{-1}$.
8. (b) $\Delta V_0 = 1240 \left(\frac{1}{300} - \frac{1}{400} \right) = 1240/1200 \approx 1.03$ V (increase).
9. (a) $\lambda = 1.227/\sqrt{100} = 0.123$ nm.
10. (c) $\lambda \propto 1/\sqrt{mq}$; $m_\alpha q_\alpha = 4m \times 2e \Rightarrow \text{ratio} = \sqrt{8} = 2\sqrt{2}$.
11. (a) $\lambda = h/p \Rightarrow$ equal λ means equal p ; the lighter electron then has more KE and speed.
12. (c) $\lambda \propto 1/\sqrt{K} \Rightarrow \lambda_2 = \lambda_1/\sqrt{4} = \lambda_1/2$.
13. (b) Photon: $hc/\lambda = 1240$ eV. Electron: $K = \frac{h^2}{2m\lambda^2} = 2.4 \times 10^{-19} \text{ J} \approx 1.5$ eV.
14. (a) Photon energy = $1240/400 = 3.1$ eV; greater than 2.0 eV but less than 5.0 eV.
15. (a) No energy has to be accumulated over time, because the whole quantum $h\nu$ is given at once – so emission starts within $\sim 10^{-9}$ s.
16. (c) $K_{max} = h\nu - \phi_0$ is independent of intensity (A false); more photons only give more electrons (R true).
17. (d) $K_1 = 0.5h\nu_0, K_2 = 2h\nu_0 \Rightarrow K_2/K_1 = 4$.
18. (c) $p = E/c = 3.3 \times 10^{-19} / 3 \times 10^8 = 1.1 \times 10^{-27} \text{ kg m s}^{-1}$.
19. (c) $F = 2P/c = 60 / (3 \times 10^8) = 2 \times 10^{-7} \text{ N}$.
20. (c) Common stopping potential $\Rightarrow$ same ν ; different saturation currents $\Rightarrow$ different intensities.

CHAPTER

12

Atoms

Day 19

Weightage ★★☆☆

NCERT Class 12

 α -scattering

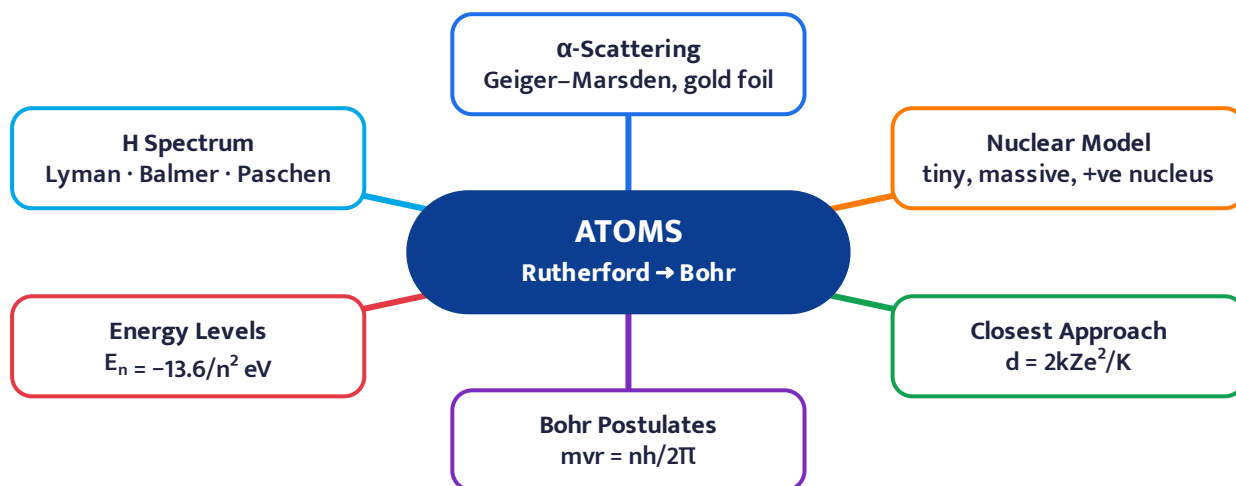
Nuclear model

Closest approach

Hydrogen spectrum

Bohr model

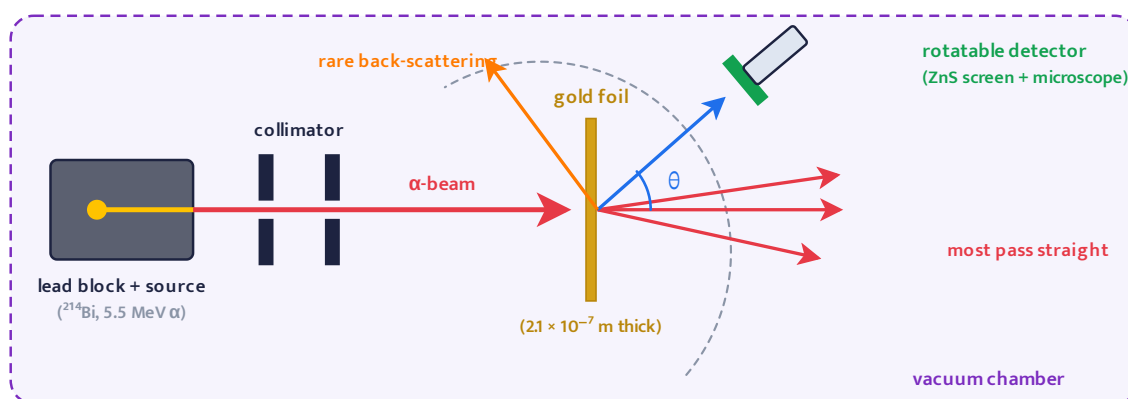
Energy levels



Chapter at a glance

12.1 Alpha-Particle Scattering – Geiger–Marsden Experiment

Thomson's model (1898) pictured the atom as a sphere of positive charge with electrons embedded like seeds in a watermelon ("plum-pudding"). To test it, **Geiger and Marsden (1911)**, on Rutherford's suggestion, fired a collimated beam of **5.5 MeV α -particles** (from $^{214}_{83}\text{Bi}$) at a thin **gold foil** and counted the scattered particles with a rotatable ZnS scintillation detector.



Geiger–Marsden scattering experiment (top view; the whole set-up is in vacuum)

Observations

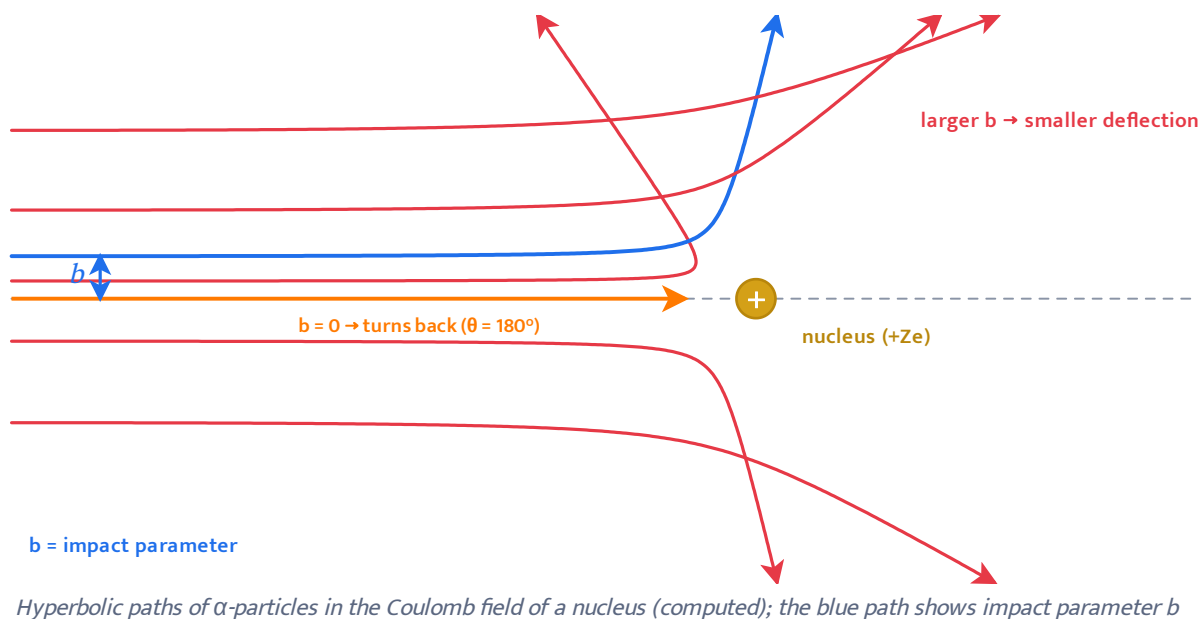
- **Most** α -particles pass straight through the foil.
- Only about **0.14%** are scattered by more than 1°.
- About **1 in 8000** is deflected by more than 90°; a few bounce back.

Rutherford's nuclear model

- All + charge and almost all mass in a tiny **nucleus**, 10^{-15} – 10^{-14} m.
- Atom $\sim 10^{-10}$ m: mostly **empty space**.
- Electrons orbit it; Coulomb force = centripetal force.

12.2 Alpha-Particle Trajectory: Impact Parameter and Closest Approach

The **impact parameter b** is the perpendicular distance of the initial velocity of the α -particle from the centre of the nucleus. Small $b \rightarrow$ close approach $\rightarrow$ large deflection; $b = 0$ (head-on) $\rightarrow$ the α -particle stops momentarily and **rebounds** ($\theta = 180^\circ$). Since the nucleus is tiny, very few particles have small b – hence very few large-angle deflections.



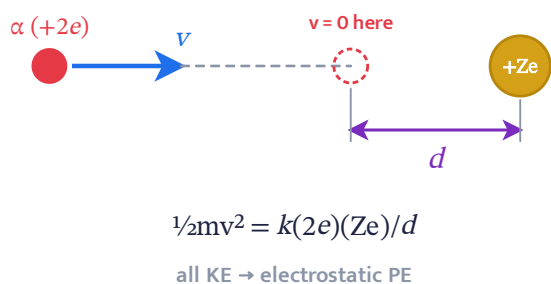
Scattering formulas

$$d = \frac{1}{4\pi\epsilon_0} \cdot \frac{2Ze^2}{K}$$

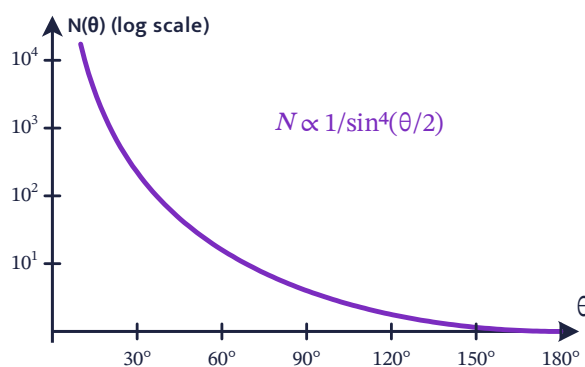
Distance of closest approach (head-on), K = initial KE

$$N(\theta) \propto \frac{1}{\sin^4(\theta/2)}$$

Number scattered at angle θ (Rutherford)



Head-on approach: KE is fully converted into PE at distance d



Scattered count falls steeply with angle



Solved Example

Find the distance of closest approach of a 7.7 MeV α -particle to a gold nucleus ($Z = 79$). ($\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$)

Solution: $d = \frac{9 \times 10^9 \times 2 \times 79 \times (1.6 \times 10^{-19})^2}{7.7 \times 10^6 \times 1.6 \times 10^{-19}} = 2.95 \times 10^{-14} \text{ m} \approx 30 \text{ fm}$. The nucleus must be smaller than this.

! Common Trap

- $d \propto 1/K$: doubling KE **halves** d . $d \propto Z$.
- In eV units: $K = 7.7$ MeV means $K = 7.7 \times 10^6 \times e$ joule – one e cancels.
- d is an **upper limit** of nuclear size, not the nuclear radius itself.

**JEE Main Extension**

- Impact parameter and scattering angle:

$$b = \frac{1}{4\pi\epsilon_0} \cdot \frac{Ze^2 \cot(\theta/2)}{K} \quad (K = \frac{1}{2}mv^2).$$

- $b = 0 \Rightarrow \theta = 180^\circ$; large $b \Rightarrow \theta \rightarrow 0$.

12.3 Electron Orbits and the Failure of Rutherford's Model

For an electron in a circular orbit of radius r in hydrogen, the Coulomb attraction provides the centripetal force:

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}. \text{ Hence } K = \frac{e^2}{8\pi\epsilon_0 r}, U = -\frac{e^2}{4\pi\epsilon_0 r} \text{ and total energy } E = -\frac{e^2}{8\pi\epsilon_0 r} \text{ (negative } \Rightarrow \text{ bound).}$$

! Two failures of the Rutherford model

- **Stability**: an orbiting (accelerated) electron must radiate energy (classical EM theory), spiral inward and fall into the nucleus – atoms would not be stable.
- **Spectrum**: the frequency of revolution would change continuously, giving a **continuous** spectrum – but atoms emit **line** spectra.

12.4 Atomic Spectra – Spectral Series of Hydrogen

A heated gas (or one excited by electric discharge) emits light of only certain wavelengths – an **emission line spectrum**. When white light passes through the gas, dark lines appear at the same wavelengths – an **absorption spectrum**. Each element has its own characteristic spectrum – its "fingerprint". Balmer (1885) found an empirical formula for the visible lines of hydrogen; its general form is:

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right), \quad R = 1.097 \times 10^7 \text{ m}^{-1}$$

Series	n_f	n_i	Region	First (longest) line	Series limit
Lyman	1	2, 3, ...	ultraviolet	121.6 nm	91.2 nm
Balmer	2	3, 4, ...	visible	656.3 nm (H_α)	364.6 nm
Paschen	3	4, 5, ...	infrared	1875 nm	820.4 nm
Brackett	4	5, 6, ...	infrared	4051 nm	1458 nm
Pfund	5	6, 7, ...	infrared	7458 nm	2279 nm



The Balmer series in the visible region: lines crowd together towards the series limit

Memory Trick

"**Lazy Boys Play Basketball Poorly**" – Lyman (1), Balmer (2), Paschen (3), Brackett (4), Pfund (5). Only **Balmer** is visible; Lyman is UV; the rest are IR.

VS Sir's Exam Tip

- Series limit: $\lambda_{\infty} = \frac{n_f^2}{R}$ ($1/R = 91.2 \text{ nm}$).
- Longest $\lambda \leftrightarrow$ smallest energy gap ($n_i = n_f + 1$); shortest $\lambda \leftrightarrow n_i = \infty$.

12.5 Bohr Model of the Hydrogen Atom

- Stationary orbits:** an electron can revolve in certain stable orbits without radiating energy. Each orbit has a definite total energy.
- Quantisation of angular momentum:** allowed orbits are those for which $L = mvr = \frac{nh}{2\pi}$, $n = 1, 2, 3, \dots$ (principal quantum number).
- Frequency condition:** when an electron jumps from a higher orbit to a lower one, a photon is emitted with $h\nu = E_i - E_f$.

Bohr model results (hydrogen-like atom, atomic number Z)

$$r_n = \frac{n^2 h^2 \epsilon_0}{\pi m Z e^2} = 0.529 \frac{n^2}{Z} \text{ \AA}$$

Radius of n-th orbit ($a_0 = 0.529 \text{ \AA}$, Bohr radius)

$$v_n = \frac{Ze^2}{2\epsilon_0 nh} = 2.18 \times 10^6 \frac{Z}{n} \text{ m s}^{-1}$$

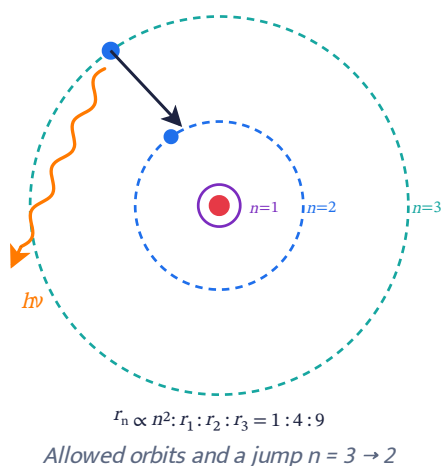
Speed in n-th orbit ($v_1 \approx c / 137$)

$$E_n = -\frac{mZ^2 e^4}{8\epsilon_0^2 n^2 h^2} = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

Total energy of n-th orbit

$$K = -E, \quad U = 2E$$

KE : PE : E = 1 : -2 : -1



Quantity	Depends on
radius r	n^2 / Z
speed v	Z / n
energy E, KE	Z^2 / n^2
angular momentum L	n
time period T	n^3 / Z^2
frequency of revolution	Z^2 / n^3
current due to electron	Z^2 / n^3

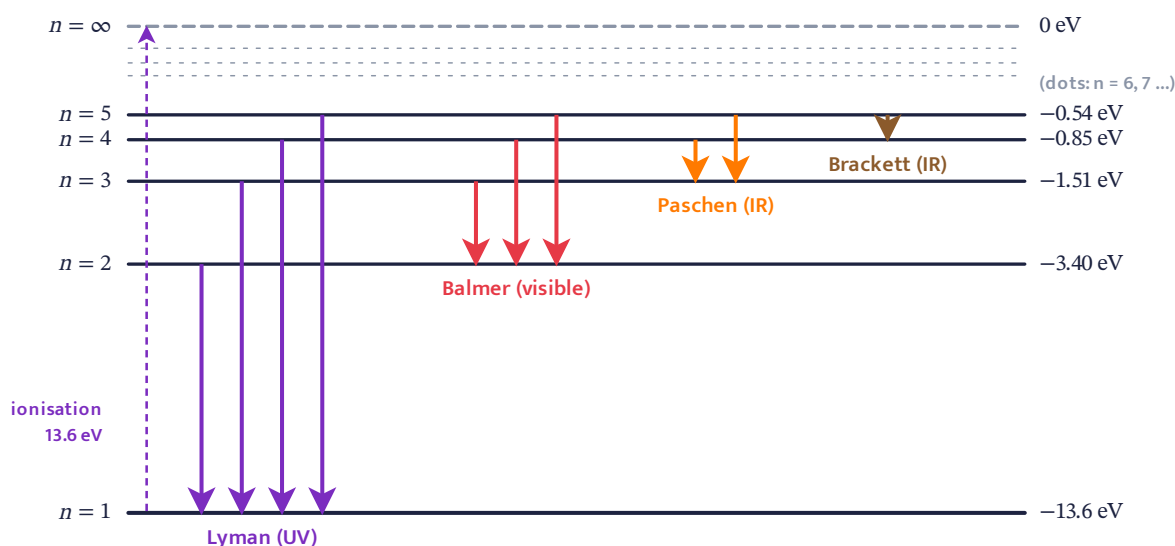
Solved Example

For a He^+ ion ($Z = 2$), find the radius, speed and energy of the electron in the ground state.

Solution: $r_1 = 0.529 / 2 = 0.265 \text{ \AA}$; $v_1 = 2.18 \times 10^6 \times 2 = 4.36 \times 10^6 \text{ m s}^{-1}$; $E_1 = -13.6 \times 4 = -54.4 \text{ eV}$ (ionisation energy of $\text{He}^+ = 54.4 \text{ eV}$).

12.6 Energy Levels and the Line Spectrum of Hydrogen

The lowest level $n = 1$ (–13.6 eV) is the **ground state**; higher levels are **excited states**. Energy needed to remove the electron from the ground state is the **ionisation energy** (13.6 eV for H). Energy to raise it to $n = 2$ is the **first excitation energy** (10.2 eV); to $n = 3$, 12.09 eV. At room temperature almost all H atoms are in the ground state.



Energy-level diagram of hydrogen with emission series (spacing not to scale)

Transitions

$$h\nu = E_i - E_f = 13.6 Z^2 \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \text{eV} \quad \text{Energy of emitted photon}$$

$$N = \frac{n(n-1)}{2} \quad \text{Max. number of spectral lines from level } n \text{ to ground (many atoms)}$$

Solved Example

An electron in a hydrogen atom jumps from $n = 4$ to $n = 2$. Find the energy and wavelength of the emitted photon. Which series and colour?

Solution: $E = 13.6 \left(\frac{1}{4} - \frac{1}{16} \right) = 2.55 \text{ eV}$; $\lambda = \frac{1240}{2.55} \approx 486 \text{ nm}$ – the H_β line of the **Balmer** series (blue-green).

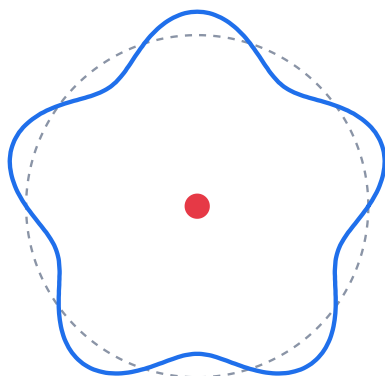
Common Trap

- A **single** atom excited to $n = 4$ can give at most **3** lines ($4 \rightarrow 3 \rightarrow 2 \rightarrow 1$); $n(n-1)/2 = 6$ needs many atoms.
- Ground-state H **absorbs** only Lyman lines – absorption spectrum at room temperature shows Lyman series only.
- Energy levels get **closer** as n increases, while radii get **farther** apart.

VS Sir's Exam Tip

- Use $\lambda(\text{nm}) = \frac{1240}{\Delta E(\text{eV})}$ – fastest route for every transition question.
- Photon energy absorbed must **exactly** equal a level difference (except for ionisation: any $E \geq 13.6 \text{ eV}$).

12.7 de Broglie's Explanation of Bohr's Quantisation



$$2\pi r_n = n\lambda \quad (n = 5 \text{ here})$$

$$\lambda = h/mv \Rightarrow mvr = nh/2\pi$$

Standing matter wave on an allowed orbit

de Broglie argued that the electron in an orbit behaves as a **standing wave**. A stable (non-radiating) wave on a circle of radius r_n requires the circumference to hold a **whole number** of wavelengths:

$$2\pi r_n = n\lambda = \frac{nh}{mv_n} \Rightarrow mv_n r_n = \frac{nh}{2\pi}$$

This is exactly **Bohr's second postulate** – quantisation of angular momentum follows from the wave nature of the electron.



Limitations of the Bohr model

- Works only for **hydrogenic** (one-electron) atoms – H, He^+ , Li^{2+} ; cannot be extended even to two-electron atoms (it ignores electron–electron interaction).
- Cannot explain the **relative intensities** of spectral lines.



Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- NCERT based** In the Geiger–Marsden experiment most α -particles passed through the gold foil undeflected. This shows that

(a) most of the atom is empty space
 (b) positive charge is spread uniformly
 (c) the nucleus is negatively charged
 (d) electrons are very heavy
- NCERT based** The radius of the first Bohr orbit of hydrogen is $0.53 \text{ \AA}$. The radius of the third orbit is

(a) $2.12 \text{ \AA}$ (b) $1.59 \text{ \AA}$ (c) $4.77 \text{ \AA}$ (d) $1.06 \text{ \AA}$
- NEET pattern** The energy of the electron in the $n = 3$ orbit of hydrogen is

(a) -0.85 eV (b) -4.53 eV (c) -3.40 eV (d) -1.51 eV
- NEET pattern** The ionisation energy of He^+ in its ground state is

(a) 27.2 eV (b) 54.4 eV (c) 13.6 eV (d) 40.8 eV
- NEET pattern** The ratio of the longest wavelength of the Lyman series to that of the Balmer series is

(a) $5 : 27$ (b) $27 : 5$ (c) $9 : 4$ (d) $4 : 9$

- 6 **NEET pattern** Hydrogen atoms are excited to the $n = 5$ level. The maximum number of spectral lines in the emission spectrum is
(a) 5 (b) 4 (c) 15 (d) 10
- 7 **NCERT based** Which series of the hydrogen spectrum lies in the visible region?
(a) Pfund (b) Balmer (c) Paschen (d) Lyman
- 8 **NEET pattern** The speed of the electron in the first Bohr orbit of hydrogen is about
(a) $1.1 \times 10^6 \text{ m s}^{-1}$ (b) $2.2 \times 10^6 \text{ m s}^{-1}$ (c) $2.2 \times 10^4 \text{ m s}^{-1}$ (d) $3 \times 10^8 \text{ m s}^{-1}$
- 9 **JEE Main pattern** The distance of closest approach of an α -particle to a nucleus is d . If the kinetic energy of the α -particle is doubled, the distance of closest approach becomes
(a) $d / \sqrt{2}$ (b) $d/4$ (c) $d/2$ (d) $2d$
- 10 **NEET pattern** In a Bohr orbit, the ratio of kinetic energy : potential energy : total energy of the electron is
(a) $1 : -1 : -2$ (b) $1 : -2 : -1$ (c) $1 : 2 : 1$ (d) $-1 : 2 : 1$
- 11 **NEET pattern** Using $R = 1.097 \times 10^7 \text{ m}^{-1}$, the wavelength of the H_α line (first Balmer line) is
(a) 122 nm (b) 486 nm (c) 656 nm (d) 365 nm
- 12 **JEE Main pattern** If the shortest wavelength of the Lyman series of hydrogen is λ , the shortest wavelength of the Paschen series is
(a) 3λ (b) 9λ (c) 4λ (d) $\lambda/9$
- 13 **NEET pattern** The angular momentum of the electron in the third Bohr orbit is
(a) $\frac{9h}{2\pi}$ (b) $\frac{3h}{2\pi}$ (c) $\frac{h}{2\pi}$ (d) $\frac{3h}{\pi}$
- 14 **NCERT based** According to de Broglie's explanation, the number of de Broglie wavelengths that fit in the circumference of the third Bohr orbit is
(a) 9 (b) 1 (c) 3 (d) 6
- 15 **Assertion-Reason** Assertion (A): Bohr's model cannot explain the spectrum of the helium atom. Reason (R): Bohr's model ignores the electrostatic interaction between electrons.
(a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 16 **Assertion-Reason** Assertion (A): Since a hydrogen atom has only one electron, its emission spectrum has only one line. Reason (R): A transition between two energy levels emits a photon of a single definite frequency.
(a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false

17. **JEE Main pattern** Ground-state H atoms absorb 12.75 eV photons. The number of spectral lines then emitted is
(a) 10 (b) 3 (c) 6 (d) 4
18. **NEET pattern** The ratio of the orbital periods of the electron in the 1st and 2nd Bohr orbits of H is
(a) 1 : 8 (b) 8 : 1 (c) 1 : 4 (d) 1 : 2
19. **JEE Main pattern** Which He^+ transition gives the same λ as the H line $n = 3 \rightarrow 2$ (first Balmer line)?
(a) $n = 6 \rightarrow 3$ (b) $n = 4 \rightarrow 2$ (c) $n = 3 \rightarrow 2$ (d) $n = 6 \rightarrow 4$
20. **NCERT based** In the energy-level diagram shown, which transition emits the photon of the **longest** wavelength?
-
- (a) A ($4 \rightarrow 1$) (b) B ($3 \rightarrow 1$) (c) C ($4 \rightarrow 2$) (d) D ($3 \rightarrow 2$)

✓ Answer Key & Quick Solutions

- 1(a) 2(c) 3(d) 4(b) 5(a) 6(d) 7(b) 8(b) 9(c) 10(b) 11(c) 12(b) 13(b) 14(c) 15(a)
16(c) 17(c) 18(a) 19(d) 20(d)

1. (a) Undelected passage $\Rightarrow$ the atom is mostly empty; the nucleus is tiny.
2. (c) $r \propto n^2$: $0.53 \times 9 = 4.77 \text{ \AA}$.
3. (d) $E_3 = -13.6 / 9 = -1.51 \text{ eV}$.
4. (b) $13.6 Z^2 = 13.6 \times 4 = 54.4 \text{ eV}$.
5. (a) Lyman ($2 \rightarrow 1$): $\frac{1}{\lambda} = \frac{3R}{4}$; Balmer ($3 \rightarrow 2$): $\frac{1}{\lambda} = \frac{5R}{36} \Rightarrow \text{ratio} = \frac{5/36}{3/4} = \frac{5}{27}$.
6. (d) $n(n-1)/2 = 5 \times 4 / 2 = 10$.
7. (b) Balmer lines (656.3 nm to 364.6 nm) are visible (limit in near UV).
8. (b) $v_1 = c / 137 \approx 2.18 \times 10^6 \text{ m s}^{-1}$.
9. (c) $d = \frac{2kZe^2}{K} \propto \frac{1}{K} \Rightarrow d/2$.
10. (b) $K = -E$ and $U = 2E \Rightarrow K : U : E = 1 : -2 : -1$.
11. (c) $\frac{1}{\lambda} = R \left(\frac{1}{4} - \frac{1}{9} \right) = \frac{5R}{36} \Rightarrow \lambda = 36 / (5 \times 1.097 \times 10^7) = 656 \text{ nm}$.
12. (b) Series limit $\lambda_\infty = n_f^2 / R$: Lyman $1/R$, Paschen $9/R \Rightarrow 9\lambda$.
13. (b) $L = nh / 2\pi$ with $n = 3$.
14. (c) $2\pi r_n = n\lambda \Rightarrow n = 3$ wavelengths.
15. (a) He has two electrons; their mutual repulsion is not included in Bohr's model, so it applies only to one-electron atoms.
16. (c) Many atoms in a sample undergo different transitions $\Rightarrow$ many lines (A false). R is true.
17. (c) $-13.6 + 12.75 = -0.85 \text{ eV} \Rightarrow n = 4$; lines $= 4 \times 3 / 2 = 6$.
18. (a) $T = 2\pi r / v \propto n^2 / (1/n) = n^3 \Rightarrow 1 : 8$.
19. (d) $4 \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) = \frac{1}{4} - \frac{1}{9} \Rightarrow \frac{1}{n_f^2} - \frac{1}{n_i^2} = \frac{1}{16} - \frac{1}{36} \Rightarrow 6 \rightarrow 4$.
20. (d) Longest $\lambda \Leftrightarrow$ smallest energy gap: $3 \rightarrow 2$ (1.89 eV) is smaller than $4 \rightarrow 2$ (2.55 eV) and the jumps to $n = 1$.

CHAPTER

13

Nuclei

Day 20

Weightage ★★☆☆

NCERT Class 12

Size & density

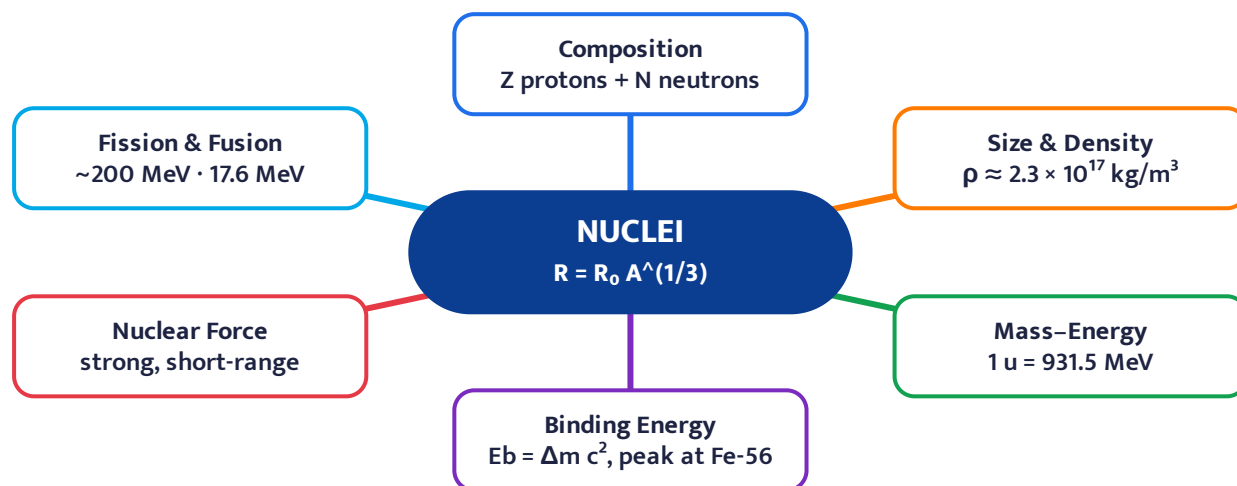
Isotopes, isobars

Mass defect

B.E. per nucleon

Nuclear force

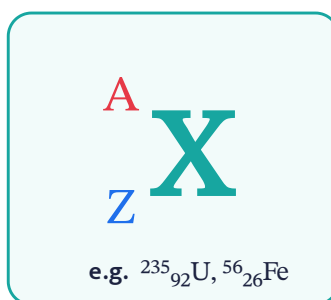
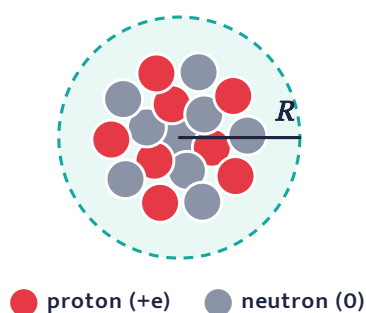
Fission & fusion



Chapter at a glance

13.1 Composition of the Nucleus & Atomic Masses

Rutherford showed that almost all the mass and all the positive charge of an atom sit in a tiny **nucleus**. A nucleus contains **Z protons** and **N neutrons** (together called **nucleons**). The neutron (mass $\approx$ proton mass, zero charge) was discovered by **Chadwick (1932)** by bombarding beryllium with α -particles.

 $A = \text{mass number} = Z + N$ $Z = \text{atomic number (protons)}$ $N = A - Z \text{ (neutrons)}$ nucleus $\approx 10^{-15} \text{ m (fm)}$ atom $\approx 10^{-10} \text{ m}$ → atom is 10^5 times biggerA nucleus ${}^A_Z X$: Z protons + (A - Z) neutrons, radius of a few fermi

Atomic masses

$$1 \text{ u} = \frac{1}{12} m({}^{12}\text{C}) \quad \text{Atomic mass unit (unified)}$$

$$1 \text{ u} = 1.6605 \times 10^{-27} \text{ kg} = 931.5 \text{ MeV}/c^2$$

$$m_p = 1.00728 \text{ u} \quad \text{proton}$$

$$m_n = 1.00866 \text{ u} \quad \text{neutron (slightly heavier)}$$

$$m({}^1\text{H}) = 1.00783 \text{ u} \quad \text{H atom} = m_p + m_e$$

Term	Same	Example
Isotopes	Z	${}^1_1\text{H}, {}^2_1\text{H}, {}^3_1\text{H}$
Isobars	A	${}^3_1\text{H}, {}^3_2\text{He}$
Isotones	$N = A - Z$	${}^{198}_{80}\text{Hg}, {}^{197}_{79}\text{Au}$

NCERT Line to Remember

- Isotopes have the **same chemical properties** (same Z, same electrons) but different masses. Chlorine: 75.4% ${}^{35}\text{Cl}$ (34.98 u) + 24.6% ${}^{37}\text{Cl}$ (36.98 u) → average 35.47 u.
- Hydrogen isotopes: protium ${}^1\text{H}$ (99.985%), deuterium ${}^2\text{H}$ and **tritium** ${}^3\text{H}$ (unstable, not natural in large amounts).
- A free neutron is **unstable** (mean life ≈ 1000 s): $n \rightarrow p + e^- + \bar{\nu}$. Inside a nucleus it is stable.

13.2 Size of the Nucleus & Nuclear Density

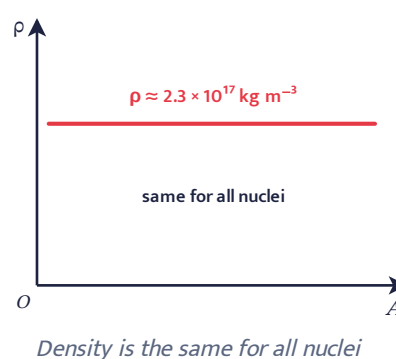
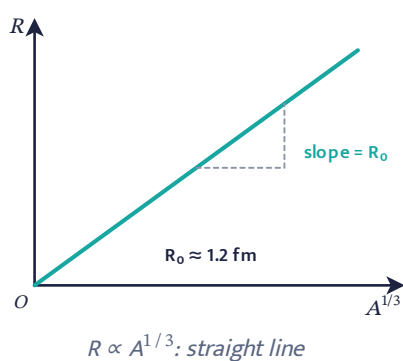
α -scattering and electron-scattering experiments show that the **volume of a nucleus is proportional to A**, so its radius grows as the cube root of A:

Size and density

$$R = R_0 A^{1/3}, \quad R_0 = 1.2 \text{ fm} = 1.2 \times 10^{-15} \text{ m} \quad \text{Nuclear radius}$$

$$V = \frac{4}{3}\pi R_0^3 A \propto A \quad \text{Nuclear volume}$$

$$\rho = \frac{A m_u}{\frac{4}{3}\pi R_0^3 A} = \frac{3m_u}{4\pi R_0^3} \approx 2.3 \times 10^{17} \text{ kg m}^{-3} \quad \text{Nuclear density – independent of A}$$



Solved Example

The radius of ${}^{27}_{13}\text{Al}$ is 3.6 fm. Find the radius of ${}^{125}_{52}\text{Te}$.

Solution: $\frac{R_{\text{Te}}}{R_{\text{Al}}} = \left(\frac{125}{27}\right)^{1/3} = \frac{5}{3} \Rightarrow R_{\text{Te}} = \frac{5}{3} \times 3.6 = 6.0 \text{ fm}.$

! Common Trap

- Nuclear density $\approx 2.3 \times 10^{17} \text{ kg m}^{-3}$ is about 10^{14} times the density of water, yet it is the **same** for ${}^4\text{He}$ and ${}^{238}\text{U}$. Ratio of densities of any two nuclei = **1 : 1**.
- Doubling A does **not** double R : $R \propto A^{1/3}$, so $A \times 8 \Rightarrow R \times 2$; surface area $\propto A^{2/3}$.

13.3 Mass–Energy Relation, Mass Defect & Binding Energy

Einstein: mass is a form of energy, $E = mc^2$. The mass of a nucleus is always **less** than the sum of the masses of its free nucleons. This missing mass, the **mass defect** Δm , appears as the **binding energy** E_b – the energy needed to pull the nucleus apart into free nucleons (equally, the energy released when it is formed).

Mass defect & binding energy

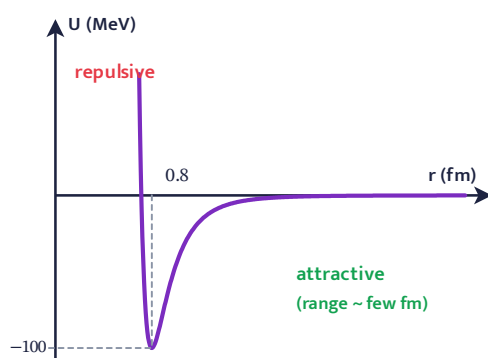
$E = mc^2$	$1 \text{ kg} \equiv 9 \times 10^{16} \text{ J}; 1 \text{ g} \equiv 9 \times 10^{13} \text{ J}$
$1 \text{ u} \times c^2 = 931.5 \text{ MeV}$	Use this for every nuclear numerical
$\Delta m = [Zm_p + (A - Z)m_n] - M_N$	Mass defect (nuclear masses)
$\Delta m = [Zm_H + (A - Z)m_n] - M_{atom}$	Same, using atomic masses (electron masses cancel)
$E_b = \Delta m c^2 = \Delta m(\text{u}) \times 931.5 \text{ MeV}$	Binding energy
$E_{bn} = \frac{E_b}{A}$	Binding energy per nucleon – measures stability

🔧 Solved Example

Find the binding energy and B.E. per nucleon of ${}^{56}_{26}\text{Fe}$. Atomic mass of ${}^{56}\text{Fe} = 55.934939 \text{ u}$, $m_H = 1.007825 \text{ u}$, $m_n = 1.008665 \text{ u}$.

Solution: $\Delta m = 26(1.007825) + 30(1.008665) - 55.934939 = 56.463400 - 55.934939 = 0.528461 \text{ u}$. $E_b = 0.528461 \times 931.5 = \mathbf{492.3 \text{ MeV}}$; $E_{bn} = 492.3/56 = \mathbf{8.79 \text{ MeV}}$.

13.4 Nuclear Force



Potential energy of a pair of nucleons



Properties of the nuclear force

- Strongest** force – far stronger than the Coulomb force.
- Short range** – a few fermi; negligible beyond.
- Attractive** for $r > 0.8 \text{ fm}$, **repulsive** for $r < 0.8 \text{ fm}$.
- Charge-independent:** $n\text{--}n \approx p\text{--}p \approx n\text{--}p$.
- Saturating;** not an inverse-square law; no simple formula.

! Common Trap

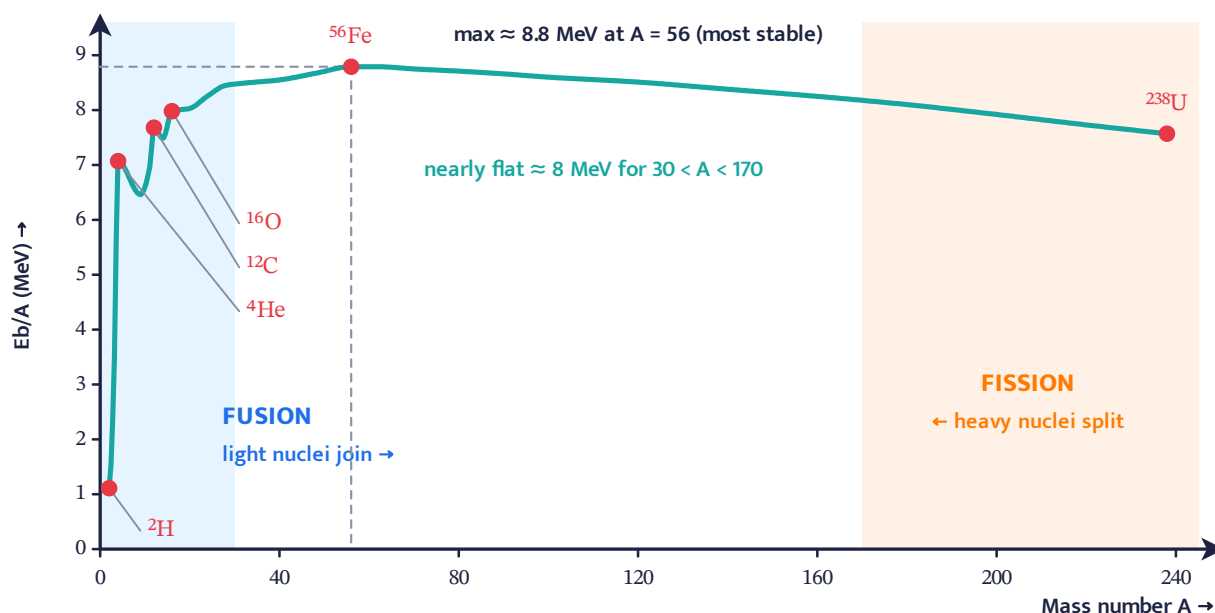
- Don't mix atomic and nuclear masses: with an **atomic** mass use m_H (1.007825 u), not m_p .
- Keep 5–6 decimals in u – Δm is a small difference of big numbers.

🧠 Memory Trick

Key B.E./nucleon values: ${}^2\text{H}$ 1.1, ${}^4\text{He}$ 7.07, ${}^{12}\text{C}$ 7.68, ${}^{16}\text{O}$ 7.98, ${}^{56}\text{Fe}$ 8.79 (max), ${}^{238}\text{U}$ 7.6 MeV. Deuteron: $E_b = 2.22$ MeV.

13.5 Binding Energy per Nucleon vs Mass Number**📖 NCERT Line to Remember**

- E_{bn} is practically **constant** (≈ 8 MeV) for $30 < A < 170$ – because the nuclear force is **short-ranged**: a nucleon feels only its nearest neighbours (**saturation**).
- Maximum ≈ 8.75 MeV at $A = 56$ (Fe) – the most stable nuclei.
- E_{bn} is **lower for light** ($A < 30$) and **heavy** ($A > 170$) nuclei. Heavy $\rightarrow$ middle (**fission**) and light $\rightarrow$ middle (**fusion**) both increase E_{bn} and so **release energy**.



The B.E./nucleon curve – the single most important graph of the chapter

📐 Q-value of a nuclear reaction

$$Q = \left(\sum E_b \right)_{\text{products}} - \left(\sum E_b \right)_{\text{reactants}} \quad \text{Energy released from binding energies}$$

$$Q = \left(\sum m_{\text{reactants}} - \sum m_{\text{products}} \right) \times 931.5 \text{ MeV} \quad \text{Energy released from masses (} Q > 0: \text{ exothermic)}$$

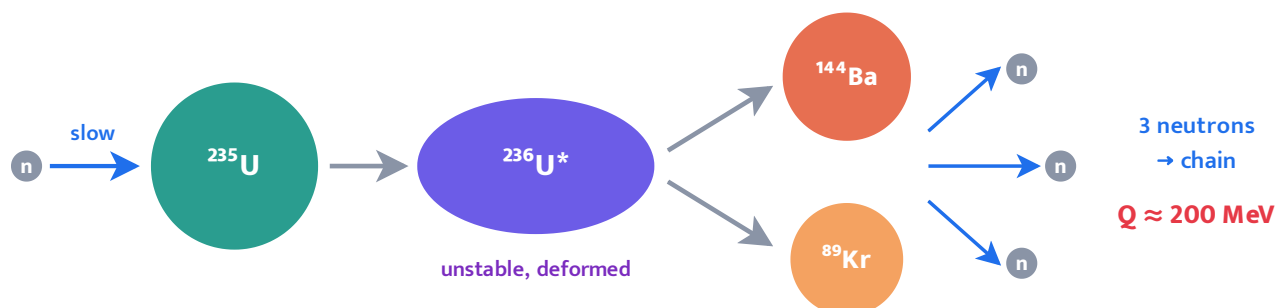
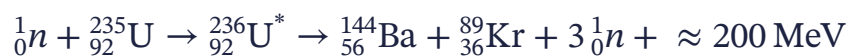
🔧 Solved Example

A nucleus with $A = 240$ ($E_{bn} = 7.6$ MeV) splits into two nuclei of $A = 120$ ($E_{bn} = 8.5$ MeV). Find the energy released.

Solution: $Q = 2(120 \times 8.5) - 240 \times 7.6 = 2040 - 1824 = 216$ MeV (≈ 0.9 MeV per nucleon).

13.6 Nuclear Fission

When a heavy nucleus like ^{235}U absorbs a **slow (thermal) neutron**, it forms the highly unstable $^{236}\text{U}^*$, which splits into two medium-mass fragments plus 2–3 fast neutrons:



Fission of U-235: the emitted neutrons can trigger further fissions (chain reaction)

Key Concept

- Energy per fission $\approx 200 \text{ MeV}$ (mostly as KE of fragments). Other channel: ${}_{51}^{133}\text{Sb} + {}_{41}^{99}\text{Nb} + 4n$.
- Always check: **A and Z are conserved** on both sides ($236 = 144 + 89 + 3$; $92 = 56 + 36$).
- Fragments are neutron-rich and radioactive.

Nuclear reactor (idea)

- **Moderator** (heavy water, graphite) slows neutrons.
- **Control rods** (cadmium, boron) absorb neutrons; keep multiplication factor $K = 1$.
- $K > 1$ uncontrolled $\rightarrow$ atom bomb.

Solved Example

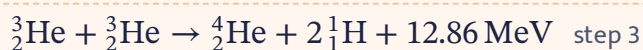
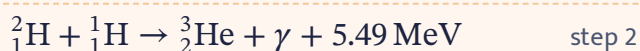
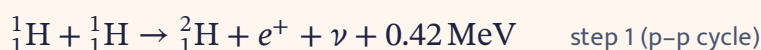
Find the energy released by complete fission of 1 kg of ^{235}U (200 MeV per fission).

Solution: Nuclei $= \frac{1000}{235} \times 6.02 \times 10^{23} = 2.56 \times 10^{24}$. $E = 2.56 \times 10^{24} \times 200 \times 1.6 \times 10^{-13} \text{ J} \approx 8.2 \times 10^{13} \text{ J}$ (≈ 20 kilotonne TNT).

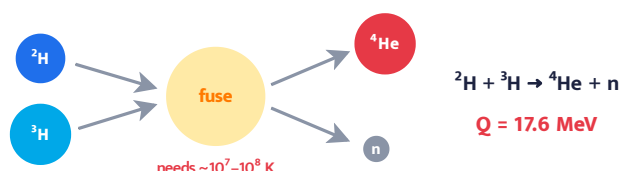
13.7 Nuclear Fusion – Energy of the Stars

Two light nuclei combine to form a heavier one. They must overcome the **Coulomb barrier**, so fusion needs temperatures of about 10^7 – 10^8 K (**thermonuclear fusion**). The Sun burns hydrogen into helium by the proton–proton cycle.

Proton–proton cycle in the Sun



Feature	Fission	Fusion
Process	heavy nucleus splits	light nuclei combine
Energy per event	≈ 200 MeV	≈ 17.6 MeV (D-T), 26.7 MeV (Sun)
Energy per nucleon	≈ 0.9 MeV	≈ 3.5 MeV (D-T) – larger
Condition	slow neutron, critical mass	$\sim 10^7 - 10^8$ K, high density
Use	nuclear reactor, atom bomb	stars, hydrogen bomb

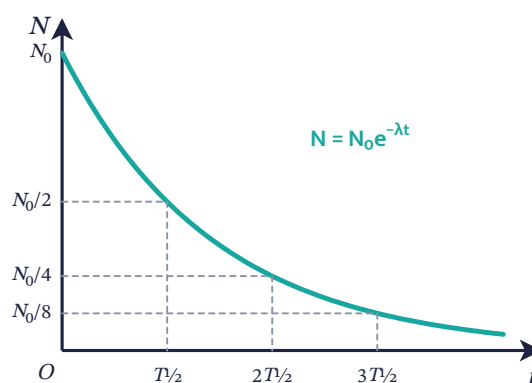


D-T fusion: the reaction planned for fusion reactors

VS Sir's Exam Tip

Per event, fission gives more energy (200 MeV vs 17.6 MeV); **per unit mass (per nucleon), fusion gives more**. NEET loves this contrast. In any energy question, find where E_{bn} increases – that direction releases energy.

13.8 Radioactive Decay – Bridge Box



Exponential decay: equal fractions in equal times



Decay law & half-life (compact)

- Radioactivity was trimmed from the rationalised NCERT text, but decay-law numericals still appear in JEE Main-level practice. Revise these five lines:
- $N = N_0 e^{-\lambda t}$; activity $R = \lambda N = R_0 e^{-\lambda t}$ (1 Bq = 1 decay/s; 1 Ci = 3.7×10^{10} Bq)
- $T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$; mean life $\tau = \frac{1}{\lambda} = 1.44 T_{1/2}$
- After n half-lives: $N = N_0 \left(\frac{1}{2}\right)^n$, $n = t / T_{1/2}$
- α : $A - 4, Z - 2$ β^- : A same, $Z + 1$ γ : no change in A, Z



Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1 **NCERT based** The ratio of the nuclear radii of ${}_{13}^{27}\text{Al}$ and ${}_{52}^{125}\text{Te}$ is
 (a) 13 : 52 (b) 9 : 25 (c) 3 : 5 (d) 27 : 125

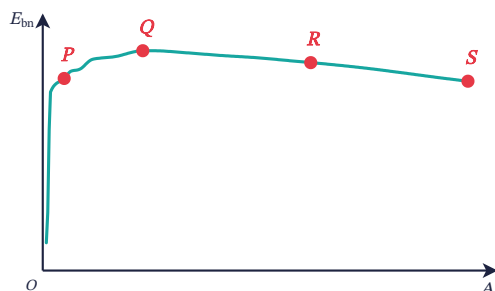
- 2 **NCERT based** The order of magnitude of nuclear density is
 (a) $10^{27} \text{ kg m}^{-3}$ (b) 10^3 kg m^{-3}
 (c) $10^{10} \text{ kg m}^{-3}$ (d) $10^{17} \text{ kg m}^{-3}$

- 3 **NEET pattern** Which pair of nuclides are isotones?
 (a) ${}_{6}^{14}\text{C}, {}_{8}^{16}\text{O}$ (b) ${}_{18}^{40}\text{Ar}, {}_{20}^{40}\text{Ca}$ (c) ${}_{6}^{12}\text{C}, {}_{6}^{14}\text{C}$ (d) ${}_{1}^3\text{H}, {}_{2}^3\text{He}$

- 4 **NEET pattern** Atomic mass of ${}^4_2\text{He} = 4.002603 \text{ u}$, $m_H = 1.007825 \text{ u}$, $m_n = 1.008665 \text{ u}$. The binding energy per nucleon of ${}^4\text{He}$ is about
- (a) 7.07 MeV (b) 3.54 MeV (c) 28.3 MeV (d) 14.1 MeV

- 5 **NCERT based** The energy equivalent of 1 g of matter is
- (a) $3 \times 10^5 \text{ J}$ (b) $9 \times 10^{10} \text{ J}$ (c) $9 \times 10^{13} \text{ J}$ (d) $9 \times 10^{16} \text{ J}$

- 6 **NEET pattern** On the B.E./nucleon curve shown, which marked nucleus is the most stable?



- (a) P (b) Q (c) R (d) S
- 7 **NEET pattern** Which statement about the nuclear force is **incorrect**?
- (a) It is charge-independent
(b) It becomes repulsive below about 0.8 fm
(c) It is short-ranged
(d) It obeys the inverse-square law

- 8 **NEET pattern** A nucleus of mass number 240 ($E_{bn} = 7.6 \text{ MeV}$) breaks into two nuclei of mass number 120 each ($E_{bn} = 8.5 \text{ MeV}$). Energy released is
- (a) 216 MeV (b) 0.9 MeV (c) 2040 MeV (d) 108 MeV

- 9 **JEE Main pattern** The radius of a nucleus is 3.6 fm. Taking $R_0 = 1.2 \text{ fm}$, its mass number is
- (a) 9 (b) 81 (c) 3 (d) 27

- 10 **NEET pattern** B.E./nucleon of ${}^2\text{H}$ is 1.1 MeV and of ${}^4\text{He}$ is 7.0 MeV. The energy released when two deuterons fuse to form one ${}^4\text{He}$ nucleus is
- (a) 28.0 MeV (b) 32.4 MeV (c) 5.9 MeV (d) 23.6 MeV

- 11 **NCERT based** In every nuclear reaction, which quantities are conserved?
- (a) Binding energy
(b) Total rest mass only
(c) Number of protons and number of neutrons separately
(d) Charge and total number of nucleons

- 12 **Assertion-Reason** Assertion: Energy is released when a heavy nucleus undergoes fission. Reason: The binding energy per nucleon of the fragments is greater than that of the heavy nucleus.
- (a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 13 **Assertion-Reason** Assertion: The nuclear force between two protons is always attractive. Reason: The nuclear force is charge-independent.
- (a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 14 **JEE Main pattern** For ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^3_2\text{He} + {}^1_0\text{n}$, masses: ${}^2_1\text{H} = 2.014102 \text{ u}$, ${}^3_2\text{He} = 3.016029 \text{ u}$, $\text{n} = 1.008665 \text{ u}$. The Q-value is
- (a) 3.27 MeV (b) 2.22 MeV (c) 17.6 MeV (d) 4.03 MeV
- 15 **NEET pattern** The ratio of nuclear densities of ${}^{16}_8\text{O}$ and ${}^4_2\text{He}$ is
- (a) 16 : 1 (b) 4 : 1 (c) 1 : 1 (d) 2 : 1
- 16 **NEET pattern** Using $m_H = 1.007825 \text{ u}$, $m_n = 1.008665 \text{ u}$ and $m({}^2_1\text{H}) = 2.014102 \text{ u}$, the binding energy of the deuteron is
- (a) 4.45 MeV (b) 1.11 MeV (c) 0.93 MeV (d) 2.22 MeV
- 17 **NCERT based** The approximate energy released in the fission of one ${}^{235}\text{U}$ nucleus by a slow neutron is
- (a) 2000 MeV (b) 20 MeV (c) 2 MeV (d) 200 MeV
- 18 **JEE Main pattern** A radioactive sample has half-life 20 min. The time between 33% decay and 67% decay is about
- (a) 10 min (b) 20 min (c) 40 min (d) 30 min
- 19 **NEET pattern** Fusion reactions need extremely high temperatures because
- (a) the binding energy must be reduced
(b) neutrons must be slowed down
(c) nuclei must overcome their Coulomb repulsion to come within nuclear-force range
(d) electrons must be removed from the atoms
- 20 **JEE Main pattern** The energy released by complete fission of 1 kg of ${}^{235}\text{U}$ (200 MeV per fission) is about
- (a) $8.2 \times 10^{13} \text{ J}$ (b) $8.2 \times 10^{10} \text{ J}$ (c) $5.1 \times 10^{26} \text{ J}$ (d) $2.0 \times 10^{13} \text{ J}$

✓ Answer Key & Quick Solutions

- 1(c) 2(d) 3(a) 4(a) 5(c) 6(b) 7(d) 8(a) 9(d) 10(d) 11(d) 12(a) 13(c) 14(a) 15(c)
16(d) 17(d) 18(b) 19(c) 20(a)

1. (c) $R \propto A^{1/3} \Rightarrow (27/125)^{1/3} = 3/5$.

2. (d) $\rho = 3m_u / (4\pi R_0^3) \approx 2.3 \times 10^{17} \text{ kg m}^{-3}$.
3. (a) Both ^{14}C and ^{16}O have $N = 8$ neutrons. (H-3/He-3 and Ar-40/Ca-40 are isobars; C-12/C-14 isotopes.)
4. (a) $\Delta m = 2(1.007825) + 2(1.008665) - 4.002603 = 0.030377 \text{ u} \rightarrow 28.3 \text{ MeV}; \div 4 = 7.07 \text{ MeV}$.
5. (c) $E = mc^2 = 10^{-3} \times (3 \times 10^8)^2 = 9 \times 10^{13} \text{ J}$.
6. (b) Highest E_{bn} (peak, $A \approx 56$) $\Rightarrow$ most stable: Q.
7. (d) The nuclear force has no simple distance law; it is not inverse-square.
8. (a) $Q = 240 \times 8.5 - 240 \times 7.6 = 240 \times 0.9 = 216 \text{ MeV}$.
9. (d) $A = (R / R_0)^3 = 3^3 = 27$.
10. (d) $Q = 4(7.0) - 2 \times 2(1.1) = 28.0 - 4.4 = 23.6 \text{ MeV}$.
11. (d) Charge (Z) and nucleon number (A) are conserved; rest mass is not ($\Delta m \rightarrow \text{energy}$).
12. (a) Fragments ($A \approx 90\text{--}140$) have $E_{bn} \approx 8.5 \text{ MeV} > 7.6 \text{ MeV}$ of U; the increase is released.
13. (c) Below $\approx 0.8 \text{ fm}$ the nuclear force is repulsive, so A is false; R is true.
14. (a) $\Delta m = 4.028204 - 4.024694 = 0.003510 \text{ u}; Q = 0.00351 \times 931.5 = 3.27 \text{ MeV}$.
15. (c) Nuclear density is independent of A .
16. (d) $\Delta m = 2.016490 - 2.014102 = 0.002388 \text{ u} \rightarrow 2.22 \text{ MeV}$.
17. (d) $\approx 0.9 \text{ MeV per nucleon} \times 236 \approx 200 \text{ MeV}$.
18. (b) Remaining falls from 67% to 33% $\approx$ halves $\Rightarrow$ one half-life $\approx 20 \text{ min}$.
19. (c) High KE lets positively charged nuclei cross the Coulomb barrier and fuse.
20. (a) $N = (1000/235)(6.02 \times 10^{23}) = 2.56 \times 10^{24}$; $E = N \times 200 \times 1.6 \times 10^{-13} \text{ J} \approx 8.2 \times 10^{13} \text{ J}$.

CHAPTER

14

Semiconductor Electronics

Day 21–22

Weightage ★★★

NCERT Class 12

Energy bands

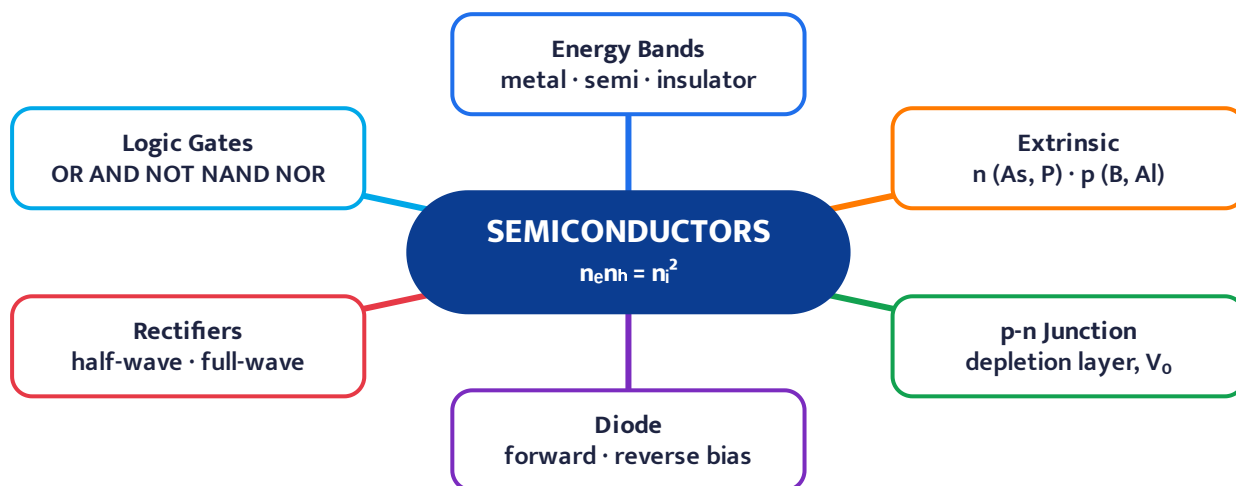
n- and p-type

p-n junction

Diode & rectifiers

Zener, LED, solar cell

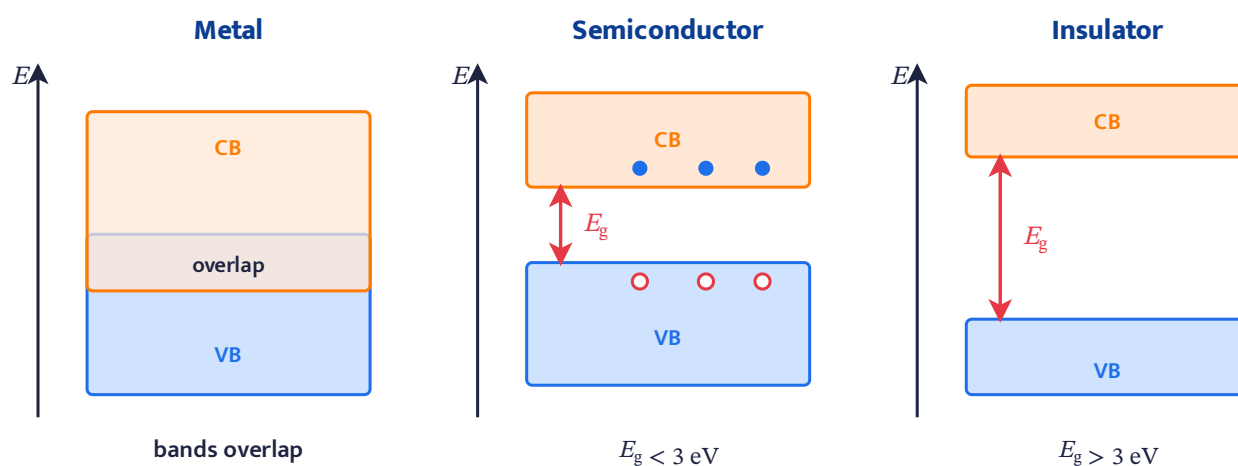
Logic gates



Chapter at a glance

14.1 Metals, Semiconductors and Insulators

Solids are classified by **resistivity ρ** (or conductivity $\sigma = 1/\rho$) and, more fundamentally, by their **energy bands**. In a crystal, the discrete atomic levels spread into bands: the **valence band (VB)** (filled with valence electrons) and the **conduction band (CB)**, separated by the **energy gap E_g** .



Energy band diagrams at a glance

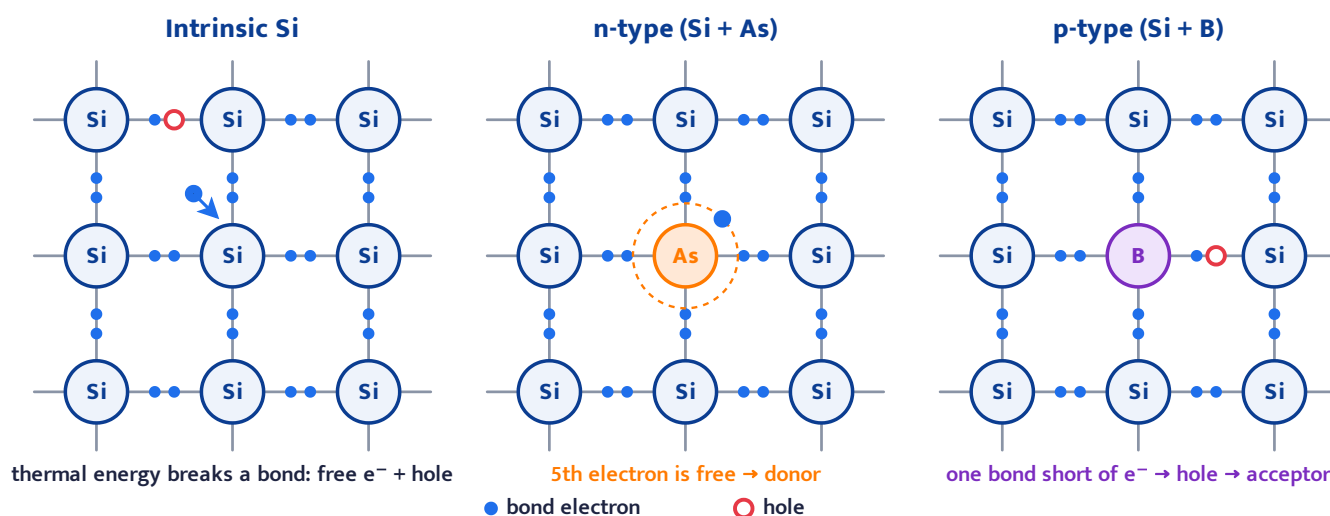
Property	Metal	Semiconductor	Insulator
ρ ($\Omega \text{ m}$)	$10^{-2} - 10^{-8}$	$10^{-5} - 10^6$	$10^{11} - 10^{19}$
Energy gap	0 (overlap)	small, $< 3 \text{ eV}$	large, $> 3 \text{ eV}$
Examples	Cu, Ag, Al	Si (1.1 eV), Ge (0.7 eV), GaAs	diamond C (5.4 eV), glass
On heating, ρ	increases	decreases	–

NCERT Line to Remember

- Semiconductors may be **elemental** (Si, Ge) or **compound** – inorganic (CdS, GaAs, CdSe, InP) or organic (anthracene, polymers such as polypyrrole).
- At $T = 0\text{ K}$ a semiconductor behaves as an **insulator** (VB full, CB empty). A semiconductor has a **negative temperature coefficient of resistance**.

14.2 Intrinsic Semiconductors

A **pure** semiconductor (Si or Ge, valency 4) is **intrinsic**. Each atom forms four covalent bonds. Thermal energy breaks a few bonds: the freed electron goes to the CB and leaves a **hole** (vacancy with effective charge $+e$) in the VB. Holes move when neighbouring bond electrons jump into them.



2-D picture of the Si lattice: intrinsic, n-type and p-type

Intrinsic semiconductor

$$n_e = n_h = n_i$$

Intrinsic: electrons = holes = intrinsic carrier concentration

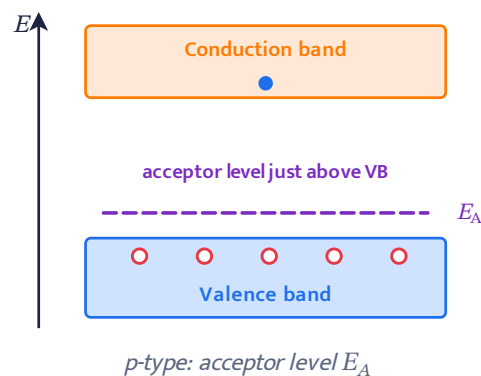
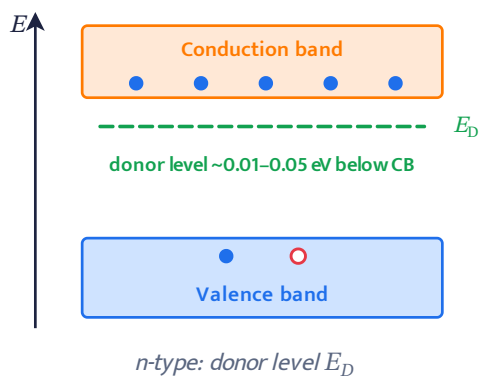
$$I = I_e + I_h$$

Total current = electron current + hole current

14.3 Extrinsic Semiconductors – Doping

Adding a tiny amount ($\approx 1\text{ ppm}$) of a suitable impurity – **doping** – raises conductivity enormously. The dopant must be of similar size so the lattice is not distorted.

Feature	n-type	p-type
Dopant	pentavalent (P, As, Sb) – donor	trivalent (B, Al, In, Ga) – acceptor
Majority carriers	electrons	holes
Minority carriers	holes	electrons
Concentrations	$n_e \approx N_D \gg n_h$	$n_h \approx N_A \gg n_e$
Impurity level	donor level just below CB	acceptor level just above VB



Carrier concentration

$$n_e n_h = n_i^2$$

Mass-action law – holds in thermal equilibrium for any (intrinsic or doped) sample

$$\sigma = e(n_e \mu_e + n_h \mu_h)$$

Conductivity (μ = mobility; $\mu_e > \mu_h$)



Solved Example

Pure Si has $n_i = 1.5 \times 10^{16} \text{ m}^{-3}$ and $5 \times 10^{28} \text{ atoms m}^{-3}$. It is doped with 1 ppm of arsenic. Find n_e and n_h .

Solution: $N_D = 5 \times 10^{28} \times 10^{-6} = 5 \times 10^{22} \text{ m}^{-3} \Rightarrow n_e \approx 5 \times 10^{22} \text{ m}^{-3}$. $n_h = n_i^2 / n_e = (2.25 \times 10^{32}) / (5 \times 10^{22}) = 4.5 \times 10^9 \text{ m}^{-3}$.



Common Trap

- An n-type crystal is **electrically neutral**, not negative – every extra electron is balanced by a fixed positive donor ion. Same for p-type.
- Doping increases one carrier and **decreases** the other (product stays n_i^2).

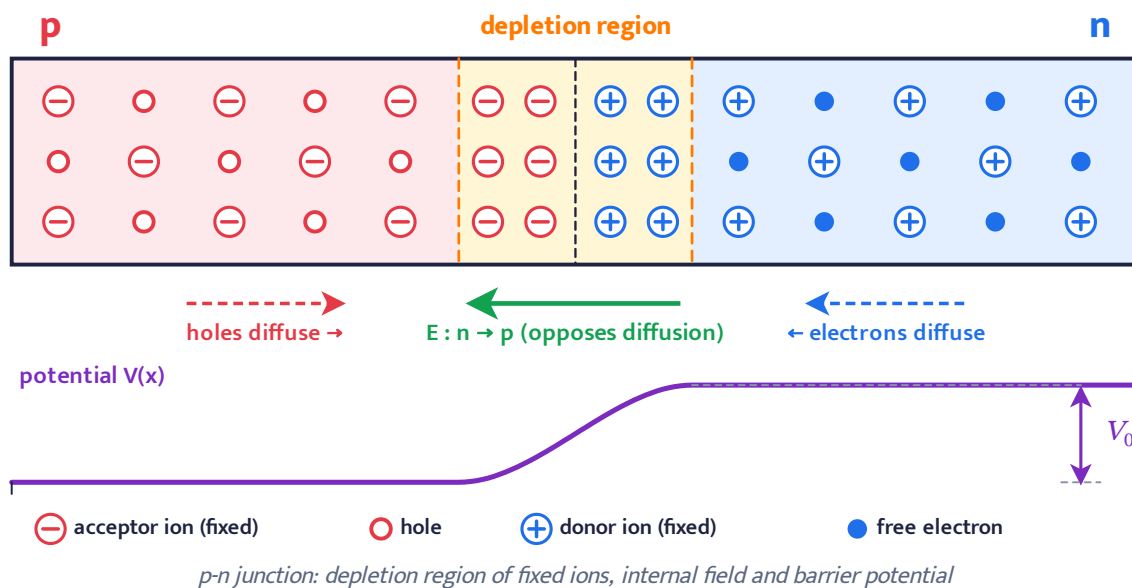


Memory Trick

Penta → **N**-type (**N**egative carriers); Tri → **P**-type (**P**ositive holes). "**PAN**: P, As, Sb → N" and "**BAIG**: B, Al, In, Ga → P".

14.4 p-n Junction Formation

When p- and n-regions are formed in one crystal, holes **diffuse** from p to n and electrons from n to p. They leave behind **immobile ions** (negative acceptor ions on the p-side, positive donor ions on the n-side). This thin layer ($\sim 0.1 \mu\text{m}$) without free carriers is the **depletion region**. Its field ($n \rightarrow p$) drives a **drift** current opposite to diffusion; at equilibrium the two are equal and a **barrier potential** V_0 exists.



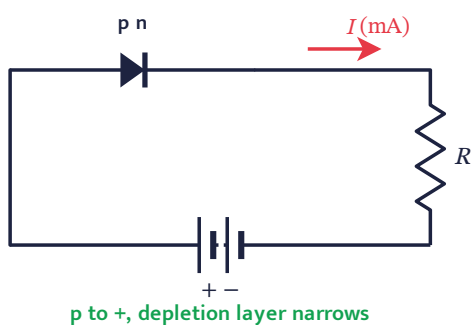
NCERT Line to Remember

- Barrier potential $\approx 0.7 \text{ V}$ for Si, $\approx 0.2\text{--}0.3 \text{ V}$ for Ge.
- Diffusion current: majority carriers ($p \rightarrow n$). Drift current: minority carriers, due to the field.
- Open junction: **no net current**.

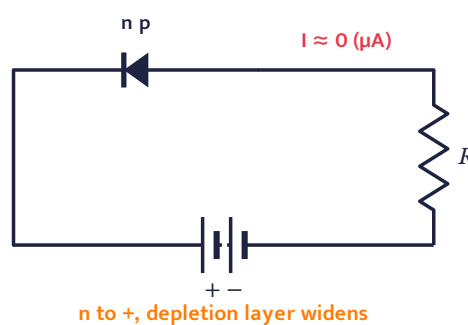
VS Sir's Exam Tip

Picture questions: the **p-side** of the depletion layer carries **negative** ions and the **n-side positive** ions – the field inside points **from n to p**. Examiners swap these.

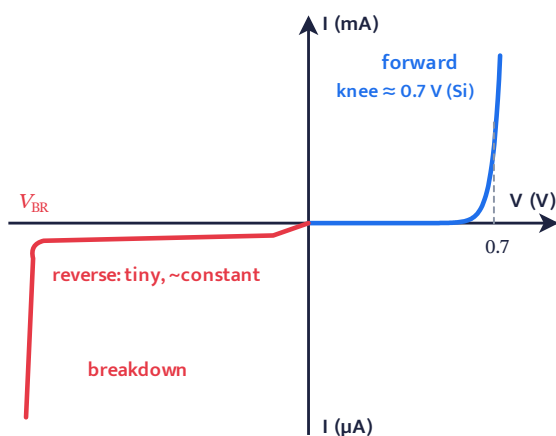
14.5 Semiconductor Diode – Biasing and I–V Characteristics



Forward bias: barrier lowers to $V_0 - V$



Reverse bias: barrier rises to $V_0 + V$



Diode I–V curve (note different scales: mA vs μA)

Key Concept

- Forward:** current is negligible until the **threshold (knee/cut-in) voltage** ($\sim 0.7 \text{ V}$ Si, $\sim 0.2 \text{ V}$ Ge), then rises **exponentially** (mA).
- Reverse:** tiny, nearly constant current (μA) of minority carriers, till **breakdown** V_{BR} , where it rises sharply.



Dynamic resistance

$$r_d = \frac{\Delta V}{\Delta I}$$

Dynamic resistance (small in forward, huge in reverse)

Solved Example

For a Si diode the current rises from 10 mA to 20 mA when the voltage changes from 0.7 V to 0.8 V. In reverse bias, -10 V gives $-1 \mu\text{A}$. Find the forward and reverse dynamic resistances.

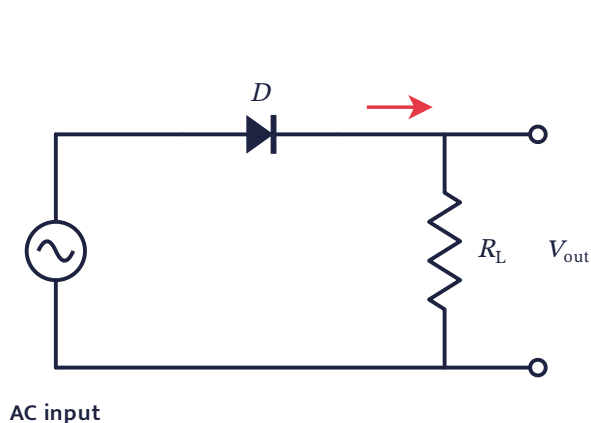
Solution: $r_{fb} = \frac{0.1 \text{ V}}{10 \text{ mA}} = 10 \Omega$; $r_{rb} = \frac{10 \text{ V}}{1 \mu\text{A}} = 10^7 \Omega$. A diode conducts one way only.

Common Trap

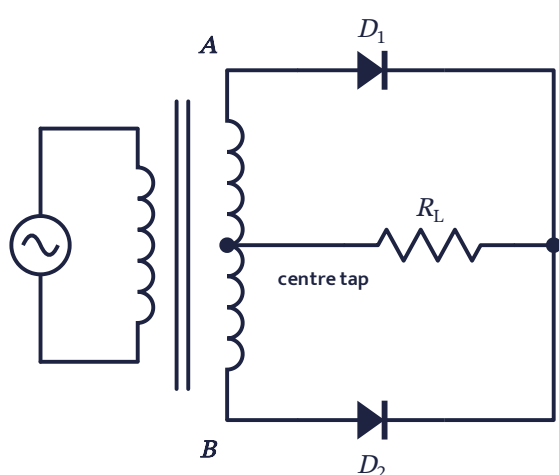
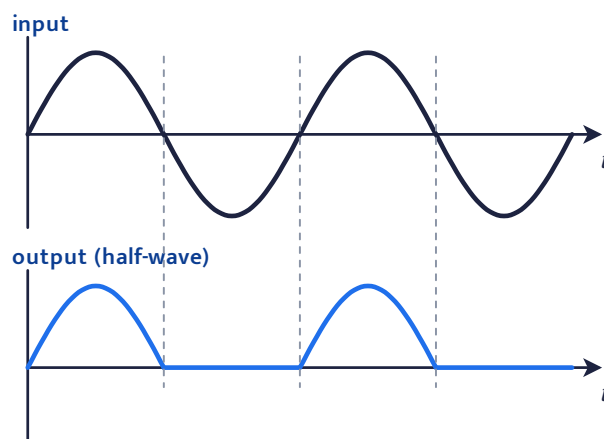
- **Forward biased** $\Leftrightarrow V_p > V_n$ (the p-side at higher potential) – even if both are negative: p at -2 V, n at -4 V is forward biased.
- **Ideal diode:** zero resistance forward (short), infinite reverse (open). For a real Si diode, subtract 0.7 V:
 $I = (V - 0.7) / R$.

14.6 Diode as a Rectifier

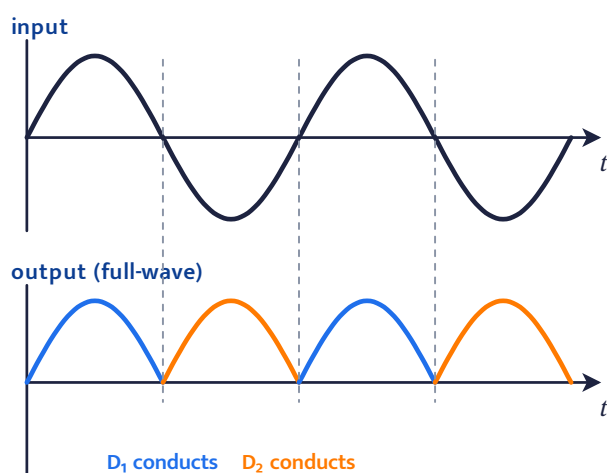
A **rectifier** converts AC into (pulsating) DC, using the fact that a diode conducts only in forward bias.



Half-wave rectifier: output only during the positive half-cycles



Full-wave (centre-tap) rectifier: D_1 and D_2 conduct alternately; current in R_L is always in the same direction



Quantity	Half-wave	Full-wave
Diodes used	1	2 (centre-tap) or 4 (bridge)
Output (ripple) frequency	f (50 Hz for 50 Hz mains)	$2f$ (100 Hz)
Average (DC) output	V_m / π	$2V_m / \pi$
Max. efficiency	40.6%	81.2%

Filter

A capacitor in parallel with R_L charges to the peak and discharges slowly through R_L between peaks, smoothing the ripple. Larger $CR_L \Rightarrow$ smoother DC.

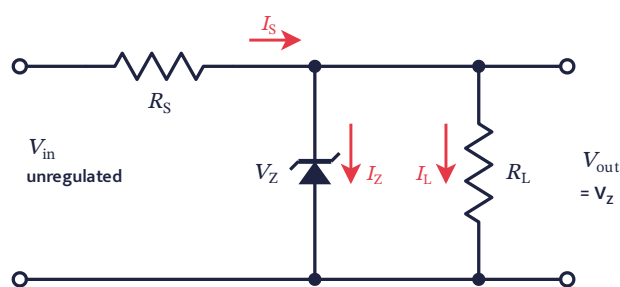
JEE Main Extension

Peak inverse voltage (PIV) across a diode: half-wave V_m ; centre-tap full-wave $2V_m$. RMS output: half-wave $V_m / 2$, full-wave $V_m / \sqrt{2}$.

14.7 Special-Purpose p-n Junction Diodes

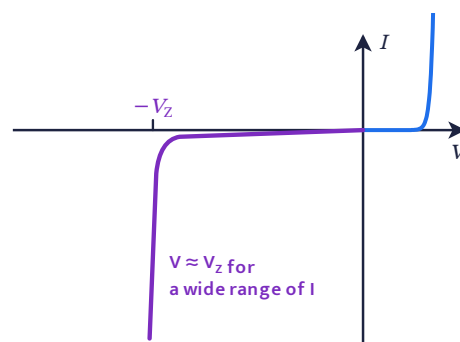
Zener diode as a voltage regulator

A **heavily doped** p-n junction with a very thin depletion layer, designed to work in **reverse breakdown**. Beyond V_Z the voltage stays constant while the current changes over a wide range – so it holds the load voltage fixed even when the input or load varies.



Zener in REVERSE bias (breakdown)

Zener regulator: $V_{out} = V_Z$, $I_S = I_Z + I_L$



Zener characteristic

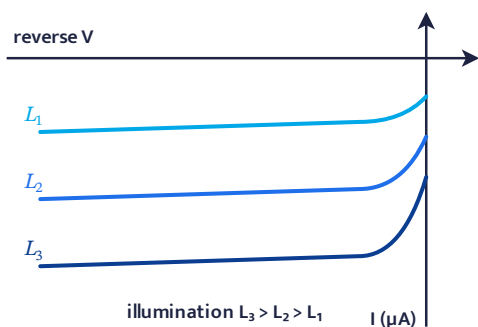
Solved Example

In a Zener regulator, $V_{in} = 15$ V, $R_S = 250$ Ω , $V_Z = 10$ V and $R_L = 1$ k Ω . Find I_S , I_L and I_Z .

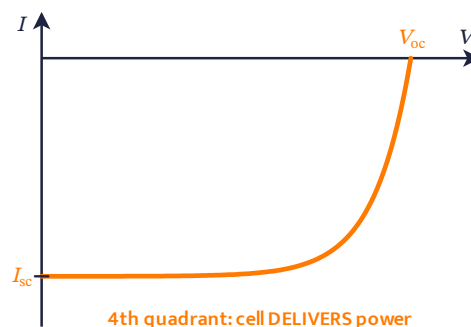
Solution: $I_S = \frac{15 - 10}{250} = 20$ mA; $I_L = \frac{10}{1000} = 10$ mA; $I_Z = I_S - I_L = 10$ mA.

Optoelectronic junction devices

Device	Bias	Principle / use
Photodiode	reverse	light with $h\nu > E_g$ creates e-h pairs near the junction $\rightarrow$ photocurrent $\propto$ intensity; light detector
LED	forward	heavily doped; e-h recombination emits photons, $h\nu \approx E_g$; $E_g \approx 1.8$ –3 eV for visible light
Solar cell	none (self)	large-area junction; generates an emf (photovoltaic) – works in the 4th quadrant; ideal $E_g \approx 1.5$ eV (Si, GaAs)



Photodiode: reverse current grows with intensity



Solar cell I-V: open-circuit voltage and short-circuit current

VS Sir's Exam Tip

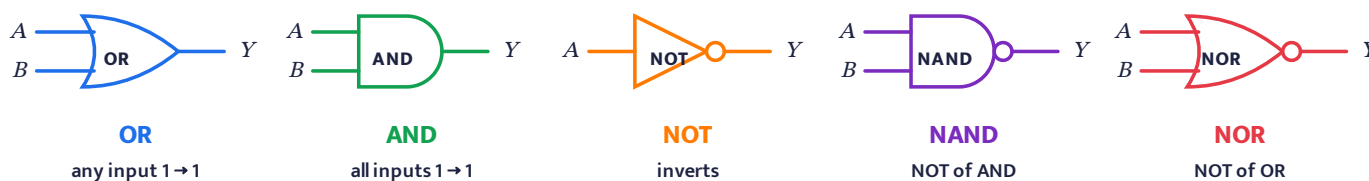
- $\lambda(\text{nm}) \approx \frac{1240}{E_g(\text{eV})}$ – for an LED (emitted) or a photodiode (maximum detectable λ).
- Photodiode is used in reverse bias because the **fractional change** in (minority-carrier) current due to light is much larger than in forward bias.

NCERT Line to Remember

- LEDs vs lamps: low voltage, low power, fast on-off, long life, no warm-up.
- Solar cell steps: **generation** (light), **separation** (junction field), **collection** (electrodes).

14.8 Digital Electronics – Logic Gates

Digital signals take only two levels: **1** (high) and **0** (low). A **logic gate** is a circuit whose output follows a fixed logical rule of its inputs, written as a **Boolean expression** and shown by a **truth table**.



Symbols of the five basic gates (a bubble means NOT)

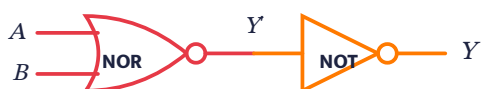
A	B	OR $A + B$	AND $A \cdot B$	NAND $\overline{A \cdot B}$	NOR $\overline{A + B}$
0	0	0	0	1	1
0	1	1	0	1	0
1	0	1	0	1	0
1	1	1	1	0	0

Memory Trick

- **NOT:** $Y = \bar{A}$ ($0 \rightarrow 1, 1 \rightarrow 0$), single input.
- OR = "any 1 gives 1"; AND = "all 1 gives 1". NAND/NOR are just their **complements**.
- **NAND and NOR are universal gates** – any gate can be built from them alone.

De Morgan & NAND tricks

- $\overline{A + B} = \bar{A} \cdot \bar{B}$ and $\overline{A \cdot B} = \bar{A} + \bar{B}$
- NAND with inputs joined = NOT; NAND followed by NOT = AND; NOTs on both inputs of NAND = OR.
- NOR with inputs joined = NOT; NOTs on both inputs of NOR = AND.



A NOR gate followed by a NOT gate

Solved Example

Identify the gate equivalent to the circuit on the left.

Solution: $Y' = \overline{A + B}$, then $Y = \overline{Y'} = \overline{\overline{A + B}} = A + B$. Truth table 00 → 0, 01 → 1, 10 → 1, 11 → 1: an **OR gate**.

**Practice Questions**

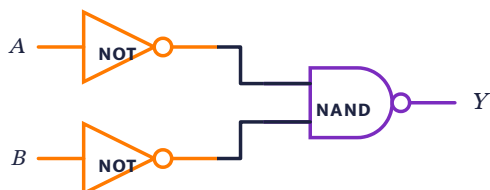
Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1 **NCERT based** The energy gap of silicon at room temperature is about
 (a) 5.4 eV (b) 0.7 eV (c) 0.01 eV (d) 1.1 eV
- 2 **NCERT based** To make n-type silicon, the dopant should be
 (a) boron (b) aluminium (c) indium (d) phosphorus
- 3 **NEET pattern** Si with $n_i = 1.5 \times 10^{16} \text{ m}^{-3}$ is doped so that $n_e = 4.5 \times 10^{22} \text{ m}^{-3}$. The hole concentration is
 (a) $3 \times 10^6 \text{ m}^{-3}$ (b) $5 \times 10^9 \text{ m}^{-3}$ (c) $1.5 \times 10^{16} \text{ m}^{-3}$ (d) $4.5 \times 10^{22} \text{ m}^{-3}$
- 4 **NCERT based** The depletion region of a p-n junction contains
 (a) both mobile electrons and holes
 (b) immobile ions only
 (c) free electrons only
 (d) holes only
- 5 **NEET pattern** When a p-n junction is forward biased
 (a) the depletion layer widens and the barrier increases
 (b) there is no change
 (c) the depletion layer narrows and the barrier decreases
 (d) the depletion layer widens but the barrier decreases
- 6 **NCERT based** As the temperature of an intrinsic semiconductor increases, its resistivity
 (a) first increases then decreases (b) remains the same
 (c) increases (d) decreases
- 7 **NEET pattern** The input to a full-wave rectifier is 50 Hz AC. The fundamental frequency of the output ripple is
 (a) 100 Hz (b) 25 Hz (c) 200 Hz (d) 50 Hz
- 8 **NEET pattern** In which case is the diode forward biased? (potentials of p and n ends given)
 (a) p: +3 V, n: +5 V (b) p: -10 V, n: -12 V
 (c) p: 0 V, n: +5 V (d) p: -5 V, n: 0 V
- 9 **JEE Main pattern** In a Zener regulator, $V_{in} = 20 \text{ V}$, $R_S = 500 \Omega$, $V_Z = 10 \text{ V}$, $R_L = 2 \text{ k}\Omega$. The Zener current is
 (a) 15 mA (b) 25 mA (c) 20 mA (d) 5 mA
- 10 **NEET pattern** An LED is made of a semiconductor of band gap 2.0 eV. The wavelength of emitted light is about
 (a) 310 nm (b) 620 nm (c) 420 nm (d) 1240 nm

- 11 **NEET pattern** A photodiode is operated in reverse bias because
- (a) light cannot enter in forward bias
 - (b) the current is larger in reverse bias
 - (c) the fractional change in current due to light is larger in reverse bias
 - (d) the junction does not heat up

- 12 **NCERT based** A gate gives output 1 only when both inputs are 0. The gate is
- (a) OR
 - (b) AND
 - (c) NOR
 - (d) NAND

- 13 **JEE Main pattern** The circuit shown is equivalent to



- (a) NAND gate
- (b) AND gate
- (c) OR gate
- (d) NOR gate

- 14 **NEET pattern** A NAND gate has both its inputs joined together as shown. It works as a



- (a) NOT gate
- (b) NOR gate
- (c) AND gate
- (d) OR gate

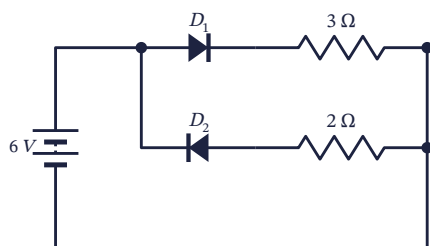
- 15 **NEET pattern** A gate has outputs 0, 1, 1, 1 for inputs AB = 00, 01, 10, 11. It is
- (a) OR
 - (b) AND
 - (c) NAND
 - (d) NOR

- 16 **JEE Main pattern** For a diode, the current increases from 10 mA to 30 mA when the forward voltage rises from 0.70 V to 0.80 V. Its dynamic resistance is
- (a) $2.5\ \Omega$
 - (b) $10\ \Omega$
 - (c) $20\ \Omega$
 - (d) $5\ \Omega$

- 17 **NEET pattern** A Si diode (cut-in 0.7 V) is connected in forward bias in series with $1\ \text{k}\Omega$ across a 5 V battery. The current is
- (a) zero
 - (b) 5.0 mA
 - (c) 4.3 mA
 - (d) 5.7 mA

- 18 **NCERT based** A p-type semiconductor is
- (a) electrically neutral
 - (b) positively charged
 - (c) negatively charged
 - (d) free of electrons

- 19 **Assertion-Reason** Assertion: At 0 K, pure silicon behaves as an insulator. Reason: At 0 K the conduction band is empty and the valence band is completely filled.
- (a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 20 **Assertion-Reason** Assertion: A Zener diode is used as a voltage regulator in forward bias. Reason: In the breakdown region the Zener voltage stays constant over a wide range of current.
- (a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false
- 21 **JEE Main pattern** Ge is doped with Al so that $N_A = 10^{21} \text{ m}^{-3}$. If $n_i = 2.5 \times 10^{19} \text{ m}^{-3}$, the electron concentration is
- (a) 10^{21} m^{-3} (b) $6.25 \times 10^{17} \text{ m}^{-3}$ (c) $2.5 \times 10^{19} \text{ m}^{-3}$ (d) $4 \times 10^{16} \text{ m}^{-3}$
- 22 **NEET pattern** The I–V characteristic of an illuminated solar cell lies in the
- (a) first quadrant (b) third quadrant
(c) fourth quadrant (d) second quadrant
- 23 **JEE Main pattern** The peak voltage of the input to a half-wave rectifier is 20 V. The average (DC) output voltage is about
- (a) 14.1 V (b) 12.7 V (c) 10 V (d) 6.4 V
- 24 **NEET pattern** In the circuit shown, the diodes are ideal. The current drawn from the 6 V battery is



- (a) 3 A (b) 2 A (c) 0 A (d) 5 A

✓ Answer Key & Quick Solutions

- 1(d) 2(d) 3(b) 4(b) 5(c) 6(d) 7(a) 8(b) 9(a) 10(b) 11(c) 12(c) 13(c) 14(a) 15(a)
16(d) 17(c) 18(a) 19(a) 20(c) 21(b) 22(c) 23(d) 24(b)

- (d) $\text{Si} \approx 1.1 \text{ eV}$ ($\text{Ge} \approx 0.7 \text{ eV}$, diamond $\approx 5.4 \text{ eV}$).
- (d) n-type needs a pentavalent donor: P (B, Al, In are trivalent).
- (b) $n_h = n_i^2 / n_e = 2.25 \times 10^{32} / 4.5 \times 10^{22} = 5 \times 10^9 \text{ m}^{-3}$.
- (b) Free carriers have diffused/recombined; only fixed donor and acceptor ions remain.
- (c) Applied V opposes V_0 : effective barrier $V_0 - V$, thinner depletion layer.

6. (d) More bonds break $\rightarrow$ more carriers $\rightarrow$ lower ρ (negative temperature coefficient).
7. (a) Both half-cycles give a pulse: $2f = 100$ Hz.
8. (b) Forward bias needs $V_p > V_n$: $-10\text{ V} > -12\text{ V}$.
9. (a) $I_S = 10/500 = 20\text{ mA}$; $I_L = 10/2000 = 5\text{ mA}$; $I_Z = 15\text{ mA}$.
10. (b) $\lambda \approx 1240/2.0 = 620\text{ nm}$ (red-orange).
11. (c) The small minority-carrier current changes by a large fraction with illumination – easy to detect.
12. (c) $\overline{A + B} = 1$ only for $A = B = 0$.
13. (c) $Y = \overline{\bar{A} \cdot \bar{B}} = A + B$ (De Morgan) – OR.
14. (a) $Y = \overline{A \cdot A} = \bar{A}$.
15. (a) Output 1 when any input is 1 – OR.
16. (d) $r_d = 0.10 / 0.020 = 5\ \Omega$.
17. (c) $I = (5 - 0.7)/1000 = 4.3\text{ mA}$.
18. (a) Each hole is balanced by a fixed negative acceptor ion – net charge zero.
19. (a) No thermal energy to cross $E_g \Rightarrow$ no free carriers; R is the correct explanation.
20. (c) Zener regulates in **reverse** breakdown, so A is false; R is true.
21. (b) $n_h \approx 10^{21}$; $n_e = n_i^2 / n_h = 6.25 \times 10^{38} / 10^{21} = 6.25 \times 10^{17}\text{ m}^{-3}$.
22. (c) The cell supplies power: V positive, current negative (drawn from the cell) – 4th quadrant.
23. (d) $V_{dc} = V_m / \pi = 20/3.14 \approx 6.4\text{ V}$ (12.7 V would be full-wave).
24. (b) D_1 is forward (conducts), D_2 reverse (open): $I = 6/3 = 2\text{ A}$.

THE FINAL SPRINT

DAYS
23–26**Last-Week Toolkit**

Days 23–24 • Formula sheets

Day 25 • Experimental skills

Top 50 traps

14 graphs at a glance

NEET + JEE Main strategy

Error log

Everything you need in the final stretch on a few pages: every Class 12 formula for NEET and JEE Main, the lab experiments both exams ask about, the traps where marks are lost, the graphs examiners love, a minute-by-minute exam plan and your personal error log.

A • Formula Sheets (Days 23–24)

How to use: Day 23 – Chapters 1–7 (electricity & magnetism); Day 24 – Chapters 8–14 (optics & modern physics). Cover the right-hand meaning and say each formula aloud with its condition. Tick the ones you had to look up and revise them again at night.

Day 23 – Electricity & Magnetism **Constants to keep at your fingertips**

$$e = 1.6 \times 10^{-19} \text{ C} \cdot k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2} \cdot \epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2} \cdot \mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1} \cdot$$

$$c = 3 \times 10^8 \text{ m/s} \cdot h = 6.63 \times 10^{-34} \text{ J s} \cdot hc \approx 1240 \text{ eV nm} \cdot m_e = 9.1 \times 10^{-31} \text{ kg} \cdot 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J} \cdot 1 \text{ u}$$

$$= 931.5 \text{ MeV}/c^2 \cdot R = 1.097 \times 10^7 \text{ m}^{-1}.$$

 Ch 1 • Electric Charges and Fields

$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$	Coulomb's law (vacuum)	$F_m = \frac{F}{K}$	In a medium of dielectric constant K
$q = ne$	Quantisation of charge	$\vec{E} = \frac{\vec{F}}{q_0}, E = \frac{kq}{r^2}$	Field of a point charge
$p = q(2a)$	Dipole moment, from – to +	$E_{ax} = \frac{2kp}{r^3}, E_{eq} = \frac{kp}{r^3}$	Short dipole; E_{eq} opposite to $\vec{p}$
$\tau = pE \sin\theta$	Torque in a uniform field (net force 0)	$U = -pE \cos\theta$	PE of dipole; $W_{0 \rightarrow \theta} = pE(1 - \cos\theta)$
$\Phi = \oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$	Gauss's law	$E = \frac{\lambda}{2\pi\epsilon_0 r}$	Infinite line charge
$E = \frac{\sigma}{2\epsilon_0}$	Infinite plane sheet	$E = \frac{\sigma}{\epsilon_0}$	Just outside a conductor; between plates
$E = \frac{kQ}{r^2} (r \geq R), 0 (r < R)$	Thin spherical shell	$E = \frac{kQr}{R^3} (r < R)$	Inside a uniformly charged solid sphere



Ch 2 • Electrostatic Potential and Capacitance

$V = \frac{W}{q_0}, V = \frac{kq}{r}$	Potential; point charge	$V = \frac{kpcos\theta}{r^2}$	Dipole (zero on the equatorial line)
$E = -\frac{dV}{dr}$	Field points to decreasing V	$U = \frac{kq_1q_2}{r}$	PE of a pair (add over all pairs)
$C = \frac{Q}{V}, C = 4\pi\epsilon_0 R$	Definition; isolated sphere	$C = \frac{\epsilon_0 A}{d}, C_K = KC$	Parallel plate; filled with dielectric
$C = \frac{\epsilon_0 A}{d - t + t/K}$	Slab of thickness t (metal: $K \rightarrow \infty$)	$\frac{1}{C_s} = \sum \frac{1}{C_i}, C_p = \sum C_i$	Series (same Q); parallel (same V)
$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$	Energy stored	$u = \frac{1}{2}\epsilon_0 E^2$	Energy density
$V_c = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$	Common potential on joining	$\Delta U = \frac{C_1C_2(V_1 - V_2)^2}{2(C_1 + C_2)}$	Energy lost as heat on joining

Dielectric K inserted

	C	Q	V	E	U
Battery connected (V fixed)	KC	KQ	V	E	KU
Battery removed (Q fixed)	KC	Q	V / K	E / K	U / K



Ch 3 • Current Electricity

$I = \frac{dq}{dt} = neAv_d$	Current and drift speed	$v_d = \frac{eE\tau}{m}, \mu = \frac{v_d}{E}$	Drift velocity; mobility
$R = \frac{\rho l}{A}, \rho = \frac{m}{ne^2\tau}$	Resistance; resistivity	$\vec{J} = \sigma \vec{E}$	Microscopic Ohm's law ($\sigma = 1/\rho$)
$R' = n^2R$	Wire stretched n times (volume constant)	$\rho_T = \rho_0[1 + \alpha(T - T_0)]$	$\alpha > 0$ metals, $\alpha < 0$ semiconductors
$R_s = \sum R_i, \frac{1}{R_p} = \sum \frac{1}{R_i}$	Series; parallel	$I = \frac{E}{R + r}$	Cell with internal resistance r
$V = E - Ir$ (discharging)	Charging: $V = E + Ir$	$P_{max} = \frac{E^2}{4r}$ at $R = r$	Maximum power transfer
$I = \frac{nE}{R + nr}$	n identical cells in series	$I = \frac{E}{R + r/m}$	m identical cells in parallel
$P = VI = I^2R = \frac{V^2}{R}$	Power; heat $H = I^2Rt$	$\sum I = 0, \sum \Delta V = 0$	Kirchhoff's junction and loop rules
$\frac{P}{Q} = \frac{R}{S}$	Balanced Wheatstone bridge	$\frac{R}{S} = \frac{l}{100 - l}$	Metre bridge (l in cm from R side)

 Resistor colour code

- **BBROYGBVGVW** = 0 1 2 3 4 5 6 7 8 9 (Black, Brown, Red, Orange, Yellow, Green, Blue, Violet, Grey, White).
- Bands 1, 2 = digits; band 3 = power of 10; band 4 = tolerance: gold 5%, silver 10%, none 20%.
- Yellow–Violet–Red–Gold = $47 \times 10^2 \Omega \pm 5\%$.



JEE Main Extension

- Two cells in parallel: $E_{eq} = \frac{E_1/r_1 + E_2/r_2}{1/r_1 + 1/r_2}$,
 $\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2}$.
- Mixed grouping (n in series, m rows): current is maximum when $R = \frac{nr}{m}$.

 Ch 4 • Moving Charges and Magnetism

$\vec{F} = q(\vec{v} \times \vec{B})$	Magnetic force; $F = qvB\sin\theta$	$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$	Lorentz force; selector $v = E/B$
$r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$	Radius of circular path	$T = \frac{2\pi m}{qB}$	Period: independent of v and r
$p = v\cos\theta \cdot T$	Pitch of helix	$\vec{F} = I\vec{l} \times \vec{B}$	Force on a current-carrying wire
$d\vec{B} = \frac{\mu_0 Id \vec{l} \times \hat{r}}{4\pi r^2}$	Biot–Savart law	$B = \frac{\mu_0 I}{2\pi r}$	Long straight wire
$B = \frac{\mu_0 NI}{2R}$	Centre of a circular coil	$B = \frac{\mu_0 NIR^2}{2(R^2 + x^2)^{3/2}}$	On the axis of a coil
$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$	Ampere's circuital law	$B = \mu_0 nI, B = \frac{\mu_0 NI}{2\pi r}$	Long solenoid; toroid
$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$	Parallel wires: like currents attract	$\vec{\tau} = \vec{m} \times \vec{B}, m = NIA$	Torque on a current loop
$\frac{\phi}{I} = \frac{NAB}{k}, \frac{\phi}{V} = \frac{NAB}{kG}$	Current, voltage sensitivity	$S = \frac{I_g G}{I - I_g}, R = \frac{V}{I_g} - G$	Shunt for ammeter; series R for voltmeter

 Ch 5 • Magnetism and Matter

$m = q_m(2l)$	Magnetic moment (S → N)	$B_{ax} = \frac{\mu_0 2m}{4\pi r^3}, B_{eq} = \frac{\mu_0 m}{4\pi r^3}$	Short bar magnet
$\tau = mB\sin\theta, U = -mB\cos\theta$	Magnet in uniform B	$T = 2\pi\sqrt{\frac{I}{mB}}$	Small oscillations in B
$\oint \vec{B} \cdot d\vec{A} = 0$	Gauss's law in magnetism	$M = \frac{m_{net}}{V}, \chi = \frac{M}{H}$	Magnetisation; susceptibility
$B = \mu_0(H + M) = \mu_0\mu_r H$	B, H and M	$\mu_r = 1 + \chi$	Relative permeability
$\chi = \frac{C}{T}$	Curie's law (paramagnets)	$W = mB(1 - \cos\theta)$	Work to turn from 0 to θ

Property	Diamagnetic	Paramagnetic	Ferromagnetic
χ	small, negative	small, positive	very large, positive
μ_r	< 1	slightly > 1	$\gg 1$
In non-uniform B	moves to weaker field	moves to stronger field	strongly attracted
Temperature	almost independent	$\chi \propto 1/T$	paramagnetic above T_C
Examples	Bi, Cu, water, N_2 , superconductor	Al, Na, O_2 , Pt	Fe, Co, Ni, alnico



Ch 6 • Electromagnetic Induction

$\Phi = BA \cos \theta$	Magnetic flux (Wb)	$\varepsilon = -N \frac{d\Phi}{dt}$	Faraday's law + Lenz's sign
$\varepsilon = Blv$	Rod moving $\perp$ to B	$\varepsilon = \frac{1}{2} B \omega l^2$	Rod rotating about one end
$\varepsilon = NBA \omega \sin \omega t$	Rotating coil; $\varepsilon_0 = NBA \omega$	$q = \frac{N \Delta \Phi}{R}$	Induced charge (rate-independent)
$N\Phi = LI$, $\varepsilon = -L \frac{dI}{dt}$	Self-inductance	$L = \mu_0 n^2 Al$	Long solenoid
$N_2 \Phi_2 = MI_1$, $\varepsilon_2 = -M \frac{dI_1}{dt}$	Mutual inductance	$M = \mu_0 n_1 n_2 Al$	Coaxial solenoids
$U = \frac{1}{2} LI^2$	Energy in an inductor	$u_B = \frac{B^2}{2\mu_0}$	Magnetic energy density



Ch 7 • Alternating Current

$I_{rms} = \frac{I_0}{\sqrt{2}}$, $V_{rms} = \frac{V_0}{\sqrt{2}}$	rms values (meters read these)	$X_L = \omega L$, $X_C = \frac{1}{\omega C}$	Reactances (Ω)
$Z = \sqrt{R^2 + (X_L - X_C)^2}$	Impedance of series LCR	$\tan \phi = \frac{X_L - X_C}{R}$	Phase of V ahead of I
$\omega_0 = \frac{1}{\sqrt{LC}}$	Resonance: $Z = R$, I maximum	$Q = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$	Quality factor (sharpness)
$P = V_{rms} I_{rms} \cos \phi$	Average power	$\cos \phi = \frac{R}{Z}$	Power factor (pure L or C: 0)
$\omega = \frac{1}{\sqrt{LC}}$, $q = q_0 \cos \omega t$	LC oscillations	$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$	Ideal transformer
$\Delta \omega = \frac{R}{L}$	Bandwidth (JEE)	$\bar{I}_{half} = \frac{2I_0}{\pi}$	Mean over half cycle (0 over full)

 Phase: "CIVIL"

- In a Capacitor, I leads V by 90° ; in an inductor (L), V leads I by 90° .
- Pure R : in phase, $\cos\phi = 1$. Pure L or C : average power zero (wattless).
- $X_L > X_C$: inductive (V leads); $X_C > X_L$: capacitive (I leads).

 NCERT Line to Remember

- Transformer losses: flux leakage, winding (copper) resistance, eddy currents (reduced by a **laminated** core) and hysteresis (reduced by a **soft-iron** core).
- Power is transmitted at high voltage and low current to cut I^2R loss.

Day 24 – Optics, Modern Physics & Semiconductors

 Ch 8 • Electromagnetic Waves

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ Speed in vacuum (m/s)}$$

$$v = \frac{1}{\sqrt{\mu \epsilon}} = \frac{c}{n} \text{ Speed in a medium}$$

$$\frac{E_0}{B_0} = c \quad \vec{E} \perp \vec{B} \perp \text{ direction; in phase}$$

$$I_d = \epsilon_0 \frac{d\Phi_E}{dt} \text{ Displacement current}$$

$$u_{av} = \frac{1}{2} \epsilon_0 E_0^2 \text{ Average energy density (shared equally by E and B)}$$

$$I = u_{av} c = \frac{1}{2} c \epsilon_0 E_0^2 \text{ Intensity}$$

$$p = \frac{U}{c} \text{ Momentum delivered (absorbed); } 2U/c \text{ if reflected}$$

$$c = \nu \lambda \text{ Frequency unchanged in a new medium}$$

Wave	Wavelength	Produced by	Key use (NCERT)
Radio	$> 0.1 \text{ m}$	accelerating charges in aerials	radio, TV, mobile communication
Microwave	$0.1 \text{ m} - 1 \text{ mm}$	klystron, magnetron, Gunn diode	radar, microwave oven
Infrared	$1 \text{ mm} - 700 \text{ nm}$	hot bodies, molecules	remote controls, physiotherapy, greenhouse effect
Visible	$700 - 400 \text{ nm}$	atoms (electron transitions)	vision
Ultraviolet	$400 \text{ nm} - 1 \text{ nm}$	Sun, arcs, very hot bodies	water purifiers, LASIK; ozone absorbs
X-rays	$1 \text{ nm} - 10^{-3} \text{ nm}$	electrons hitting a metal target	medical diagnosis
Gamma	$< 10^{-3} \text{ nm}$	nuclear reactions	cancer treatment

 Ch 9 • Ray Optics – mirrors & refraction

$$f = \frac{R}{2} \text{ Spherical mirror (paraxial)}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}, m = -\frac{v}{u} \text{ Mirror formula; magnification}$$

$$n_{21} = \frac{n_2}{n_1} = \frac{\sin i}{\sin r} \text{ Snell's law}$$

$$\frac{\text{real depth}}{\text{apparent depth}} = n \text{ Viewed normally from air}$$

$$\text{shift} = t \left(1 - \frac{1}{n} \right) \text{ Normal shift by a slab}$$

$$\sin C = \frac{1}{n} \text{ Critical angle, denser} \rightarrow \text{rarer}$$

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R} \text{ Refraction at a spherical surface}$$

$$\frac{1}{f} = (n_{21} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \text{ Lens maker's formula}$$

**Ch 9 • Ray Optics – lenses, prism, instruments**

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}, m = \frac{v}{u} \quad \text{Thin lens formula}$$

$$P = \frac{1}{f(\text{m})} \quad \text{Power in dioptre (D)}$$

$$P = P_1 + P_2, \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} \quad \text{Lenses in contact}$$

$$A = r_1 + r_2, \delta = i + e - A \quad \text{Prism geometry}$$

$$n = \frac{\sin \frac{A+\delta_m}{2}}{\sin \frac{A}{2}} \quad \text{Minimum deviation: } i = e, r = A/2$$

$$\delta = (n - 1)A \quad \text{Thin prism}$$

$$m = 1 + \frac{D}{f} \text{ or } \frac{D}{f} \quad \text{Simple microscope (near point / } \infty)$$

$$m = \frac{L}{f_o} \cdot \frac{D}{f_e} \quad \text{Compound microscope, final image at } \infty$$

$$m = \frac{f_o}{f_e}, L = f_o + f_e \quad \text{Astronomical telescope, normal adjustment}$$

$$D = 25 \text{ cm} \quad \text{Least distance of distinct vision}$$

**Ch 10 • Wave Optics**

$$\Delta\phi = \frac{2\pi}{\lambda}\Delta x \quad \text{Phase vs path difference}$$

$$\Delta x = n\lambda / (2n + 1)\frac{\lambda}{2} \quad \text{Bright / dark}$$

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos\phi \quad \text{Resultant intensity}$$

$$I = 4I_0 \cos^2 \frac{\phi}{2} \quad \text{Two equal coherent sources}$$

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right)^2 \quad \text{Fringe contrast}$$

$$\beta = \frac{\lambda D}{d}, y_n = \frac{n\lambda D}{d} \quad \text{YDSE fringe width; } n\text{-th bright}$$

$$\beta' = \frac{\beta}{n} \quad \text{Whole set-up in a medium}$$

$$a \sin\theta = n\lambda \quad \text{Single-slit minima (} n = \pm 1, \pm 2 \dots)$$

$$W = \frac{2\lambda D}{a} \quad \text{Width of central maximum}$$

$$I = I_0 \cos^2\theta \quad \text{Malus' law (unpolarised } \rightarrow I_0 / 2)$$

$$\tan i_B = n \quad \text{Brewster's law; reflected } \perp \text{ refracted}$$

$$\text{shift} = \frac{(n-1)tD}{d} \quad \text{Thin film in one slit (JEE)}$$

**Ch 11 • Dual Nature of Radiation and Matter**

$$E = h\nu = \frac{hc}{\lambda} \quad E(\text{eV}) = \frac{1240}{\lambda(\text{nm})}$$

$$K_{\max} = h\nu - \phi_0 = eV_0 \quad \text{Einstein's photoelectric equation}$$

$$\phi_0 = h\nu_0 = \frac{hc}{\lambda_0} \quad \text{Work function; threshold}$$

$$V_0 = \frac{h\nu}{e} - \frac{\phi_0}{e} \quad \text{Slope } h/e \text{ (same for all metals)}$$

$$p = \frac{h}{\lambda} = \frac{E}{c} \quad \text{Photon momentum}$$

$$N = \frac{P}{h\nu} = \frac{P\lambda}{hc} \quad \text{Photons emitted per second}$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}} \quad \text{de Broglie wavelength}$$

$$\lambda_e = \frac{1.227}{\sqrt{V}} \text{ nm} \quad \text{Electron accelerated through } V \text{ volts}$$

Ch 12 • Atoms

$r_0 = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{K}$	α -particle: distance of closest approach	$mvr = \frac{nh}{2\pi}$	Bohr's quantisation
$r_n = 0.529 \frac{n^2}{Z} \text{ \AA}$	Radius of n-th orbit	$v_n = 2.18 \times 10^6 \frac{Z}{n} \text{ m/s}$	Speed in n-th orbit
$E_n = -\frac{13.6Z^2}{n^2} \text{ eV}$	Energy; KE = -E, PE = 2E	$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$	Rydberg formula
$N = \frac{n(n-1)}{2}$	Lines emitted from level n (many atoms)	$2\pi r_n = n\lambda$	de Broglie picture of Bohr orbits

Series	n_1	n_2	Region	Longest λ (H)
Lyman	1	2, 3, ...	ultraviolet	121.6 nm
Balmer	2	3, 4, ...	visible	656.3 nm
Paschen	3	4, 5, ...	infrared	1875 nm
Brackett, Pfund	4, 5	...	far infrared	–

Ch 13 • Nuclei

$R = R_0 A^{1/3}, R_0 = 1.2 \text{ fm}$	Nuclear radius	$\rho \approx 2.3 \times 10^{17} \text{ kg m}^{-3}$	Nuclear density: same for all A
$\Delta M = [Zm_p + (A - Z)m_n] - M$	Mass defect	$E_b = \Delta Mc^2$	1 u $\equiv$ 931.5 MeV
$Q = (\sum m_i - \sum m_f)c^2$	Q-value (> 0: energy released)	$\frac{E_b}{A} \approx 8 \text{ MeV}$	Peak $\approx$ 8.75 MeV near A = 56 (Fe)

Radioactive decay – only if your syllabus lists it

$N = N_0 e^{-\lambda t}, T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$, mean life $\tau = \frac{1}{\lambda}$, after n half-lives $N = \frac{N_0}{2^n}$. This topic was dropped from the rationalised NCERT; revise it only for older-pattern practice papers.

Ch 14 • Semiconductor Electronics

$n_e n_h = n_i^2$	Mass-action law (equilibrium)	$n_e \approx N_D \gg n_h$	n-type (pentavalent donor: P, As, Sb)
$n_h \approx N_A \gg n_e$	p-type (trivalent acceptor: B, Al, In)	$\sigma = e(n_e \mu_e + n_h \mu_h)$	Conductivity (JEE)
E_g : Si 1.1, Ge 0.7 eV	Insulator > 3 eV; conductor: bands overlap	$f_{out} = f / 2f$	Half-wave / full-wave rectifier
$I_S = \frac{V_{in} - V_Z}{R_S}, I_Z = I_S - I_L$	Zener regulator ($V_L = V_Z$)	$\lambda \text{ (nm)} \approx \frac{1240}{E_g \text{ (eV)}}$	LED colour / photodiode cut-off

A	B	AND	OR	NAND	NOR
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	1	0
1	1	1	1	0	0



Logic gates in one line each

- AND: $Y = A \cdot B$ – 1 only if all inputs are 1. OR: $Y = A + B$ – 1 if any input is 1. NOT: $Y = \bar{A}$.
- NAND = AND + NOT; NOR = OR + NOT. Both are **universal** gates.
- Truth-table check: NAND is 0 only for (1, 1); NOR is 1 only for (0, 0).

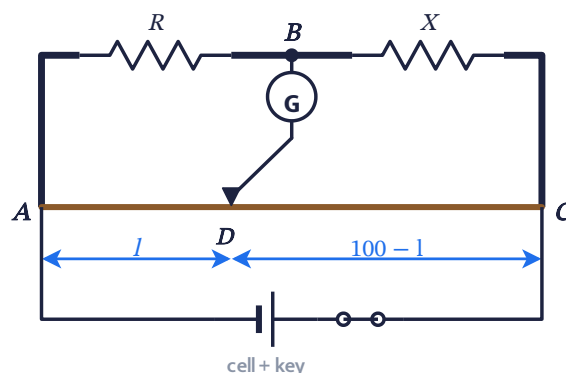
B • Experimental Skills (Day 25)

Both the NEET-UG and the JEE Main syllabus carry an **Experimental Skills** unit, and 1–2 questions come from it: a formula, a graph, a reading or a precaution. For each experiment below know the set-up, the formula, the graph and the main sources of error.



E1 • Resistivity by metre bridge

- **Set-up:** known R (resistance box) in the left gap, wire X in the right gap; slide the jockey to the null point D .
- **Formula:** $X = R \frac{100-l}{l}$, then $\rho = \frac{X\pi r^2}{L}$ (r by screw gauge).
- **Precautions:** choose R so that the null point lies near the middle (40–60 cm) – end errors matter least there; press the jockey gently, never slide it while pressed; interchange R and X and average.
- **Errors:** end resistance at the copper strips, non-uniform wire, heating by a large current.



Metre bridge: balance at D gives $\frac{R}{X} = \frac{l}{100-l}$



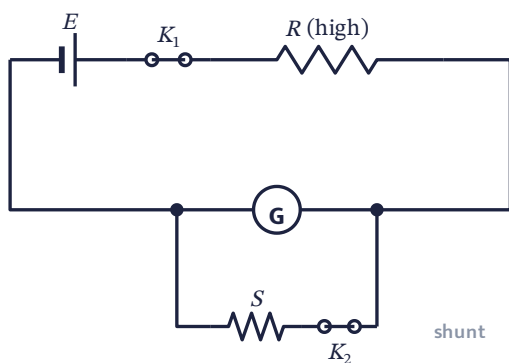
E2 • Resistance by Ohm's law (V–I)

- **Set-up:** battery, key, rheostat, ammeter in series with the wire; voltmeter in parallel across the wire.
- **Graph:** V (y-axis) vs I (x-axis) is a straight line through the origin; **R = slope**. (I vs V : slope = $1/R$.)
- **Precautions:** small current for a short time (wire must not heat); + of meters to + of battery; take readings for increasing and decreasing current.



E3 • Galvanometer: half-deflection method

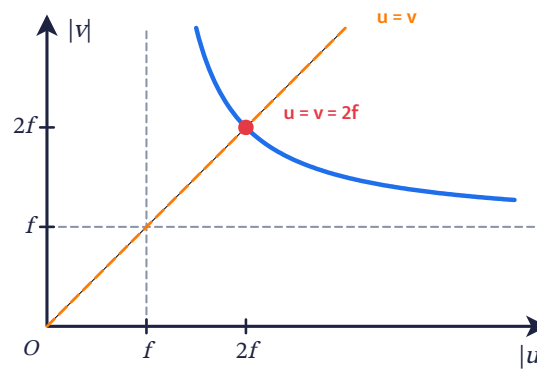
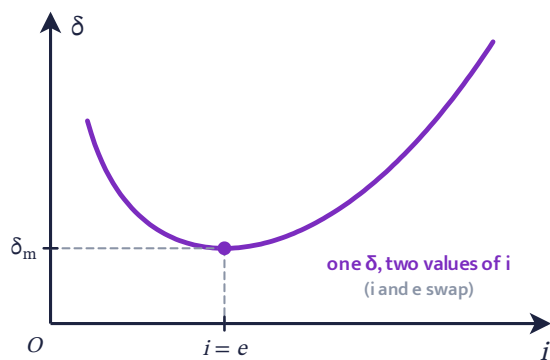
- **Step:** with K_2 open, adjust high R for an even deflection θ ; close K_2 and adjust shunt S to get $\theta/2$.
- **Formula:** $G = \frac{RS}{R-S} \approx S$ (as $R \gg S$); figure of merit $k = \frac{E}{(R+G)\theta}$ (A per division).
- **Precautions:** R must be very large (few $k\Omega$) so the main current hardly changes when S is added; use an even θ .

Half-deflection circuit: $G \approx S$ when θ halves**E4 • Conversion of a galvanometer**

- **Ammeter (0–I):** low shunt in **parallel**: $S = \frac{I_g G}{I - I_g}$; ammeter resistance $\frac{GS}{G+S}$ is very small.
- **Voltmeter (0–V):** high resistance in **series**: $R = \frac{V}{I_g} - G$; total resistance V / I_g is very large.
- **Check:** $I_g = nk$ (n = divisions on the scale, k = figure of merit). Ideal ammeter $R \rightarrow 0$, ideal voltmeter $R \rightarrow \infty$.

**E5 • Focal length: concave mirror & convex lens (u–v method)**

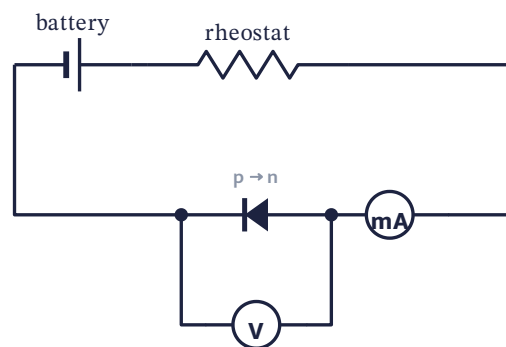
- **Method:** place the object pin at several u ; locate the real image by **removing parallax** between the image and a search pin; measure v .
- **Formula:** mirror $f = \frac{uv}{u+v}$ (magnitudes); lens $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$. Graph of $1/v$ vs $1/u$ is a straight line with intercept $1/f$ on each axis.
- **Precautions:** rough f first (image of a distant object); tips of pins at the height of the pole/optical centre; index correction for the bench.

 $|v|$ vs $|u|$: hyperbola; $u = v = 2f$ gives f  i – δ curve of a prism: minimum at $i = e$ **E6 • Prism: i – δ curve and refractive index**

- **Method:** trace the ray through the prism with pins for $i = 30^\circ$ – 60° ; measure δ for each and plot δ vs i .
- **Result:** U-shaped curve; at the minimum, $i = e$ and
$$n = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}.$$
- **Precautions:** pins upright and ≥ 8 – 10 cm apart; draw the prism outline sharply; do not take $i < 30^\circ$ (emergence becomes grazing).

E7 • p-n junction and Zener characteristics

- **Forward bias:** p to +; milliammeter; current negligible till the knee (≈ 0.7 V Si, 0.3 V Ge), then rises steeply.
- **Reverse bias:** p to –; **microammeter**; tiny, almost constant current until breakdown.
- **Zener:** in reverse bias the current rises sharply at V_Z while the voltage stays $\approx$ constant – read V_Z from the vertical part.
- **Resistance:** dynamic $r_d = \frac{\Delta V}{\Delta I}$ from the slope beyond the knee.



Forward-bias circuit (reverse: flip the battery, use a μ A meter)

E8 • Component

Look

Multimeter (resistance mode) test

Resistor	two leads, colour bands	same finite reading both ways
Diode	two leads, band = cathode	low R one way, very high R the other way (one-way conduction)
LED	two leads, longer = anode	one-way conduction and it glows when forward biased
Capacitor	two leads, may show μ F	momentary deflection, then very high (∞) resistance
IC / transistor	three or more leads	not a two-terminal device

E9 • Refractive index of a glass slab (travelling microscope)

- **Readings:** focus a mark with no slab (R_1), through the slab (R_2), and on the top of the slab (R_3).
- **Formula:** $n = \frac{\text{real}}{\text{apparent}} = \frac{R_3 - R_1}{R_3 - R_2}$.

! Lab traps asked in MCQs

- Metre bridge: null point near an end $\Rightarrow$ larger % error.
- Half deflection: $G \approx S$ only because $R \gg G$.
- Diode in reverse bias: use a μ A meter, not a mA meter.
- u–v method: $u = v$ gives $2f$, not f .

C • Top 50 Traps where students lose marks

Read this list on Day 26 and again on the morning of the exam. Each line is a mark saved.

! Ch 1 • Charges & fields

1. A body gets negatively charged by **gaining** electrons – its mass rises very slightly.
2. Flux through a closed surface depends only on the charge inside; the field on it depends on **all** charges.
3. Dipole in a uniform field: net force zero but torque $pE \sin \theta$; in a non-uniform field a net force also acts.
4. Equatorial field of a dipole points **opposite** to $\vec{p}$ and is half the axial field at the same r.

! Ch 2 • Potential & capacitance

5. Inside a charged conductor $E = 0$, but V is **not** zero – it equals the surface potential.
6. Dielectric: battery connected $\Rightarrow V$ fixed, Q and U rise K times; battery removed $\Rightarrow Q$ fixed, V and U fall K times.
7. Series capacitors carry equal charge; the **smallest** C takes the **largest** voltage.
8. Joining two charged capacitors always loses energy as heat, although charge is conserved.

! Ch 3 • Current electricity

- 9.** Stretching a wire n times (same volume) makes $R = n^2 R$; ρ does not change.
- 10.** Drift speed is only $\sim 10^{-4}$ m/s; the bulb glows at once because the field is set up almost at speed c .
- 11.** Terminal voltage is $E - Ir$ while discharging but $E + Ir$ while being charged.
- 12.** Semiconductors: resistance **decreases** on heating (α negative); metals: increases.

! Ch 5 • Magnetism & matter

- 16.** Field lines are closed loops – inside a bar magnet they run from S to N.
- 17.** Diamagnetic χ is negative and almost temperature-independent; only paramagnets obey $\chi \propto 1/T$.
- 18.** Soft iron (low coercivity, thin loop) for electromagnets and cores; steel (high retentivity and coercivity) for permanent magnets.

! Ch 7 • Alternating current

- 23.** Meters read rms; peak = $\sqrt{2} \times \text{rms}$ (220 V mains has a 311 V peak).
- 24.** Average of AC over a full cycle is zero – over a half cycle it is $2I_0 / \pi$.
- 25.** At resonance V_L and V_C can each be much larger than the supply voltage – they cancel.
- 26.** A transformer changes V and I , not power; it does not work on steady DC.

! Ch 9 • Ray optics

- 30.** A mirror's focal length does not change in water; a lens's does (it depends on n_{21}).
- 31.** TIR needs light going from denser to rarer **and** $i > C$.
- 32.** Covering half a lens gives the **full** image, only dimmer.
- 33.** Sign convention: for a concave mirror and concave lens, f is negative; real images have $v < 0$ for mirrors, $v > 0$ for lenses.

! Ch 4 • Moving charges

- 13.** Magnetic force does no work – speed and KE stay constant in a pure B field.
- 14.** Period $2\pi m / qB$ does not depend on speed or radius (basis of the cyclotron).
- 15.** Ammeter = G with a **low** shunt in parallel; voltmeter = G with a **high** R in series.

! Ch 6 • EMI

- 19.** Rotating coil: emf is **maximum** when the flux is **zero** (plane of coil parallel to B).
- 20.** Induced charge $N\Delta\Phi / R$ does not depend on how fast the flux changes.
- 21.** Lenz's law is energy conservation: the induced current always opposes its cause.
- 22.** Magnet falling through a copper ring: $a < g$ both while approaching and while leaving.

! Ch 8 • EM waves

- 27.** In a medium frequency stays the same; speed and wavelength drop by n .
- 28.** E and B oscillate **in phase**, not 90° apart; energy is shared equally between them.
- 29.** EM waves need no medium; they carry energy **and** momentum.

! Ch 10 • Wave optics

- 34.** YDSE fringes are equally wide and equally bright; single-slit central maximum is twice as wide as the others.
- 35.** Apparatus immersed in a liquid: fringe width becomes β / n (fringes come closer).
- 36.** Two independent bulbs are never coherent – no steady interference pattern.
- 37.** Unpolarised light through one polaroid: $I_0 / 2$ at every orientation; apply Malus only after that.

! Ch 11 • Dual nature

38. K_{max} and V_0 depend on frequency, not intensity; intensity controls the saturation current.
39. Photoemission is instantaneous ($< 10^{-9}$ s) – no time lag even with very dim light.
40. Below the threshold frequency no electron comes out, however intense the light.
41. Same KE: lighter particle has the longer de Broglie λ ($\lambda \propto 1/\sqrt{m}$).

! Ch 12 • Atoms

42. Energy levels are negative; in a higher orbit KE is **less** but total energy is **more**.
43. Many atoms in level n give $n(n-1)/2$ lines; a single atom gives at most $n-1$.
44. Balmer = visible, Lyman = UV, Paschen = IR – do not mix them up.

! Ch 13 • Nuclei

45. Nuclear density is the same for all nuclei because $R \propto A^{1/3}$.
46. Both fission (heavy) and fusion (light) release energy – each moves towards a higher BE per nucleon.
47. Greater BE per nucleon = more stable – not greater total BE.

! Ch 14 • Semiconductors

48. An n-type semiconductor is electrically **neutral**, not negatively charged.
49. Full-wave output ripple = $2f$ (100 Hz for 50 Hz mains); half-wave = f .
50. A Zener works as a regulator in **reverse** bias; a photodiode is also used in reverse bias.

🎯 Make this list yours

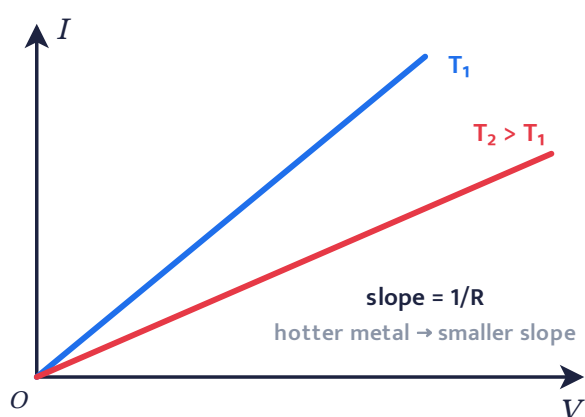
- Put a ✓ next to every trap you have actually fallen into in a test.
- Before each test, read only the ticked traps – it takes 3 minutes.
- Add your own traps from the Error Log at the end.

🧠 The 5 silent killers of Class 12

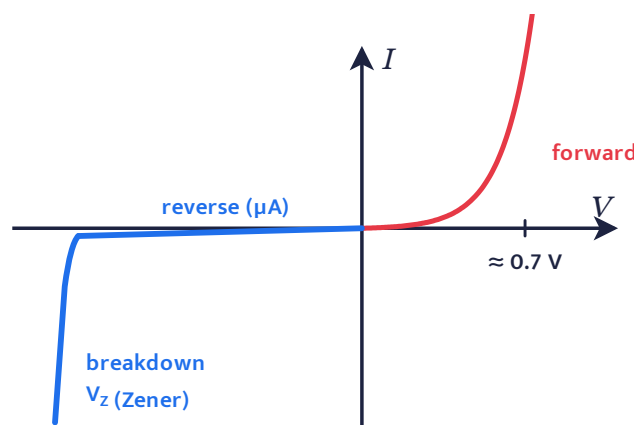
- rms read as peak (or peak as rms)
- sign convention in mirrors and lenses
- battery connected vs disconnected
- eV vs J; nm vs Å
- "incorrect statement" read as "correct"

D • Important graphs at a glance

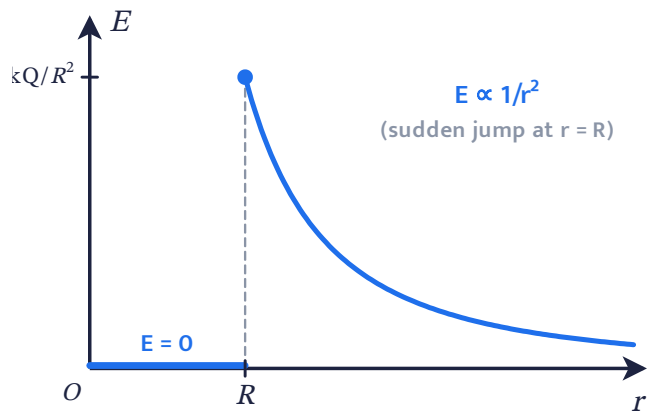
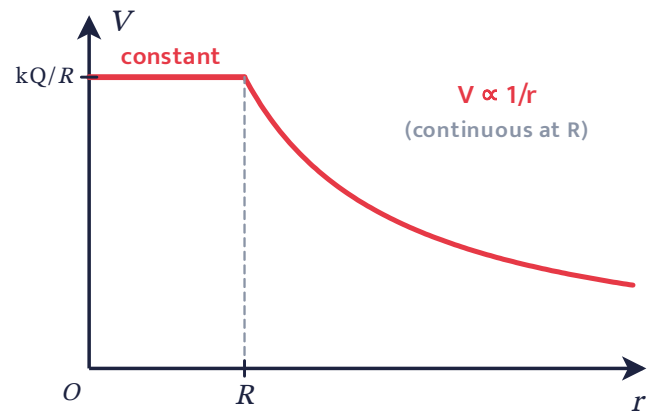
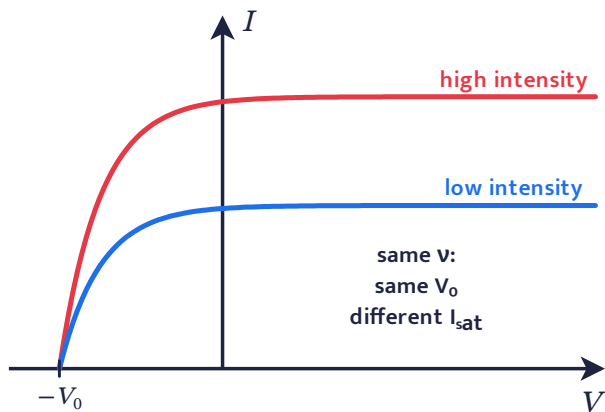
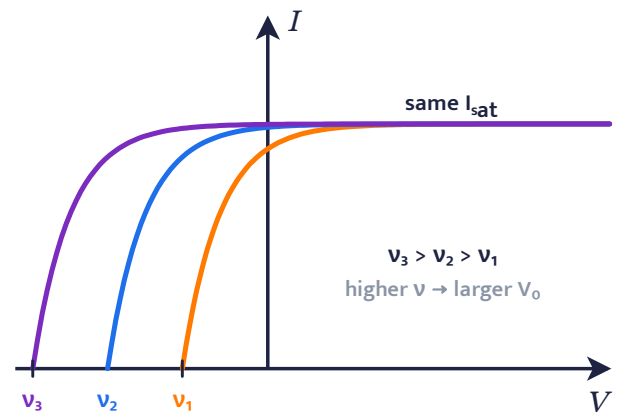
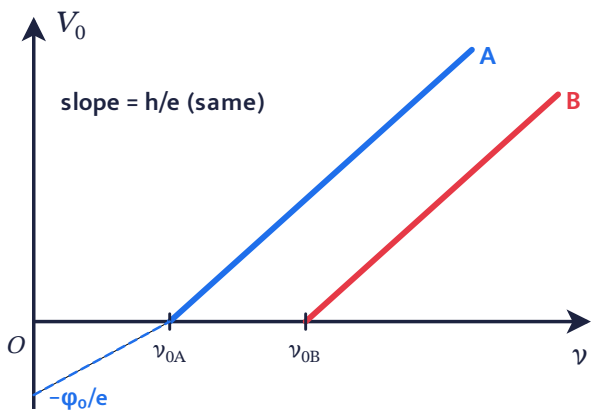
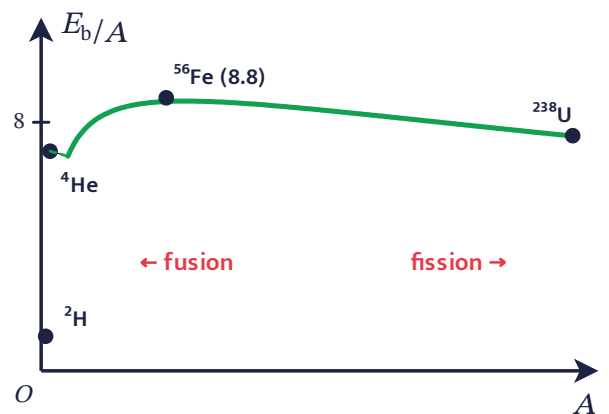
NEET and JEE Main ask 2–4 graph-based questions every year. Sketch each of these from memory once on Day 26 – shape, intercepts and slope.

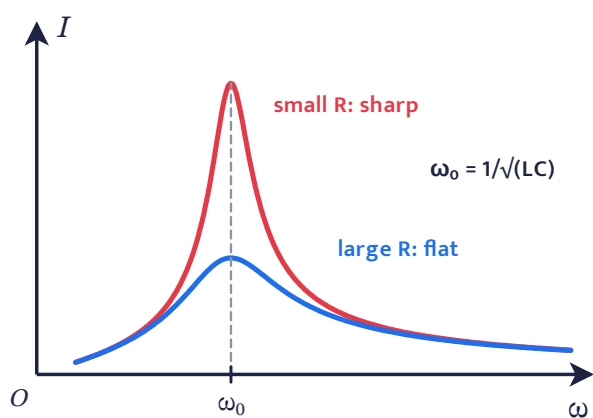
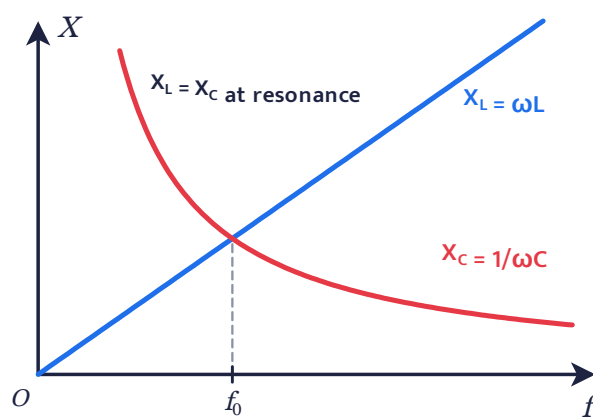
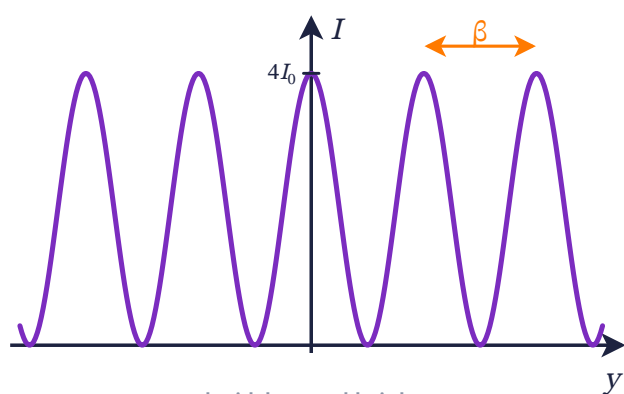
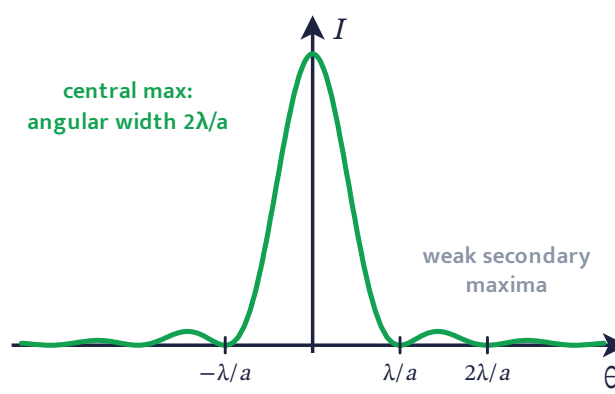


I-V of a metal: straight line, slope $1/R$

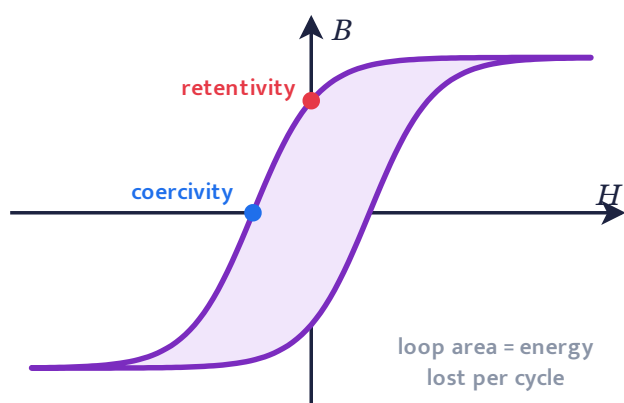
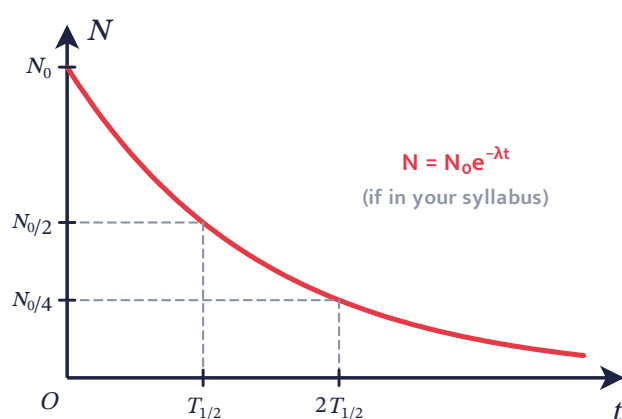


Diode I-V: knee, reverse current, Zener breakdown

 *E vs r , charged shell: zero inside* *V vs r , charged shell: constant inside**Photocurrent vs V : effect of intensity**Photocurrent vs V : effect of frequency* *V_0 vs ν : parallel lines for two metals**Binding energy per nucleon vs A*

Series LCR resonance: I vs ω  X_L, X_C vs frequencyYDSE intensity: $I = 4I_0 \cos^2(\phi / 2)$ 

Single-slit diffraction intensity

 B - H hysteresis loop of a ferromagnet

Decay curve (beyond rationalised NCERT)

E • Exam-day strategy for NEET and JEE Main Physics

Both papers give about **one minute per question** overall, and Physics usually takes longer than the other subjects. Budget **50–55 minutes** for Physics in NEET and **about 60 minutes** in JEE Main. The 3-round method below stops one hard question from eating ten easy ones.

	NEET-UG	JEE Main (Paper 1)
Physics questions	45 MCQs, all compulsory	20 MCQs + 5 numerical-value questions
Marks	180 • +4 / -1	100 • +4 / -1 (numerical answers too)
Mode	pen and OMR sheet	computer-based; mark-for-review palette
Physics time budget	50–55 min	≈ 60 min

Pattern as in recent papers – confirm it in the current NTA information bulletin.



The 3-round method (NEET timings; in JEE Main stretch each round by ~10%)

Round	Time	What to do	Target
1 • Sweep	0–20 min	NCERT facts, graphs, one-formula questions, AR, logic gates, EM spectrum. More than 1 minute – mark "?" and move on.	25–30 done
2 • Numericals	20–42 min	Return to "?" questions whose method you know. Hard cap: 2 minutes each. Approximate ($\pi^2 \approx 10$, $hc = 1240 \text{ eV nm}$).	+10–12
3 • Tough	42–52 min	Eliminate options using units, limiting cases and dimensions. Answer only if 2 options are eliminated.	+3–5
4 • Check	52–55 min	NEET: Q-number = bubble number. JEE: open the "marked for review" list; unanswered numericals – leave blank unless sure.	✓

🎯 NEET – OMR tips

- Fill the OMR **after every 10 questions** – never leave all 45 to the end.
- Darken each bubble fully with the pen provided; a marked answer cannot be changed.
- Every 10 questions, re-check that Q-number and bubble row match.
- Fill roll number, booklet code and signature first – calmly.

🎯 JEE Main – CBT tips

- Use **Save & Next** only when sure; use **Mark for Review** for "?" questions.
- Numerical answers: type exactly what is asked (integer / rounded value, correct unit).
- Numericals also carry -1 – do not type a guess.
- Glance at the on-screen timer at Q10 and Q20 of Physics.

**Guessing – the maths**

- Blind guess (1 of 4): expected gain = $\frac{1}{4}(4) - \frac{3}{4}(1) = +0.25$ – tiny, and risky.
- After eliminating 2 options: $\frac{1}{2}(4) - \frac{1}{2}(1) = +1.5$ – worth it.
- Rule: **guess only when you have eliminated two options.**

**Calm-down breathing (1 minute)**

- Breathe **in** through the nose – count 4; **hold** – count 4.
- Breathe **out** slowly through the mouth – count 6.
- Repeat 4–5 times, then say: "Next question." A hard question is hard for everyone.

**Food, sleep & the last 24 hours**

- Sleep 7–8 hours on each of the last three nights – do not stay up the night before.
- Eat a light, familiar meal 2–3 hours before the exam; no new or oily food. Drink water, but not a litre just before entering.
- Last day: only formula sheets, traps, graphs and your error log – **no new topic.**
- Keep admit card, photo ID and items listed by NTA ready the night before; reach the centre early.

Final checklist – tick before you sleep on the last night

- ☐ Formula sheets A revised (Days 23–24)
- ☐ Experimental skills B revised (Day 25)
- ☐ 50 traps + 14 graphs read and sketched (Day 26)
- ☐ Mock test done (Day 27) and analysed (Day 28)
- ☐ All wrong questions re-attempted (Day 29)
- ☐ Admit card, ID, photo, route and reporting time checked (Day 30)

**First 2 minutes in the hall**

- Fill the details, then take three slow breaths.
- Start Round 1 at Q1 and move steadily – do not hunt for "your" chapter.
- Note the start time; check it at Q15 and Q30.

**If the paper feels hard**

- It is hard for every candidate – ranks are relative.
- Wild guesses cost marks: 38 right + 2 wrong = 150, but 38 right + 7 wrong = 145.
- Never carry a bad question into the next one.

Write every mistake from the chapter practice sets and the mock here – one line each. Read this log on Day 30: these are **your** personal traps.

[illegible]

DAY

27

NEET + JEE MAIN PHYSICS • CLASS 12 • FULL SYLLABUS

Full Mock Test

45 questions

60 minutes

+4 correct • -1 wrong • 0 unattempted

One sitting – no breaks

Fill the score table after checking

Sit at a table with a timer, a pencil and rough sheets only – no book, no phone. Mark answers on a separate sheet (like an OMR). After 60 minutes, stop, check with the key, and fill the score table.

Before you start



Instructions

- 45 questions from all 14 chapters of Class 12, in chapter order – weightage like NEET and JEE Main (electricity, magnetism, optics and modern physics heavy).
- Time: **60 minutes** in one sitting. Start a timer now.
- Marking: **+4** correct, **-1** wrong, **0** unattempted.
- Do not look at the answer key until the timer stops.



VS Sir's Exam Tip

- Round 1: attempt every question you can do in < 1 min.
- Round 2: return to the numericals you skipped.
- Round 3: tough ones – guess only if you can eliminate 2 options.

Score table (fill after checking)

Chapter group	Qs	Correct	Wrong	Score = 4C – W
Ch 1–2: Electrostatics, Capacitance	8 (Q1–8)			
Ch 3–4: Current, Moving charges	9 (Q9–17)			
Ch 5–7: Magnetism, EMI, AC	8 (Q18–25)			
Ch 8–10: EM waves, Ray & Wave optics	10 (Q26–35)			
Ch 11–13: Dual nature, Atoms, Nuclei	7 (Q36–42)			
Ch 14: Semiconductors	3 (Q43–45)			
TOTAL (max 180)	45			

Time taken: _____ min • Attempted: _____ / 45 • Accuracy = Correct ÷ Attempted = _____ %



Mock Test – 45 Questions

Original NEET / JEE Main-pattern questions modelled on previous-year trends. Timer ON: 60 minutes. Answers after Q45.

1

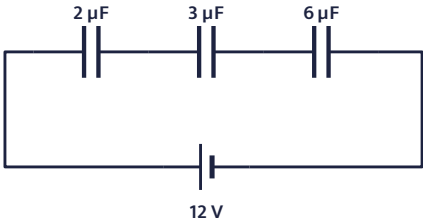
NEET pattern Point charges $+2\ \mu\text{C}$ and $+8\ \mu\text{C}$ are fixed 30 cm apart. The electric field is zero at a point on the line joining them at a distance from the $+2\ \mu\text{C}$ charge of

(a) 10 cm

(b) 6 cm

(c) 15 cm

(d) 20 cm

- 2 **NCERT based** A charge q is placed at the centre of a cube. The electric flux through **one face** of the cube is
 (a) $\frac{q}{6\epsilon_0}$ (b) $\frac{6q}{\epsilon_0}$ (c) $\frac{q}{24\epsilon_0}$ (d) $\frac{q}{\epsilon_0}$
- 3 **NEET pattern** An electric dipole of moment $4 \times 10^{-9} \text{ C m}$ is placed at 30° to a uniform field of $5 \times 10^4 \text{ N/C}$. The torque on it is
 (a) zero (b) $1 \times 10^{-4} \text{ N m}$ (c) $2 \times 10^{-4} \text{ N m}$ (d) $1.7 \times 10^{-4} \text{ N m}$
- 4 **Assertion–Reason** **Assertion (A):** In electrostatics the electric field inside a conductor is zero.
Reason (R): Two electric field lines can never intersect each other.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 5 **NEET pattern** The work needed to bring a charge of $2 \mu\text{C}$ from infinity to a point 9 cm from a fixed charge of $3 \mu\text{C}$ is
 (a) 6 J (b) 0.2 J (c) 0.06 J (d) 0.6 J
- 6 **NEET pattern** In the circuit shown, the charge on the $3 \mu\text{F}$ capacitor is
 (a) $12 \mu\text{C}$ (b) $24 \mu\text{C}$ (c) $36 \mu\text{C}$ (d) $4 \mu\text{C}$
- 
- 7 **NCERT based** A parallel-plate capacitor storing energy U is disconnected from the battery and a dielectric of $K = 4$ then fills the space between the plates. The stored energy becomes
 (a) U (b) $U / 16$ (c) $4U$ (d) $U / 4$
- 8 **JEE Main pattern** A $4 \mu\text{F}$ capacitor charged to 50 V is connected across an uncharged $6 \mu\text{F}$ capacitor. The energy lost as heat is
 (a) zero (b) 3 mJ (c) 5 mJ (d) 2 mJ
- 9 **NCERT based** A wire of resistance 10Ω is stretched uniformly to three times its length. Its new resistance is
 (a) 30Ω (b) 10Ω (c) 90Ω (d) 270Ω
- 10 **NEET pattern** A cell of emf 2 V and internal resistance 0.5Ω is connected to a 3.5Ω resistor. The terminal voltage of the cell is
 (a) 1.75 V (b) 2 V (c) 0.25 V (d) 1.5 V
- 11 **NEET pattern** In a metre bridge, a 4Ω resistance is in the left gap and the null point is 40 cm from the left end. The resistance in the right gap is
 (a) 2.67Ω (b) 6Ω (c) 10Ω (d) 1.6Ω

- 12 **NEET pattern** A current of 1.6 A flows in a copper wire of cross-section 1 mm^2 . If $n = 10^{29} \text{ m}^{-3}$, the drift speed of electrons is
(a) $1.6 \times 10^{-4} \text{ m/s}$ (b) 10^{-3} m/s (c) 10^{-2} m/s (d) 10^{-4} m/s
- 13 **JEE Main pattern** A battery of emf 12 V and internal resistance 2Ω is connected to a variable resistor R. The maximum power that can be delivered to R is
(a) 9 W (b) 36 W (c) 72 W (d) 18 W
- 14 **NEET pattern** A proton and an α -particle enter a uniform magnetic field perpendicularly with the **same velocity**. The ratio of radii $r_\alpha:r_p$ is
(a) 4 : 1 (b) 1 : 2 (c) 1 : 1 (d) 2 : 1
- 15 **NCERT based** The magnetic field at 5 cm from a long straight wire carrying 10 A is
(a) $4 \times 10^{-4} \text{ T}$ (b) $1.3 \times 10^{-4} \text{ T}$ (c) $4 \times 10^{-5} \text{ T}$ (d) $2 \times 10^{-5} \text{ T}$
- 16 **NEET pattern** A galvanometer of resistance 50Ω gives full-scale deflection at 2 mA. To convert it into an ammeter of range 0–2 A, one must connect
(a) $\approx 950 \Omega$ in series (b) $\approx 0.5 \Omega$ in parallel
(c) $\approx 0.05 \Omega$ in parallel (d) $\approx 0.05 \Omega$ in series
- 17 **JEE Main pattern** A wire bent into a single circular loop produces a field B at its centre. The same wire is now bent into a coil of **two** turns and carries the same current. The field at the centre becomes
(a) $2B$ (b) $B/2$ (c) $4B$ (d) B
- 18 **NEET pattern** A bar magnet of moment 2 A m^2 is aligned with a uniform field of 0.1 T. The work needed to turn it through 180° is
(a) 0.1 J (b) 0.2 J (c) 0.4 J (d) zero
- 19 **NCERT based** Which one of the following is diamagnetic?
(a) Bismuth (b) Cobalt (c) Aluminium (d) Oxygen
- 20 **NCERT based** A 0.5 m rod moves at 4 m/s perpendicular to its length and to a uniform field of 0.2 T. The emf across its ends is
(a) 0.1 V (b) 0.4 V (c) 4 V (d) 1.6 V
- 21 **NEET pattern** A 100-turn coil of area 0.01 m^2 is perpendicular to a field that falls uniformly from 0.5 T to zero in 0.1 s. The average induced emf is
(a) 50 V (b) 5 V (c) 0.5 V (d) 0.05 V
- 22 **Assertion–Reason** **Assertion (A):** A bar magnet dropped vertically through a horizontal copper ring falls with an acceleration less than g.
Reason (R): The current induced in the ring opposes the relative motion of the magnet.
(a) Both true, R explains A
(b) Both true, R does not explain A
(c) A false, R true
(d) Both false

- 23 **NEET pattern** In a series LCR circuit $R = 30\ \Omega$, $X_L = 80\ \Omega$ and $X_C = 40\ \Omega$. The power factor of the circuit is
(a) 0.6 (b) 0.375 (c) 0.8 (d) 1
- 24 **NCERT based** The resonant angular frequency of a series circuit with $L = 0.1\text{ H}$ and $C = 10\ \mu\text{F}$ is
(a) 159 rad/s (b) 10 000 rad/s (c) 1000 rad/s (d) 100 rad/s
- 25 **NEET pattern** An ideal transformer has 200 primary and 50 secondary turns. The primary is on 220 V mains and the secondary delivers 4 A. The primary current is
(a) 1 A (b) 4 A (c) 16 A (d) 0.25 A
- 26 **NEET pattern** In a plane EM wave in vacuum, the amplitude of the electric field is 6 V/m. The amplitude of the magnetic field is
(a) $2 \times 10^{-8}\text{ T}$ (b) $2 \times 10^{-6}\text{ T}$ (c) $6 \times 10^{-8}\text{ T}$ (d) $1.8 \times 10^9\text{ T}$
- 27 **NCERT based** The radiation used in LASIK eye surgery belongs to the
(a) microwave region (b) infrared region
(c) X-ray region (d) ultraviolet region
- 28 **NEET pattern** An object is placed 20 cm in front of a concave mirror of focal length 15 cm. The image is formed
(a) 8.6 cm in front, real, diminished
(b) 60 cm in front, real, inverted, 3 times enlarged
(c) 60 cm behind, virtual, erect, 3 times enlarged
(d) 30 cm in front, real, inverted, same size
- 29 **NEET pattern** A convex lens of focal length 20 cm is kept in contact with a concave lens of focal length 30 cm. The combination is
(a) converging, $f = 60\text{ cm}$ (b) converging, $f = 12\text{ cm}$
(c) converging, $f = 50\text{ cm}$ (d) diverging, $f = 60\text{ cm}$
- 30 **NCERT based** The critical angle for light going from glass of refractive index $\sqrt{2}$ to air is
(a) 45° (b) 30° (c) 90° (d) 60°
- 31 **NEET pattern** An equilateral glass prism gives a minimum deviation of 30° . Its refractive index is
(a) 1.5 (b) $\sqrt{2}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\sqrt{3}$
- 32 **JEE Main pattern** A glass lens ($n = 1.5$) has focal length 20 cm in air. When immersed in water ($n = 4/3$), its focal length becomes
(a) 26.7 cm (b) 20 cm (c) 80 cm (d) 40 cm
- 33 **NEET pattern** In Young's double-slit experiment $\lambda = 600\text{ nm}$, slit separation 0.5 mm and screen distance 1 m. The fringe width is
(a) 1.2 mm (b) 0.6 mm (c) 2.4 mm (d) 0.12 mm

- 34 **NEET pattern** Unpolarised light of intensity I_0 passes through two polaroids whose axes are at 60° to each other. The transmitted intensity is
 (a) $I_0 / 2$ (b) $3I_0 / 8$ (c) $I_0 / 4$ (d) $I_0 / 8$
- 35 **JEE Main pattern** Two coherent sources have intensities in the ratio 9 : 1. The ratio $I_{\max} : I_{\min}$ in their interference pattern is
 (a) 4 : 1 (b) 9 : 1 (c) 16 : 1 (d) 3 : 1
- 36 **NEET pattern** Light of wavelength 400 nm falls on a metal of work function 2.0 eV. The stopping potential is ($hc = 1240 \text{ eV nm}$)
 (a) 3.1 V (b) 5.1 V (c) 2.0 V (d) 1.1 V
- 37 **NEET pattern** The de Broglie wavelength of an electron accelerated from rest through 100 V is about
 (a) 0.0123 nm (b) 0.123 nm (c) 1.23 nm (d) 12.3 nm
- 38 **Assertion–Reason** **Assertion (A):** The stopping potential increases when the intensity of the incident light is increased.
Reason (R): The number of photoelectrons emitted per second increases with intensity.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 39 **NEET pattern** The energy needed to excite a hydrogen atom from its ground state to $n = 3$ is
 (a) 13.6 eV (b) 10.2 eV (c) 12.09 eV (d) 1.51 eV
- 40 **JEE Main pattern** The ratio of the longest wavelength of the Lyman series to the longest wavelength of the Balmer series of hydrogen is
 (a) 27 : 5 (b) 5 : 27 (c) 3 : 4 (d) 4 : 9
- 41 **NCERT based** The ratio of the nuclear radii of nuclei with mass numbers 216 and 27 is
 (a) 2 : 1 (b) 1 : 2 (c) 4 : 1 (d) 8 : 1
- 42 **NEET pattern** The mass defect of a ${}^4_2\text{He}$ nucleus is 0.0303 u. Its binding energy per nucleon is about ($1 \text{ u} = 931.5 \text{ MeV}/c^2$)
 (a) 14.1 MeV (b) 7.1 MeV (c) 3.5 MeV (d) 28.2 MeV
- 43 **NCERT based** The output of a NOR gate is fed into a NOT gate, as shown. The combination behaves as
 (a) a NAND gate (b) an AND gate (c) a NOR gate (d) an OR gate



- 44 **NCERT based** A full-wave rectifier is fed from 50 Hz mains. The fundamental frequency of the ripple in its output is
 (a) 200 Hz (b) 100 Hz (c) 25 Hz (d) 50 Hz

45. **JEE Main pattern** A 6 V Zener diode is used with a series resistor of $300\ \Omega$ across a 15 V supply; the load is $1\ \text{k}\Omega$. The current through the Zener is
- (a) 30 mA (b) 24 mA (c) 15 mA (d) 6 mA

✓ Answer Key & Quick Solutions

- 1(a) 2(a) 3(b) 4(b) 5(d) 6(a) 7(d) 8(b) 9(c) 10(a) 11(b) 12(d) 13(d) 14(d) 15(c)
 16(c) 17(c) 18(c) 19(a) 20(b) 21(b) 22(a) 23(a) 24(c) 25(a) 26(a) 27(d) 28(b) 29(a)
 30(a) 31(b) 32(c) 33(a) 34(d) 35(a) 36(d) 37(b) 38(c) 39(c) 40(b) 41(a) 42(b) 43(d)
 44(b) 45(b)

1. (a) $\frac{2}{x^2} = \frac{8}{(30-x)^2} \Rightarrow 30-x = 2x \Rightarrow x = 10\ \text{cm}$ (nearer the smaller charge).
2. (a) Total flux q / ϵ_0 is shared equally by the 6 faces by symmetry.
3. (b) $\tau = pE \sin \theta = 4 \times 10^{-9} \times 5 \times 10^4 \times 0.5 = 10^{-4}\ \text{N m}$.
4. (b) Both statements are true, but $E = 0$ inside a conductor is due to free charges rearranging, not because field lines cannot cross.
5. (d) $W = \frac{kq_1q_2}{r} = \frac{9 \times 10^9 \times 3 \times 10^{-6} \times 2 \times 10^{-6}}{0.09} = 0.6\ \text{J}$.
6. (a) Series: $\frac{1}{C} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1 \Rightarrow C = 1\ \mu\text{F}$; $Q = 12\ \mu\text{C}$ – the same on every capacitor.
7. (d) Q fixed, $C \rightarrow 4C$, so $U = \frac{Q^2}{2C}$ becomes $U/4$.
8. (b) $\Delta U = \frac{C_1C_2V^2}{2(C_1+C_2)} = \frac{24 \times 10^{-12} \times 2500}{2 \times 10^{-5}} = 3 \times 10^{-3}\ \text{J}$.
9. (c) Volume constant $\Rightarrow R' = n^2R = 9 \times 10 = 90\ \Omega$.
10. (a) $I = 2/4 = 0.5\ \text{A}$; $V = E - Ir = 2 - 0.25 = 1.75\ \text{V}$.
11. (b) $\frac{4}{X} = \frac{40}{60} \Rightarrow X = 6\ \Omega$.
12. (d) $v_d = \frac{I}{neA} = \frac{1.6}{10^{29} \times 1.6 \times 10^{-19} \times 10^{-6}} = 10^{-4}\ \text{m/s}$.
13. (d) Maximum when $R = r = 2\ \Omega$: $P = \frac{E^2}{4r} = \frac{144}{8} = 18\ \text{W}$.
14. (d) $r = \frac{mv}{qB} \propto \frac{m}{q}$: α has $m/q = 4/2 = 2$ times that of the proton.
15. (c) $B = \frac{\mu_0 I}{2\pi r} = \frac{2 \times 10^{-7} \times 10}{0.05} = 4 \times 10^{-5}\ \text{T}$.
16. (c) $S = \frac{I_g G}{I - I_g} = \frac{0.002 \times 50}{1.998} \approx 0.05\ \Omega$, as a shunt (parallel).
17. (c) Radius halves and N doubles: $B = \frac{\mu_0 NI}{2R} \Rightarrow \frac{2}{1/2} = 4$ times.
18. (c) $W = mB(1 - \cos 180^\circ) = 2mB = 0.4\ \text{J}$.
19. (a) Bi is diamagnetic; Al and O_2 are paramagnetic; Co is ferromagnetic.
20. (b) $\epsilon = Blv = 0.2 \times 0.5 \times 4 = 0.4\ \text{V}$.
21. (b) $\epsilon = N \frac{\Delta \Phi}{\Delta t} = \frac{100 \times 0.01 \times 0.5}{0.1} = 5\ \text{V}$.
22. (a) Lenz's law: the induced current opposes the motion, so a retarding force acts and a $< g$. R explains A.
23. (a) $Z = \sqrt{30^2 + 40^2} = 50\ \Omega$; $\cos \phi = R / Z = 0.6$.
24. (c) $\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-6}}} = 1000\ \text{rad/s}$.
25. (a) $V_s = 220 \times \frac{50}{200} = 55\ \text{V}$; $V_p I_p = V_s I_s \Rightarrow I_p = \frac{55 \times 4}{220} = 1\ \text{A}$.
26. (a) $B_0 = E_0 / c = 6 / (3 \times 10^8) = 2 \times 10^{-8}\ \text{T}$.
27. (d) NCERT: UV lasers are used in LASIK (ultraviolet-based eye surgery).

28. (b) $\frac{1}{v} = -\frac{1}{15} + \frac{1}{20} = -\frac{1}{60} \Rightarrow v = -60 \text{ cm}; m = -v/u = -3$ (real, inverted).

29. (a) $P = 5 - 3.33 = +1.67 \text{ D} \Rightarrow f = +60 \text{ cm}$ (converging).

30. (a) $\sin C = 1/n = 1/\sqrt{2} \Rightarrow C = 45^\circ$.

31. (b) $n = \frac{\sin \frac{60^\circ + 30^\circ}{2}}{\sin 30^\circ} = \frac{\sin 45^\circ}{0.5} = \sqrt{2}$.

32. (c) $\frac{f_w}{f_a} = \frac{n-1}{n/n_w-1} = \frac{0.5}{1.125-1} = 4 \Rightarrow f_w = 80 \text{ cm}$.

33. (a) $\beta = \frac{\lambda D}{d} = \frac{6 \times 10^{-7} \times 1}{5 \times 10^{-4}} = 1.2 \text{ mm}$.

34. (d) After the first: $I_0/2$; after the second: $\frac{I_0}{2} \cos^2 60^\circ = I_0/8$.

35. (a) $\left(\frac{\sqrt{9}+1}{\sqrt{9}-1} \right)^2 = (4/2)^2 = 4$.

36. (d) $E = 1240/400 = 3.1 \text{ eV}; eV_0 = 3.1 - 2.0 = 1.1 \text{ eV} \Rightarrow V_0 = 1.1 \text{ V}$.

37. (b) $\lambda = \frac{1.227}{\sqrt{V}} \text{ nm} = \frac{1.227}{10} \approx 0.123 \text{ nm}$.

38. (c) V_0 depends only on frequency, so A is false; R is true (saturation current rises).

39. (c) $\Delta E = 13.6 \left(1 - \frac{1}{9} \right) = 12.09 \text{ eV}$.

40. (b) $\frac{1}{\lambda_L} = R \cdot \frac{3}{4}, \frac{1}{\lambda_B} = R \cdot \frac{5}{36} \Rightarrow \frac{\lambda_L}{\lambda_B} = \frac{5/36}{3/4} = \frac{5}{27}$.

41. (a) $R \propto A^{1/3}; (216/27)^{1/3} = 8^{1/3} = 2$.

42. (b) $E_b = 0.0303 \times 931.5 \approx 28.2 \text{ MeV}$; per nucleon $\approx 28.2/4 \approx 7.1 \text{ MeV}$.

43. (d) NOR gives $A + B$; the NOT gate inverts it back to $Y = A + B$ – an OR gate.

44. (b) Both half-cycles are used, so the output repeats twice per input cycle: $2 \times 50 = 100 \text{ Hz}$.

45. (b) $I_S = \frac{15-6}{300} = 30 \text{ mA}, I_L = \frac{6}{1000} = 6 \text{ mA} \Rightarrow I_Z = 24 \text{ mA}$.

Mock analysis (Day 28)

Write **every** wrong, guessed or skipped question below. Be honest about the type of mistake – that decides the fix.

🎯 How to analyse (Day 28) and re-attempt (Day 29)

- **Concept** mistake → re-read that section and its formula box in the chapter; solve 5 questions of that topic.
- **Calculation** mistake → redo the question on paper without the solution; practise approximations ($hc \approx 1240 \text{ eV nm}$, $\sqrt{2} \approx 1.41$).
- **Silly** mistake (rms vs peak, sign convention, battery connected vs removed) → add it to the Traps list and read it before every test.
- **Time** problem → note which question type ate time; in the exam, skip it in Round 1.
- **Day 29:** cover the solutions and re-attempt every question in the table below. Target: 100% correct. Anything still wrong goes into the Error Log at the end of the book.

📊 Score bands

For this mock: **150+** excellent – keep revising traps; **110–150** good – fix calculation speed; **< 110** – go back to the formula sheets (Days 23–24) before the next test.

$$1/f = 1/v - 1/u$$

All the best for NEET & JEE!

$$F = kq_1q_2/r^2$$

Revise daily, test weekly, analyse every mistake. Physics rewards the student who practises smartly.

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$$E = h\nu$$



$$V = IR$$

$$\varepsilon = -d\Phi/dt$$

